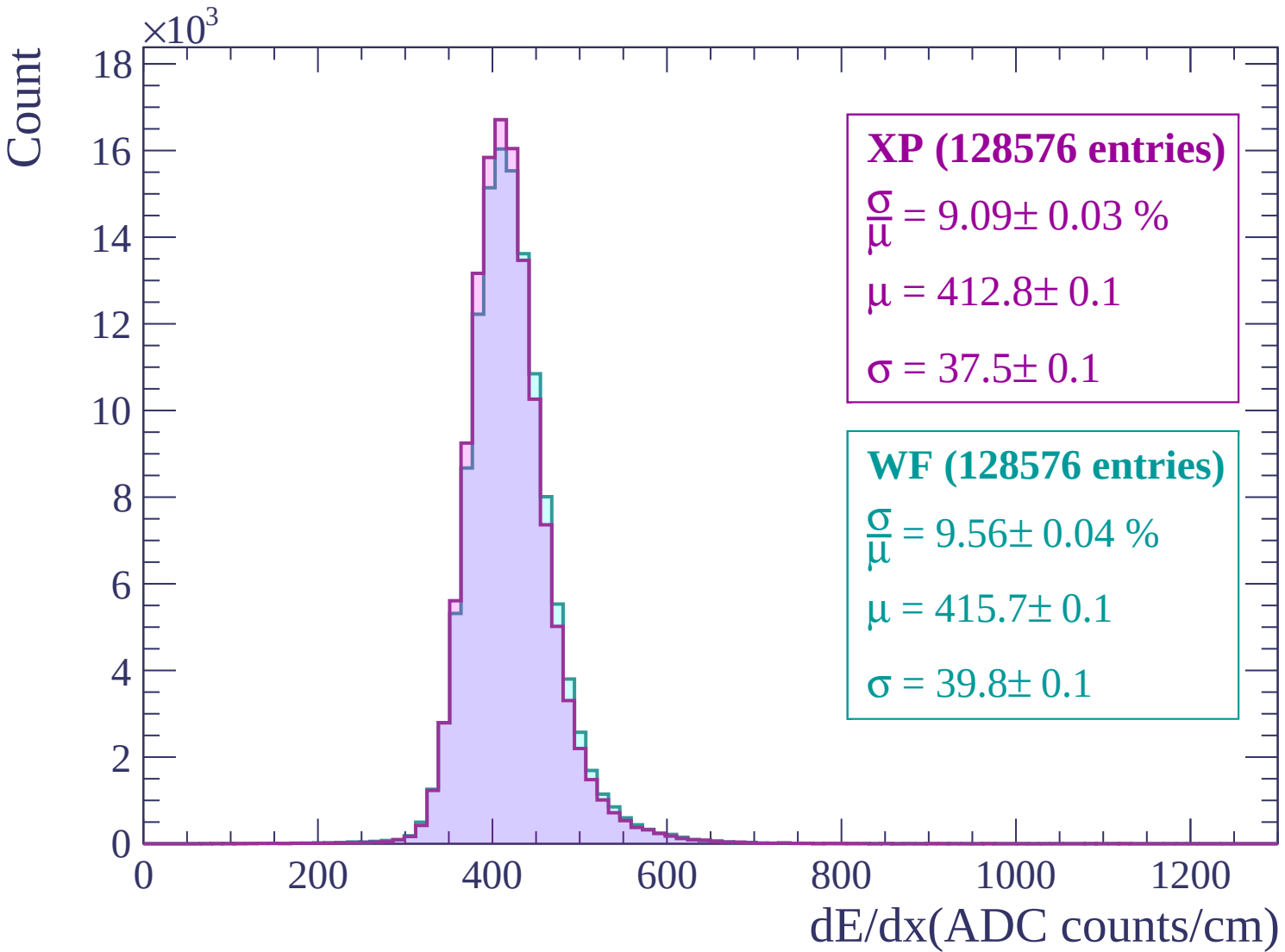
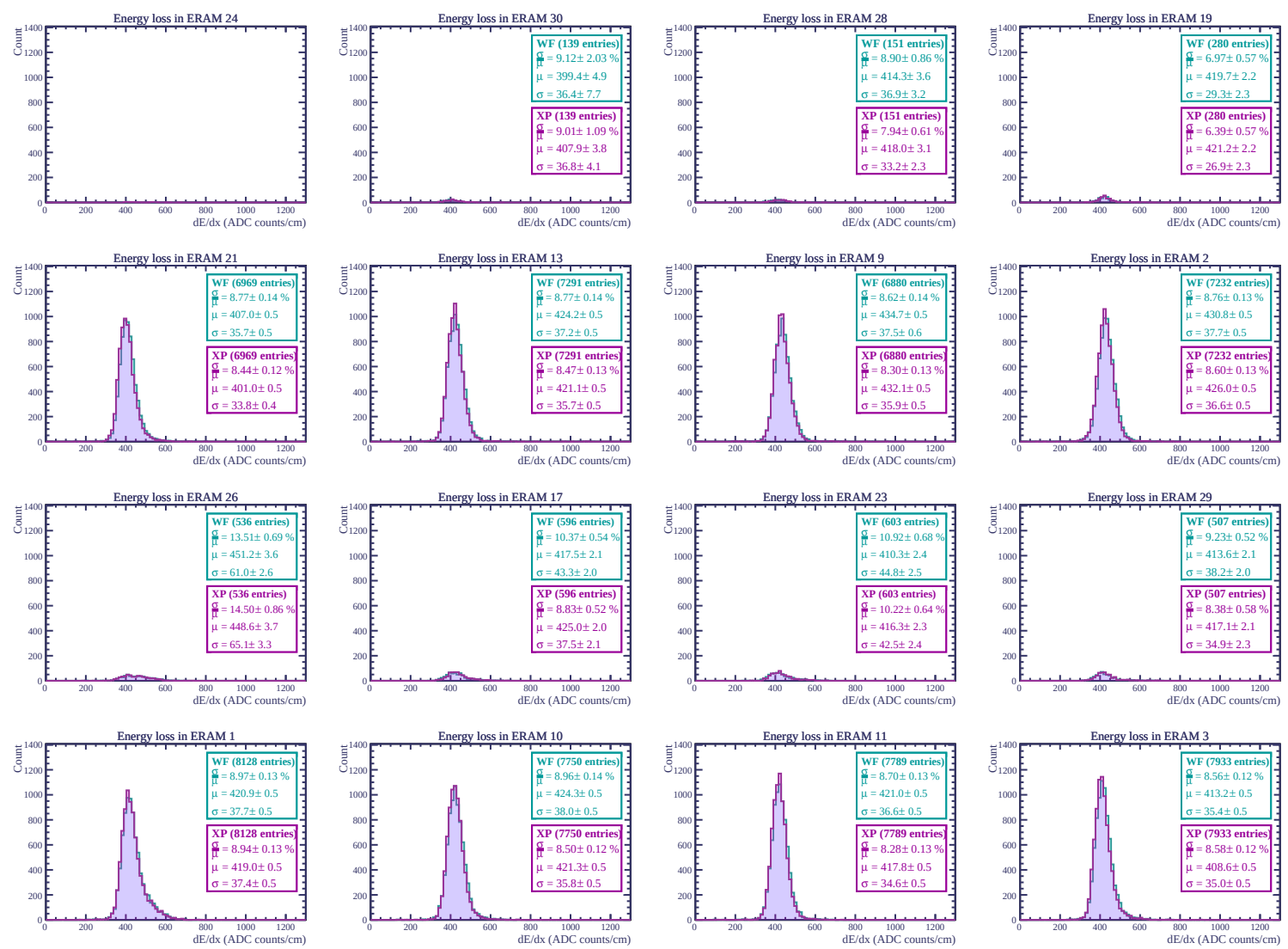
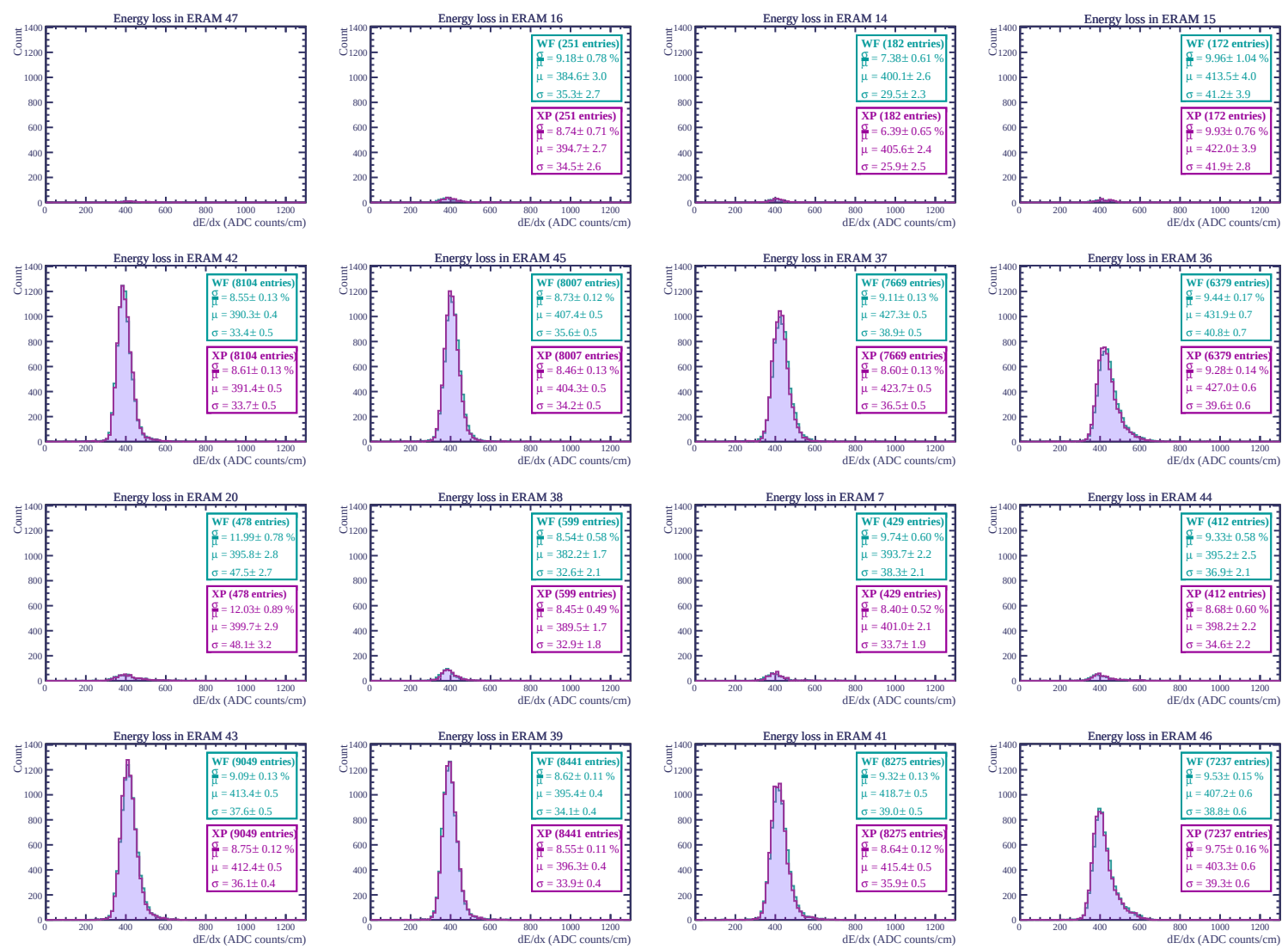
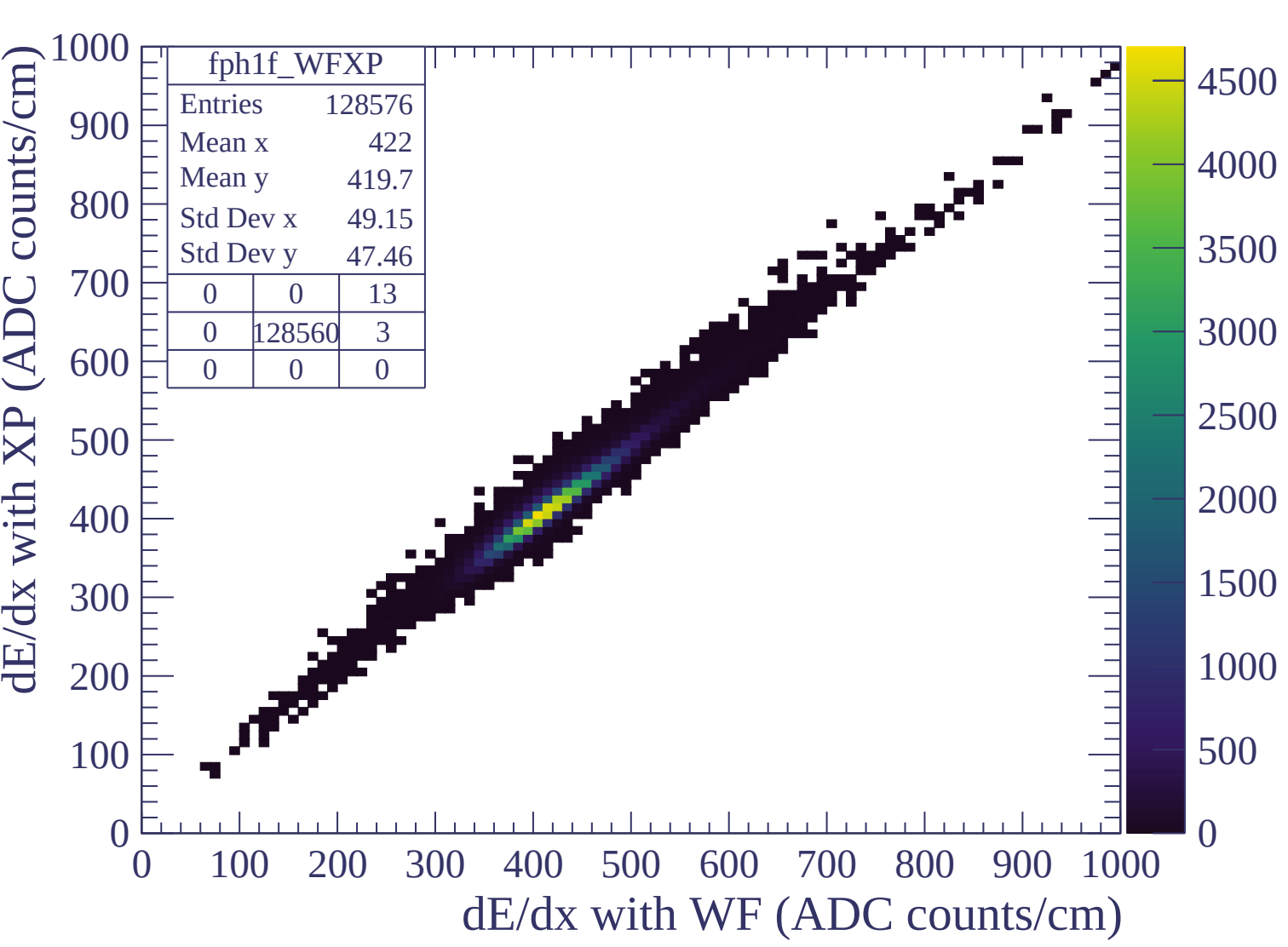
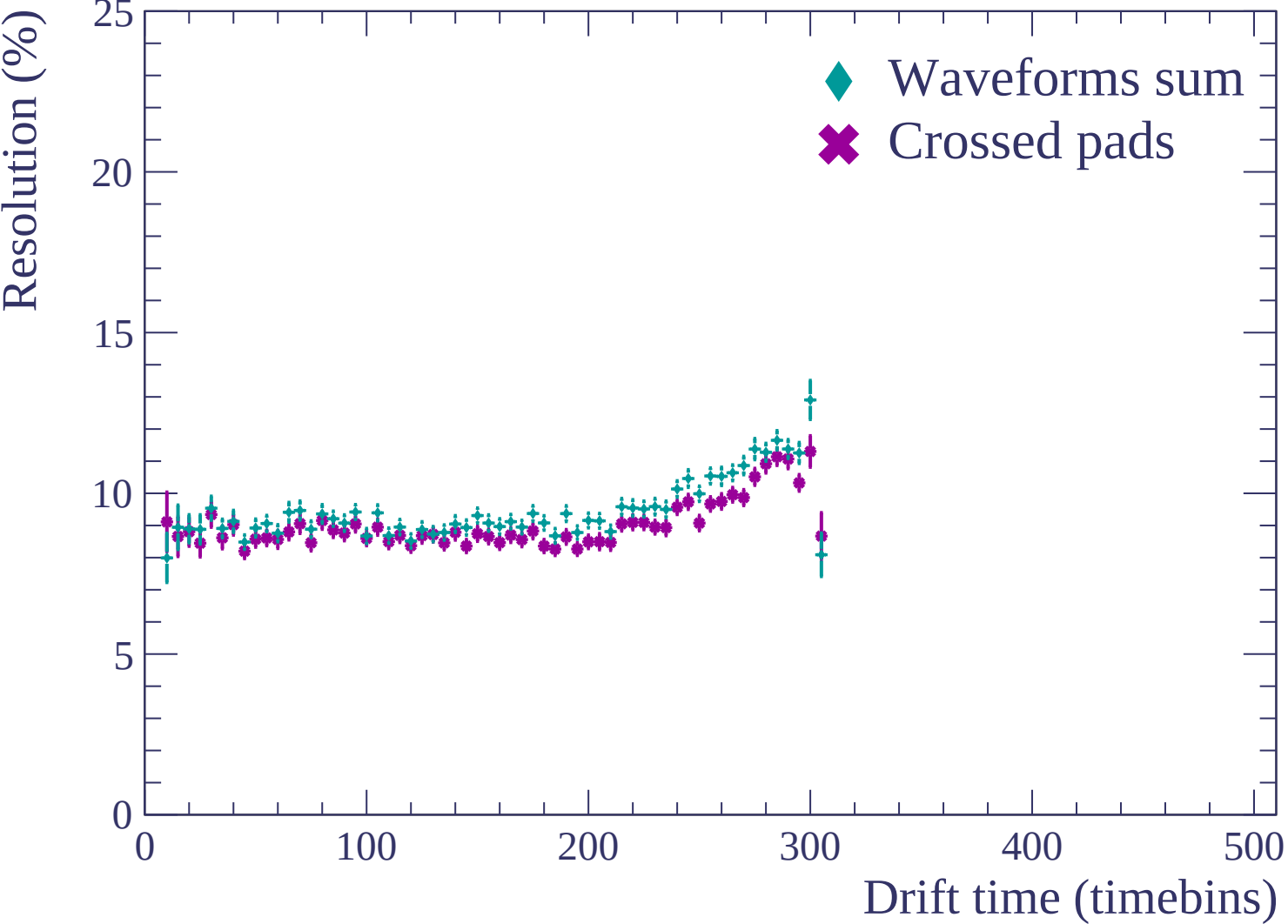


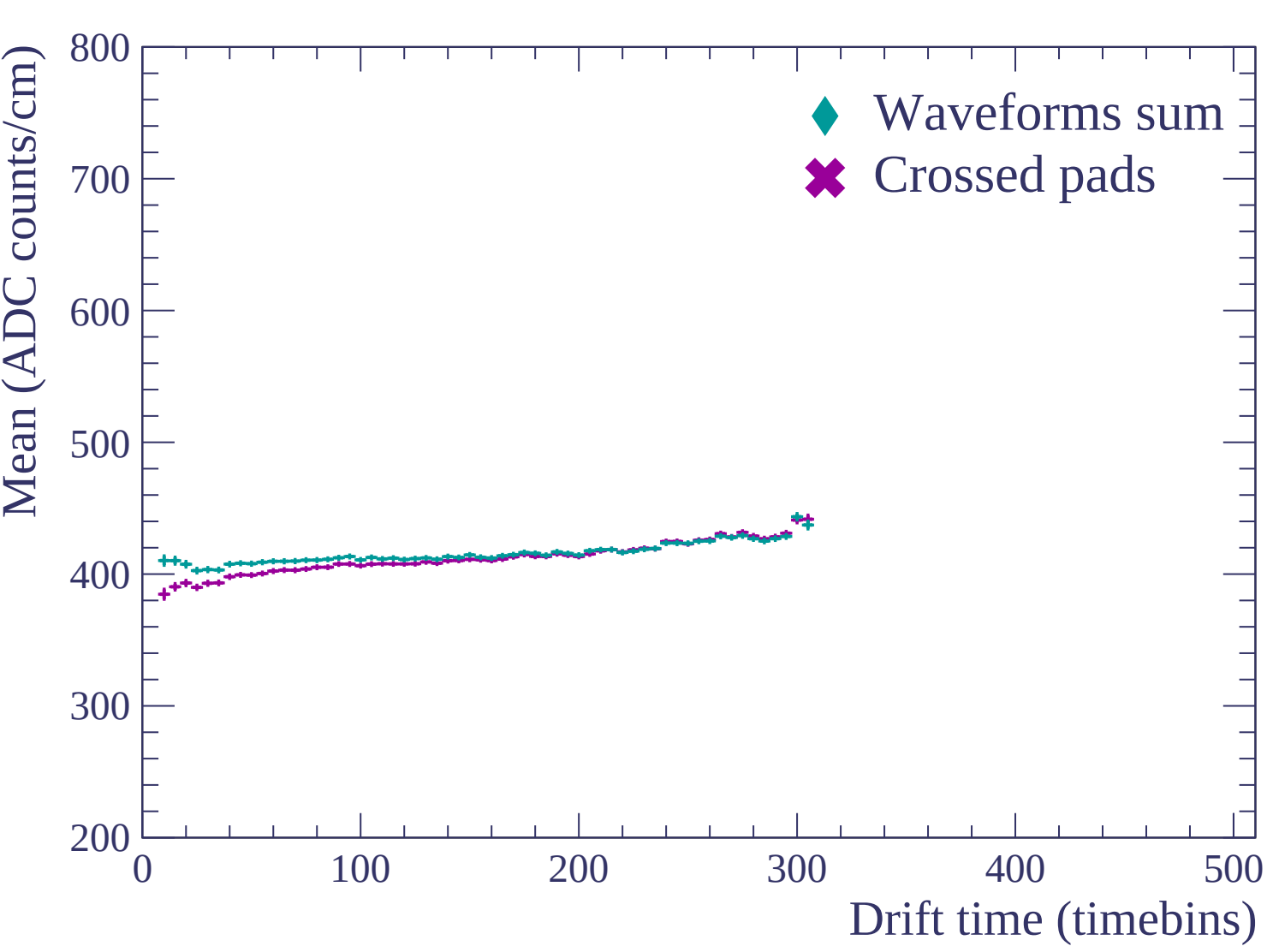


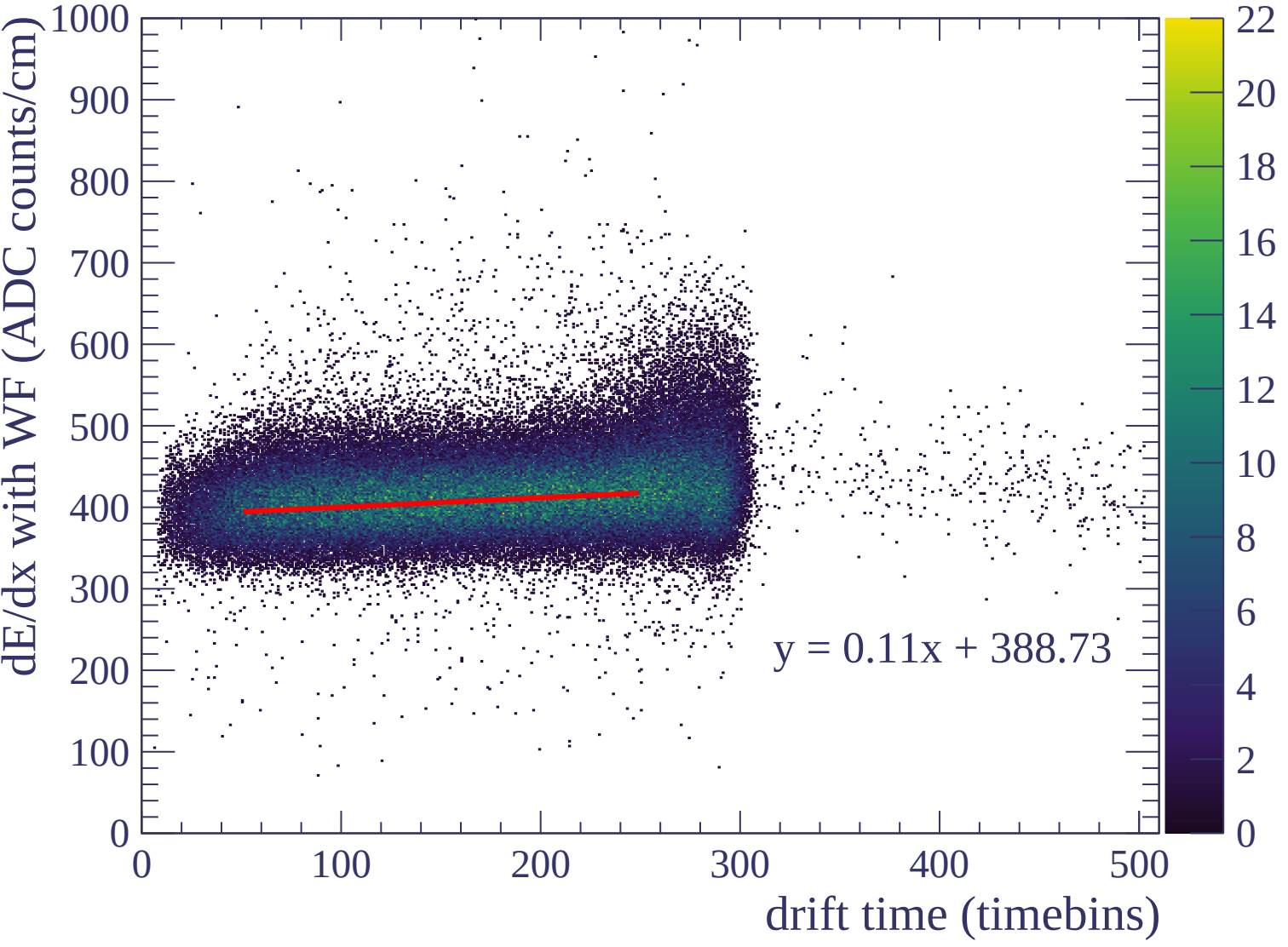


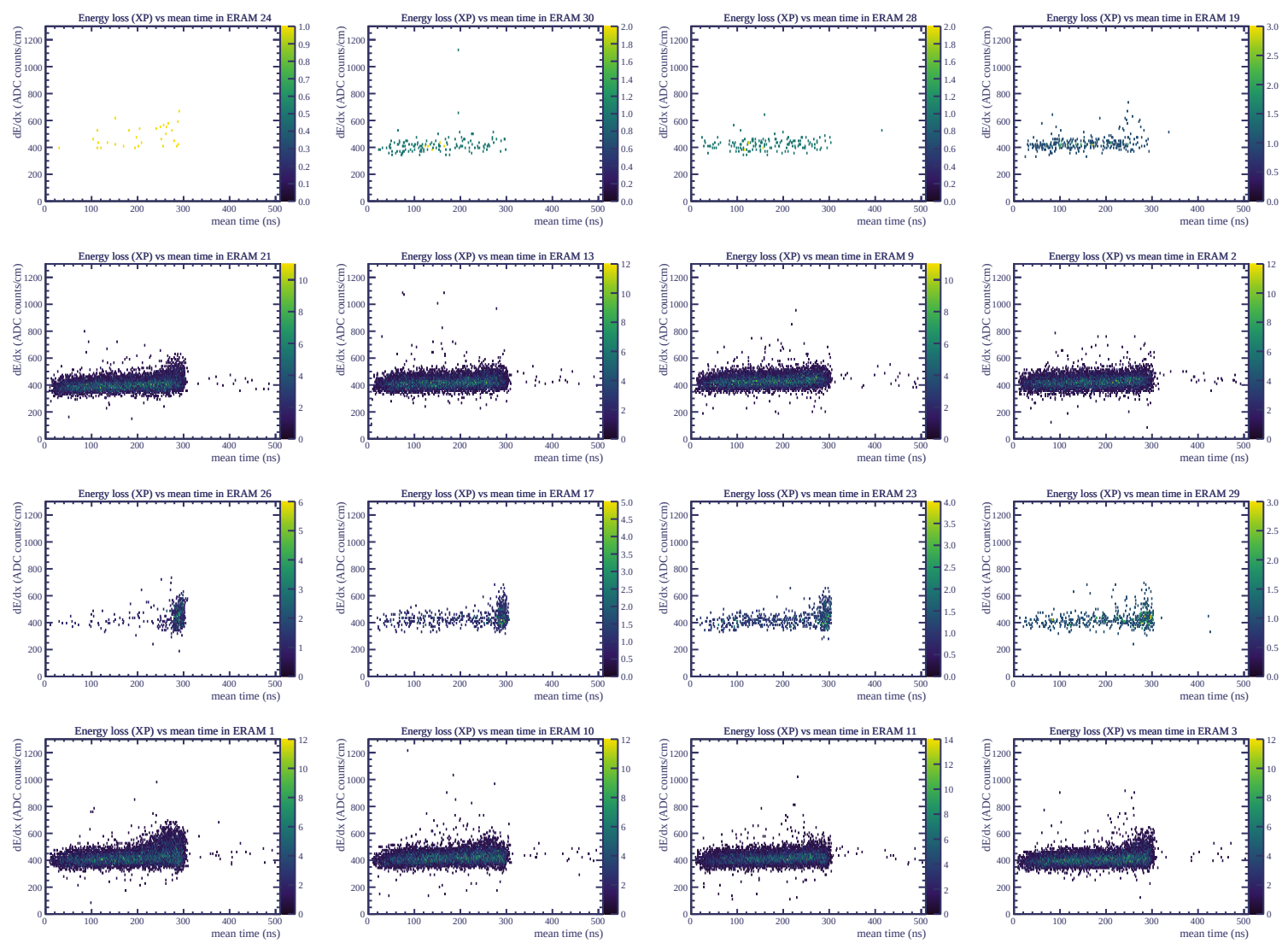


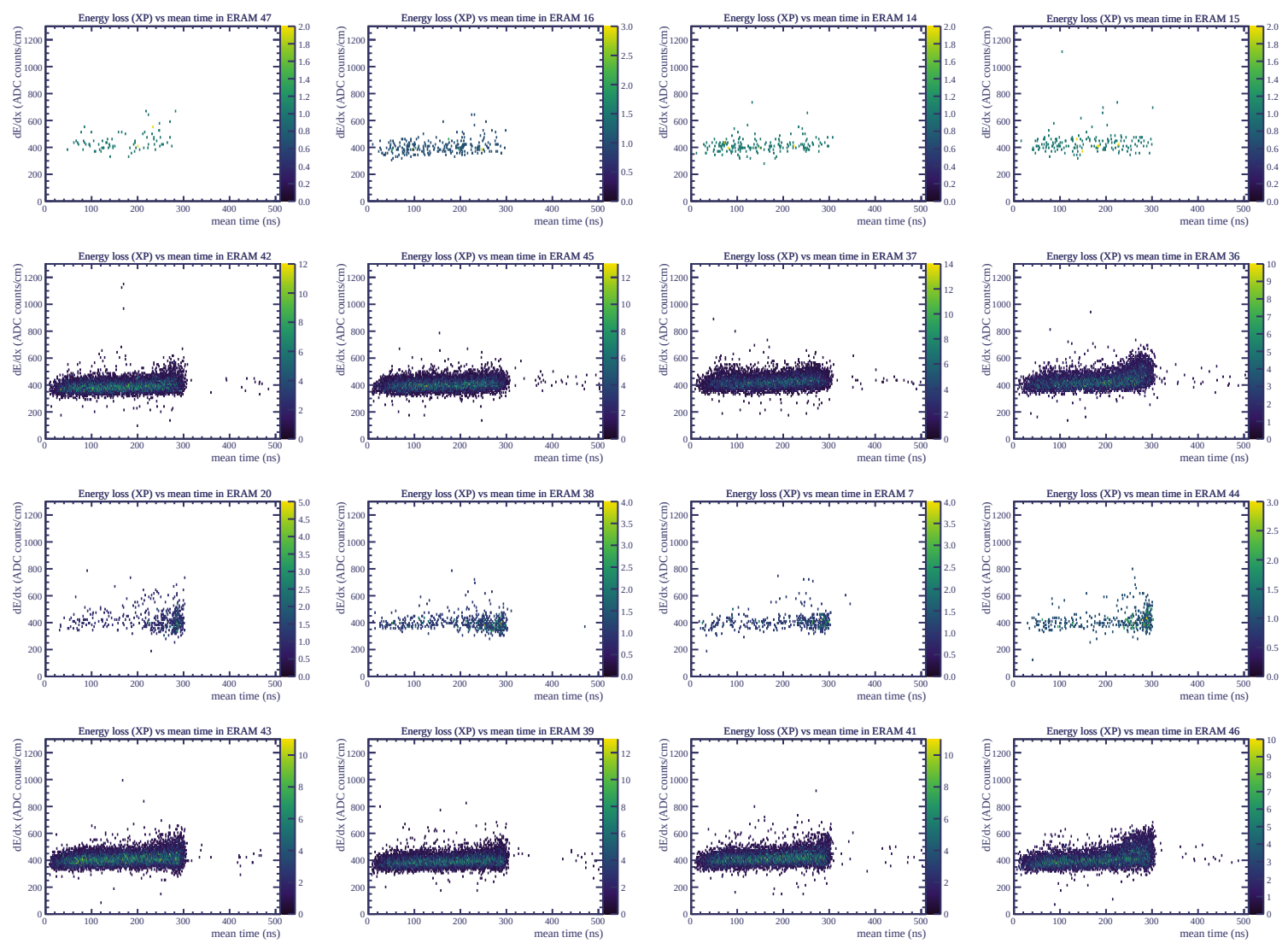


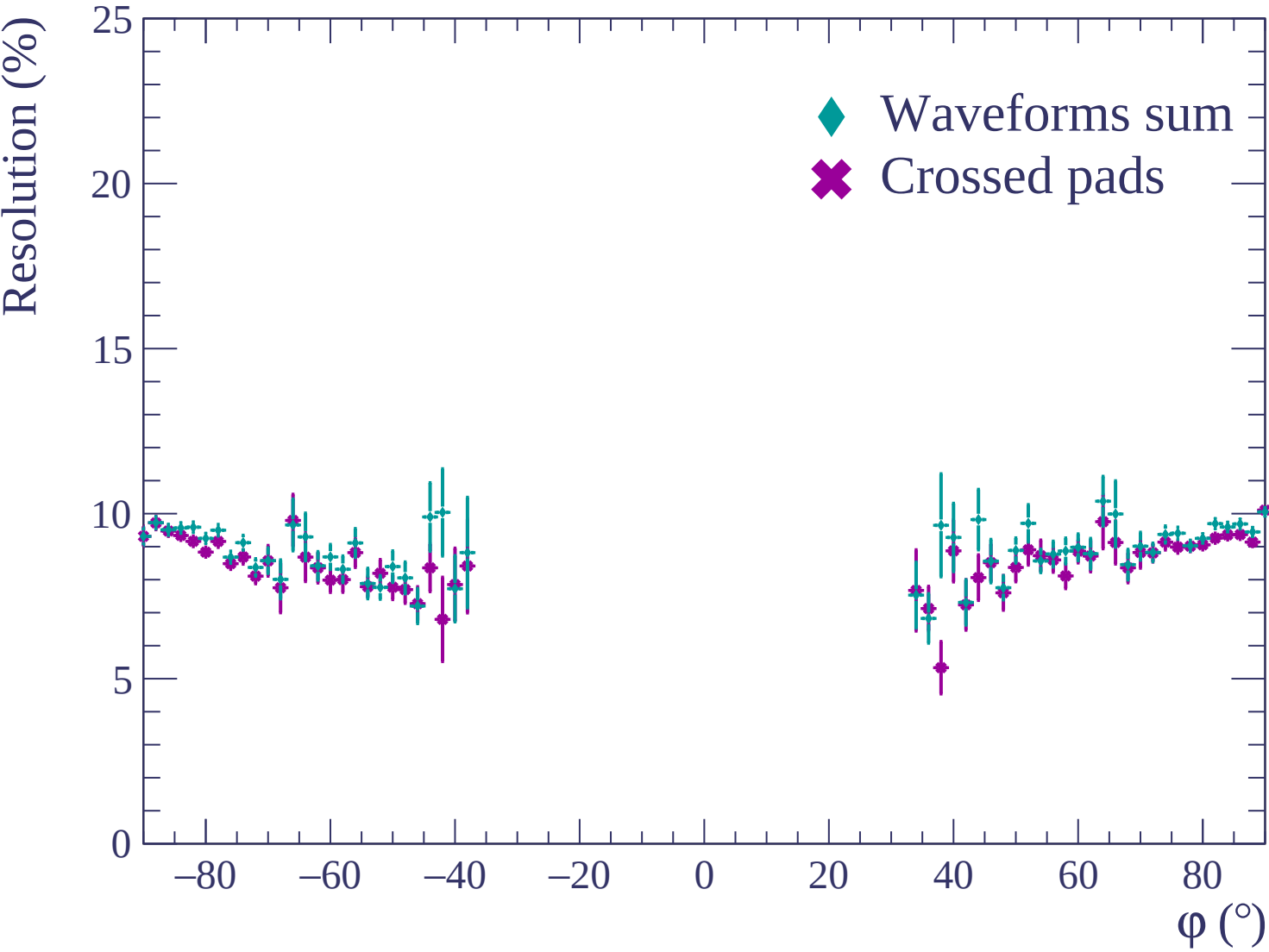


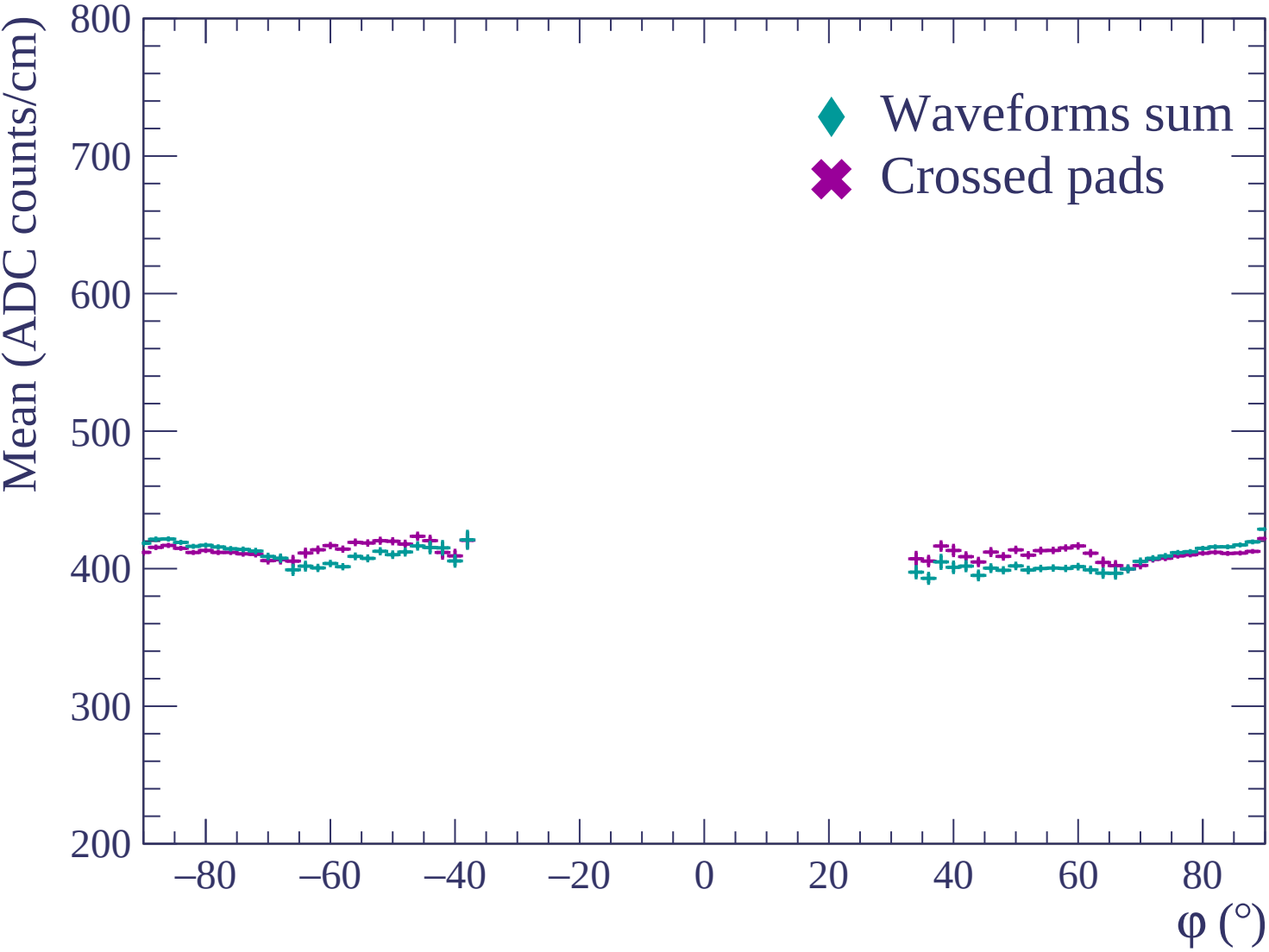


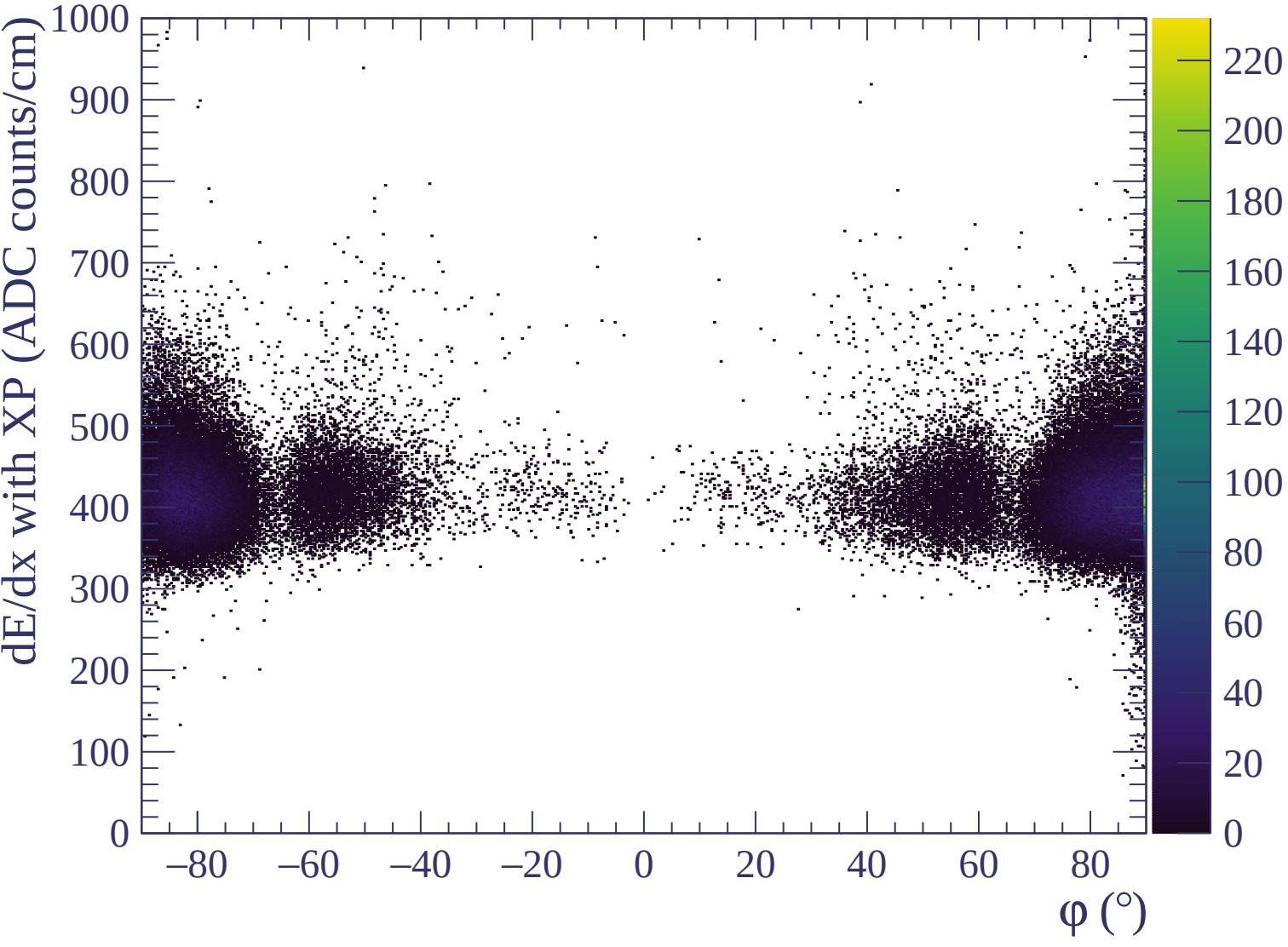


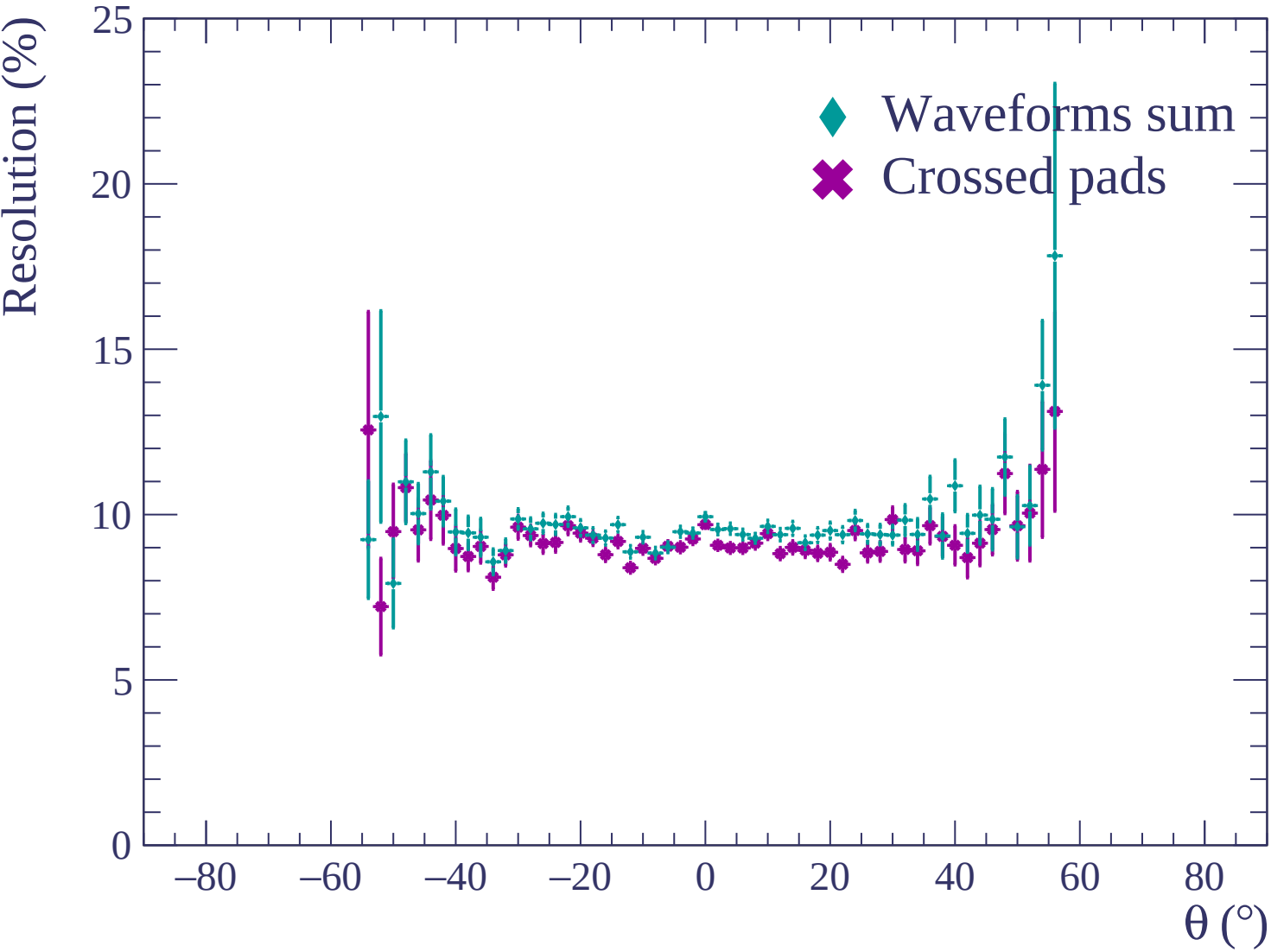


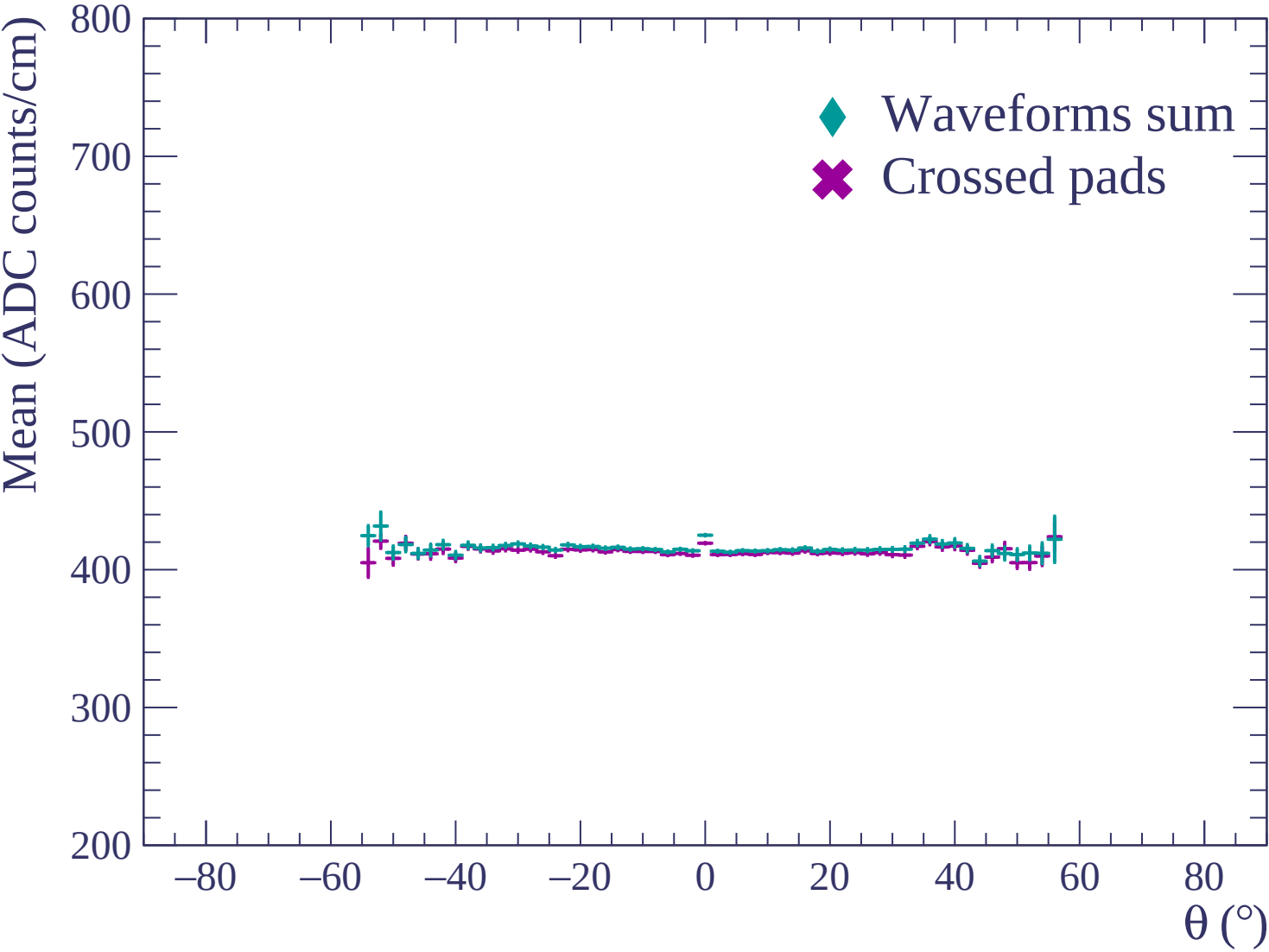


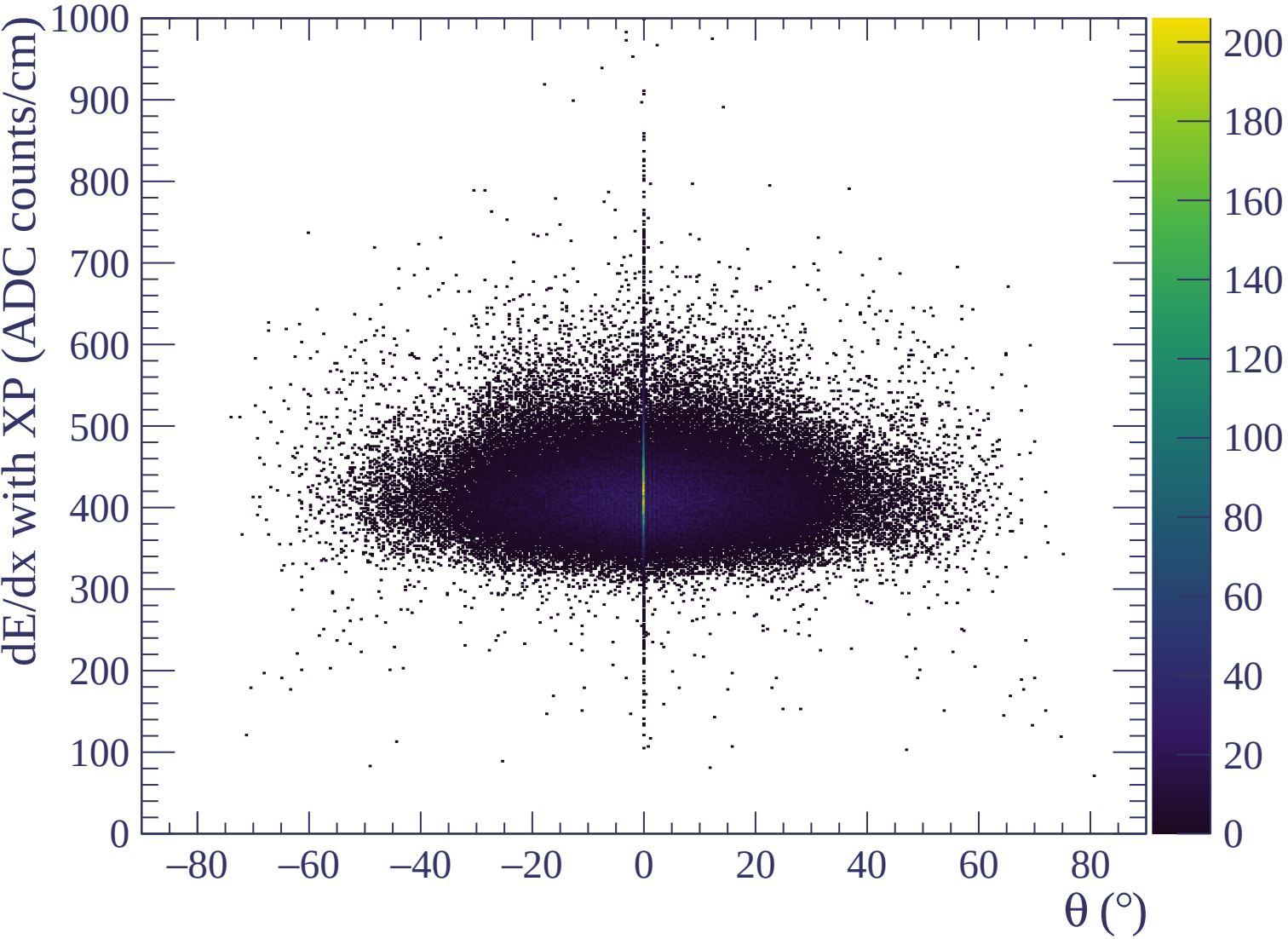


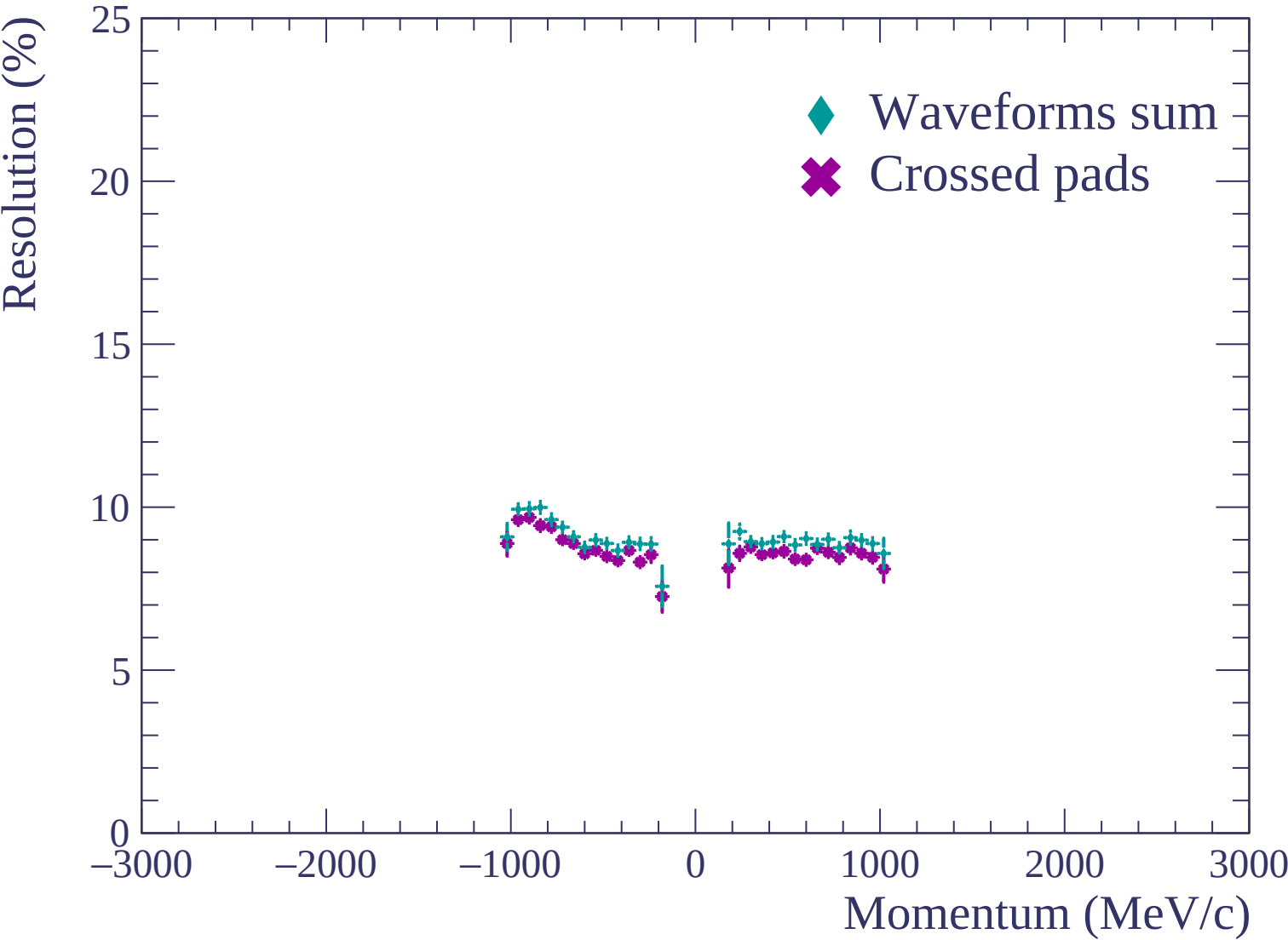


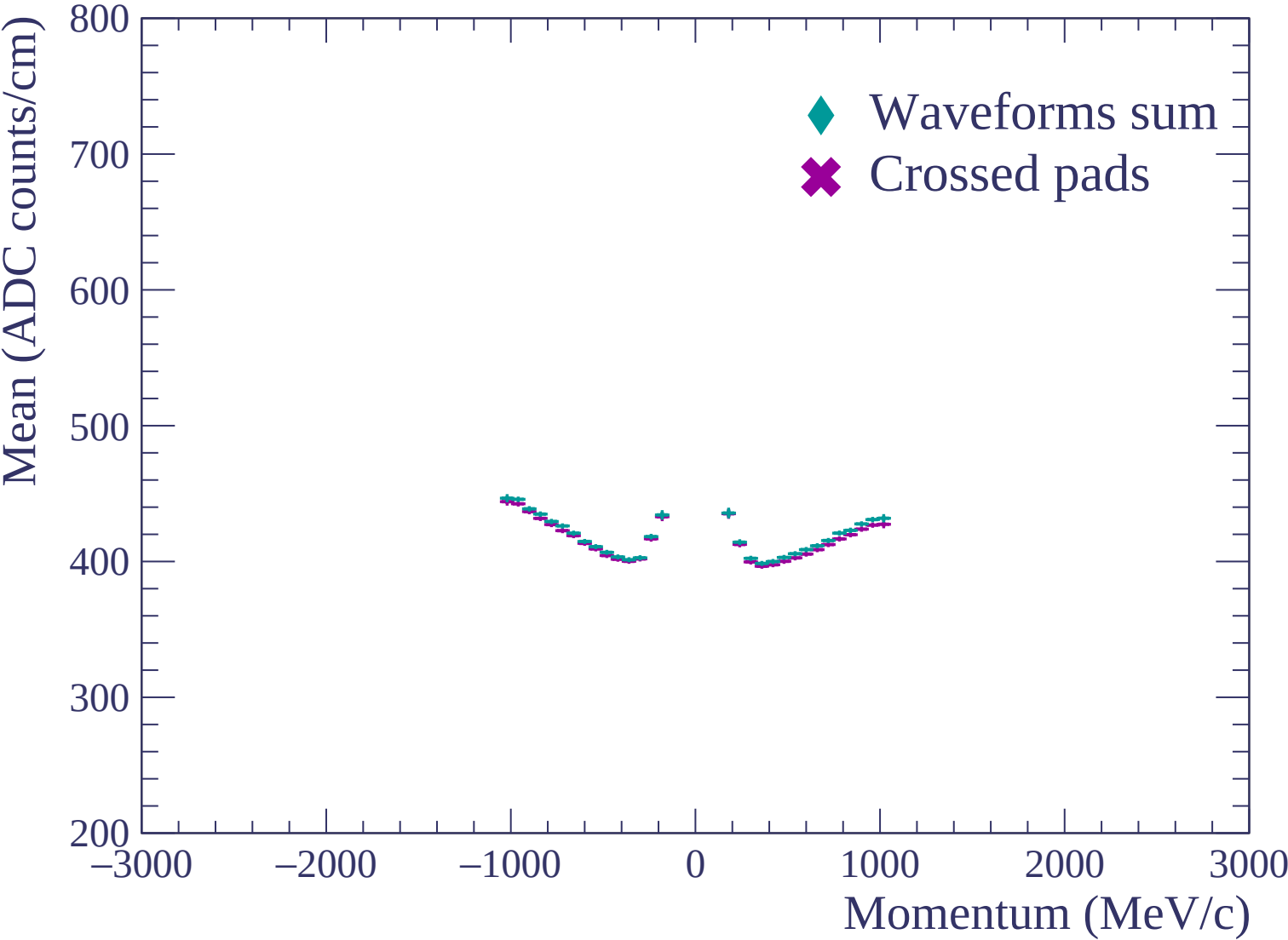


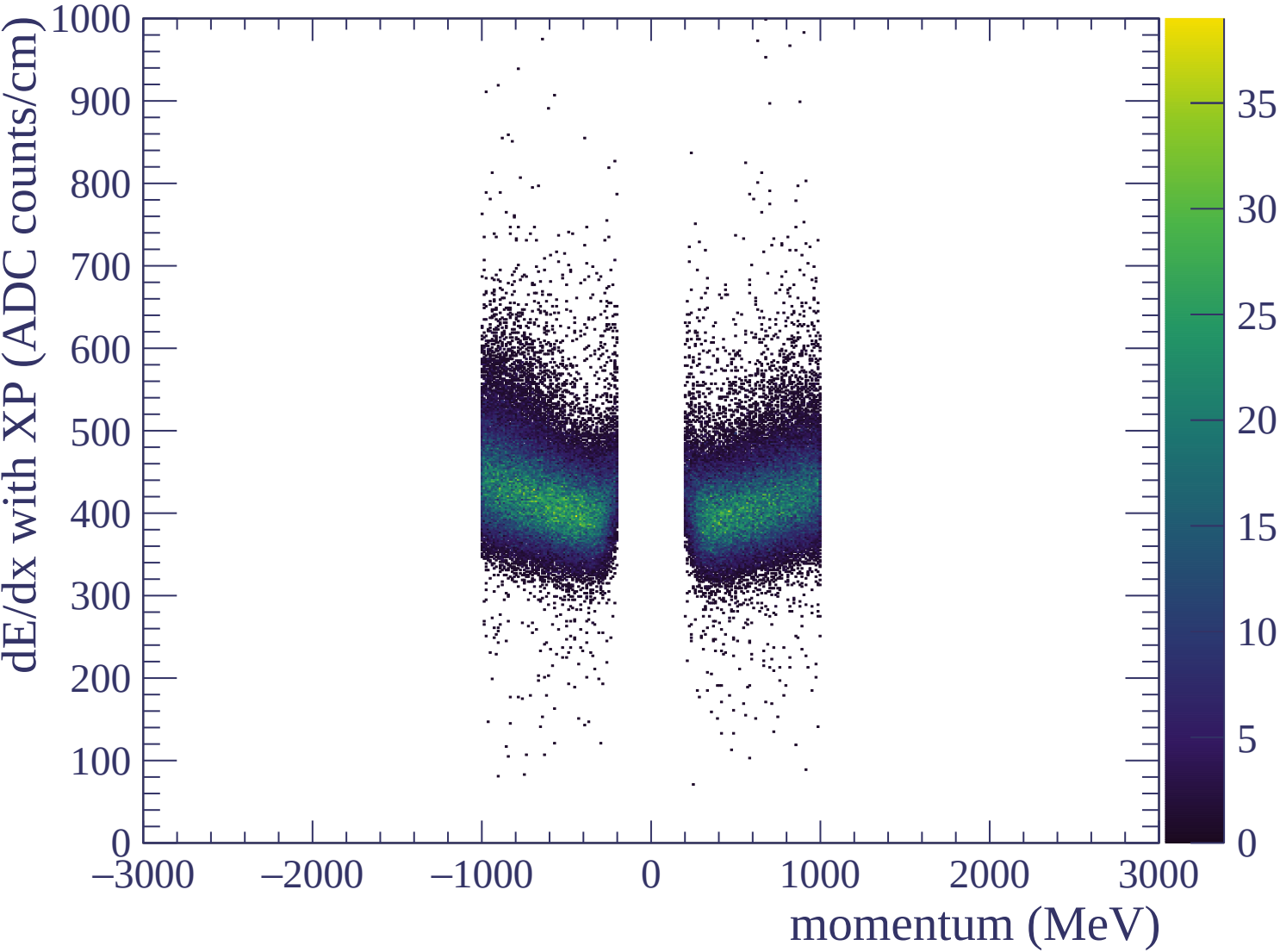


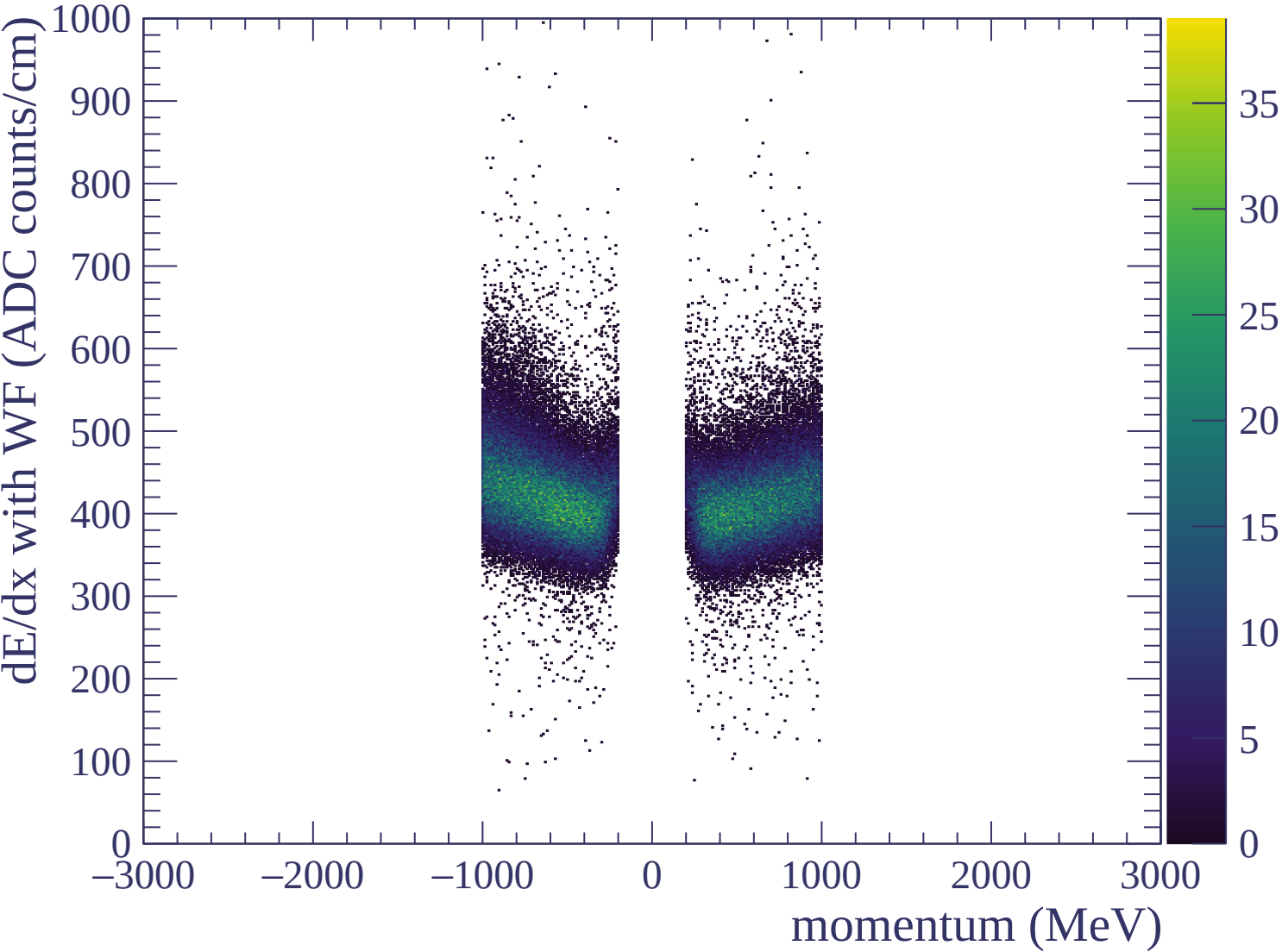


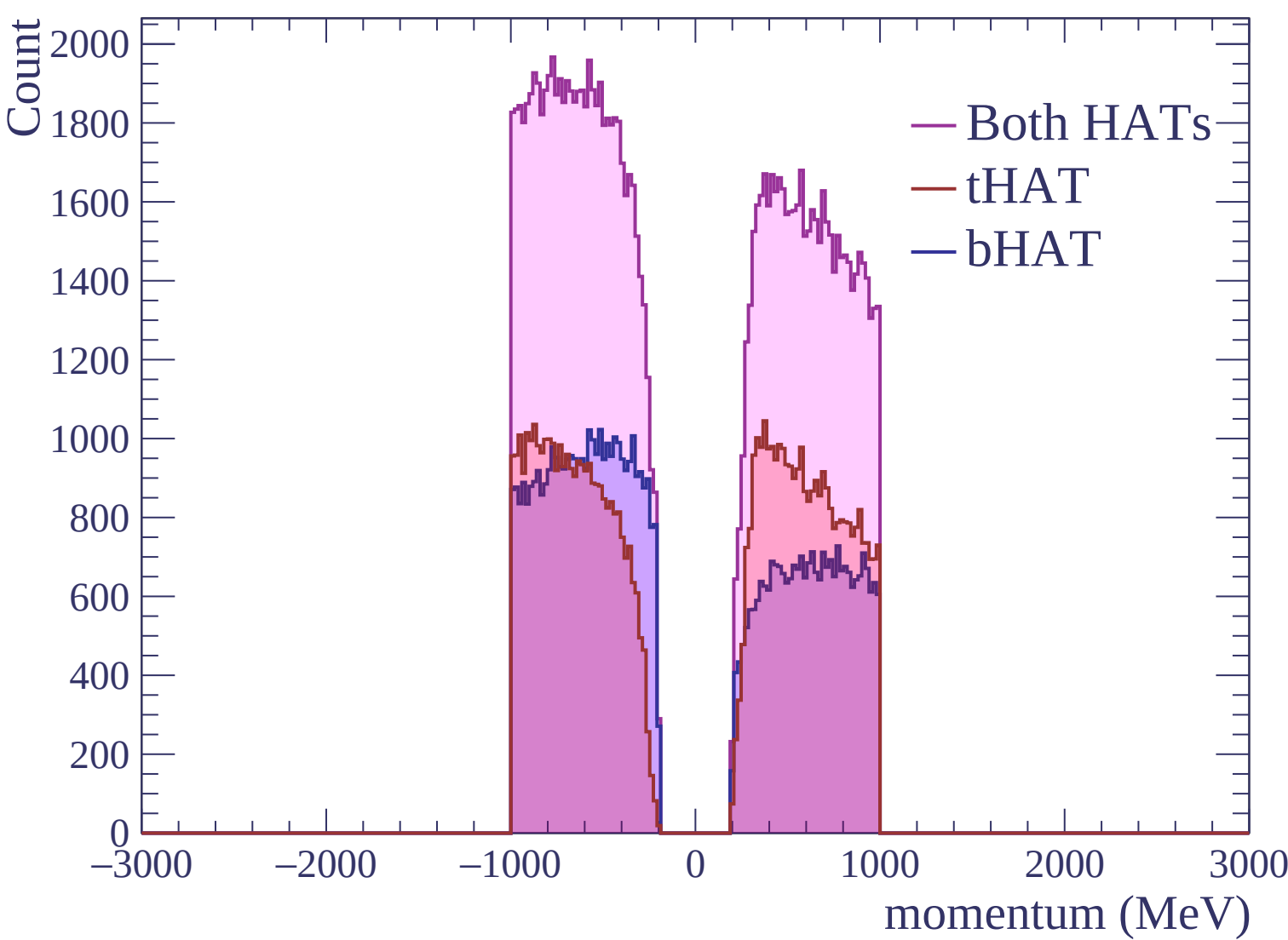


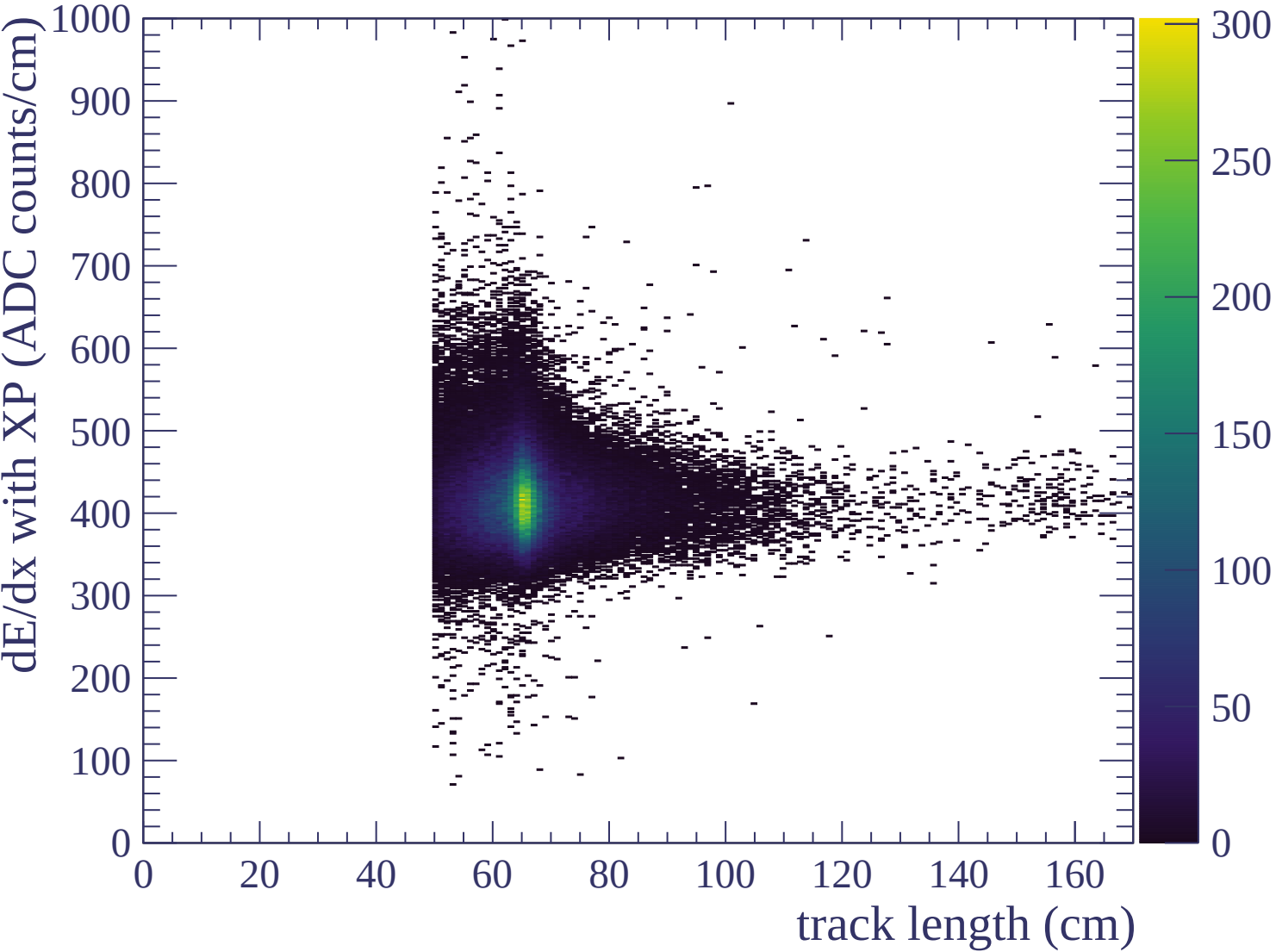


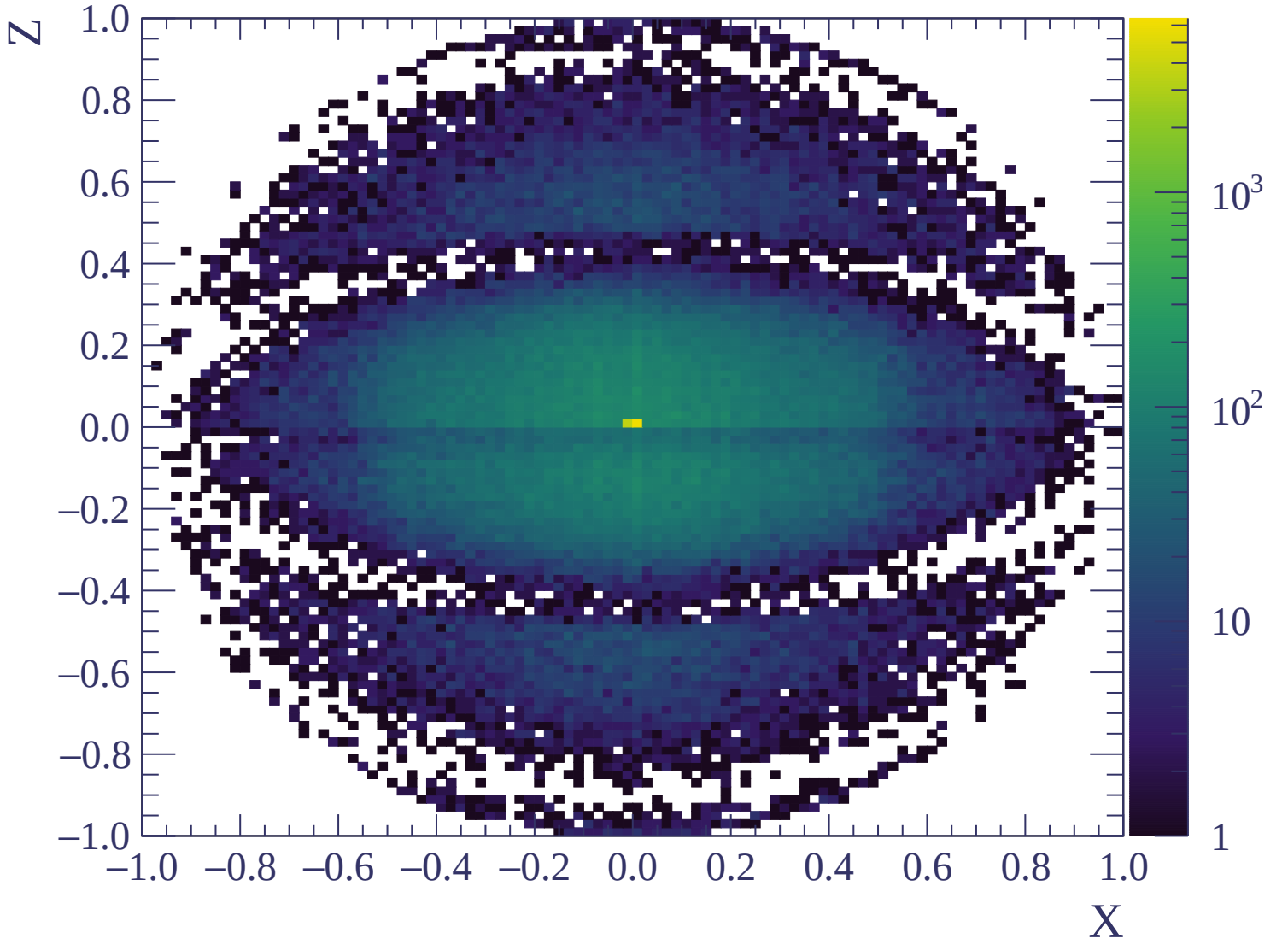


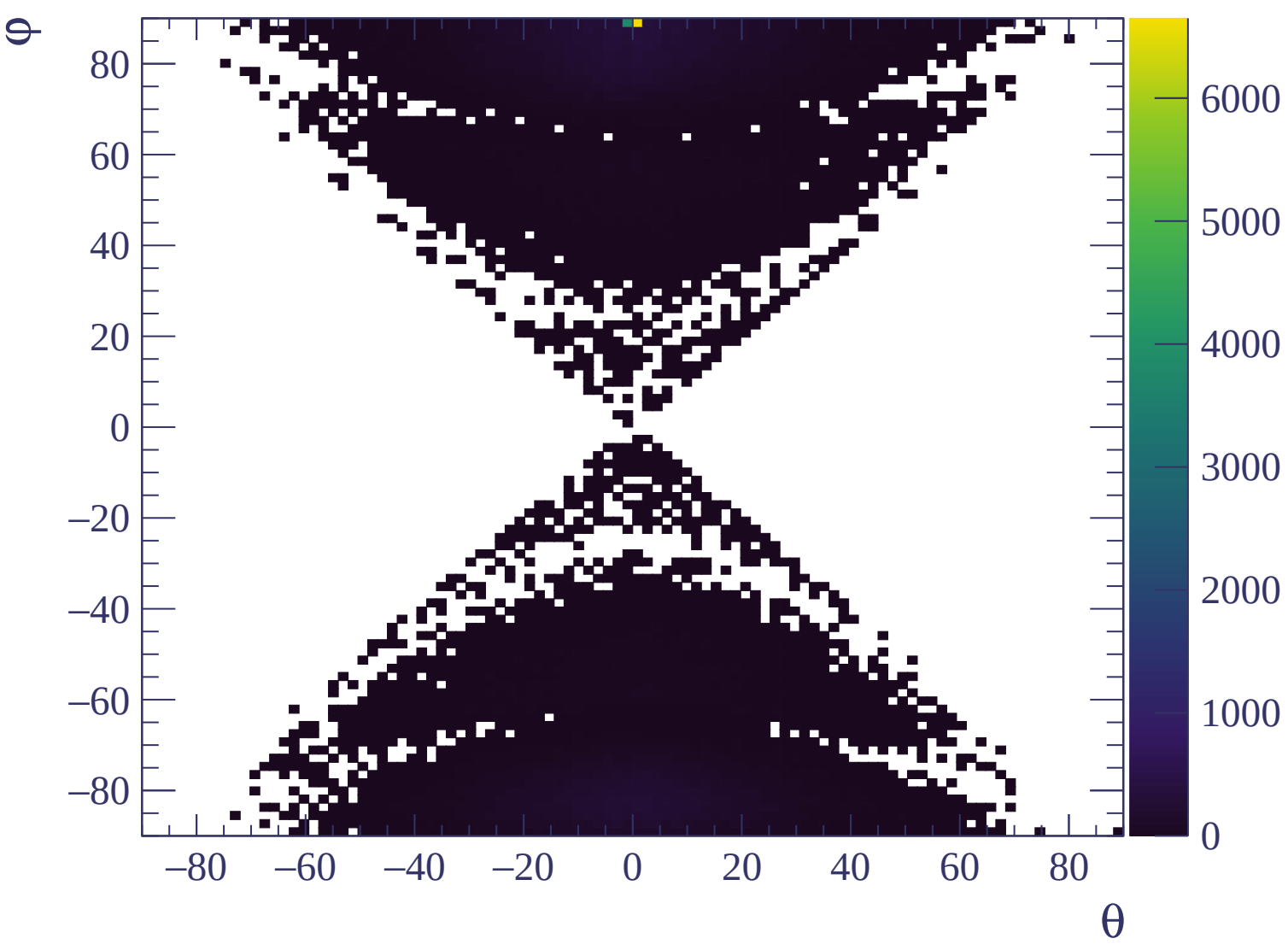


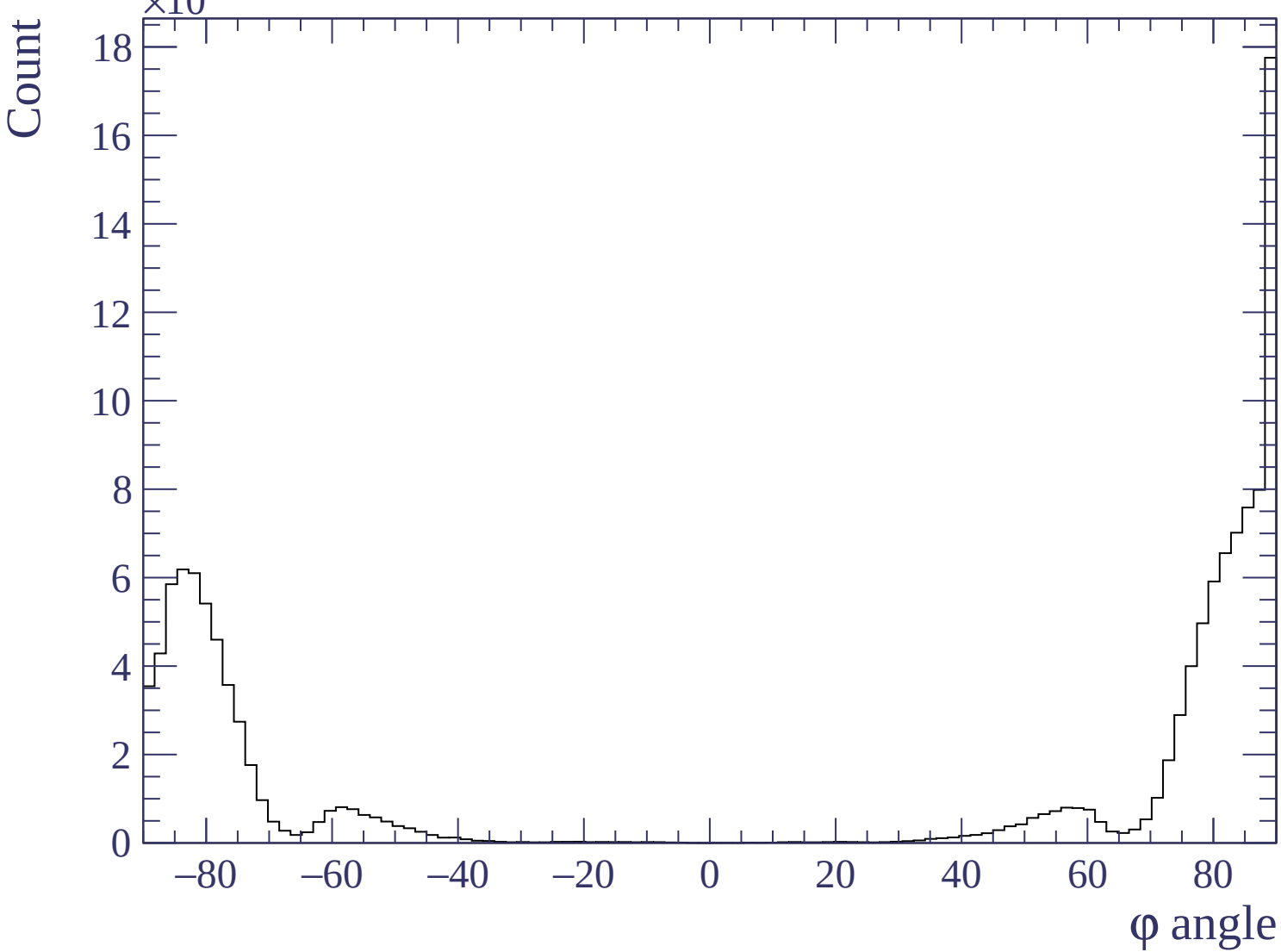


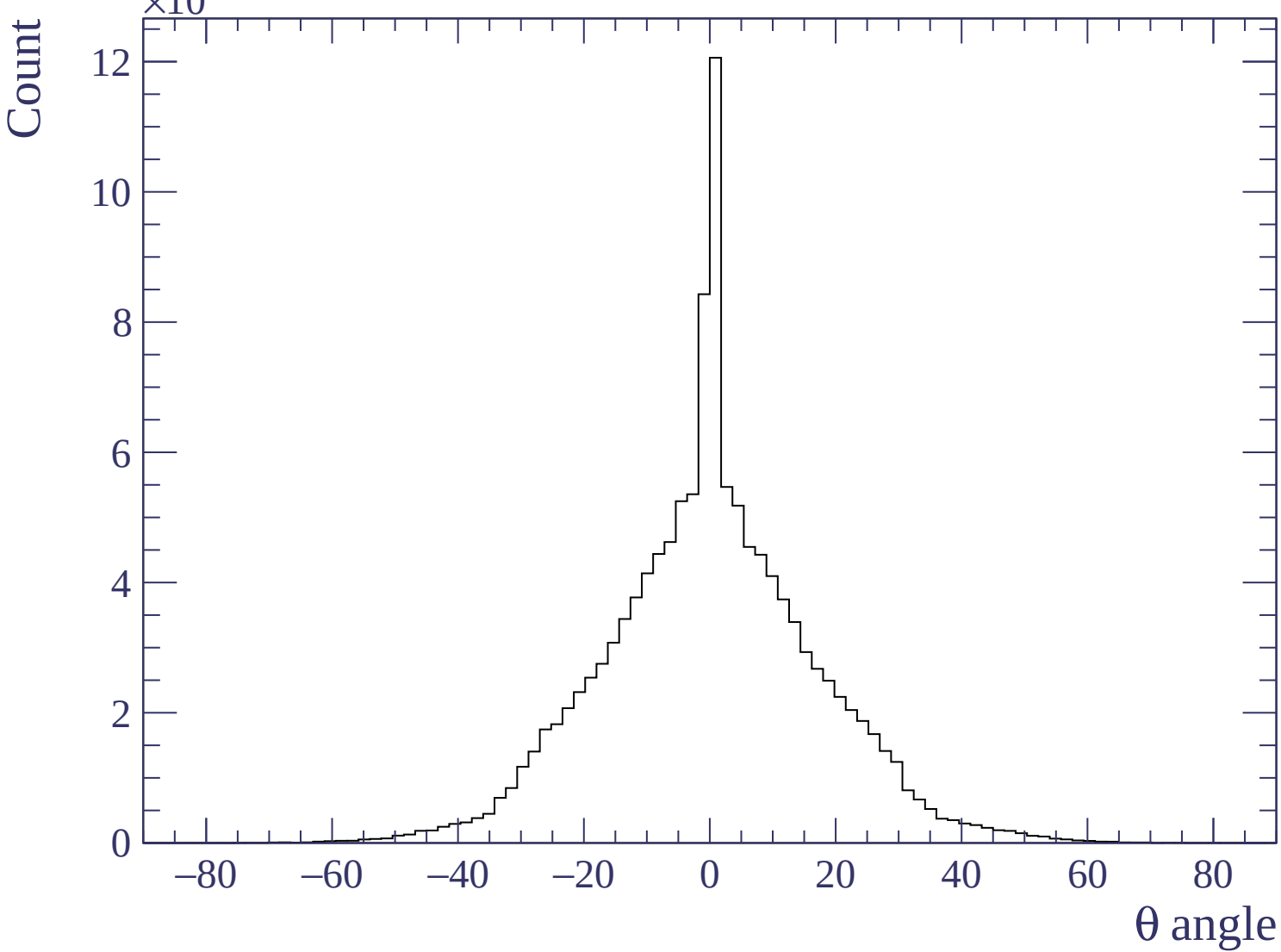


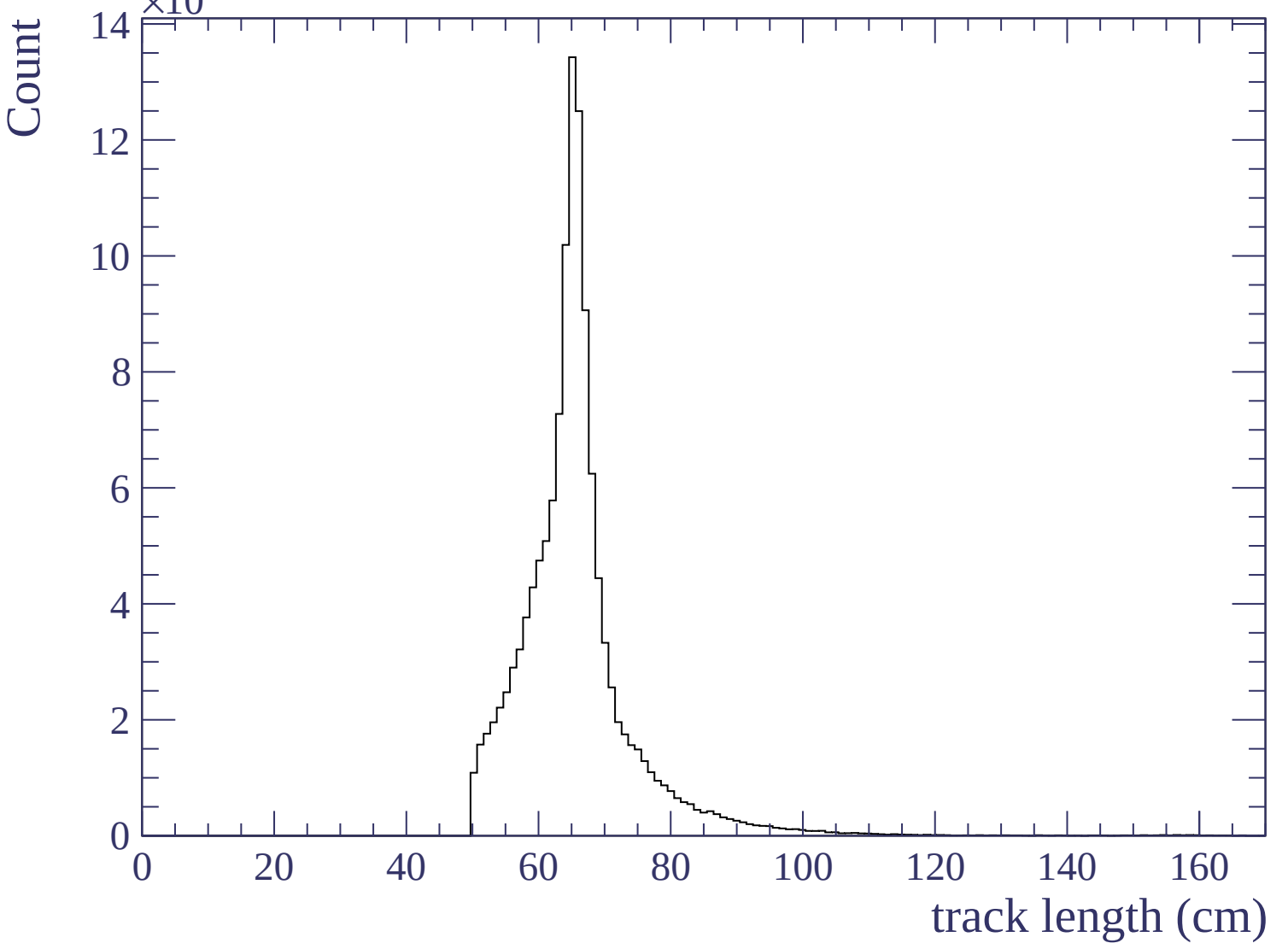


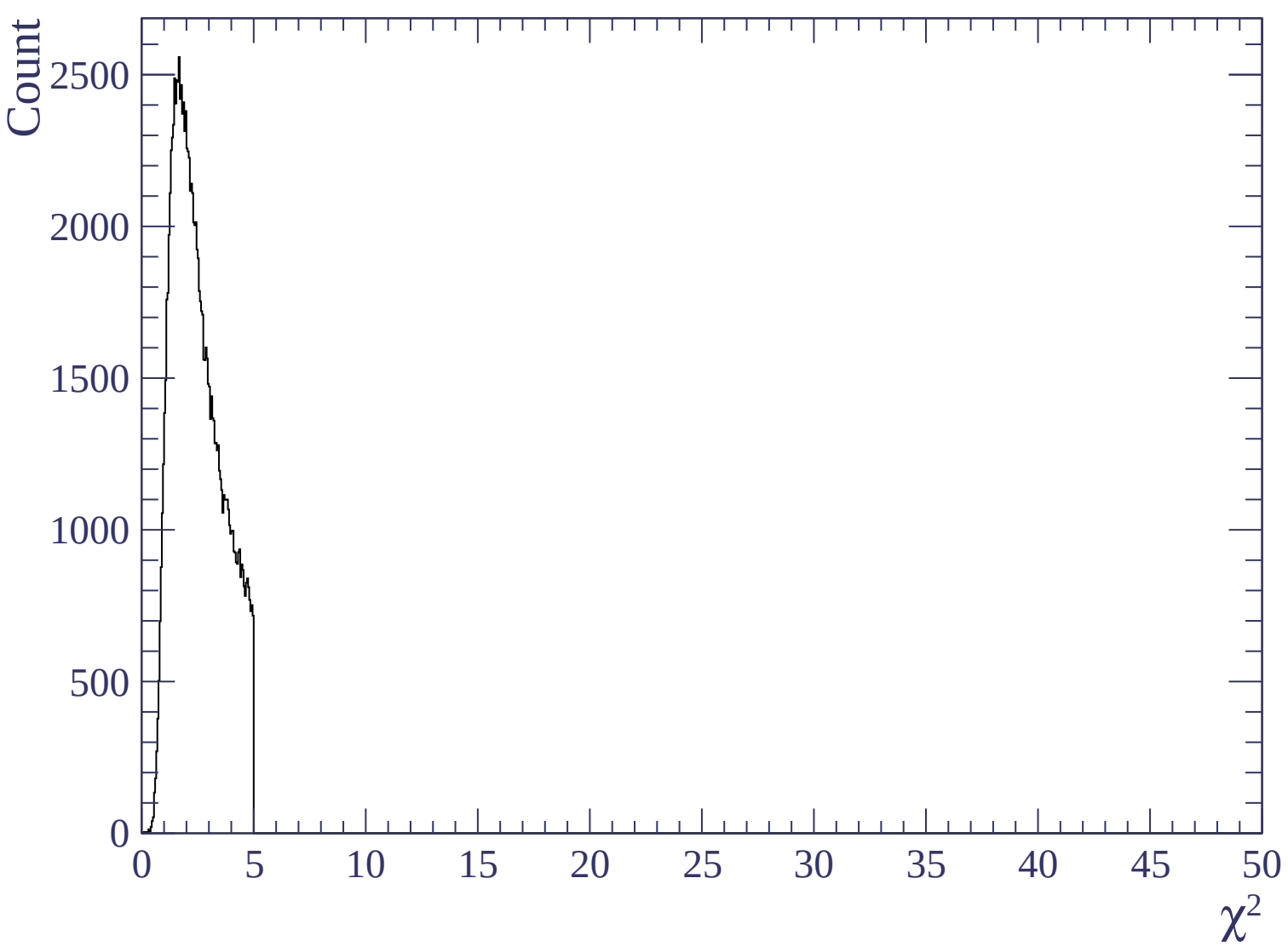


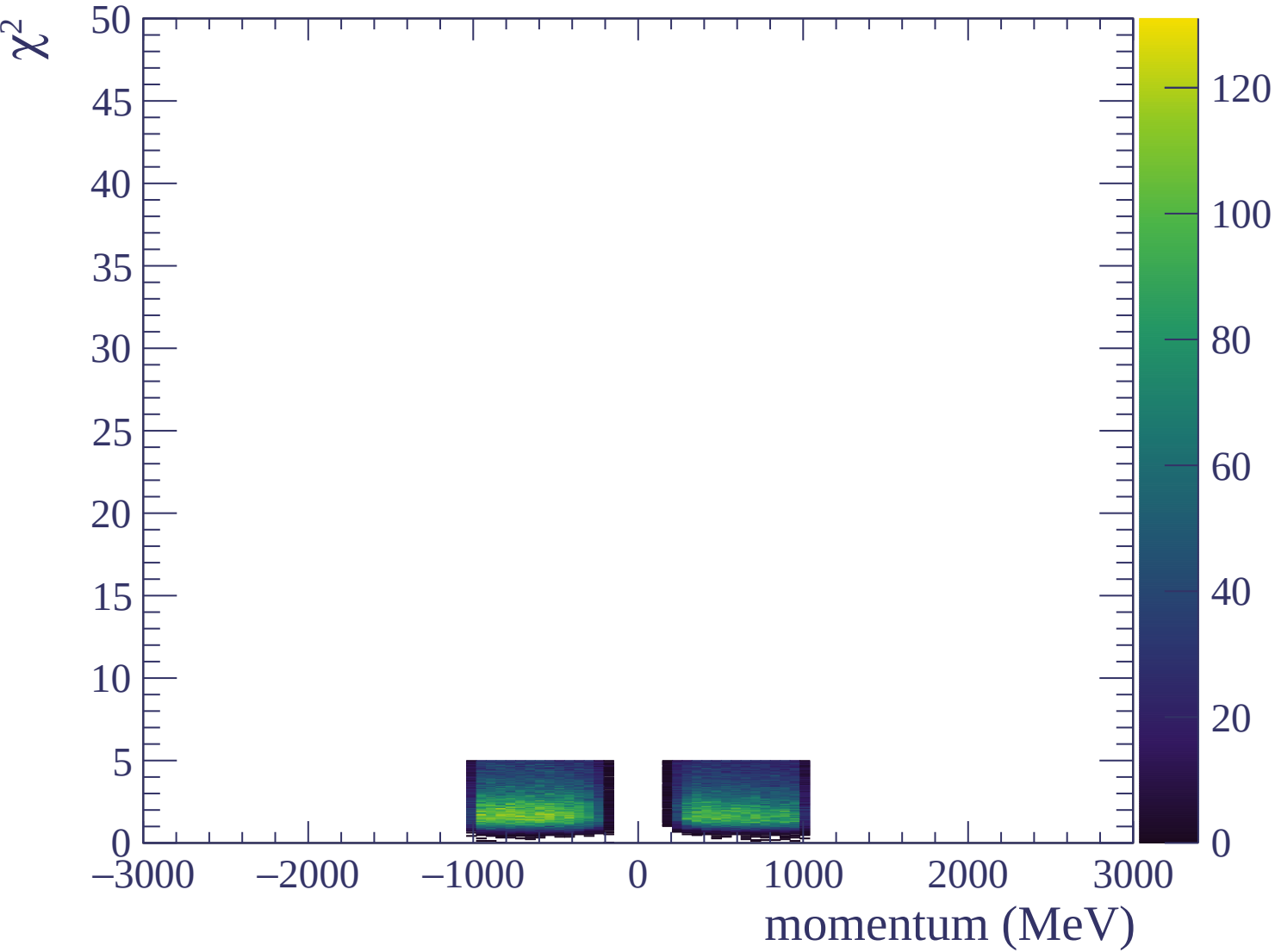


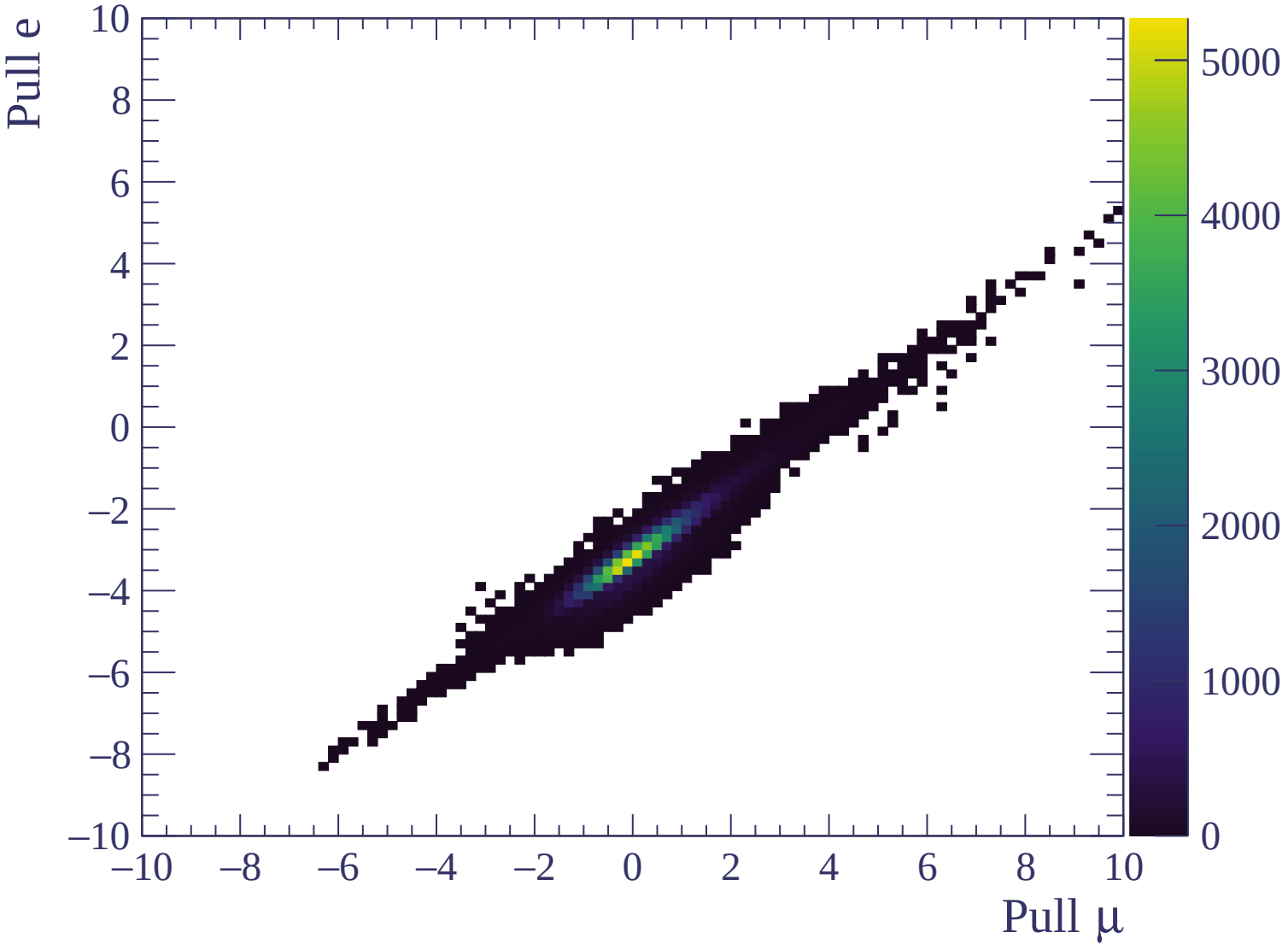


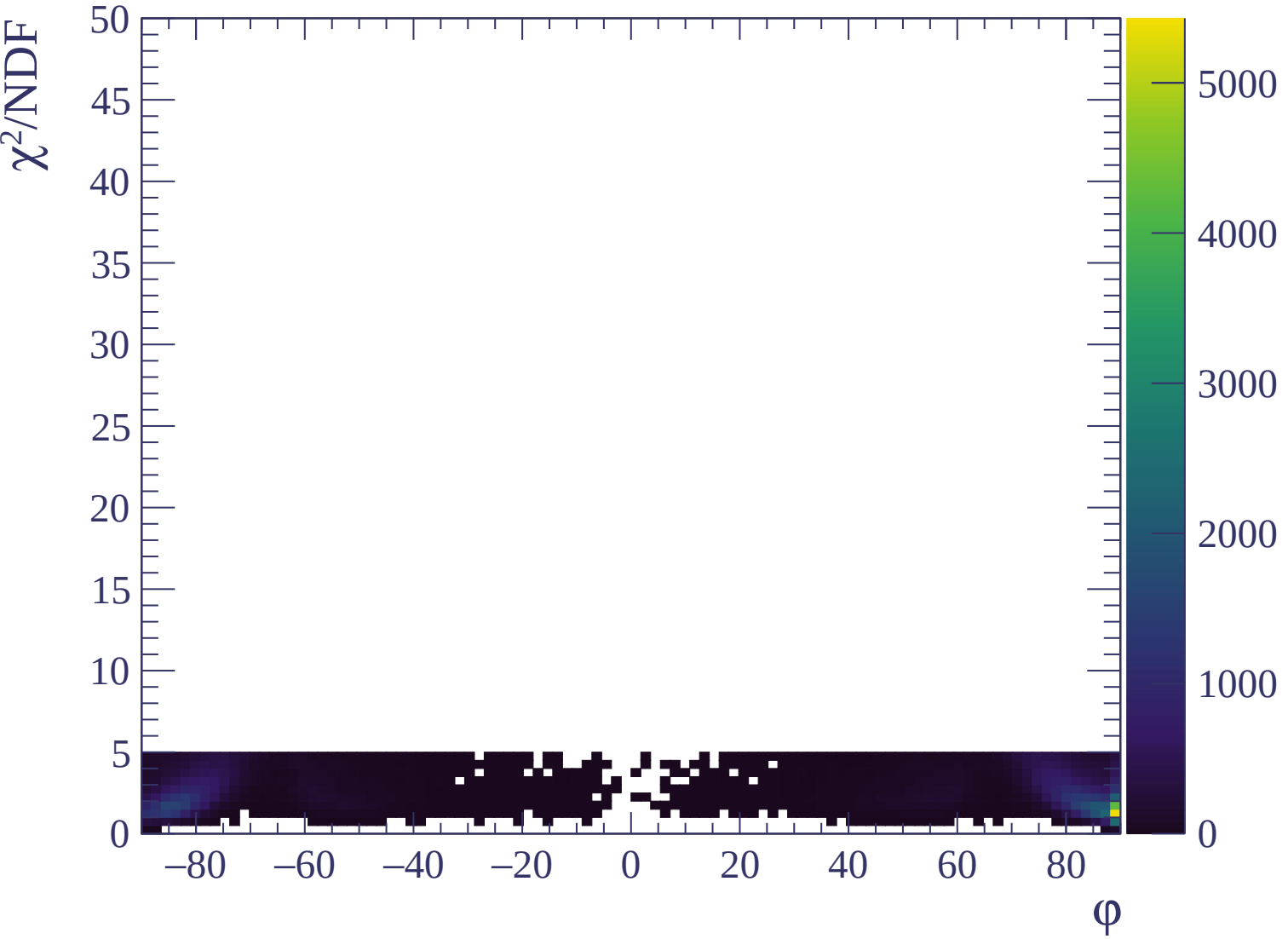


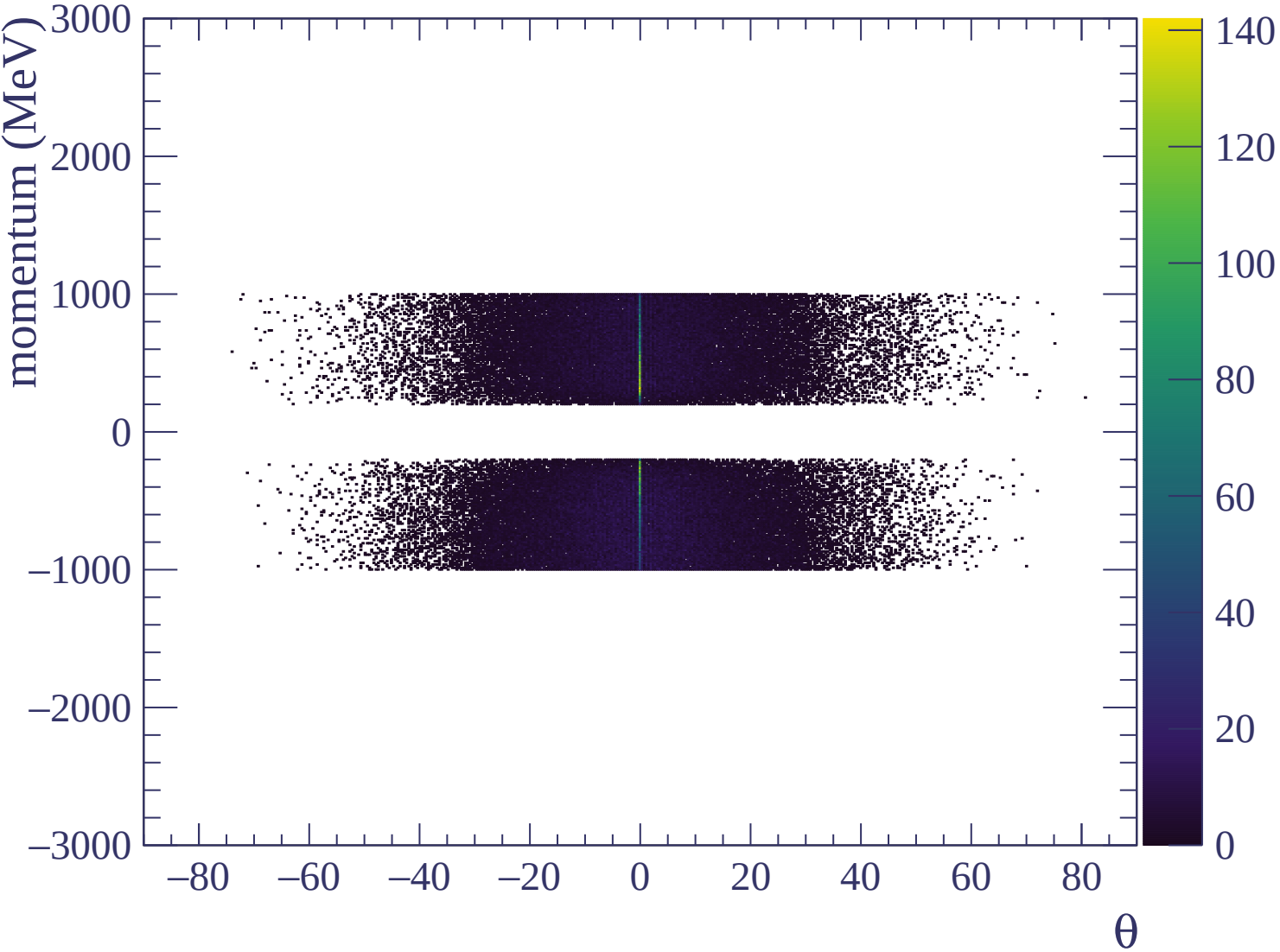


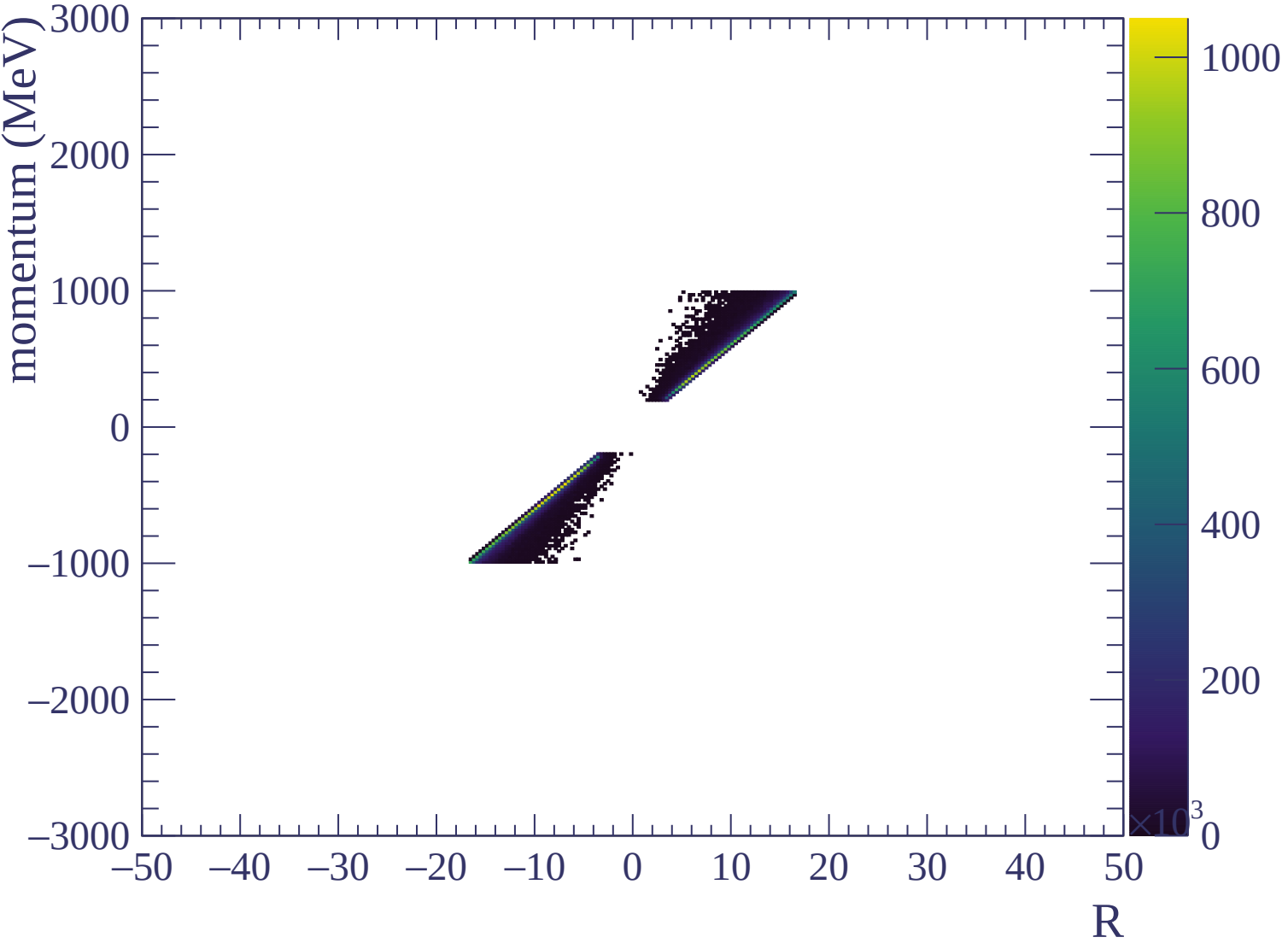


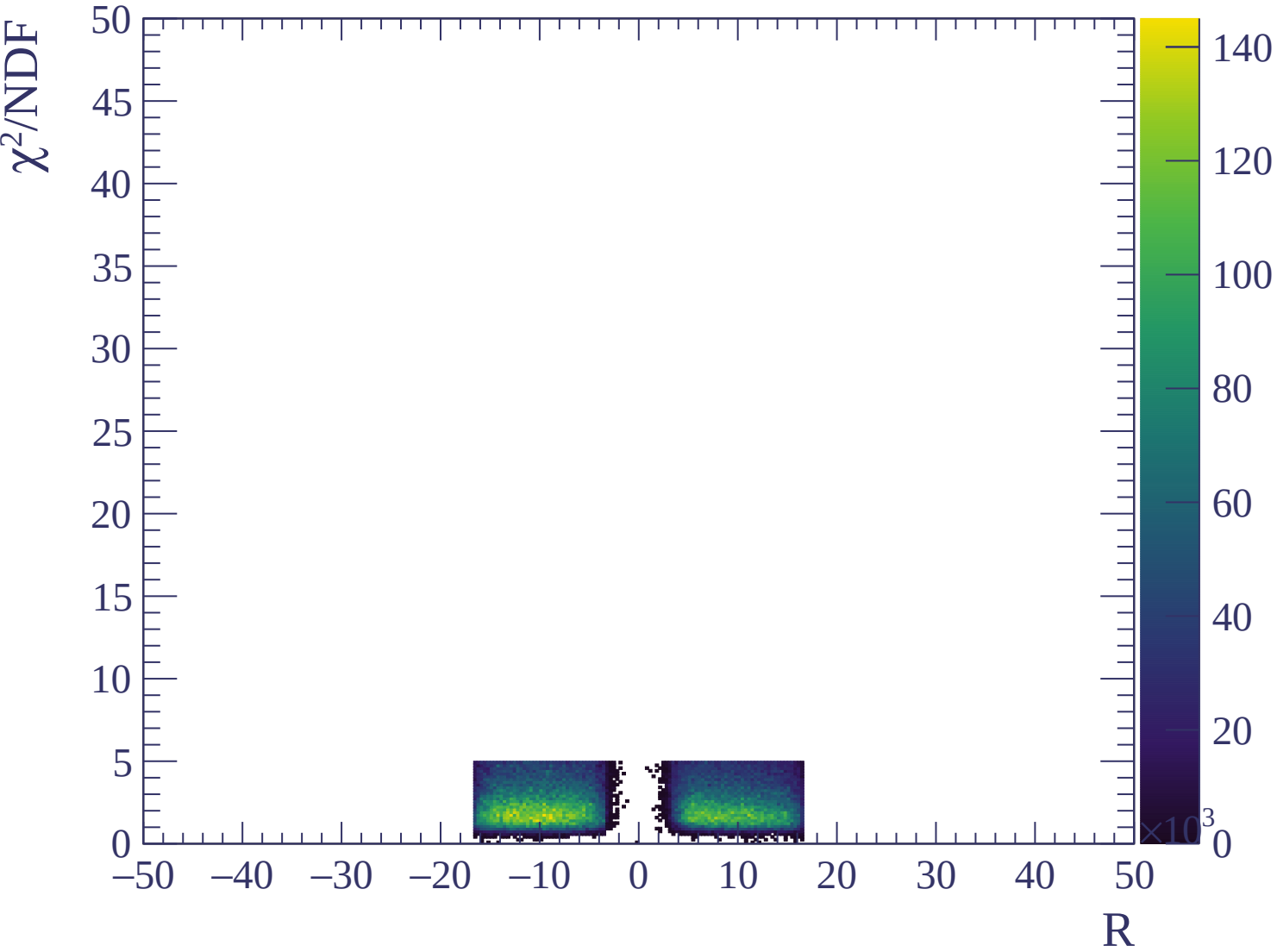


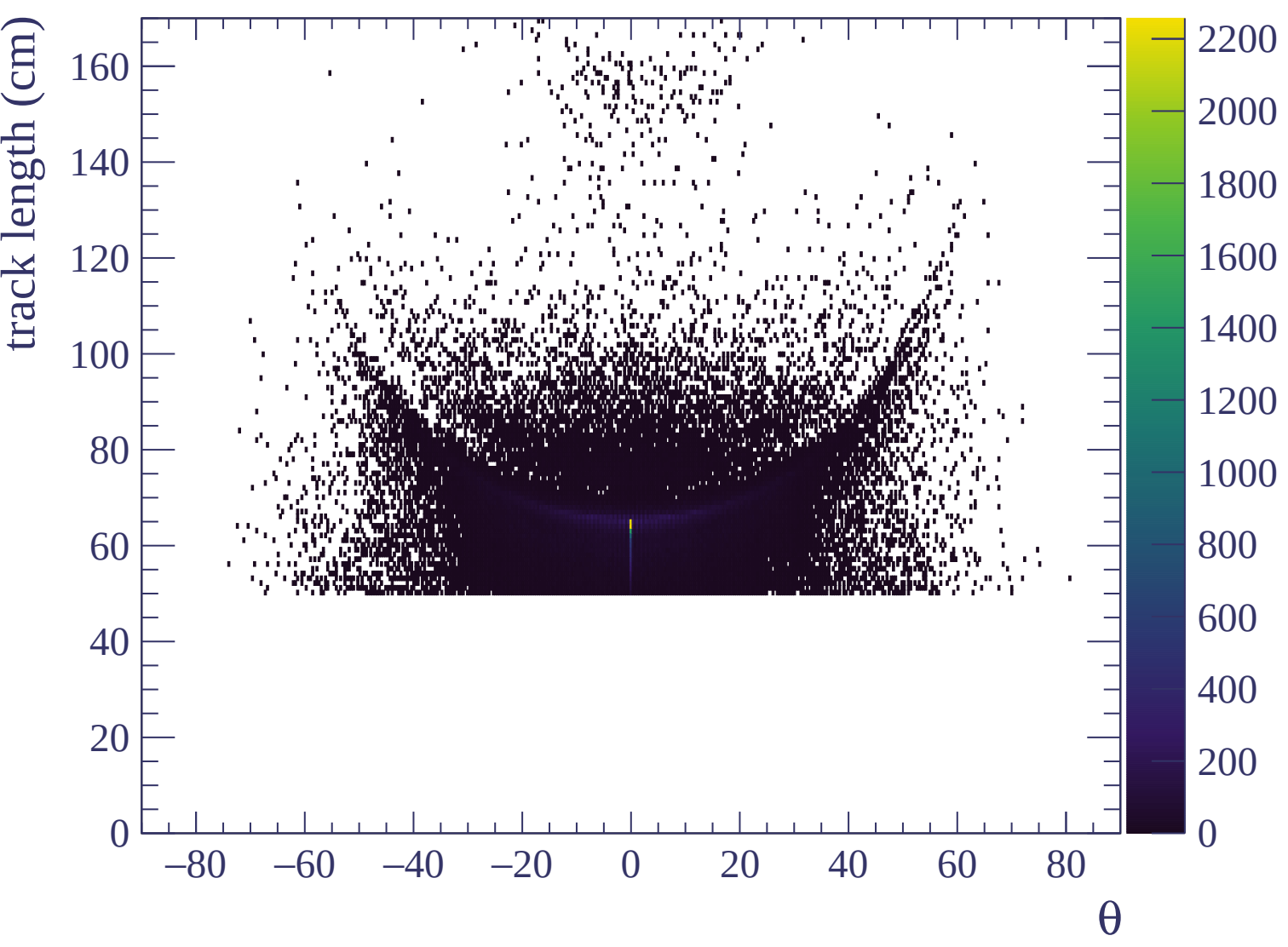


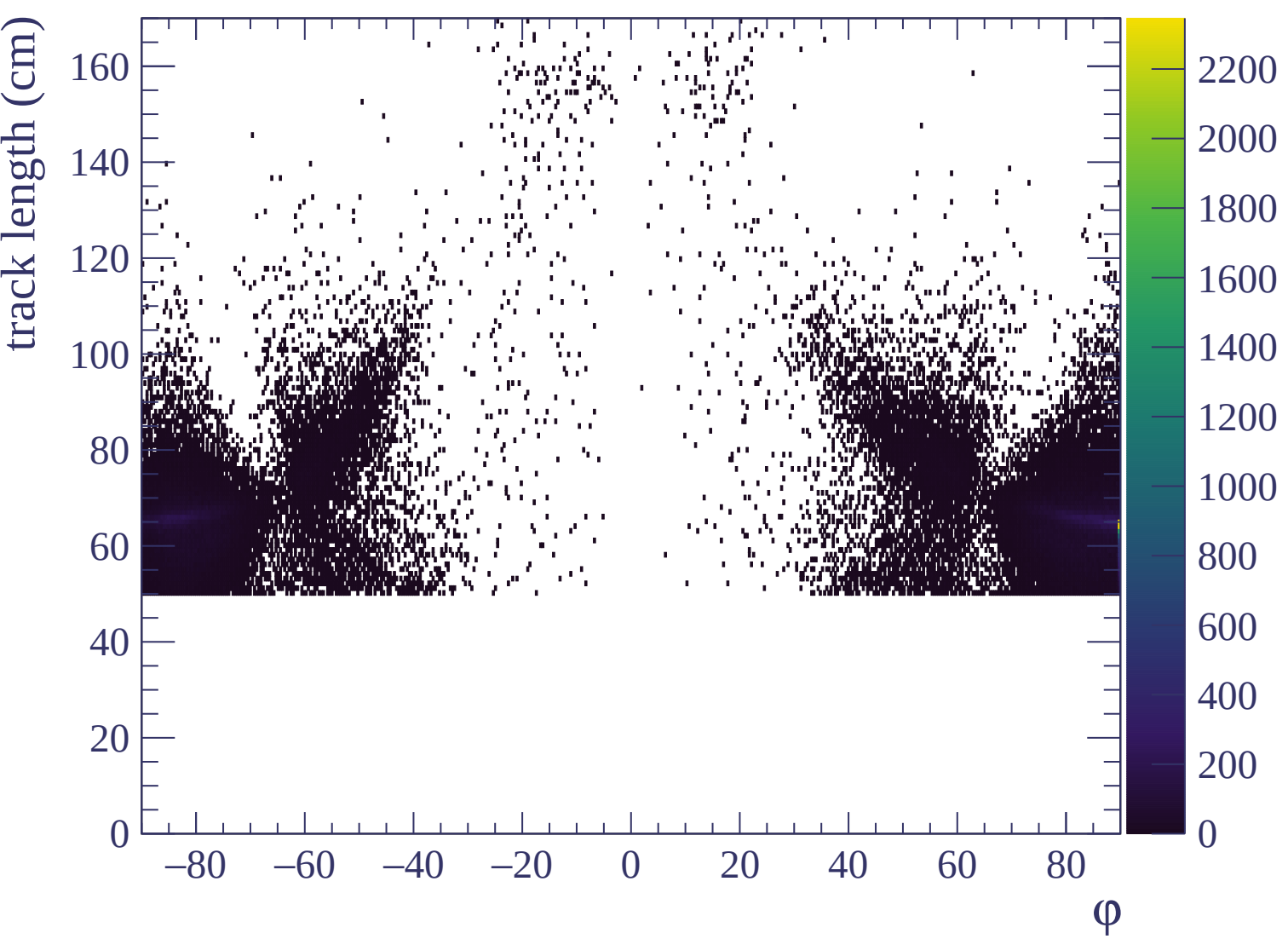












Energy loss | $-3000 < p < -2940$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2940 < p < -2880$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2880 < p < -2820

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2820 < p < -2760$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2760 < p < -2700$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2700 < p < -2640

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2640 < p < -2580$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2580 < p < -2520

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2520 < p < -2460

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2460 < p < -2400

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2400 < p < -2340

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2340 < p < -2280$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2280 < p < -2220

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2220 < p < -2160$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2160 < p < -2100

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | -2100 < p < -2040

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2040 < p < -1980$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1980 < p < -1920

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1920 < p < -1860

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1860 < p < -1800$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1800 < p < -1740

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | -1740 < p < -1680

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1680 < p < -1620$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

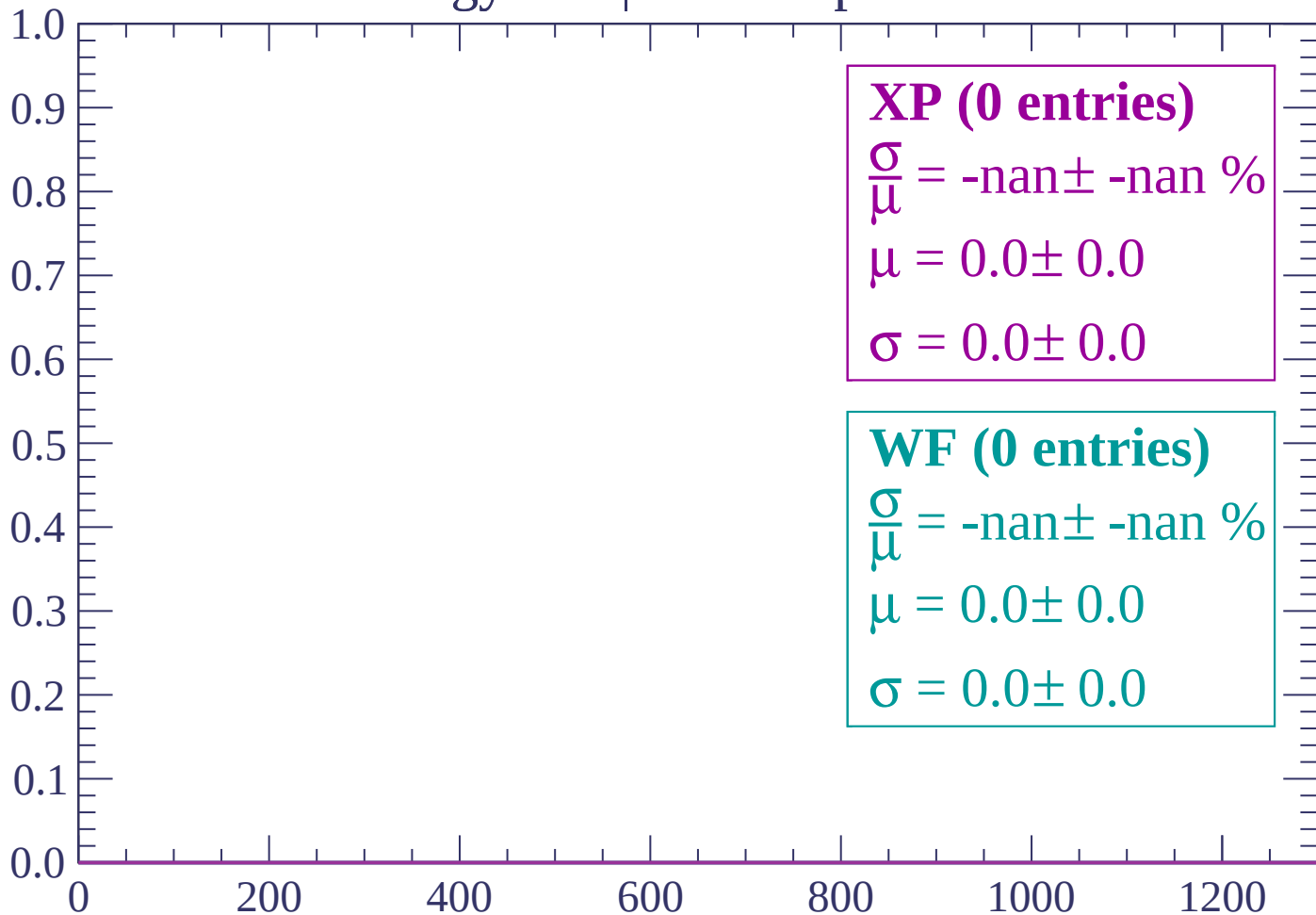
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1620 < p < -1560$

Count



dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1500$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1500 < p < -1440

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1440 < p < -1380

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1380 < p < -1320$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1320 < p < -1260$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

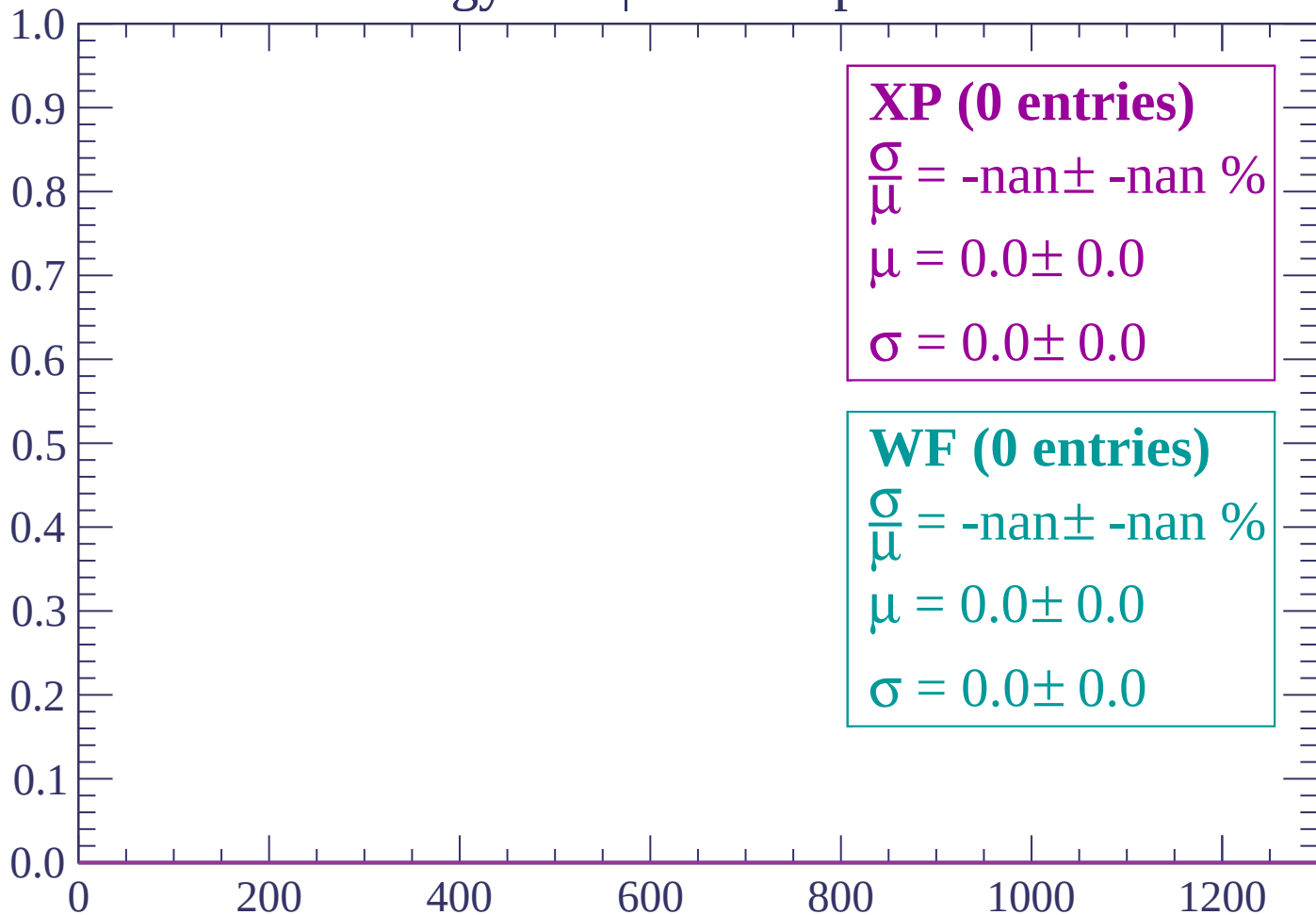
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1260 < p < -1200$

Count



dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1140 < p < -1080

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1020$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

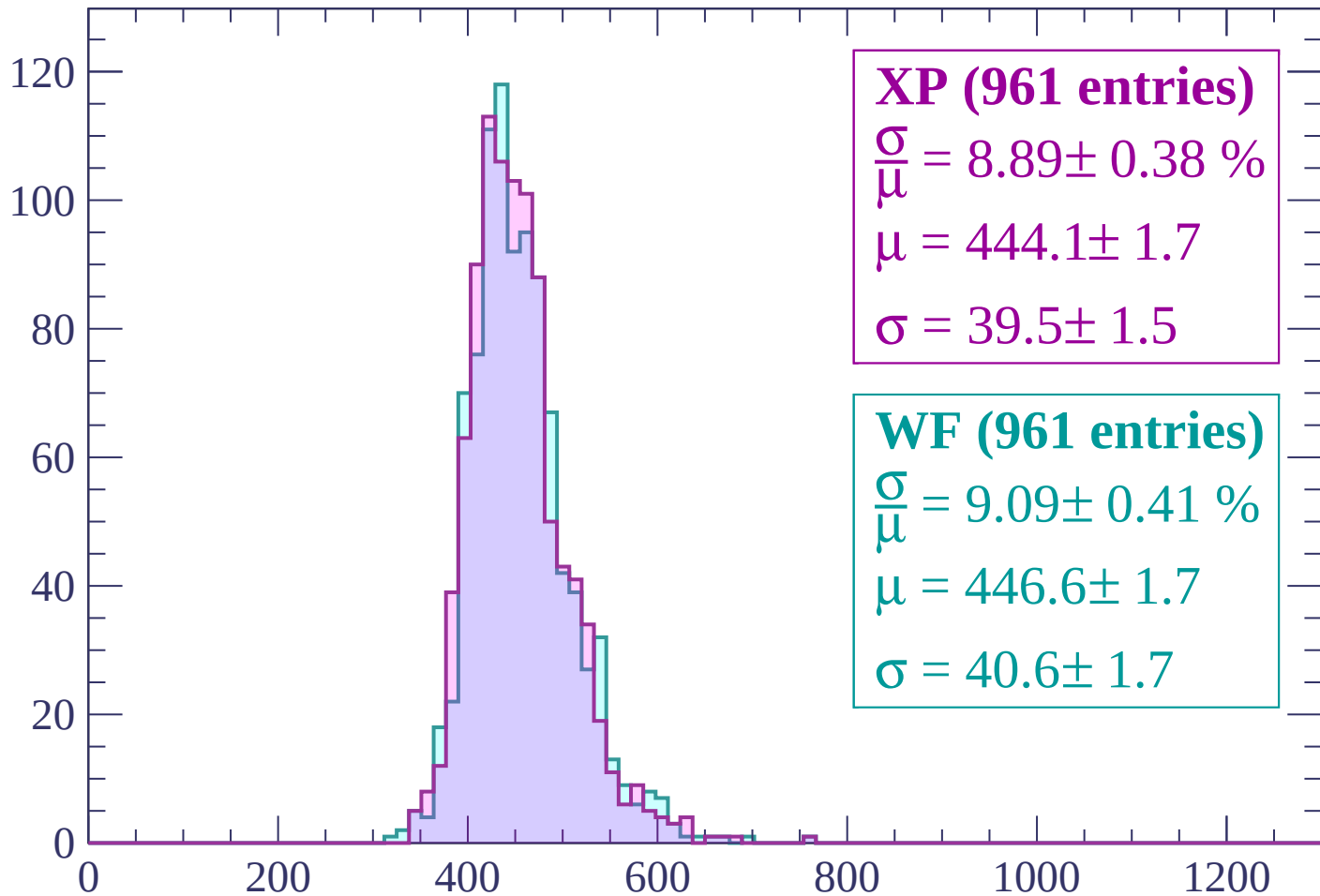
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1020 < p < -960$

Count



XP (961 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.38 \%$

$\mu = 444.1 \pm 1.7$

$\sigma = 39.5 \pm 1.5$

WF (961 entries)

$\frac{\sigma}{\mu} = 9.09 \pm 0.41 \%$

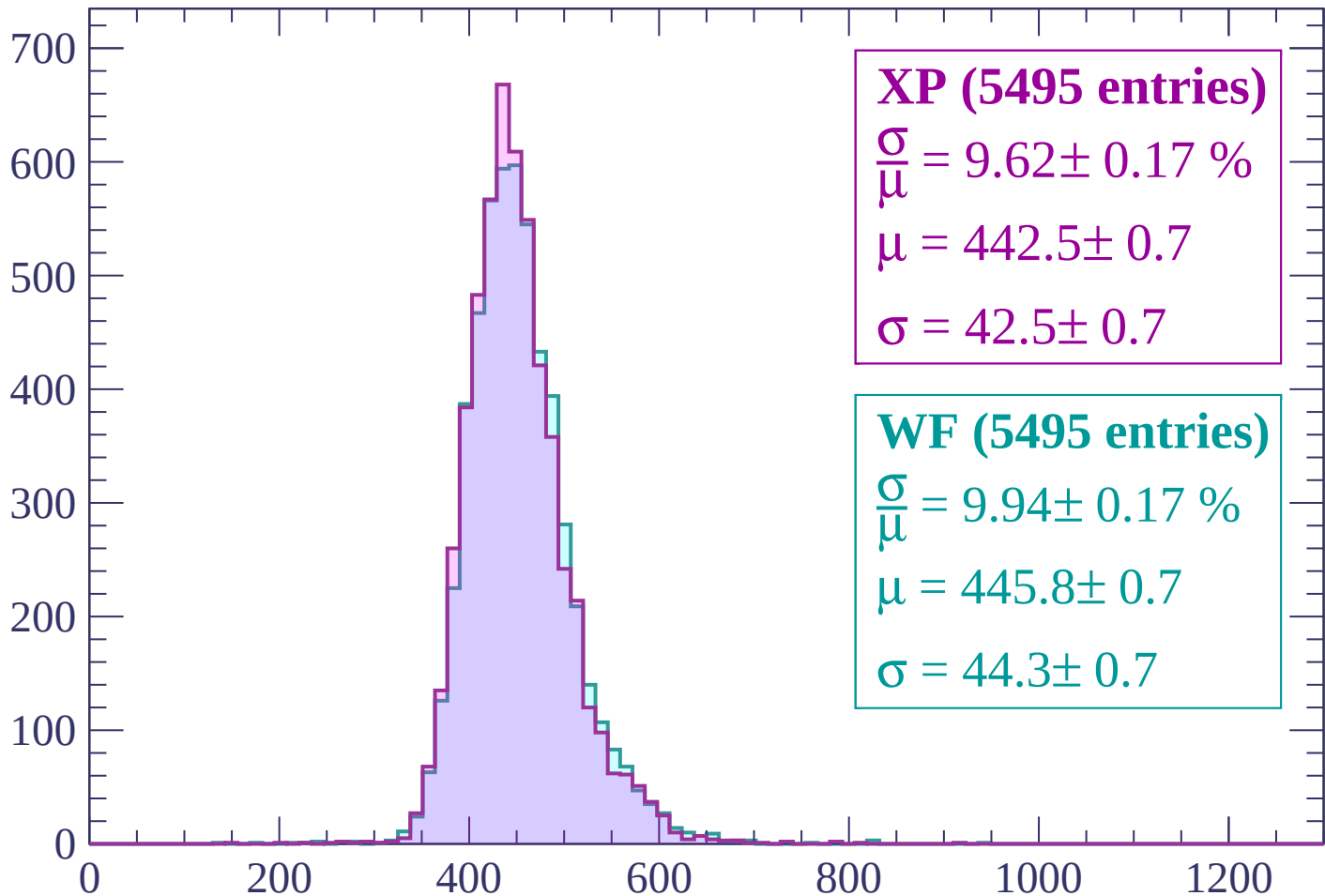
$\mu = 446.6 \pm 1.7$

$\sigma = 40.6 \pm 1.7$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (5495 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.17 \%$$

$$\mu = 442.5 \pm 0.7$$

$$\sigma = 42.5 \pm 0.7$$

WF (5495 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

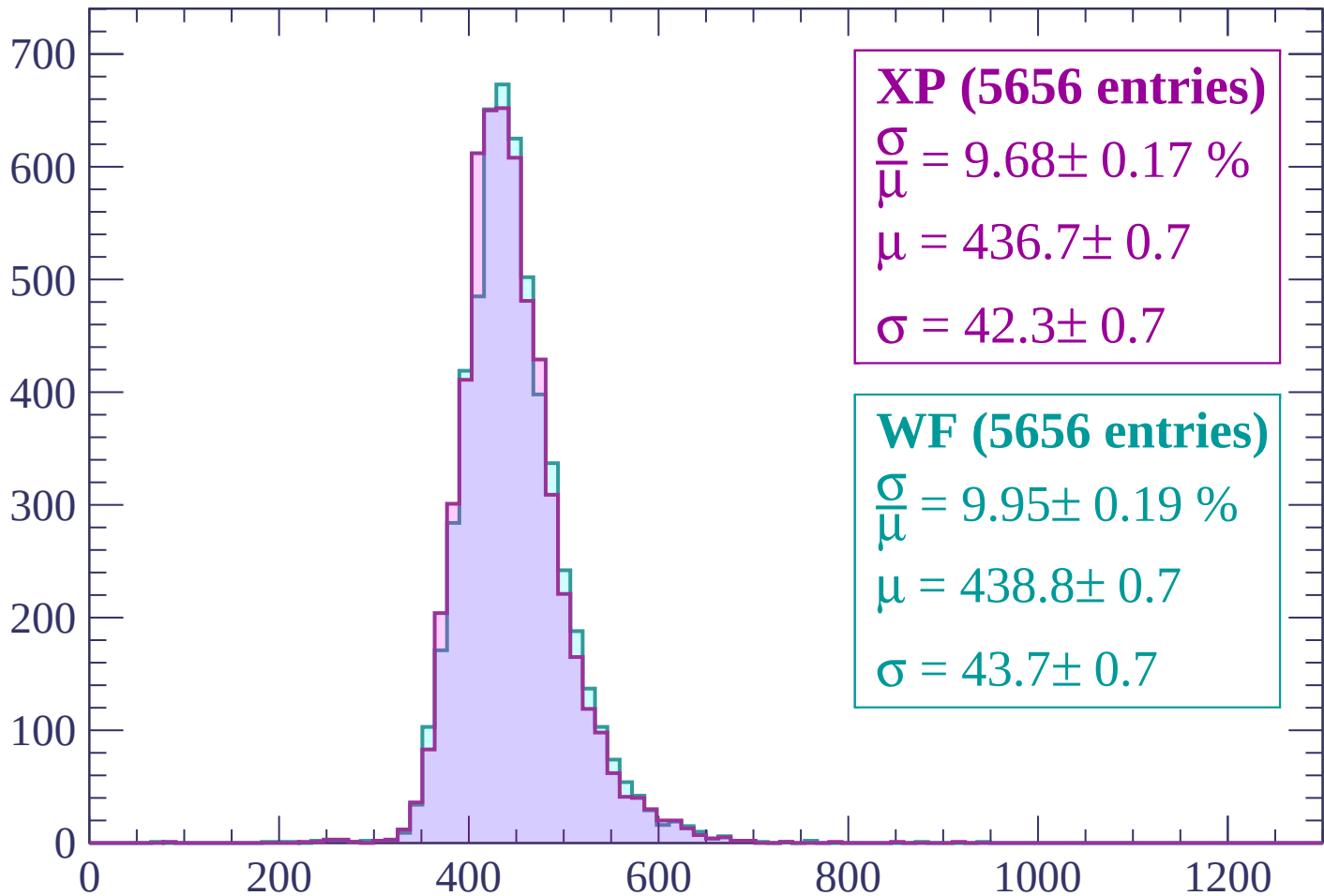
$$\mu = 445.8 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (5656 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.17 \%$$

$$\mu = 436.7 \pm 0.7$$

$$\sigma = 42.3 \pm 0.7$$

WF (5656 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.19 \%$$

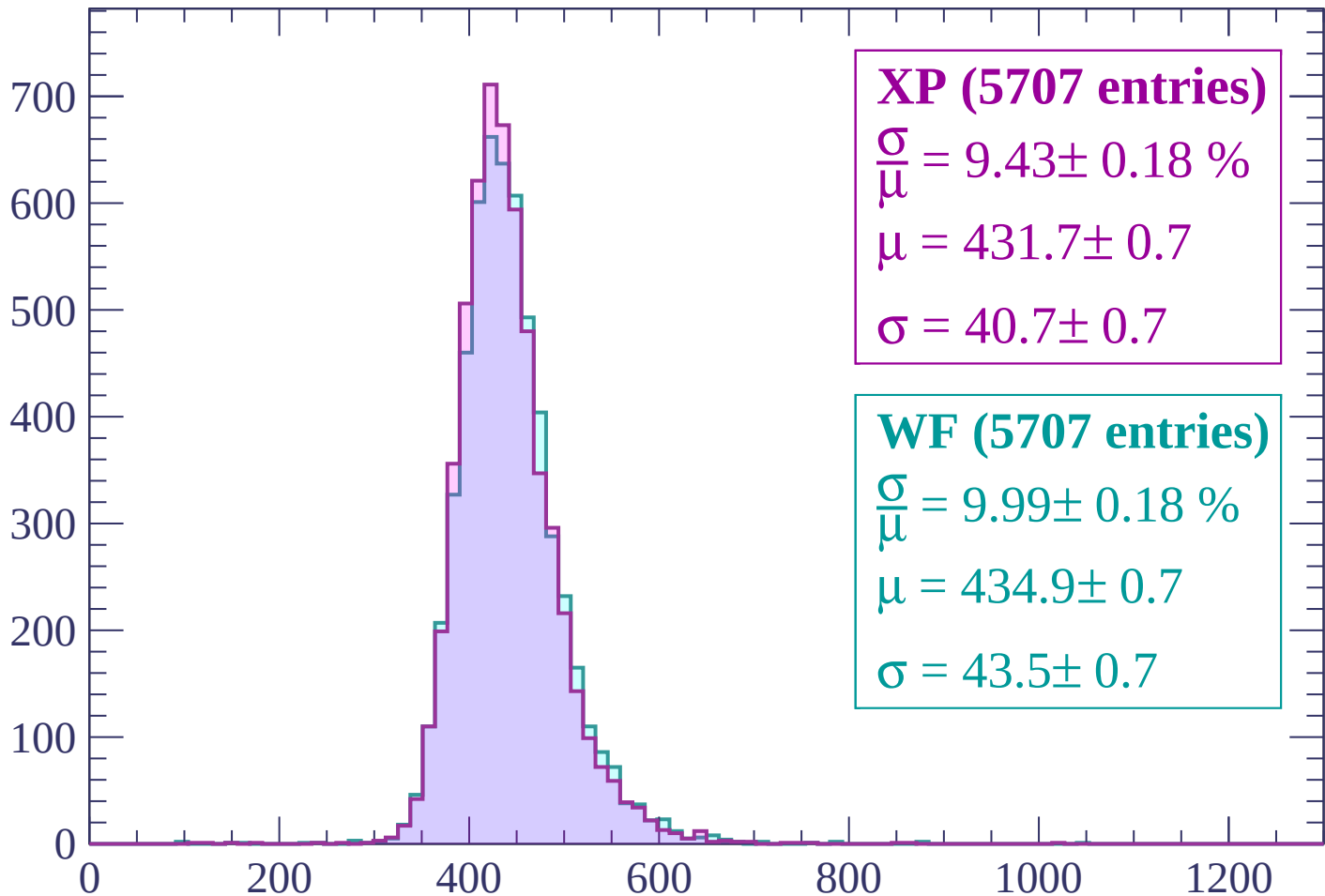
$$\mu = 438.8 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (5707 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

$$\mu = 431.7 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

WF (5707 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.18 \%$$

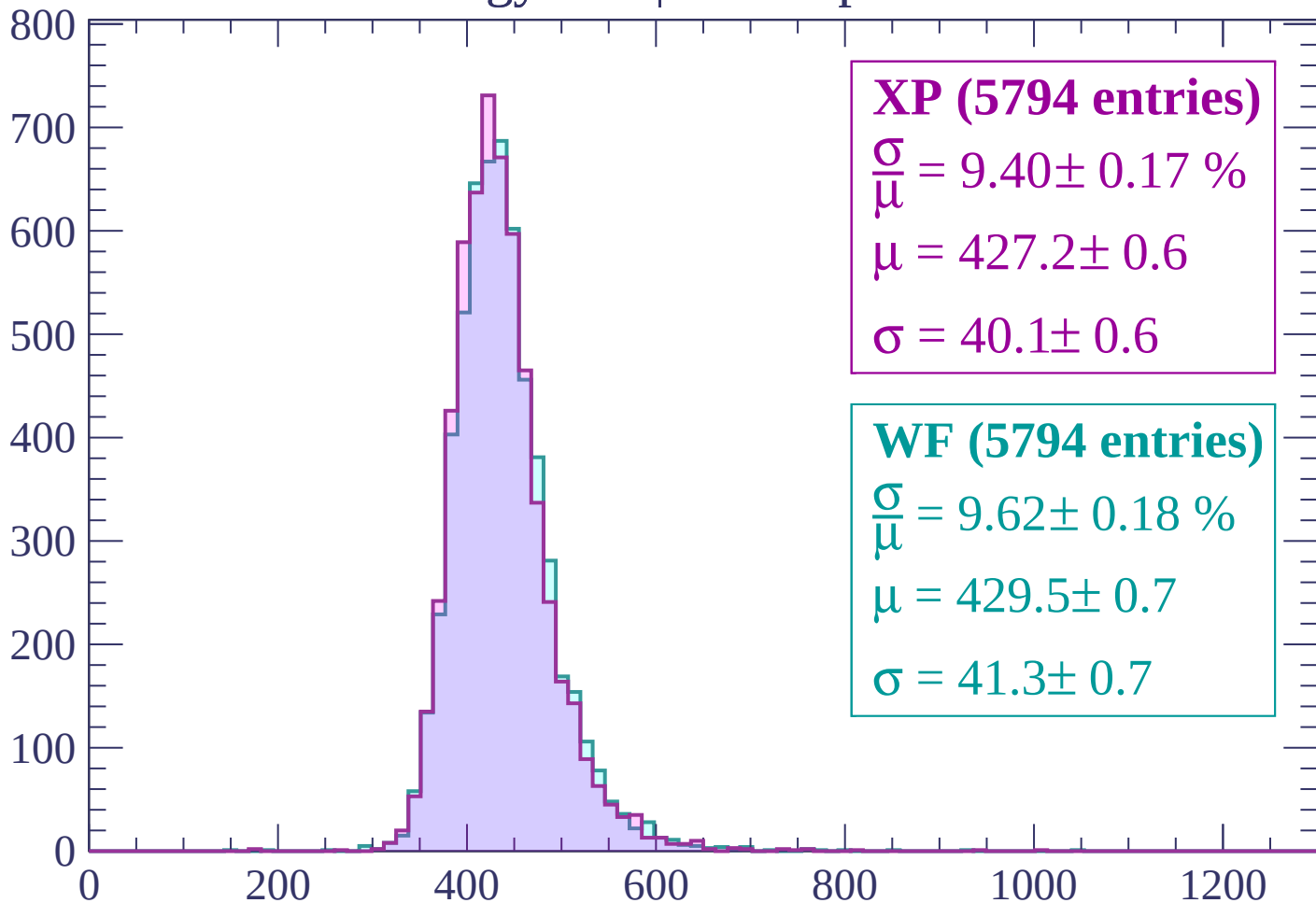
$$\mu = 434.9 \pm 0.7$$

$$\sigma = 43.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (5794 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.17 \%$$

$$\mu = 427.2 \pm 0.6$$

$$\sigma = 40.1 \pm 0.6$$

WF (5794 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

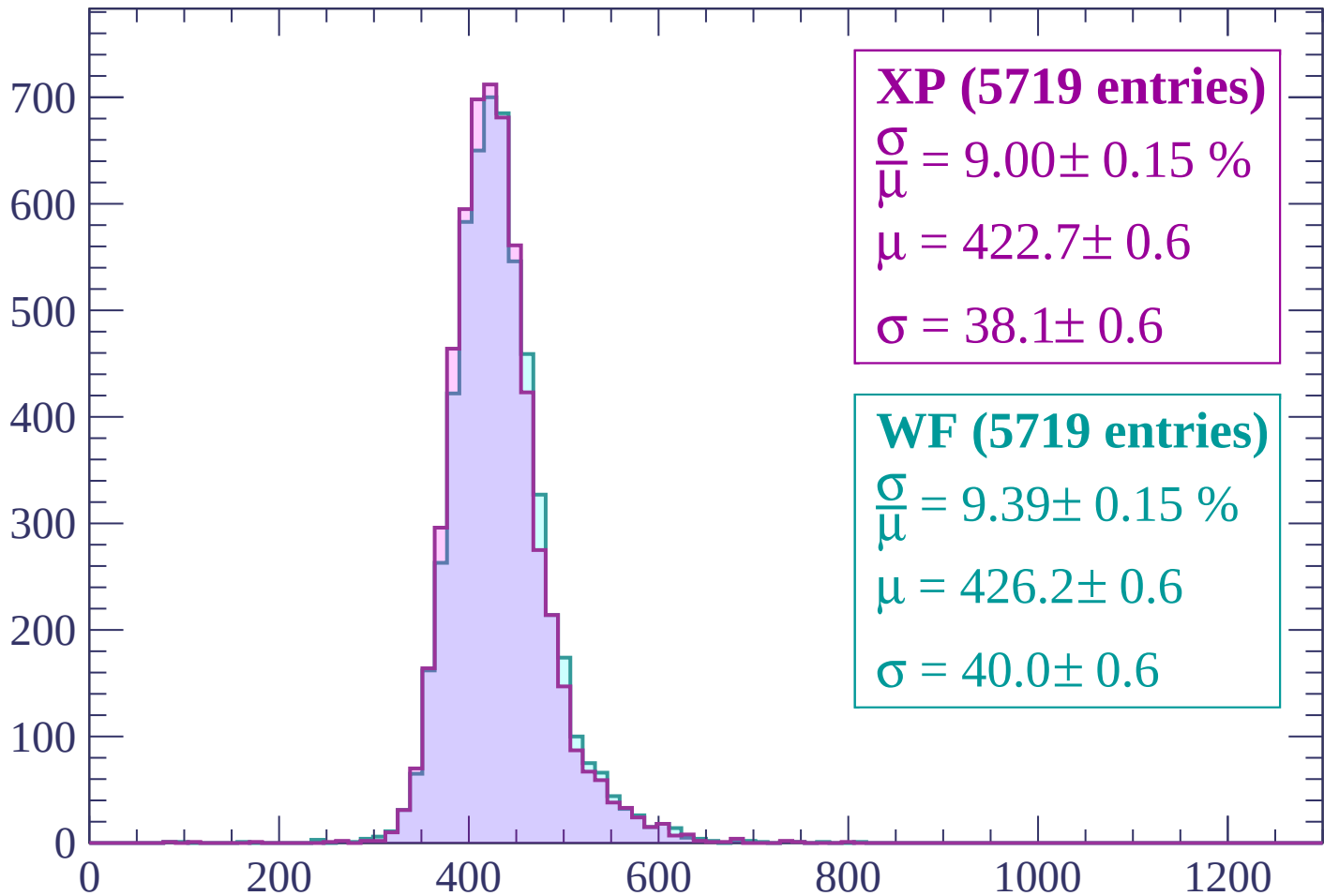
$$\mu = 429.5 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (5719 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.15 \%$

$\mu = 422.7 \pm 0.6$

$\sigma = 38.1 \pm 0.6$

WF (5719 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$

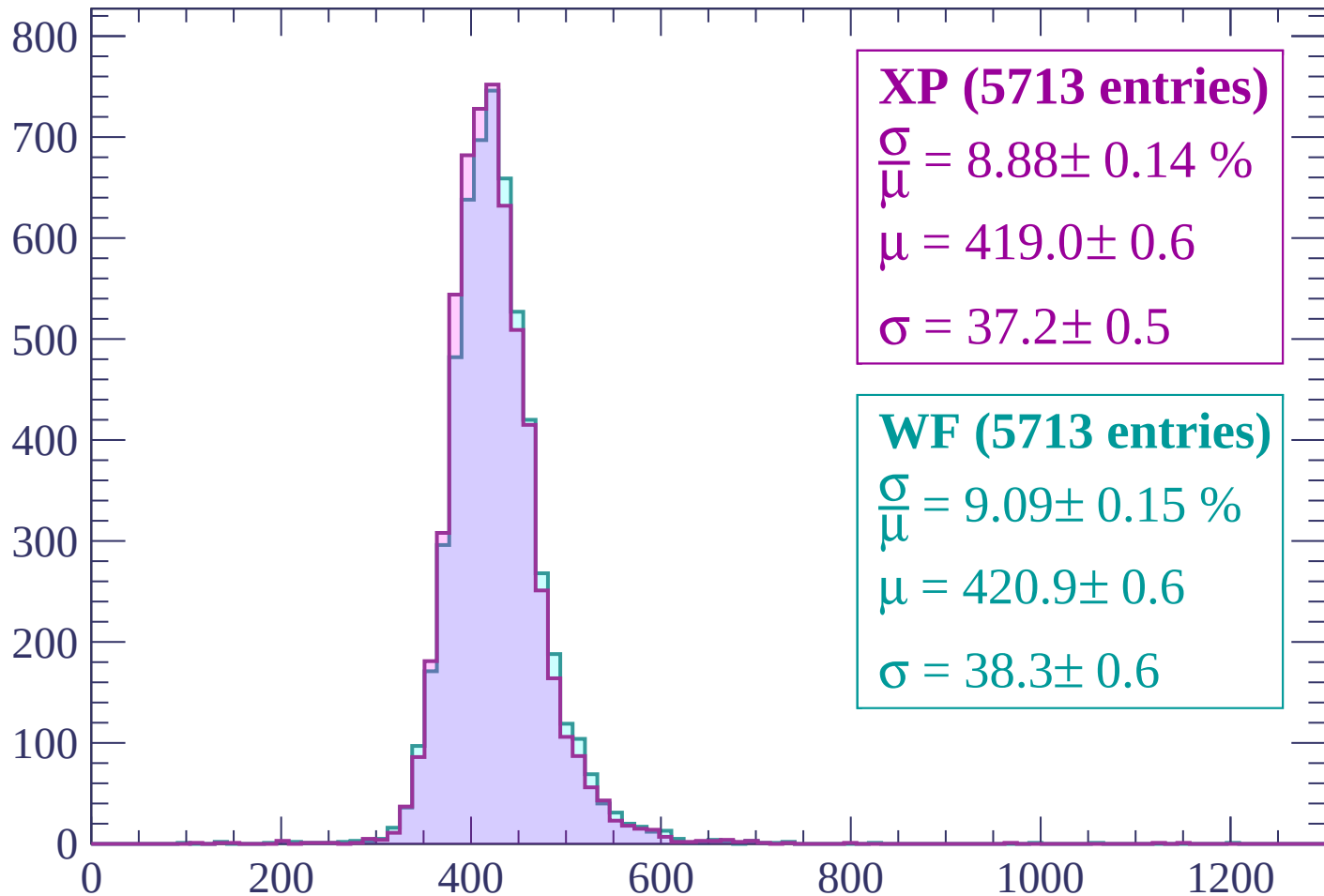
$\mu = 426.2 \pm 0.6$

$\sigma = 40.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (5713 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.14 \%$

$\mu = 419.0 \pm 0.6$

$\sigma = 37.2 \pm 0.5$

WF (5713 entries)

$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$

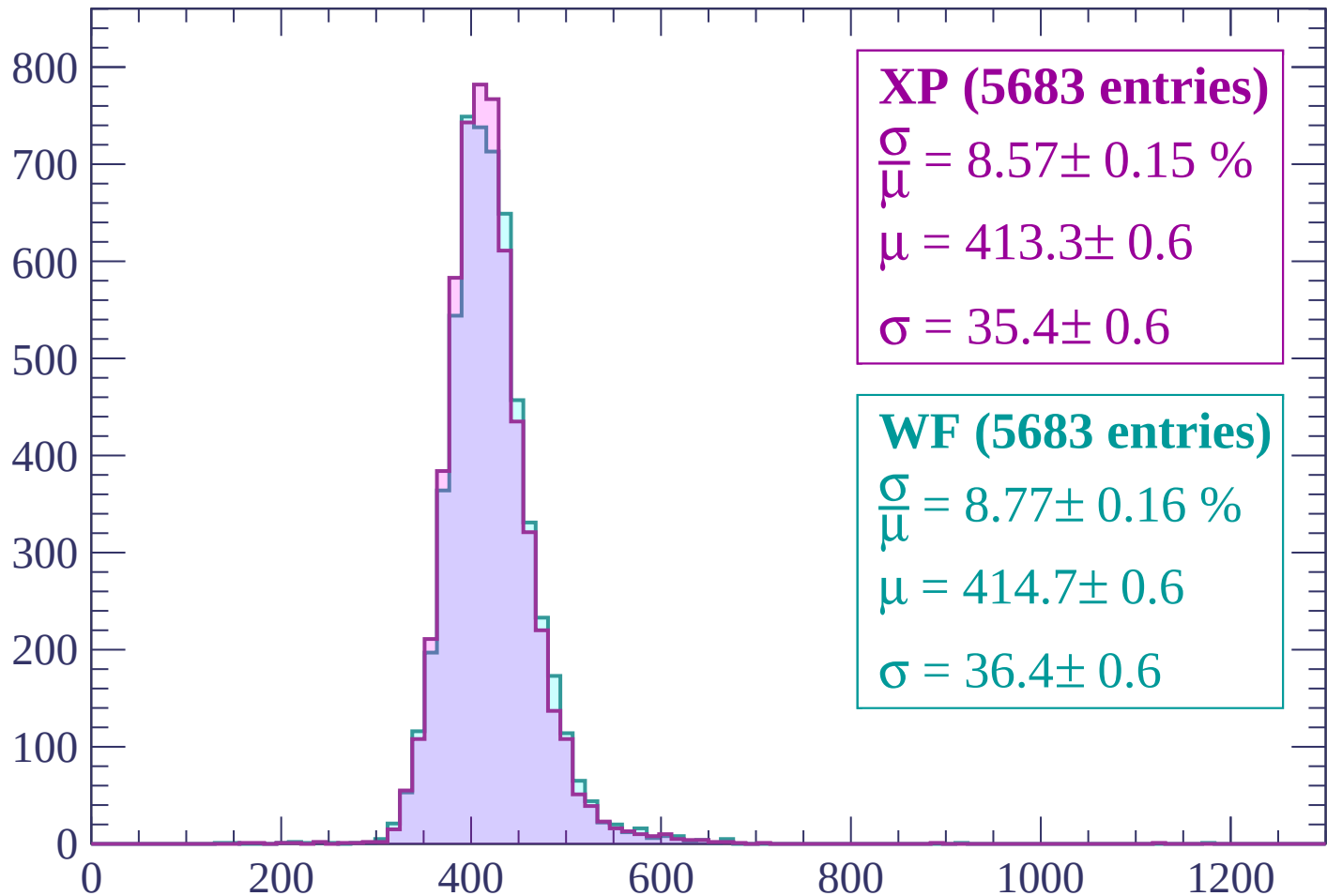
$\mu = 420.9 \pm 0.6$

$\sigma = 38.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (5683 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.15 \%$$

$$\mu = 413.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF (5683 entries)

$$\frac{\sigma}{\mu} = 8.77 \pm 0.16 \%$$

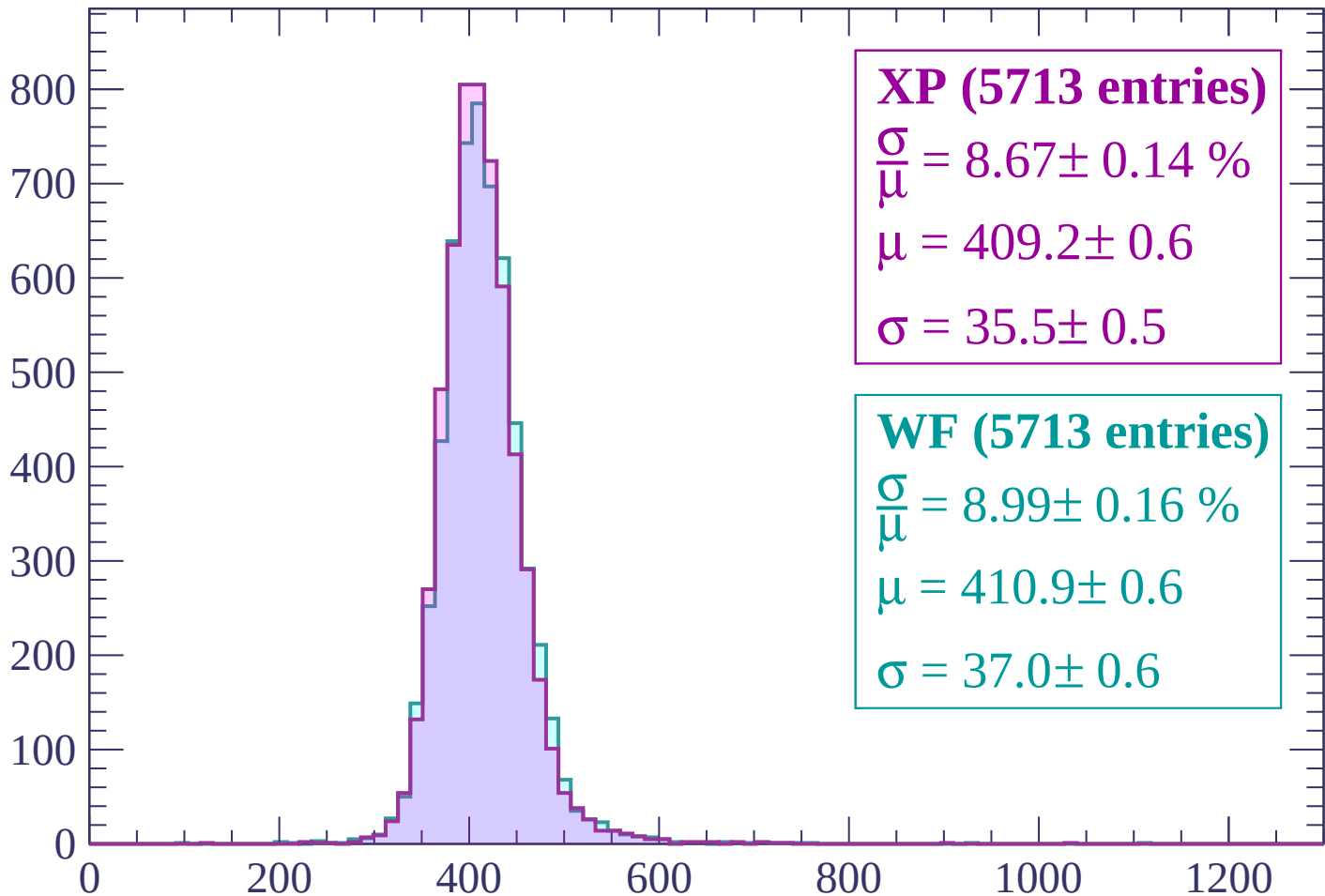
$$\mu = 414.7 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (5713 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.14 \%$

$\mu = 409.2 \pm 0.6$

$\sigma = 35.5 \pm 0.5$

WF (5713 entries)

$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$

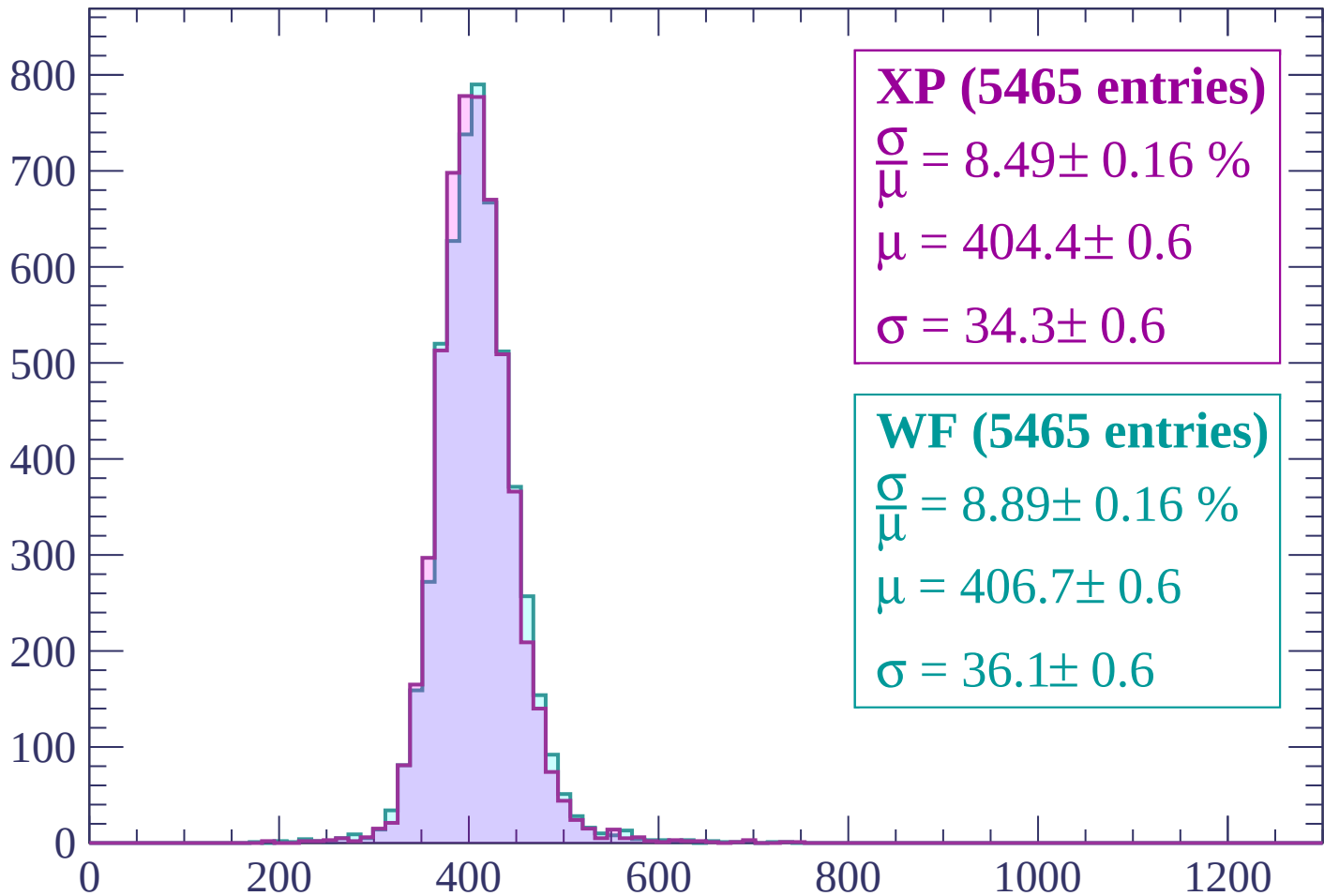
$\mu = 410.9 \pm 0.6$

$\sigma = 37.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.16 \%$$

$$\mu = 404.4 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

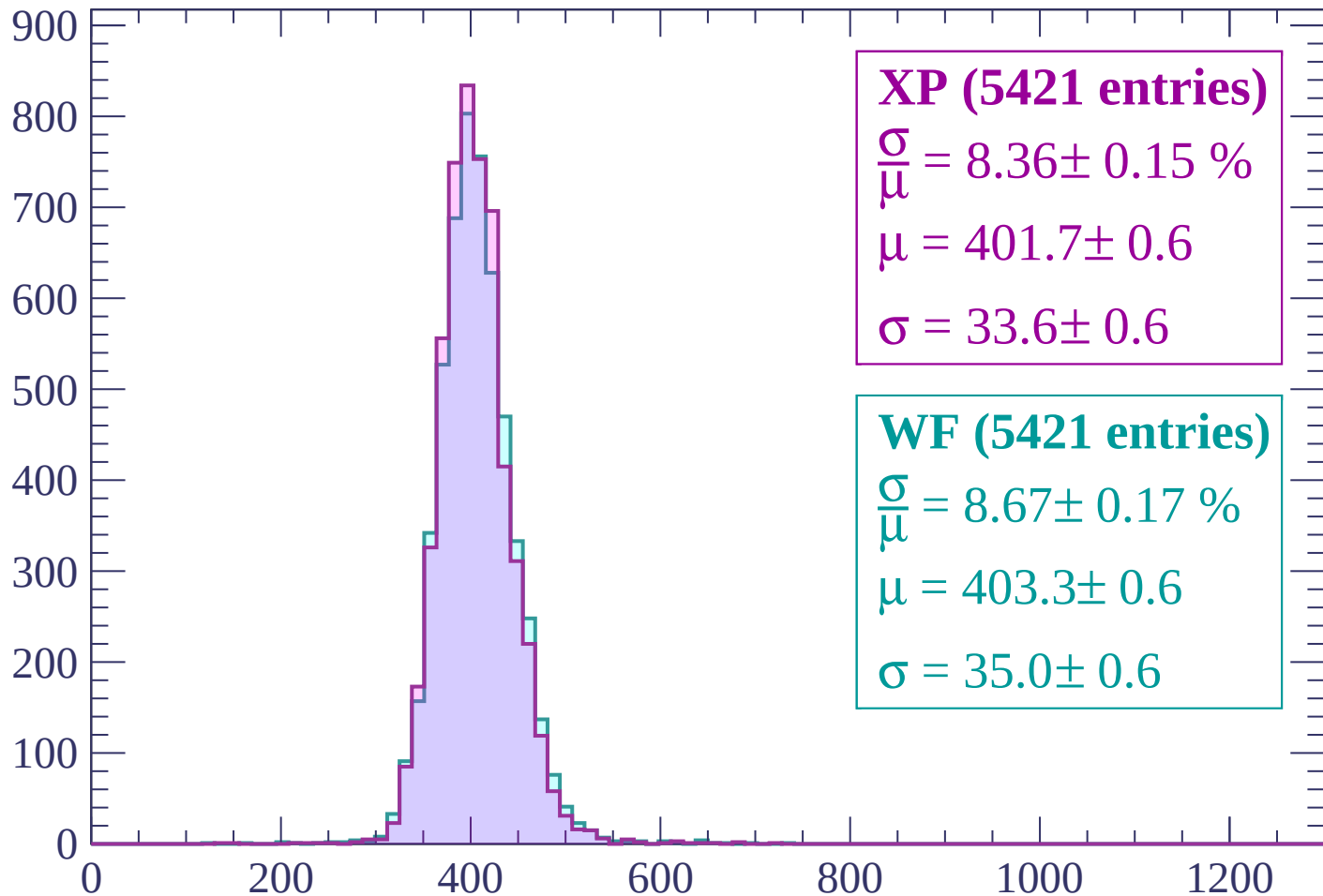
$$\mu = 406.7 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (5421 entries)

$\frac{\sigma}{\mu} = 8.36 \pm 0.15 \%$

$\mu = 401.7 \pm 0.6$

$\sigma = 33.6 \pm 0.6$

WF (5421 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.17 \%$

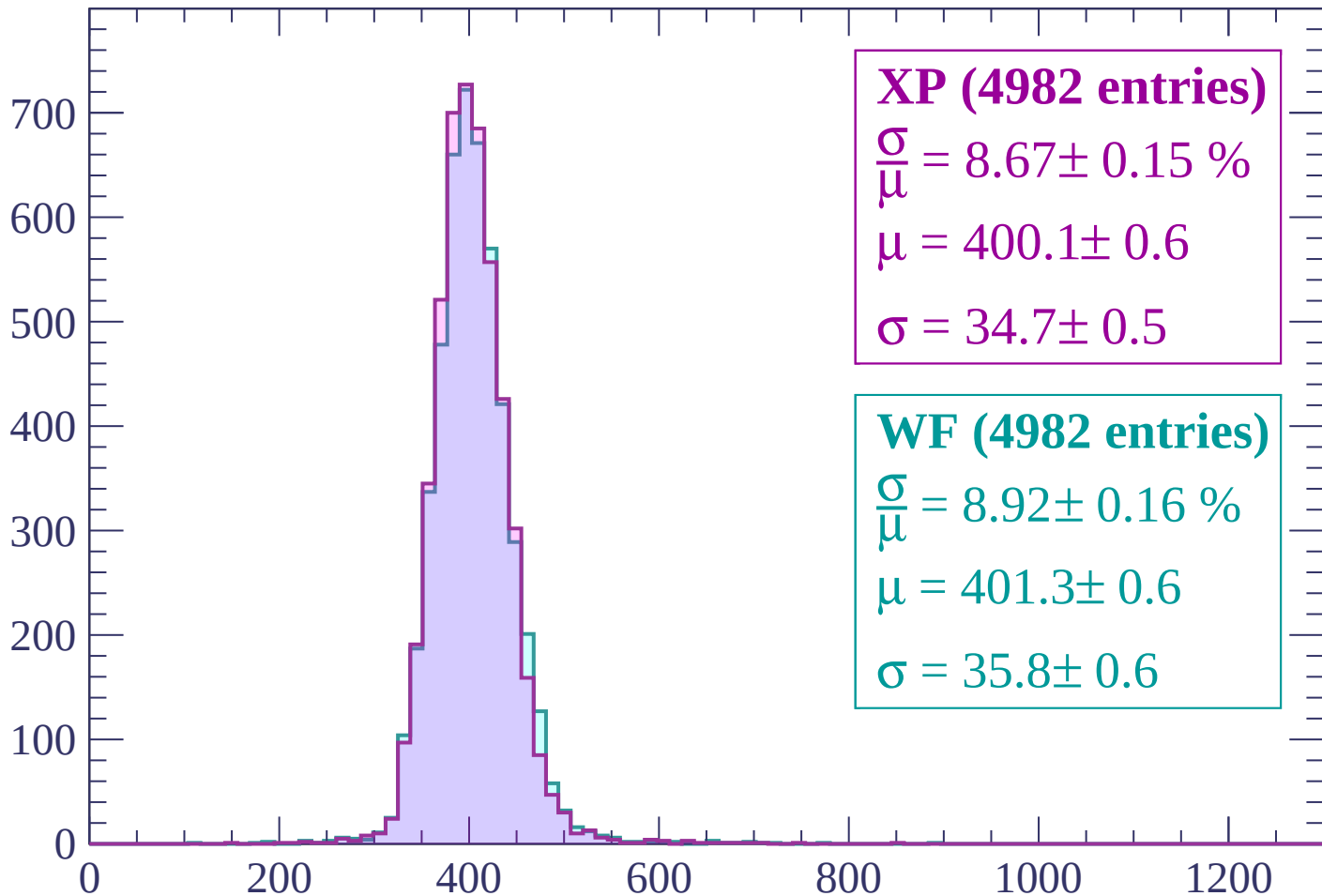
$\mu = 403.3 \pm 0.6$

$\sigma = 35.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (4982 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.15 \%$$

$$\mu = 400.1 \pm 0.6$$

$$\sigma = 34.7 \pm 0.5$$

WF (4982 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

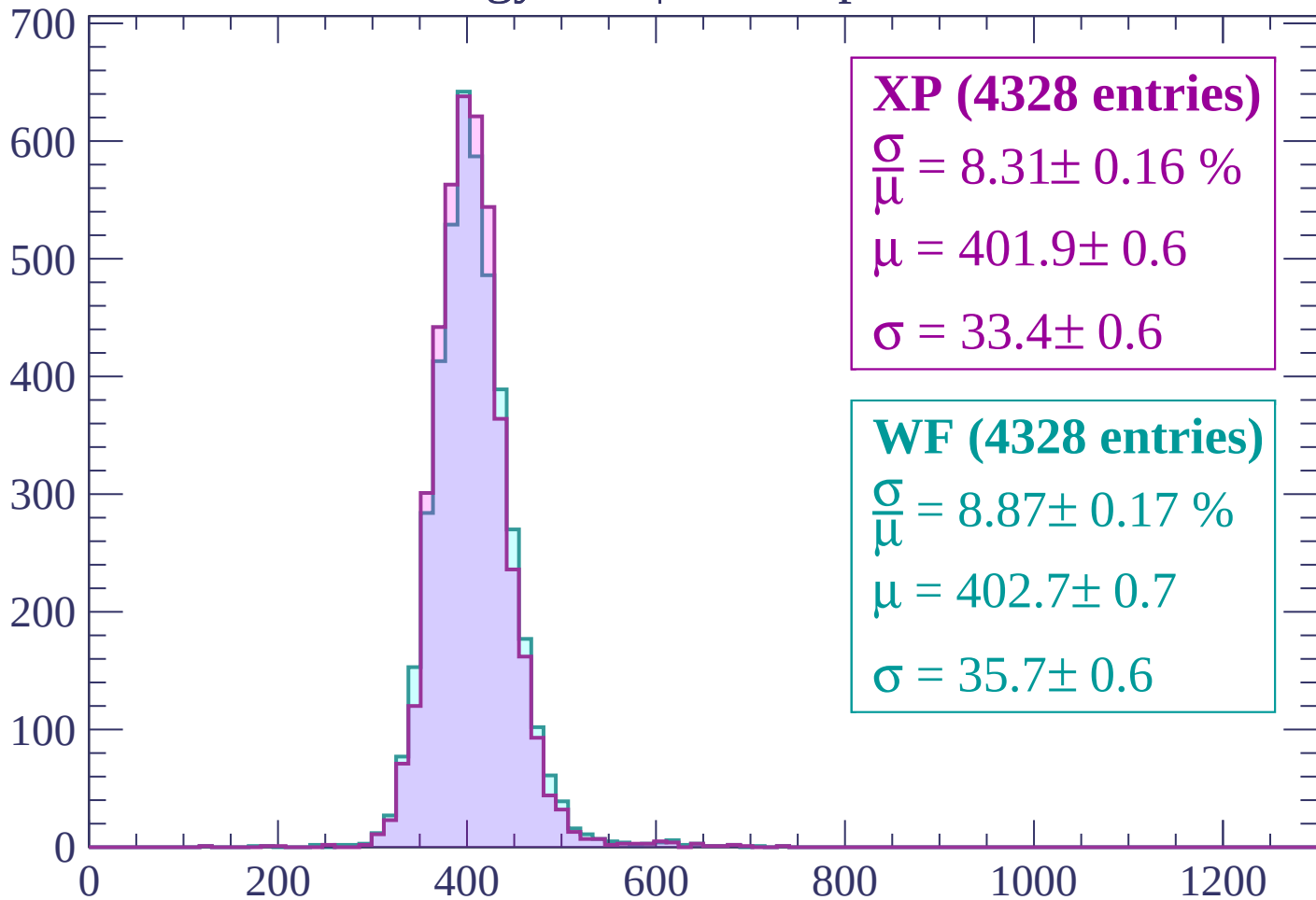
$$\mu = 401.3 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (4328 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.16 \%$$

$$\mu = 401.9 \pm 0.6$$

$$\sigma = 33.4 \pm 0.6$$

WF (4328 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.17 \%$$

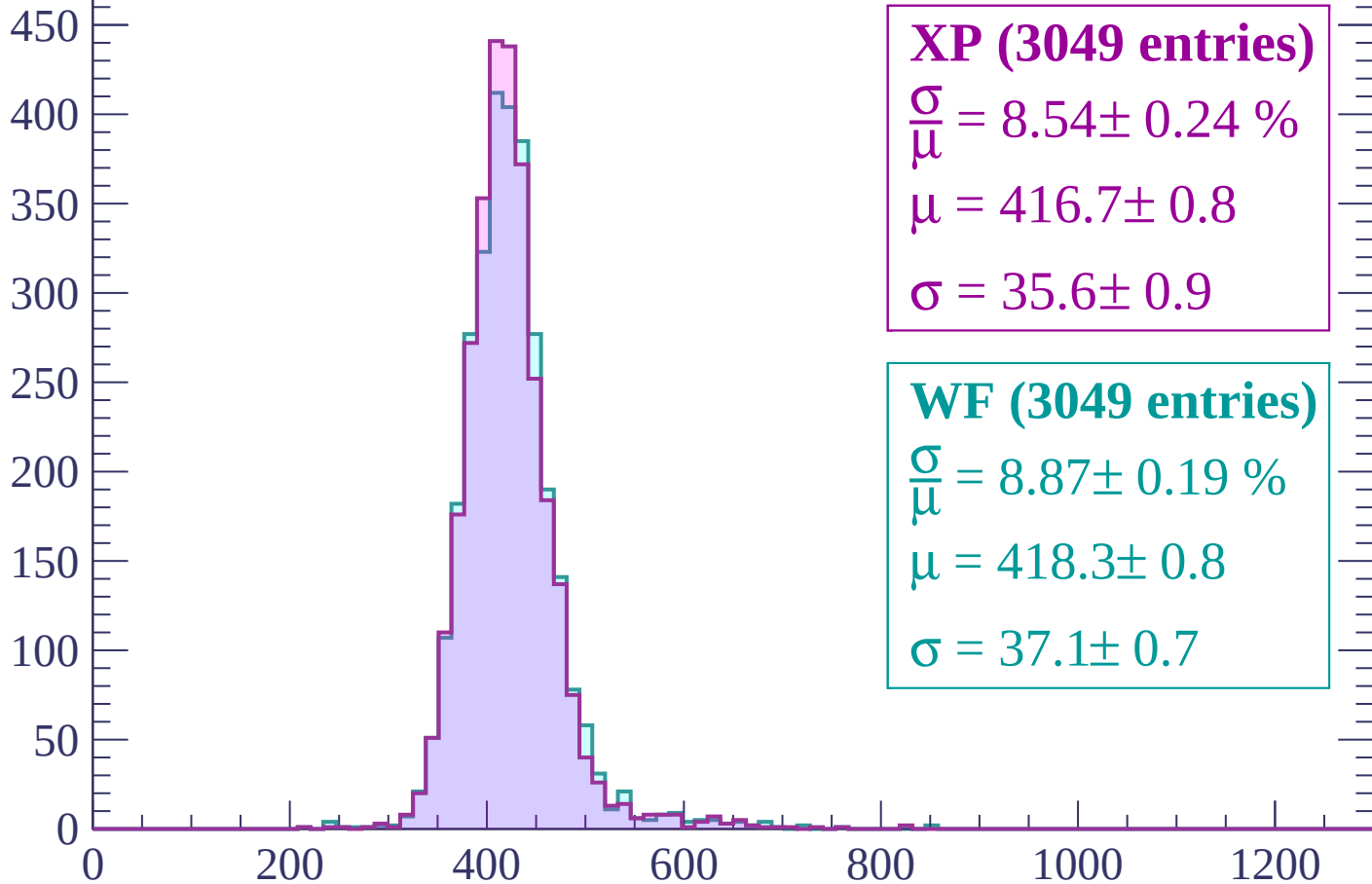
$$\mu = 402.7 \pm 0.7$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3049 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.24 \%$$

$$\mu = 416.7 \pm 0.8$$

$$\sigma = 35.6 \pm 0.9$$

WF (3049 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$$

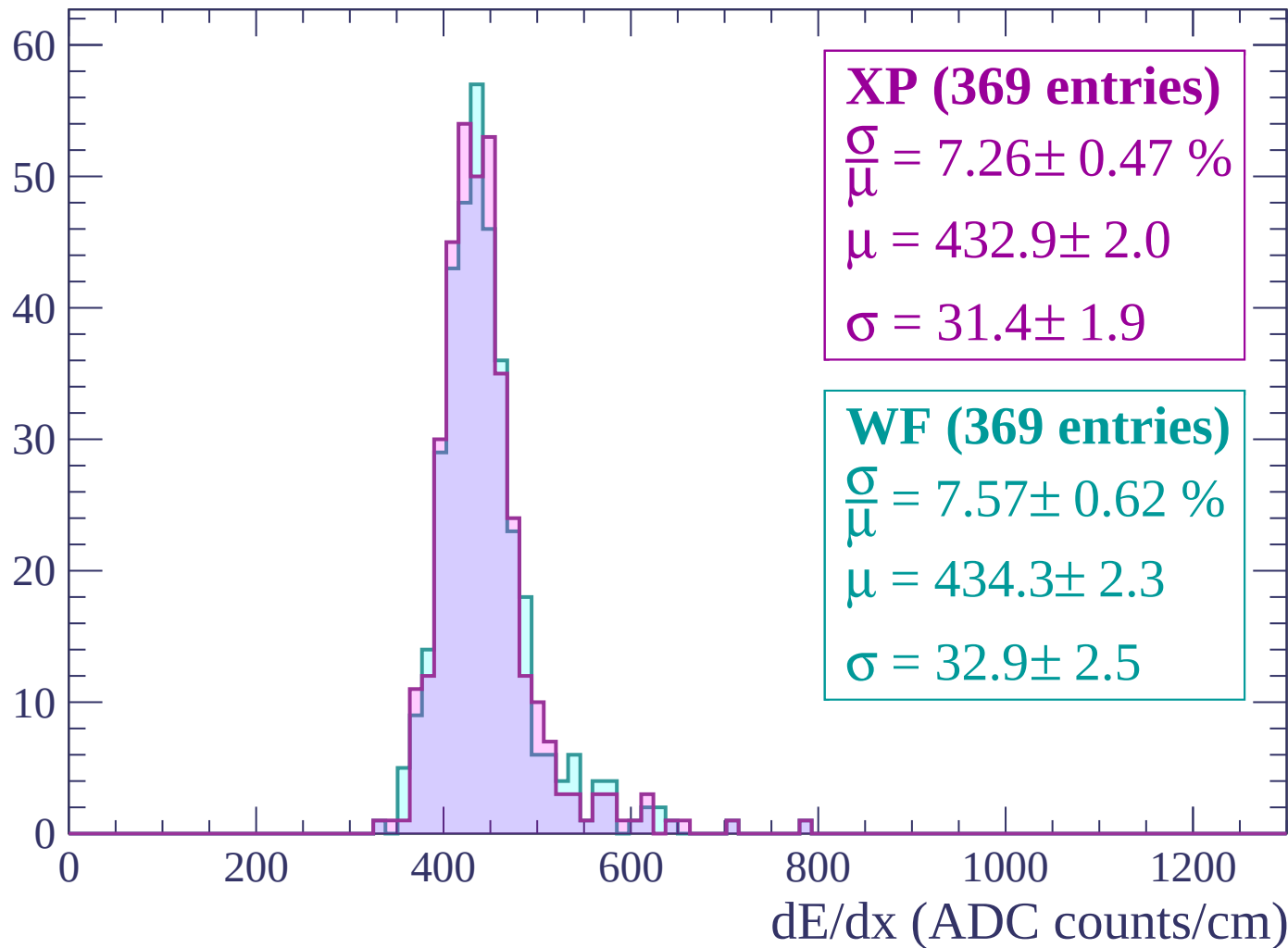
$$\mu = 418.3 \pm 0.8$$

$$\sigma = 37.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



Energy loss | $-120 < p < -60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

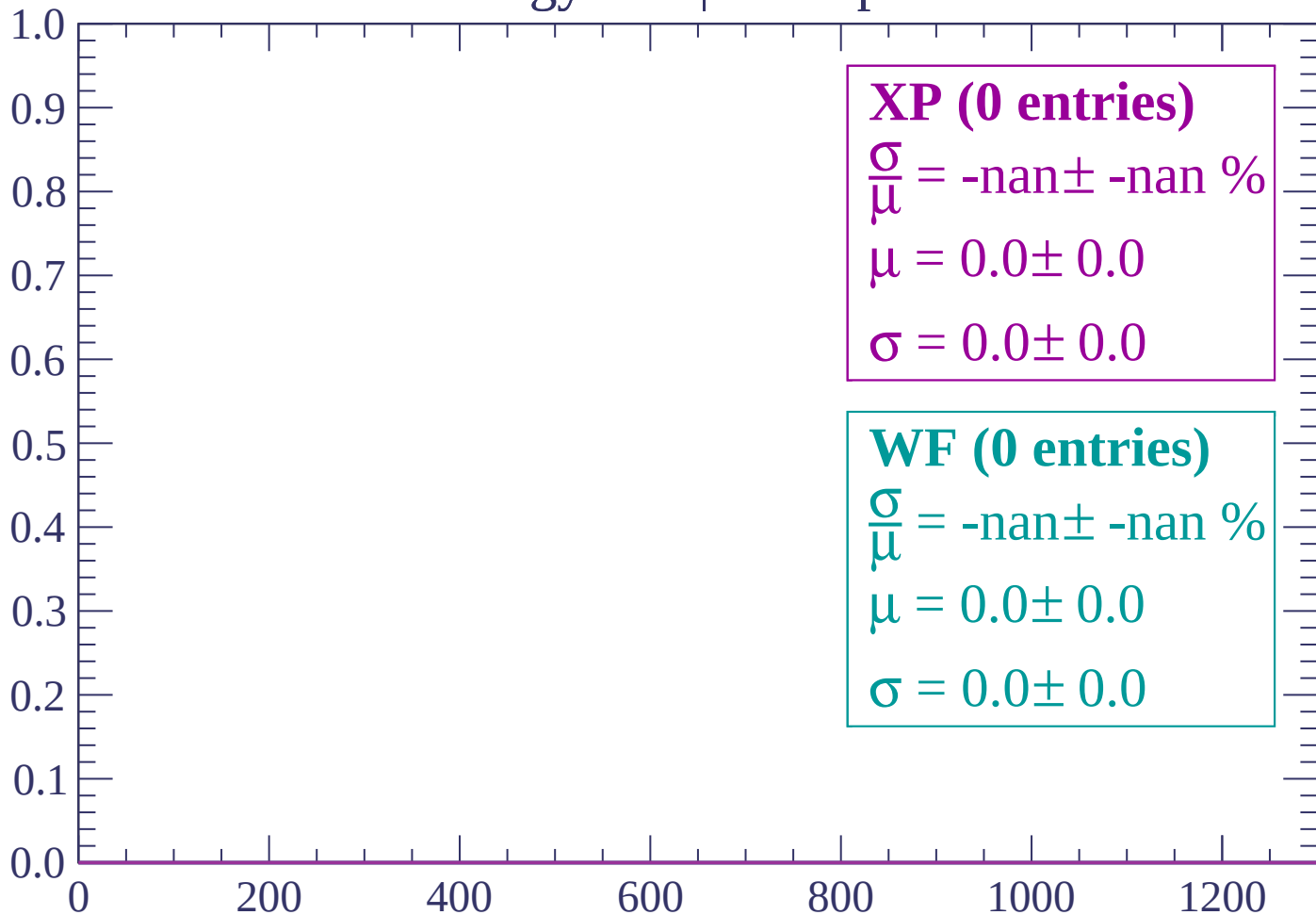
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-60 < p < 0$

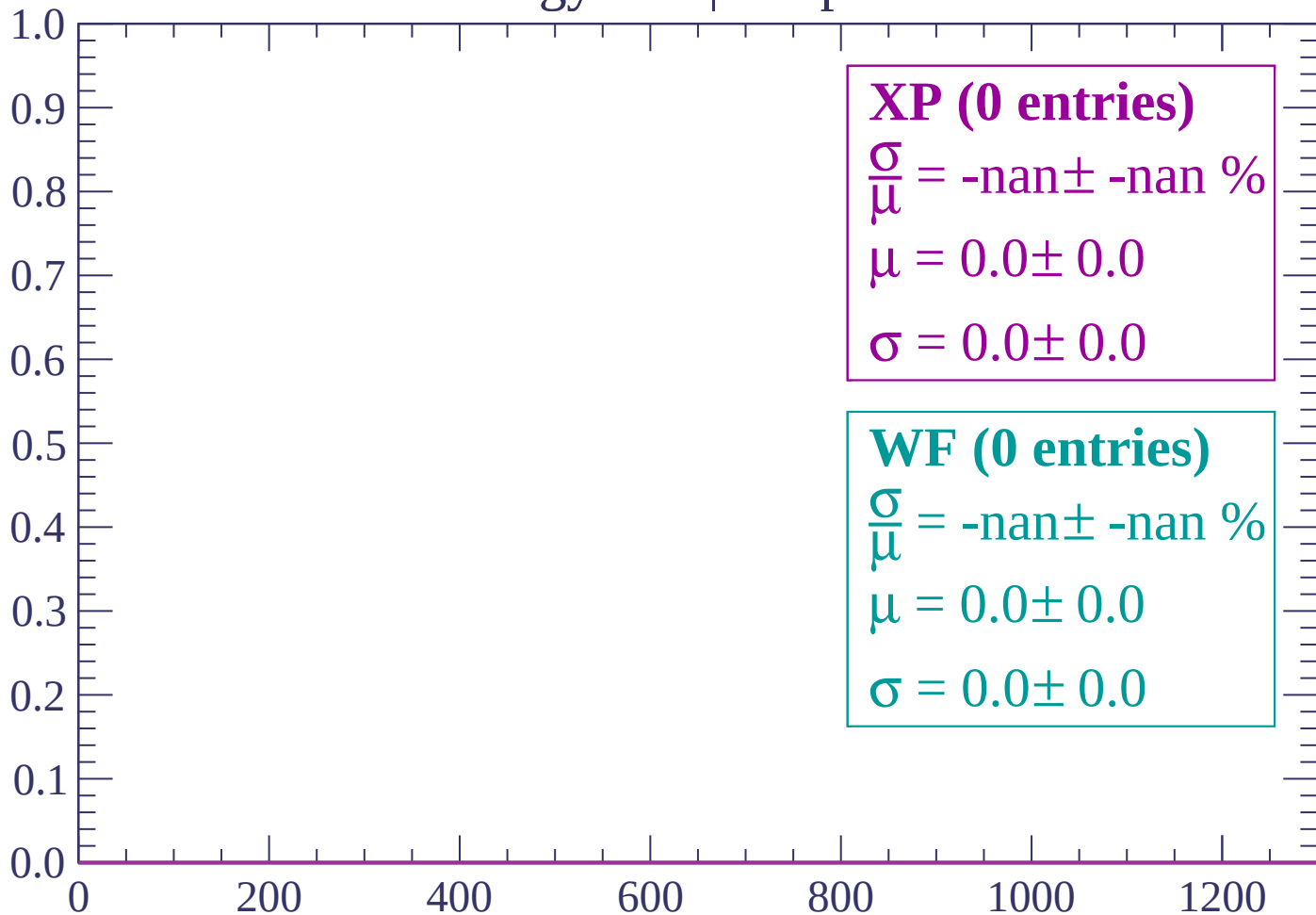
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < p < 60$

Count



dE/dx (ADC counts/cm)

Energy loss | $60 < p < 120$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $120 < p < 180$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

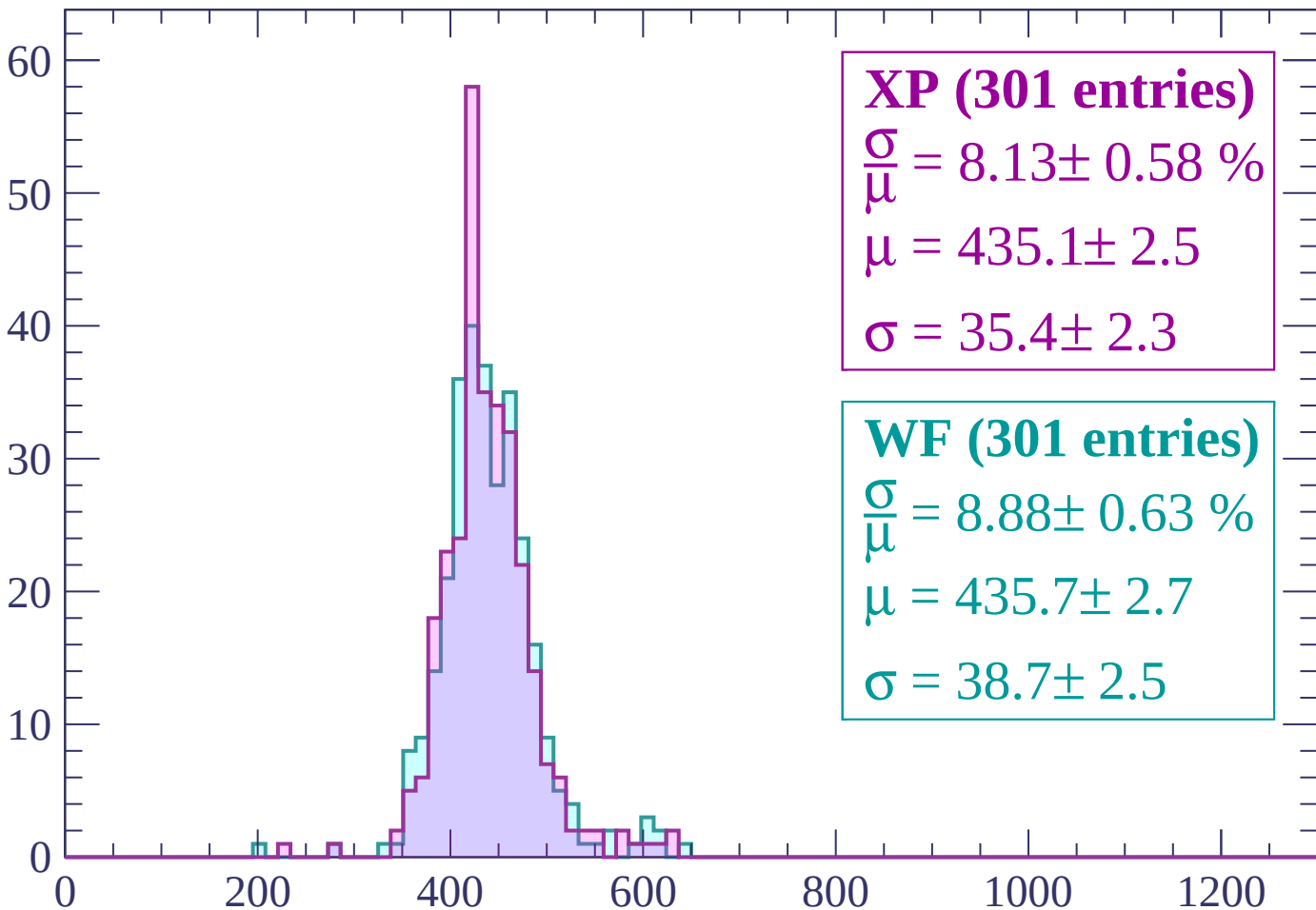
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $180 < p < 240$

Count



XP (301 entries)

$\frac{\sigma}{\mu} = 8.13 \pm 0.58 \%$

$\mu = 435.1 \pm 2.5$

$\sigma = 35.4 \pm 2.3$

WF (301 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.63 \%$

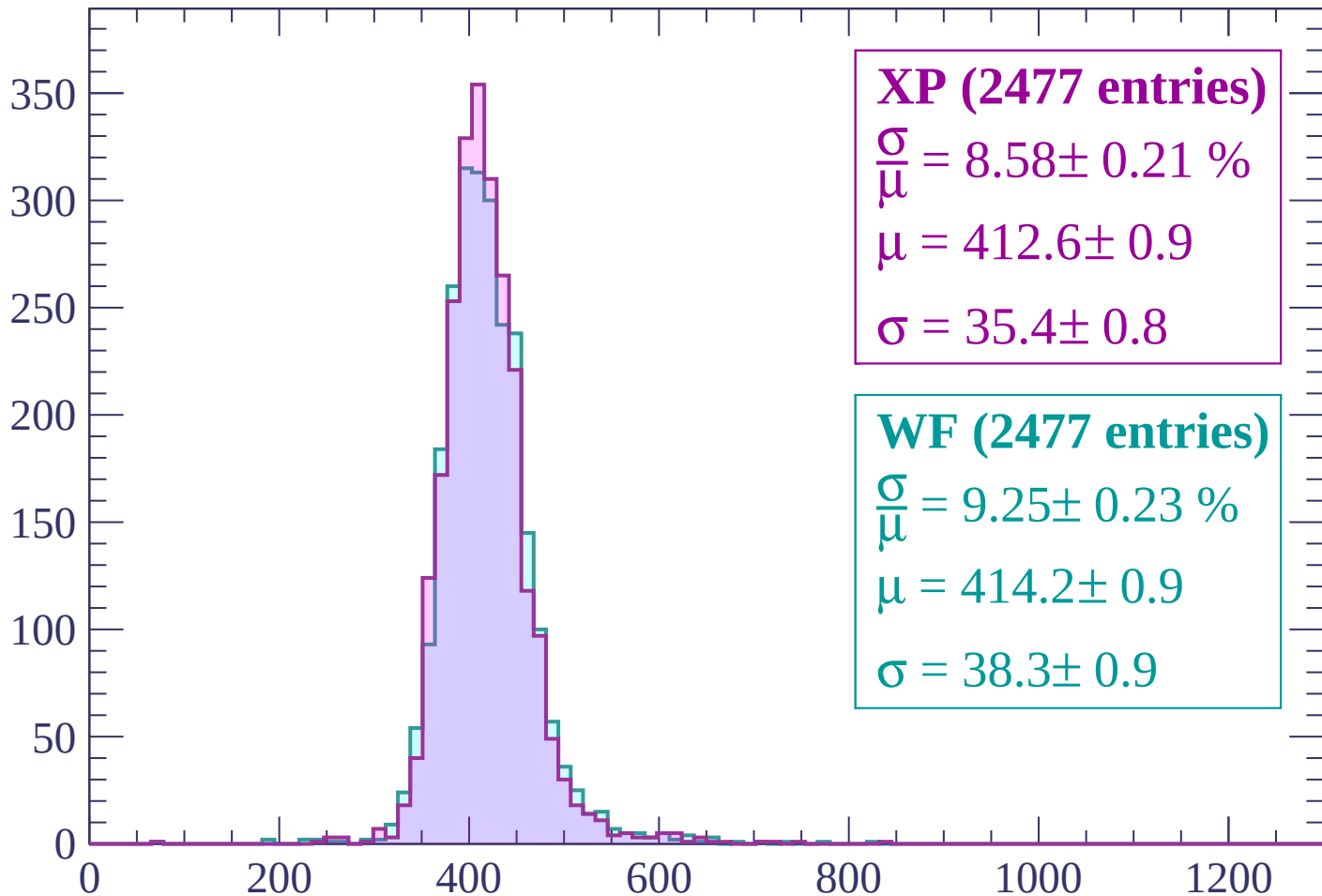
$\mu = 435.7 \pm 2.7$

$\sigma = 38.7 \pm 2.5$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (2477 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.21 \%$$

$$\mu = 412.6 \pm 0.9$$

$$\sigma = 35.4 \pm 0.8$$

WF (2477 entries)

$$\frac{\sigma}{\mu} = 9.25 \pm 0.23 \%$$

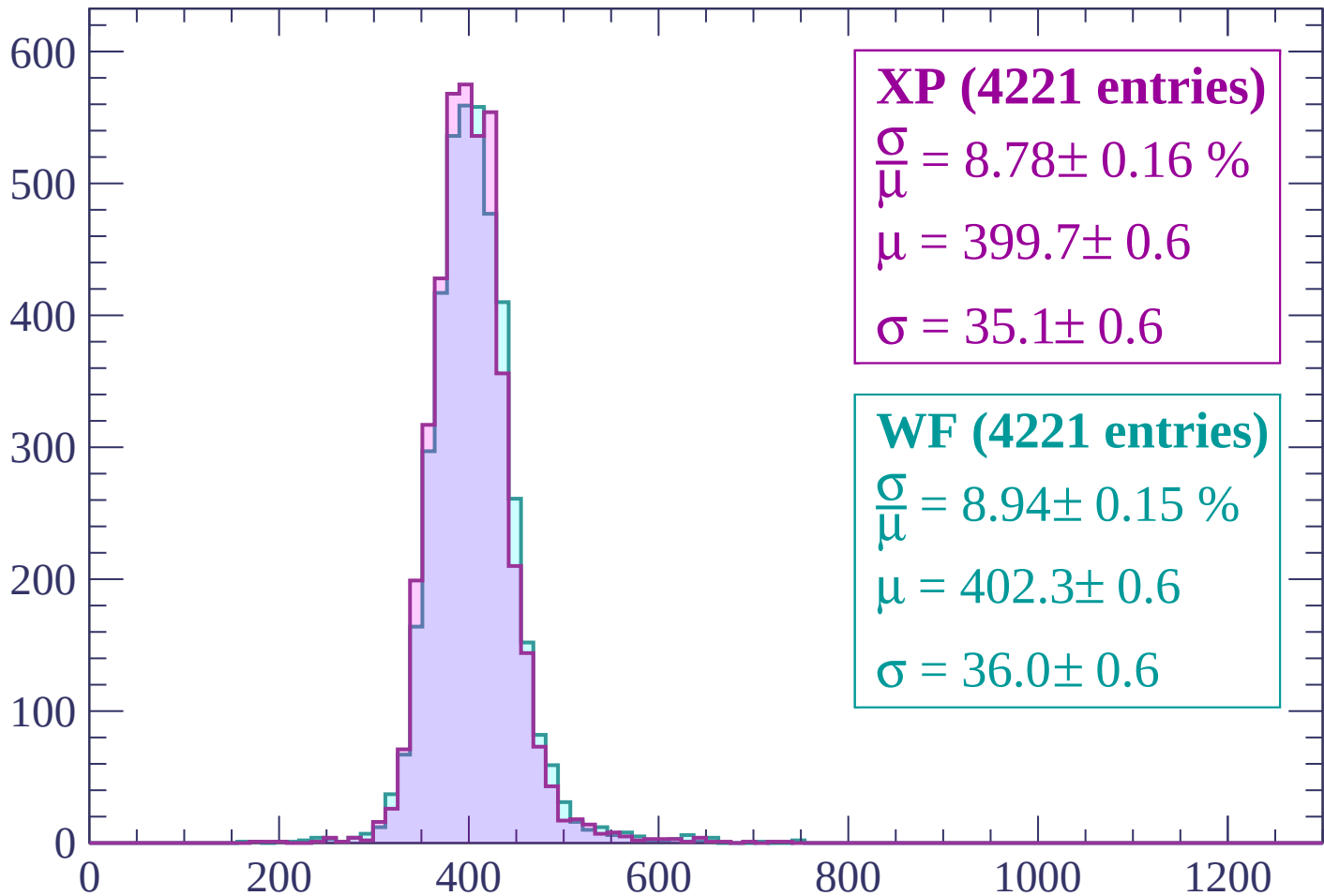
$$\mu = 414.2 \pm 0.9$$

$$\sigma = 38.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (4221 entries)

$$\frac{\sigma}{\mu} = 8.78 \pm 0.16 \%$$

$$\mu = 399.7 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

WF (4221 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.15 \%$$

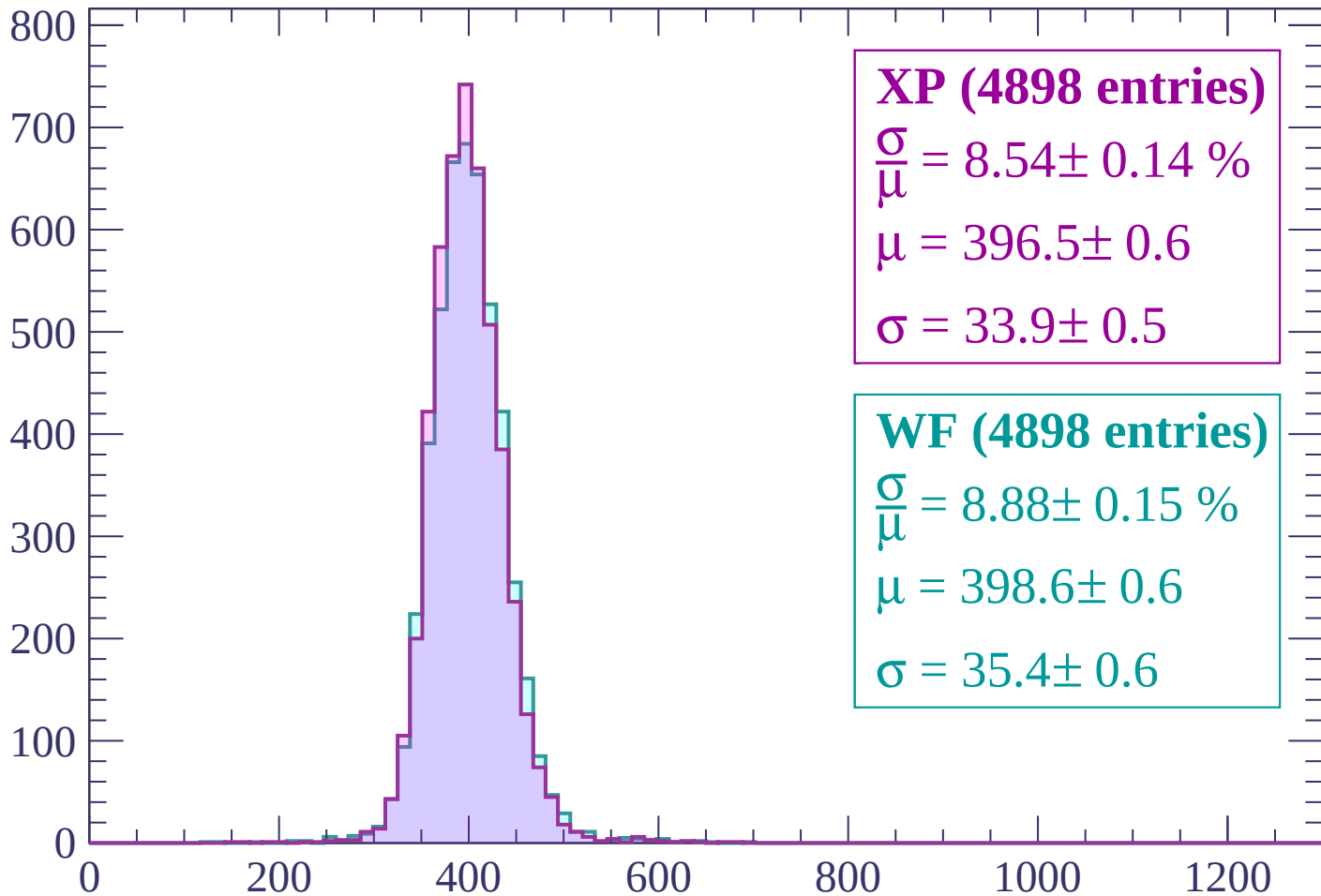
$$\mu = 402.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (4898 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.14 \%$$

$$\mu = 396.5 \pm 0.6$$

$$\sigma = 33.9 \pm 0.5$$

WF (4898 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$$

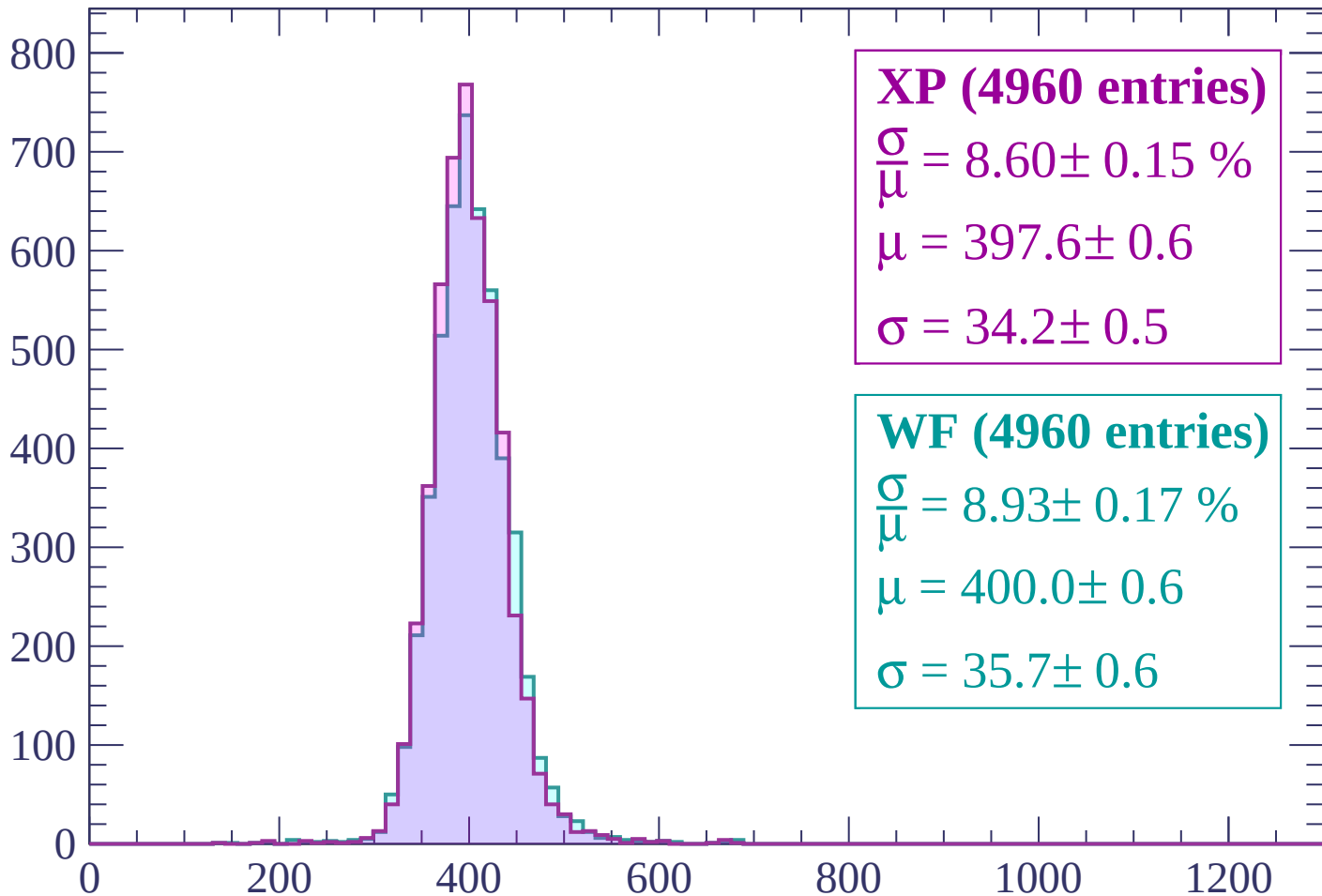
$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (4960 entries)

$$\frac{\sigma}{\mu} = 8.60 \pm 0.15 \%$$

$$\mu = 397.6 \pm 0.6$$

$$\sigma = 34.2 \pm 0.5$$

WF (4960 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.17 \%$$

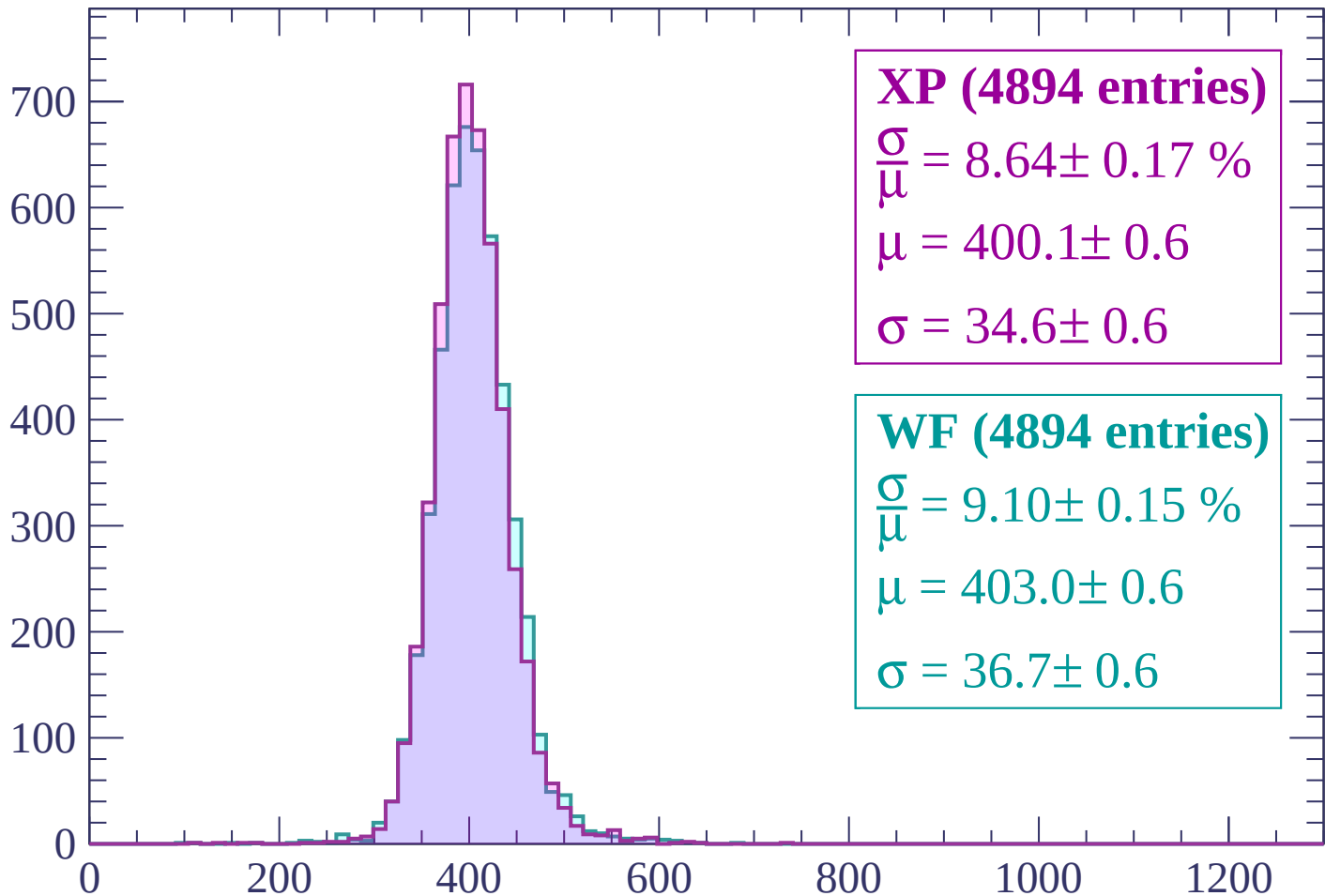
$$\mu = 400.0 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (4894 entries)

$$\frac{\sigma}{\mu} = 8.64 \pm 0.17 \%$$

$$\mu = 400.1 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

WF (4894 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$$

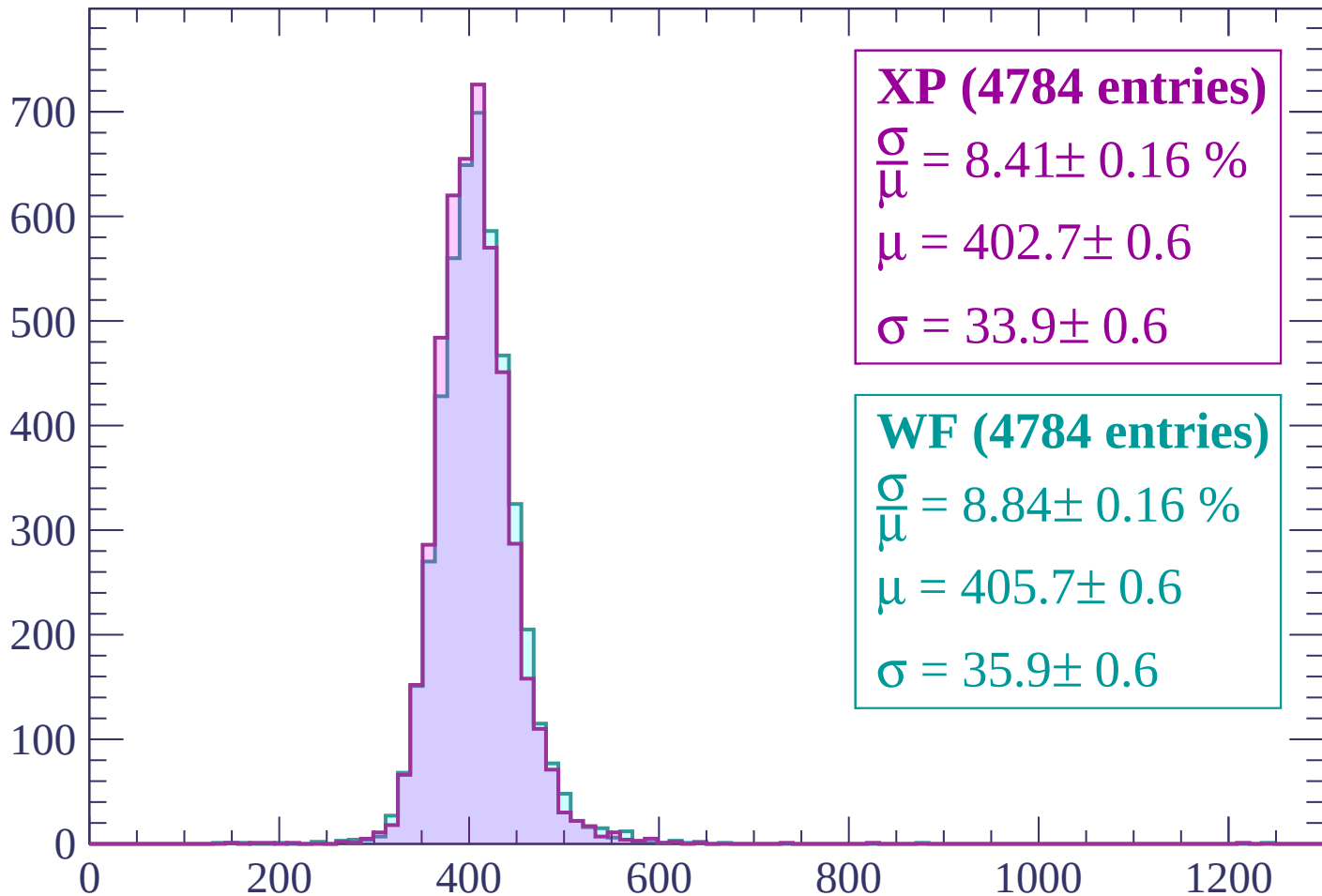
$$\mu = 403.0 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4784 entries)

$$\frac{\sigma}{\mu} = 8.41 \pm 0.16 \%$$

$$\mu = 402.7 \pm 0.6$$

$$\sigma = 33.9 \pm 0.6$$

WF (4784 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$$

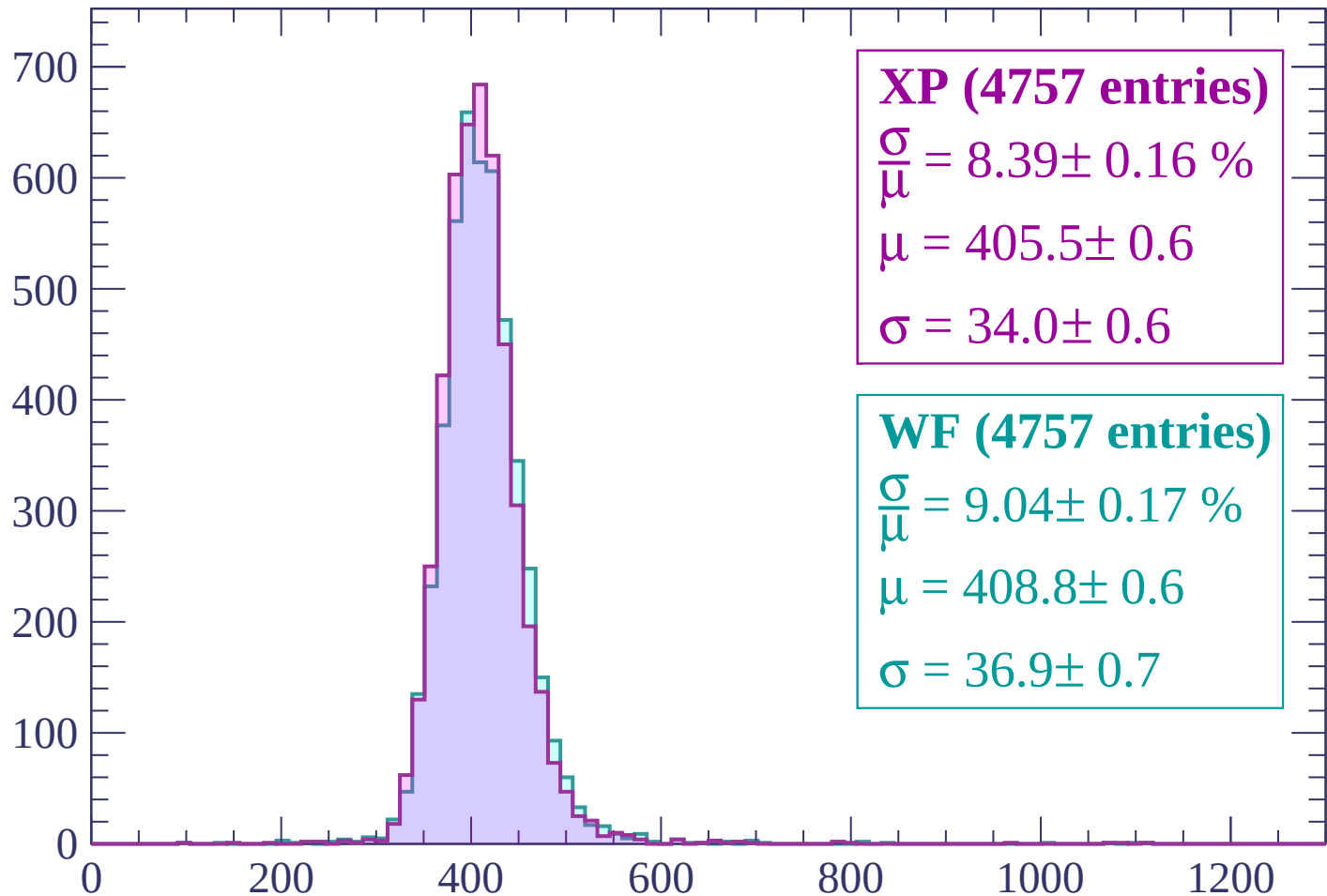
$$\mu = 405.7 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (4757 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.16 \%$$

$$\mu = 405.5 \pm 0.6$$

$$\sigma = 34.0 \pm 0.6$$

WF (4757 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

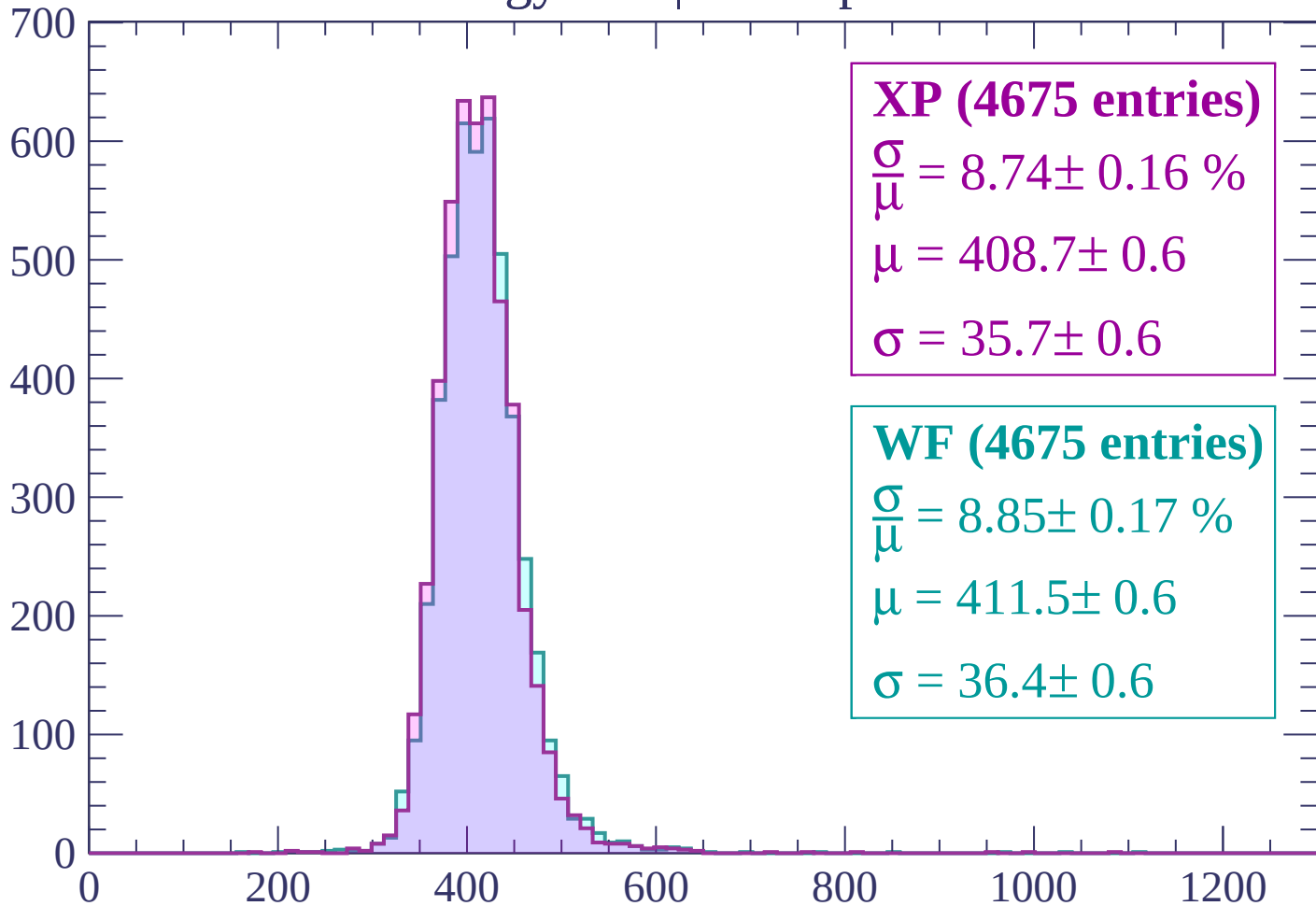
$$\mu = 408.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (4675 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.16 \%$$

$$\mu = 408.7 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

WF (4675 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

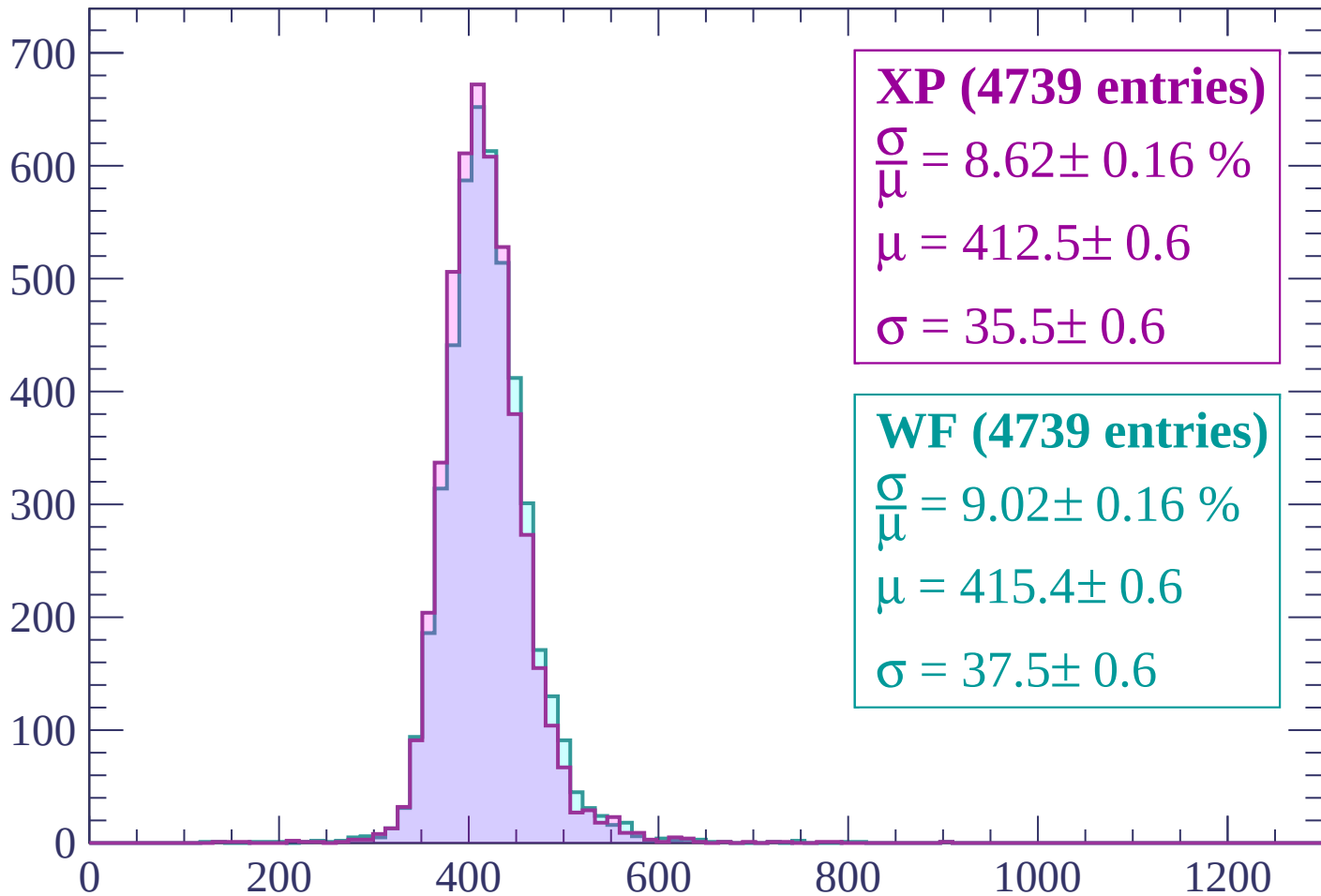
$$\mu = 411.5 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (4739 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.16 \%$$

$$\mu = 412.5 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF (4739 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

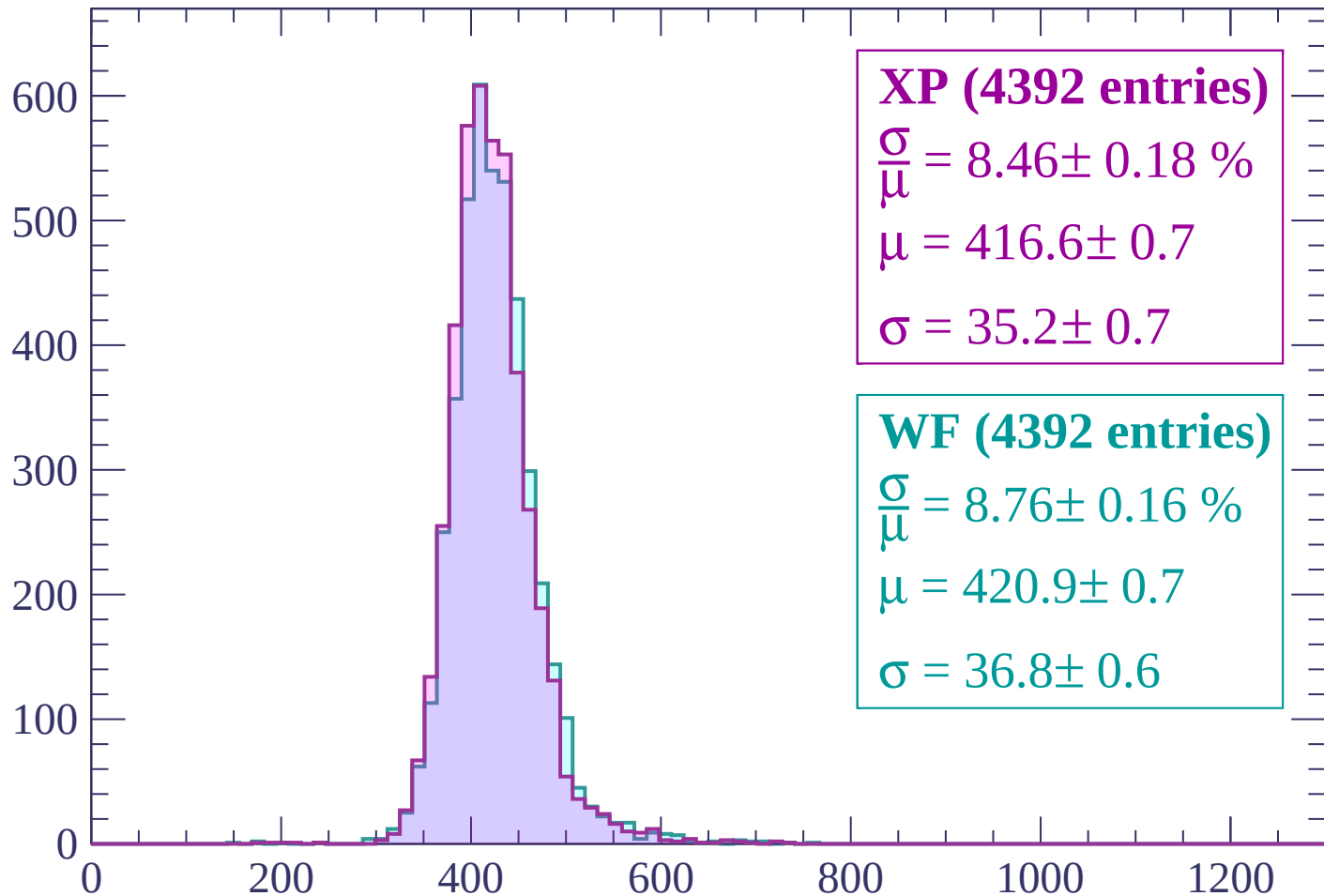
$$\mu = 415.4 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (4392 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.18 \%$$

$$\mu = 416.6 \pm 0.7$$

$$\sigma = 35.2 \pm 0.7$$

WF (4392 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.16 \%$$

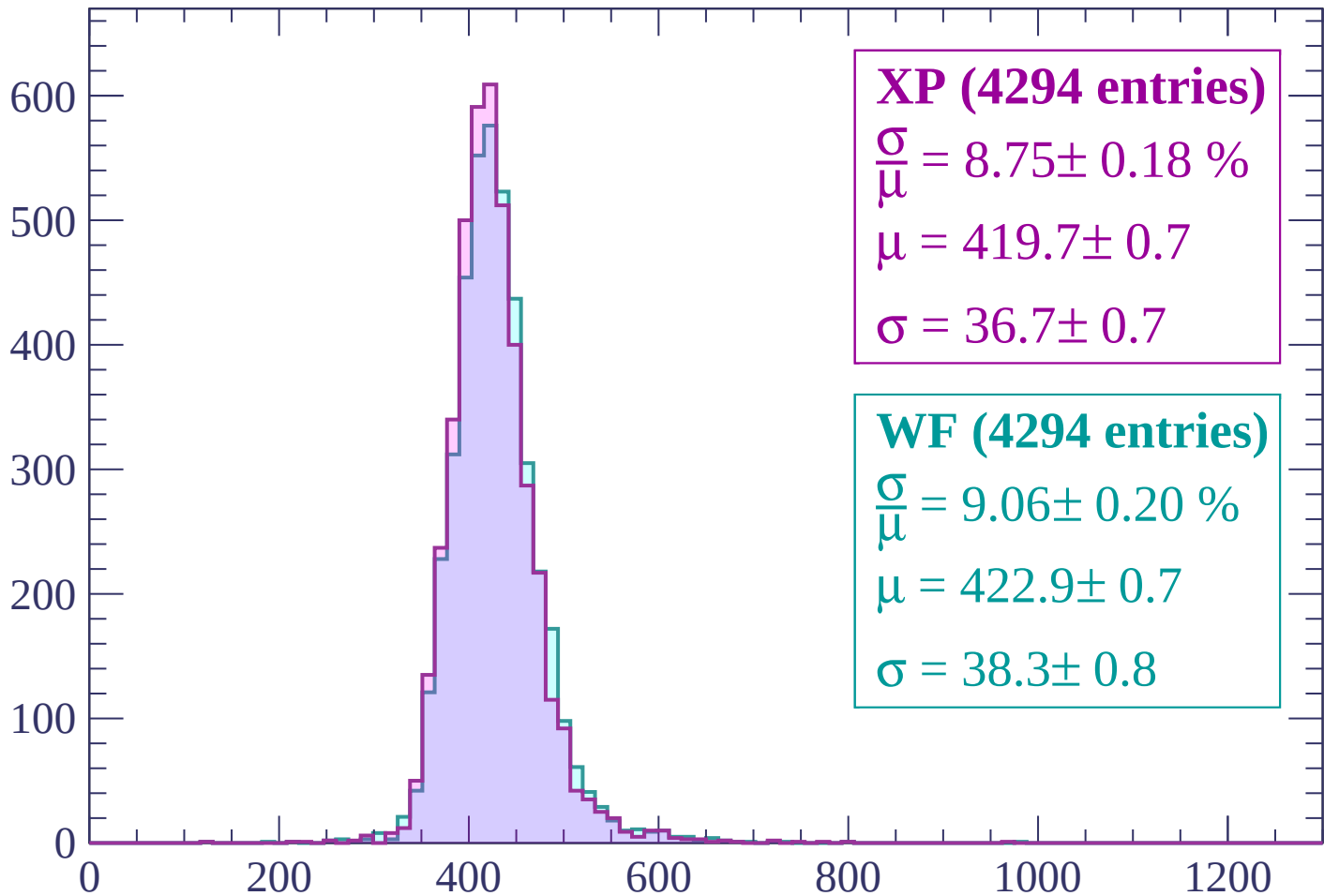
$$\mu = 420.9 \pm 0.7$$

$$\sigma = 36.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (4294 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.18 \%$$

$$\mu = 419.7 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF (4294 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.20 \%$$

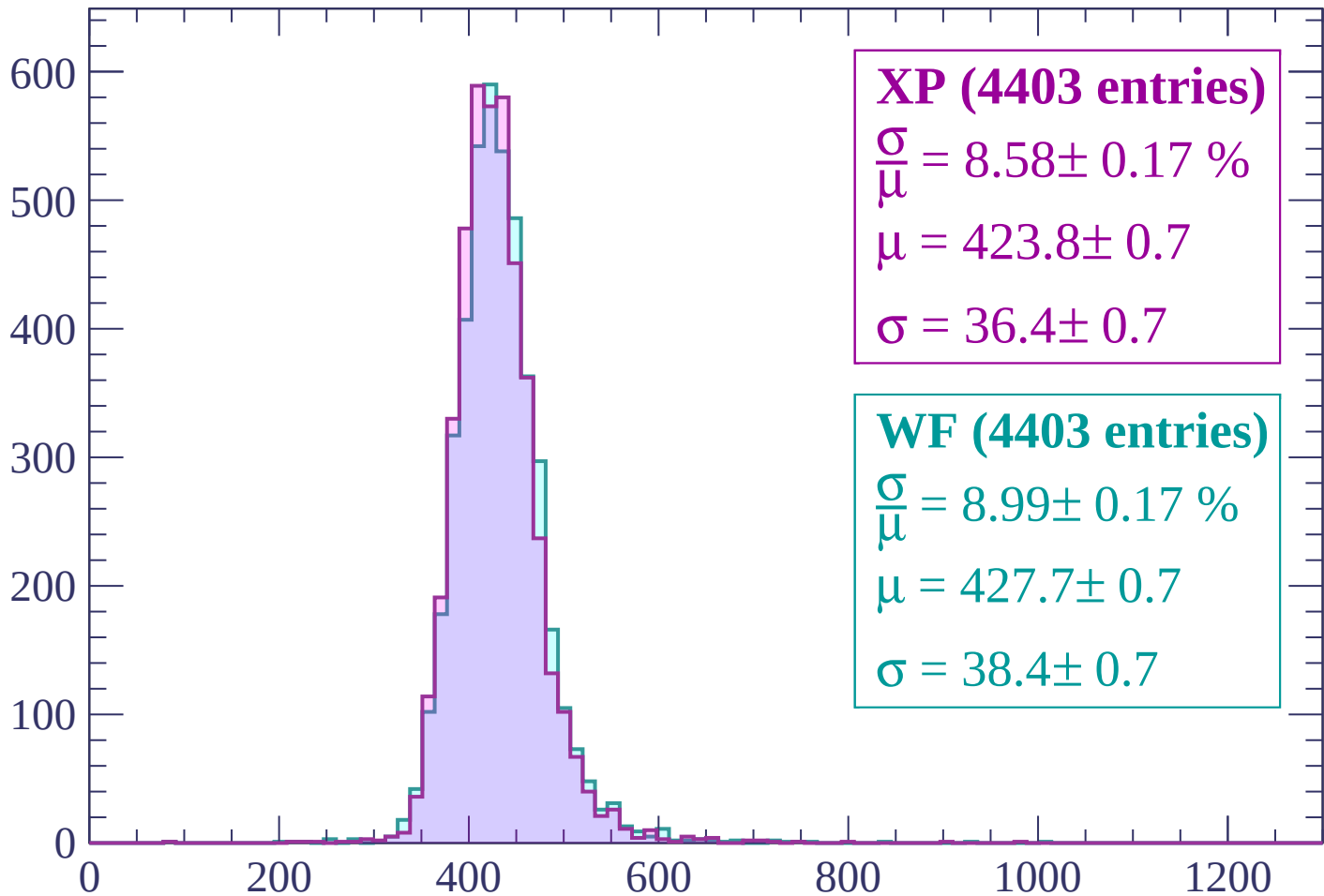
$$\mu = 422.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (4403 entries)

$$\frac{\sigma}{\mu} = 8.58 \pm 0.17 \%$$

$$\mu = 423.8 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

WF (4403 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$$

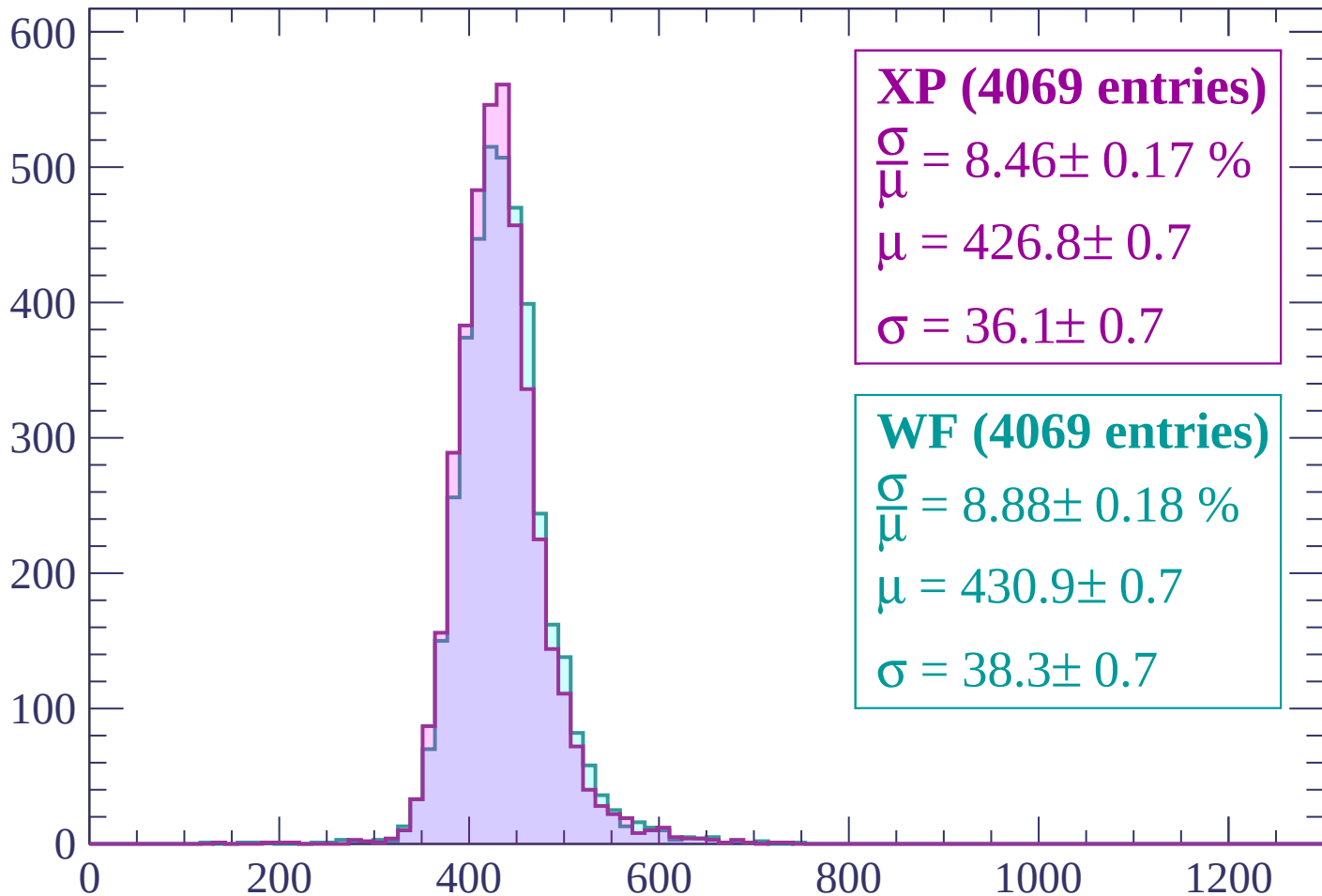
$$\mu = 427.7 \pm 0.7$$

$$\sigma = 38.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4069 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

$$\mu = 426.8 \pm 0.7$$

$$\sigma = 36.1 \pm 0.7$$

WF (4069 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.18 \%$$

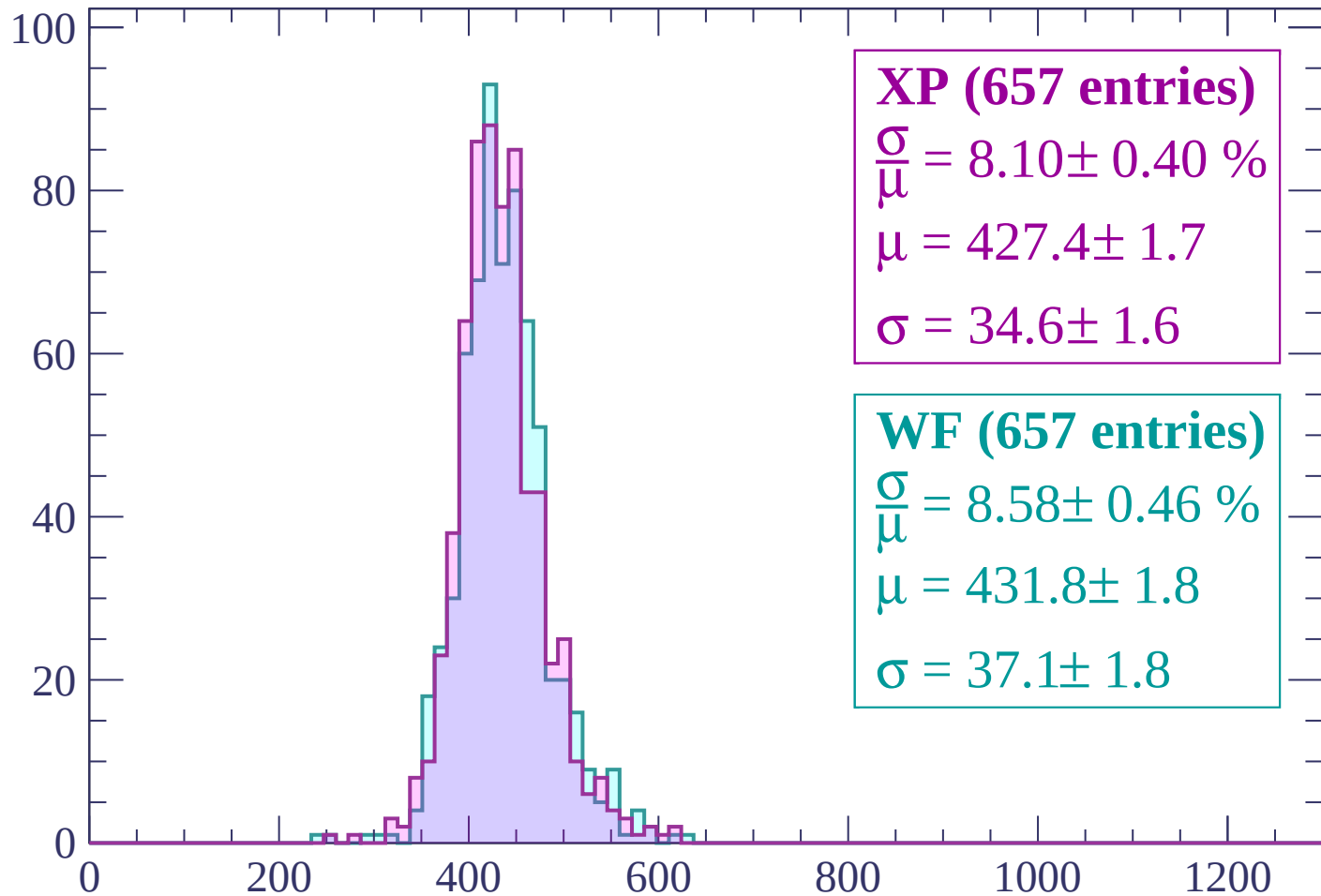
$$\mu = 430.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (657 entries)

$\frac{\sigma}{\mu} = 8.10 \pm 0.40 \%$

$\mu = 427.4 \pm 1.7$

$\sigma = 34.6 \pm 1.6$

WF (657 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.46 \%$

$\mu = 431.8 \pm 1.8$

$\sigma = 37.1 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1140 < p < 1200$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1200 < p < 1260$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1260 < p < 1320

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1380 < p < 1440$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1440 < p < 1500$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1500 < p < 1560

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1560 < p < 1620

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1680 < p < 1740

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1740 < p < 1800

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1800 < p < 1860

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1860 < p < 1920

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1920 < p < 1980

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1980 < p < 2040

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2040 < p < 2100$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2100 < p < 2160

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2160 < p < 2220

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2220 < p < 2280

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2280 < p < 2340

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2340 < p < 2400

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2400 < p < 2460$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2460 < p < 2520

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2520 < p < 2580

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2580 < p < 2640

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2640 < p < 2700

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2700 < p < 2760

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2760 < p < 2820

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2820 < p < 2880$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2880 < p < 2940

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2940 < p < 3000$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-74 < \theta < -72$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

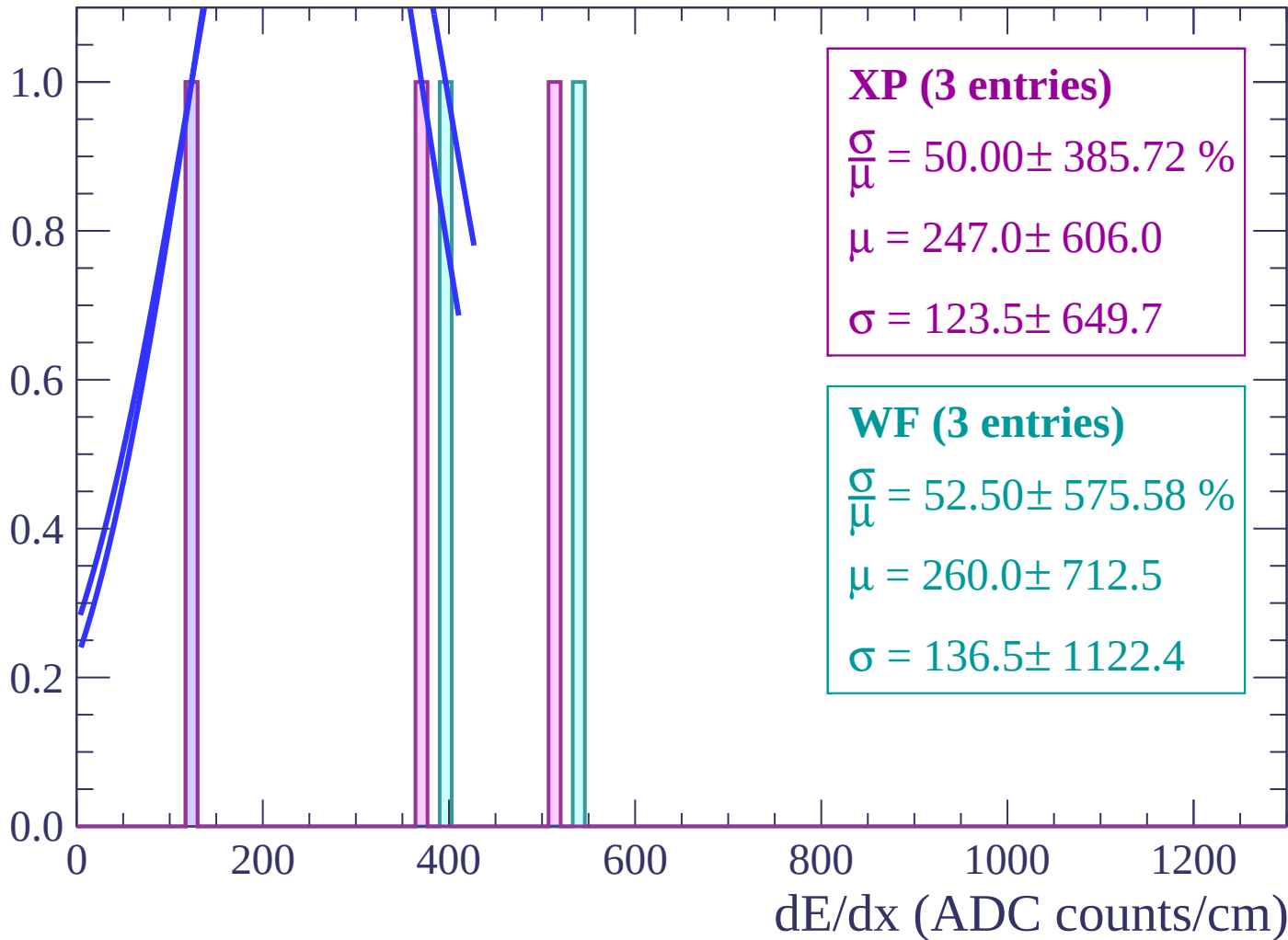
$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

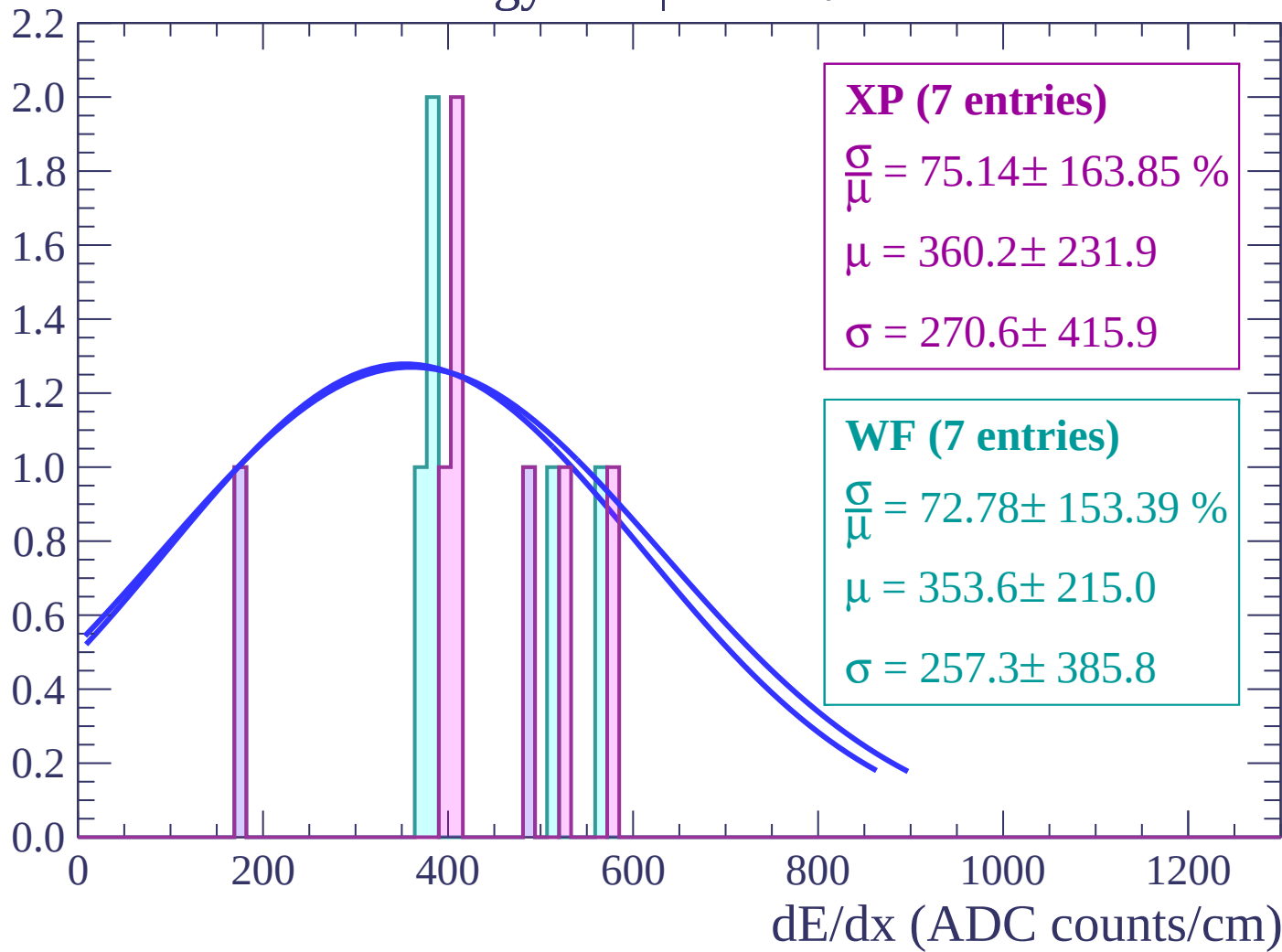
Energy loss | $-72 < \theta < -70$

Count



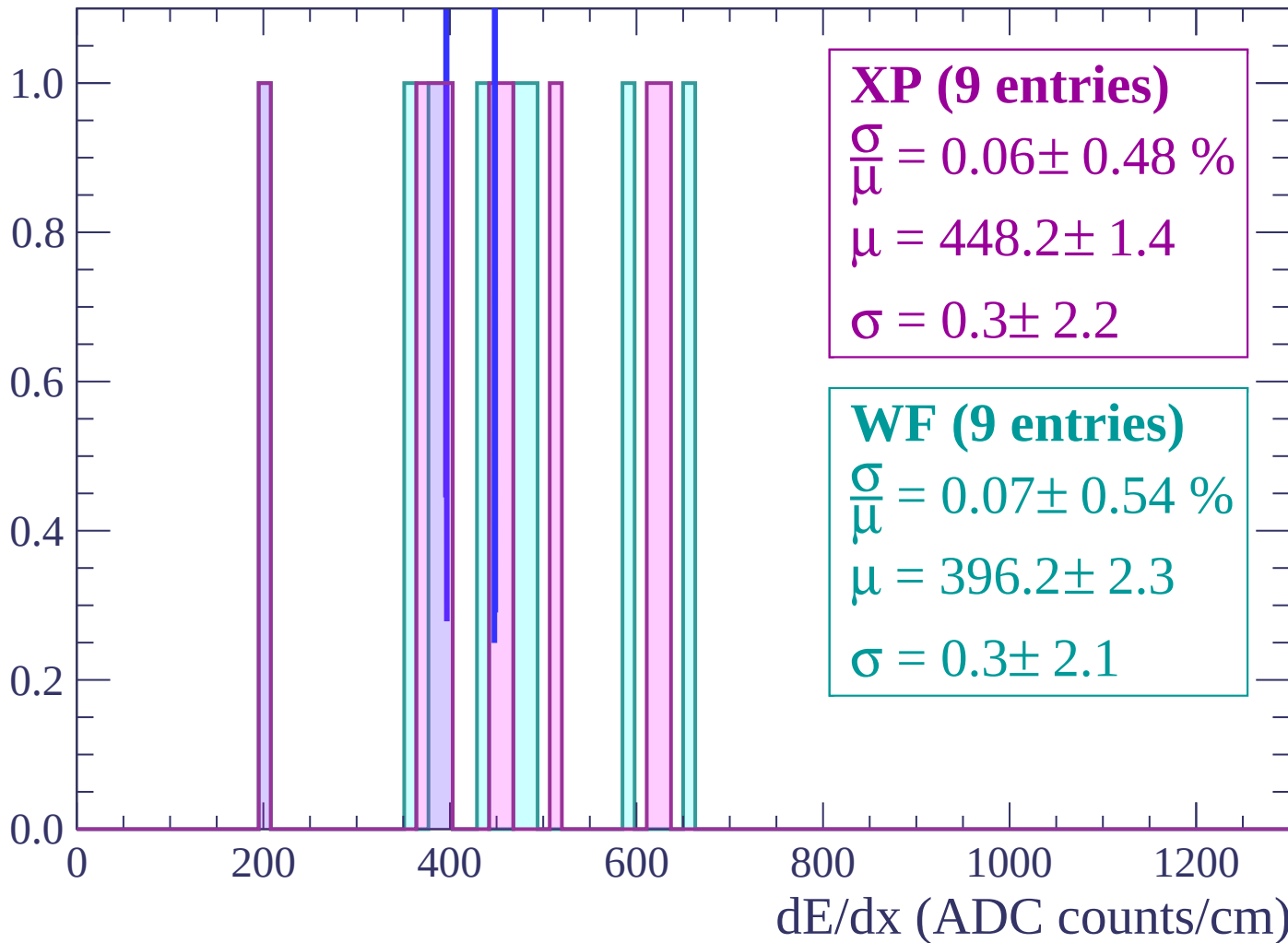
Energy loss | $-70 < \theta < -68$

Count



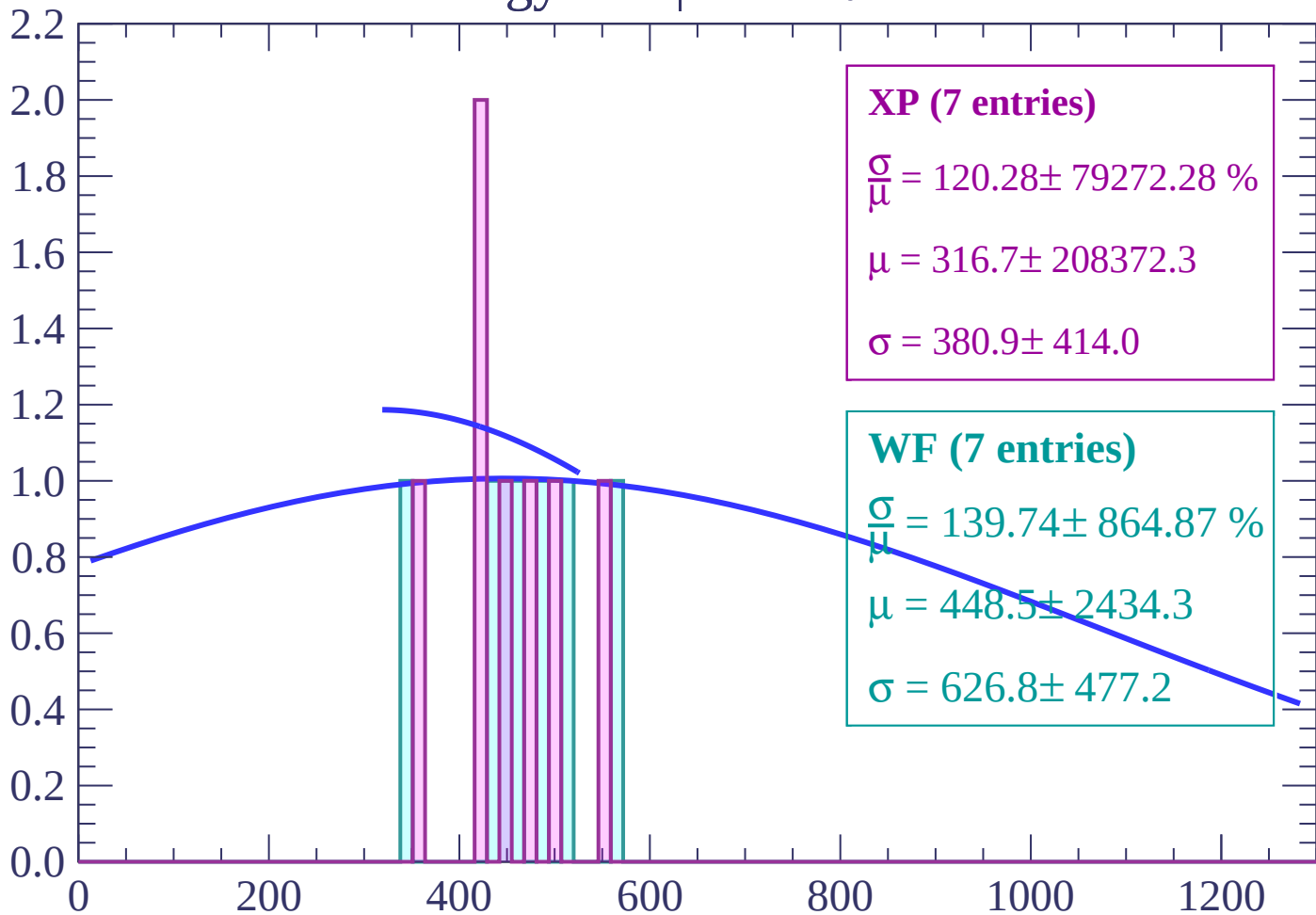
Energy loss | $-68 < \theta < -66$

Count



Energy loss | $-66 < \theta < -64$

Count



XP (7 entries)

$\frac{\sigma}{\mu} = 120.28 \pm 79272.28 \%$

$\mu = 316.7 \pm 208372.3$

$\sigma = 380.9 \pm 414.0$

WF (7 entries)

$\frac{\sigma}{\mu} = 139.74 \pm 864.87 \%$

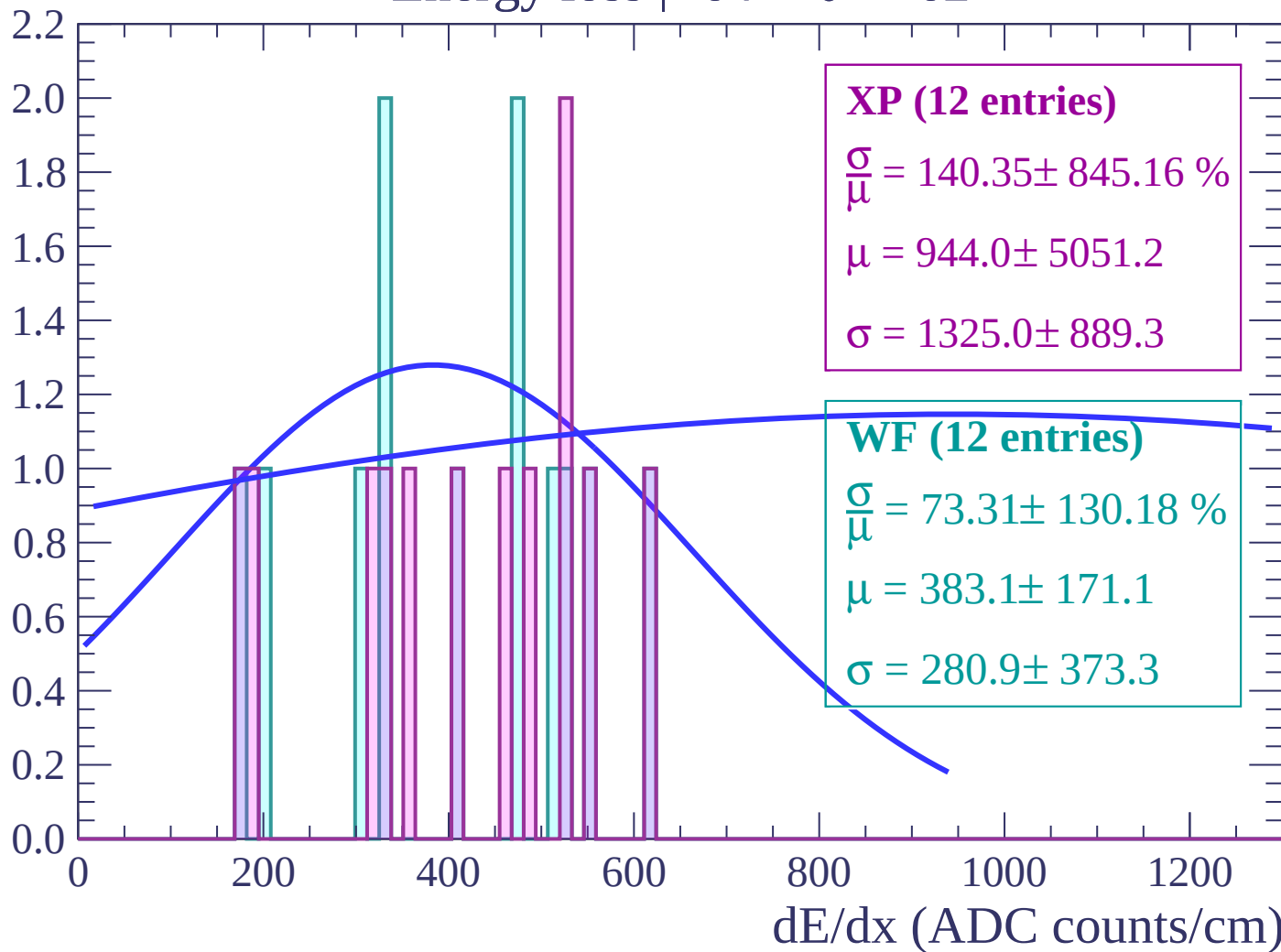
$\mu = 448.5 \pm 2434.3$

$\sigma = 626.8 \pm 477.2$

dE/dx (ADC counts/cm)

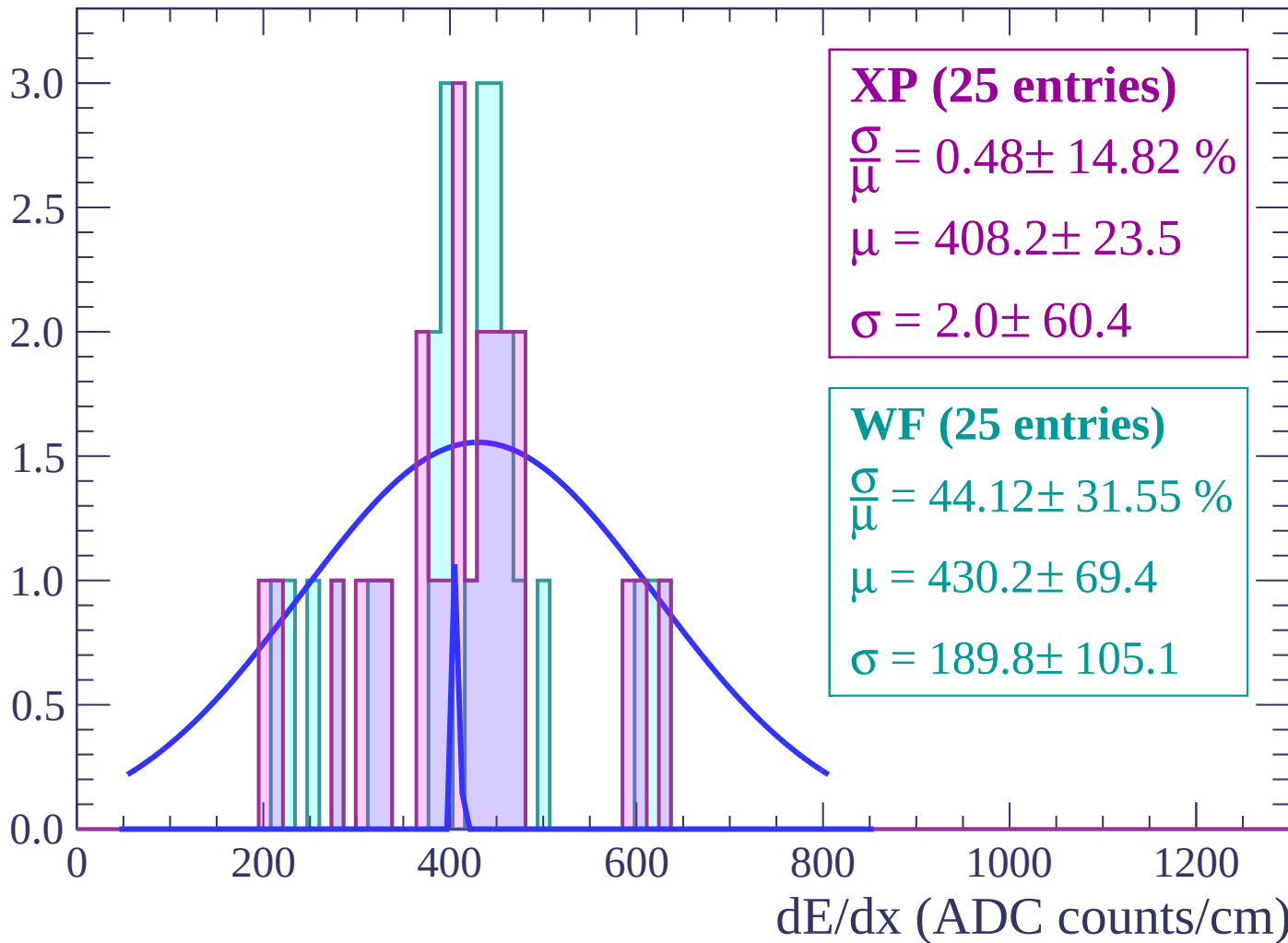
Energy loss | $-64 < \theta < -62$

Count



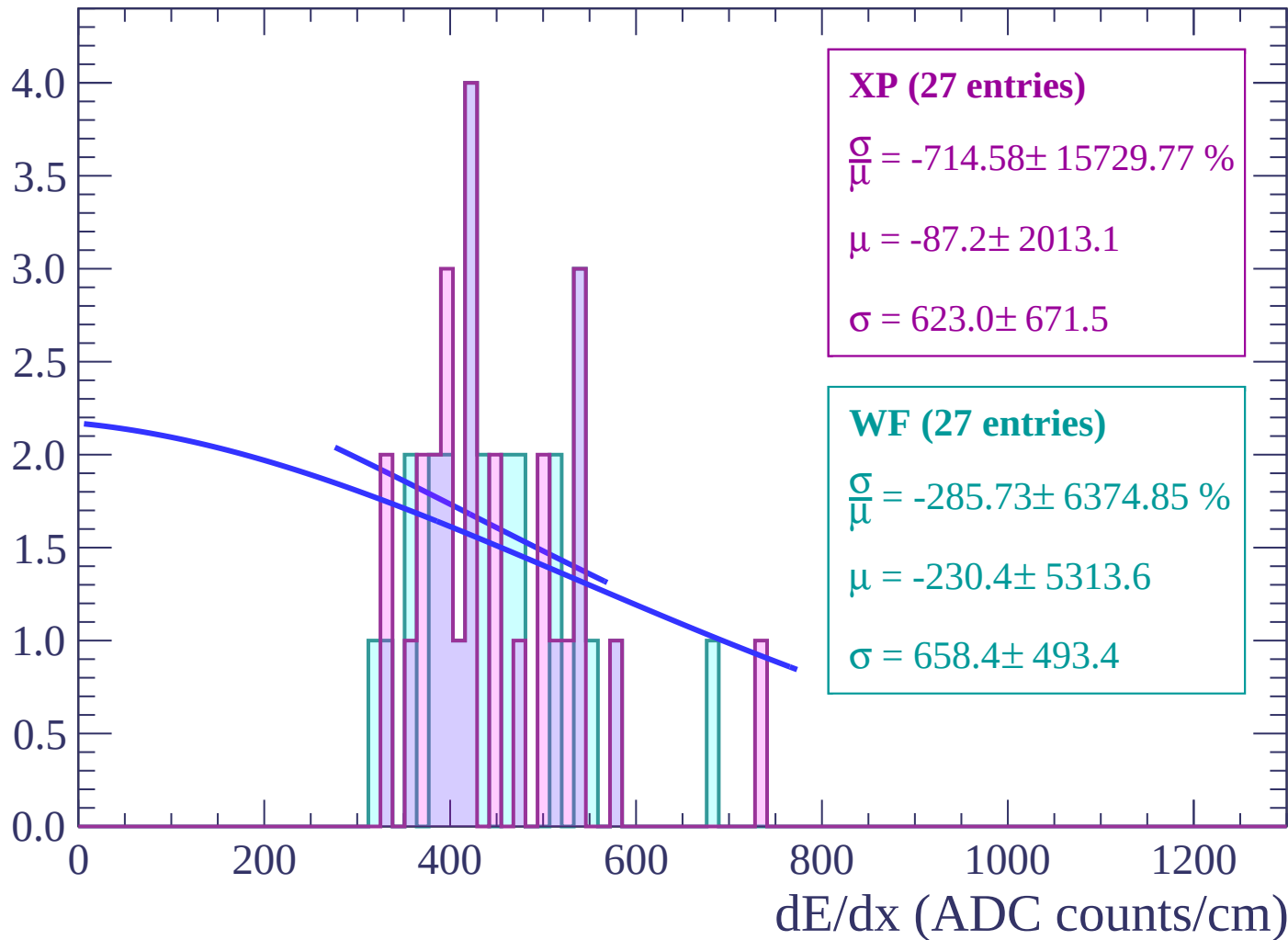
Energy loss | $-62 < \theta < -60$

Count



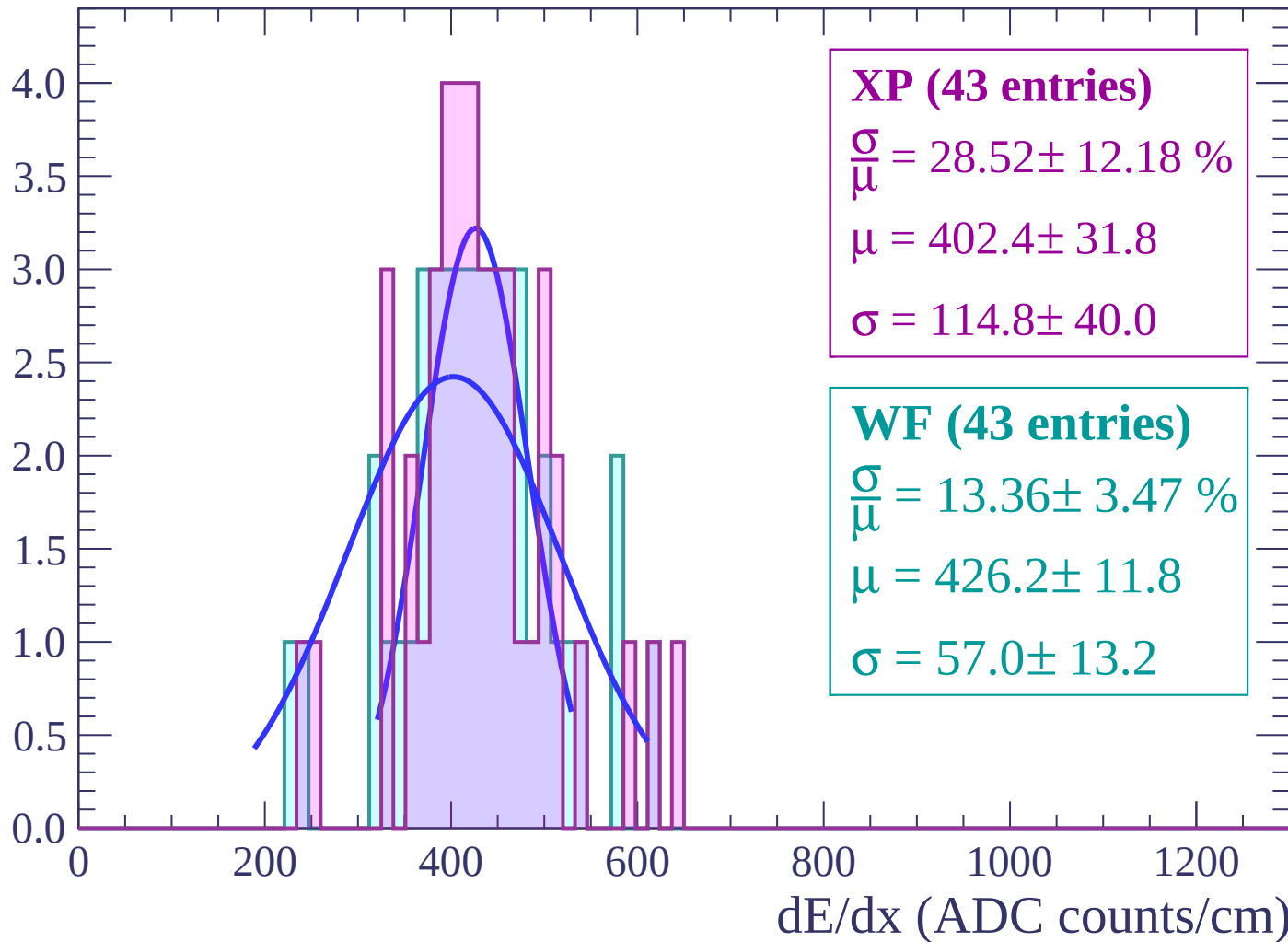
Energy loss | $-60 < \theta < -58$

Count



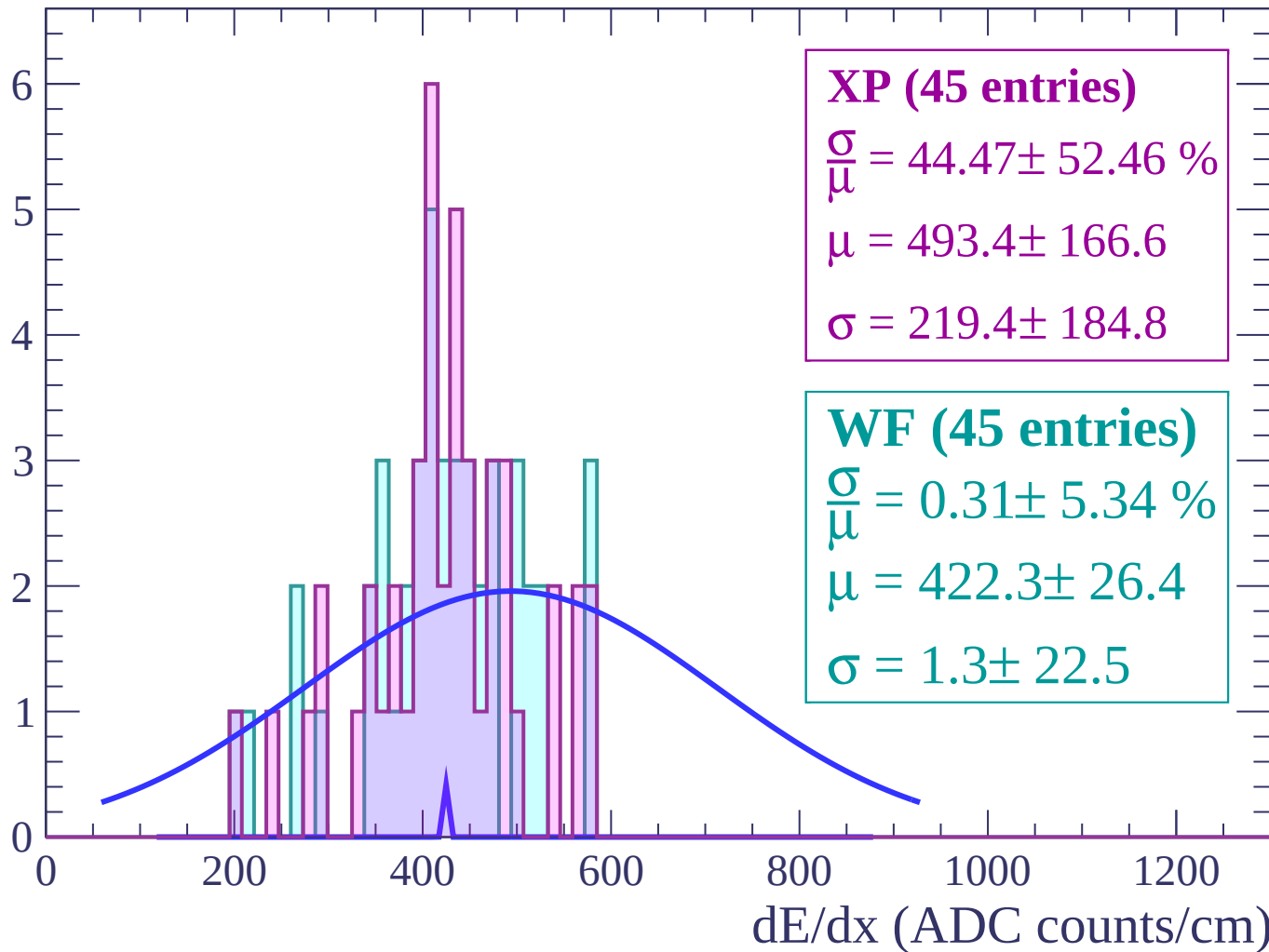
Energy loss | $-58 < \theta < -56$

Count



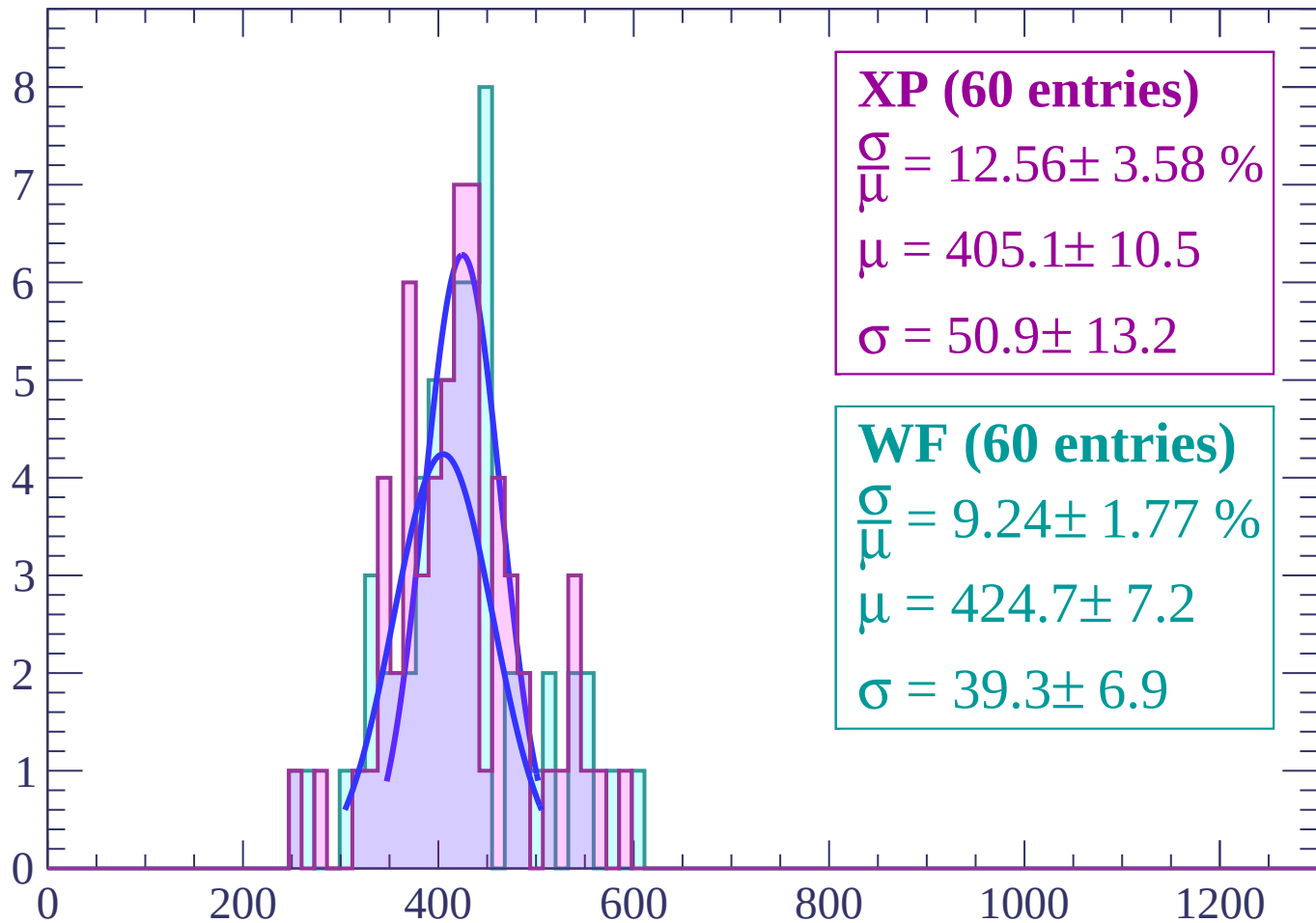
Energy loss | $-56 < \theta < -54$

Count



Energy loss | $-54 < \theta < -52$

Count



XP (60 entries)

$$\frac{\sigma}{\mu} = 12.56 \pm 3.58 \%$$

$$\mu = 405.1 \pm 10.5$$

$$\sigma = 50.9 \pm 13.2$$

WF (60 entries)

$$\frac{\sigma}{\mu} = 9.24 \pm 1.77 \%$$

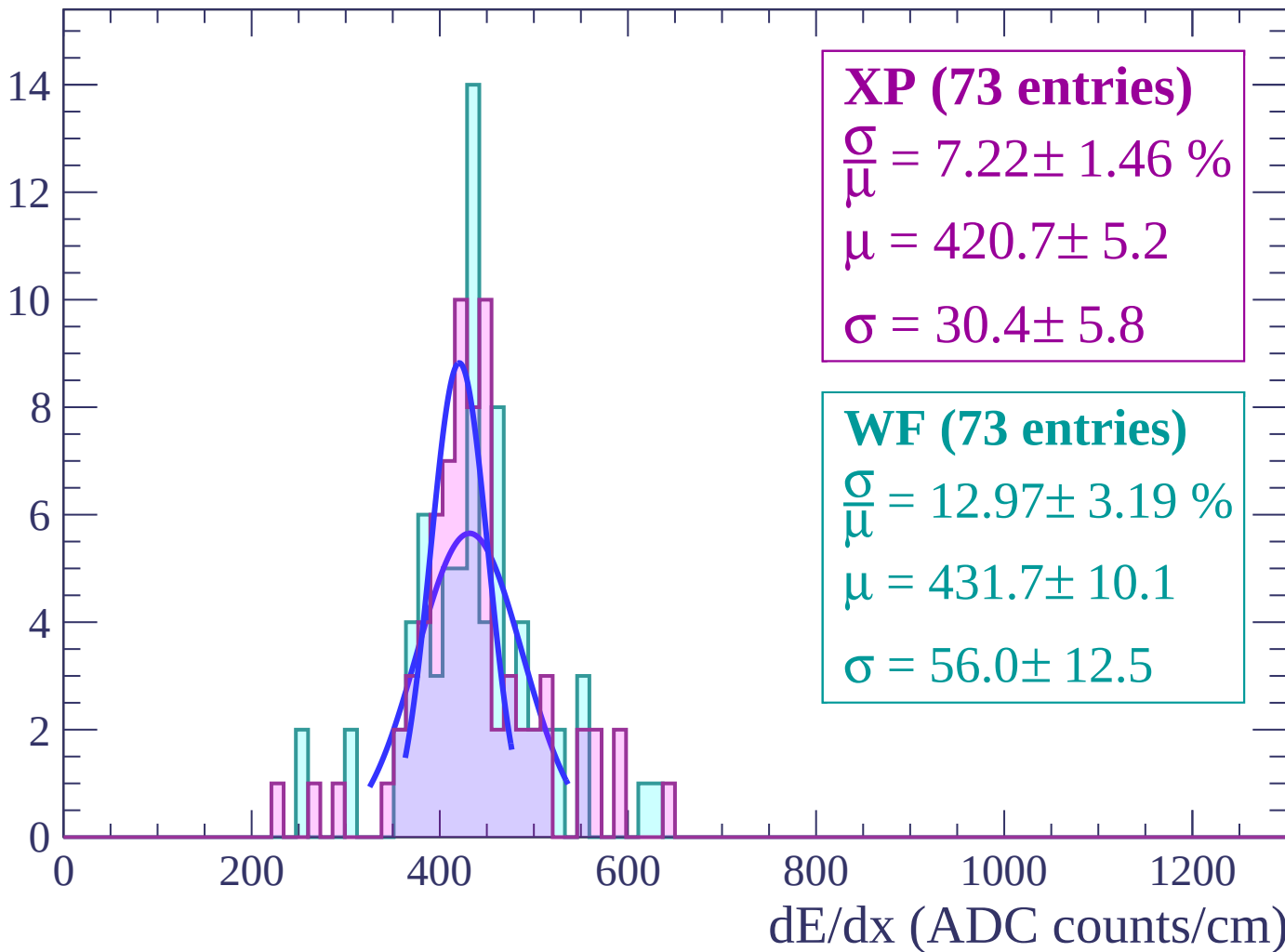
$$\mu = 424.7 \pm 7.2$$

$$\sigma = 39.3 \pm 6.9$$

dE/dx (ADC counts/cm)

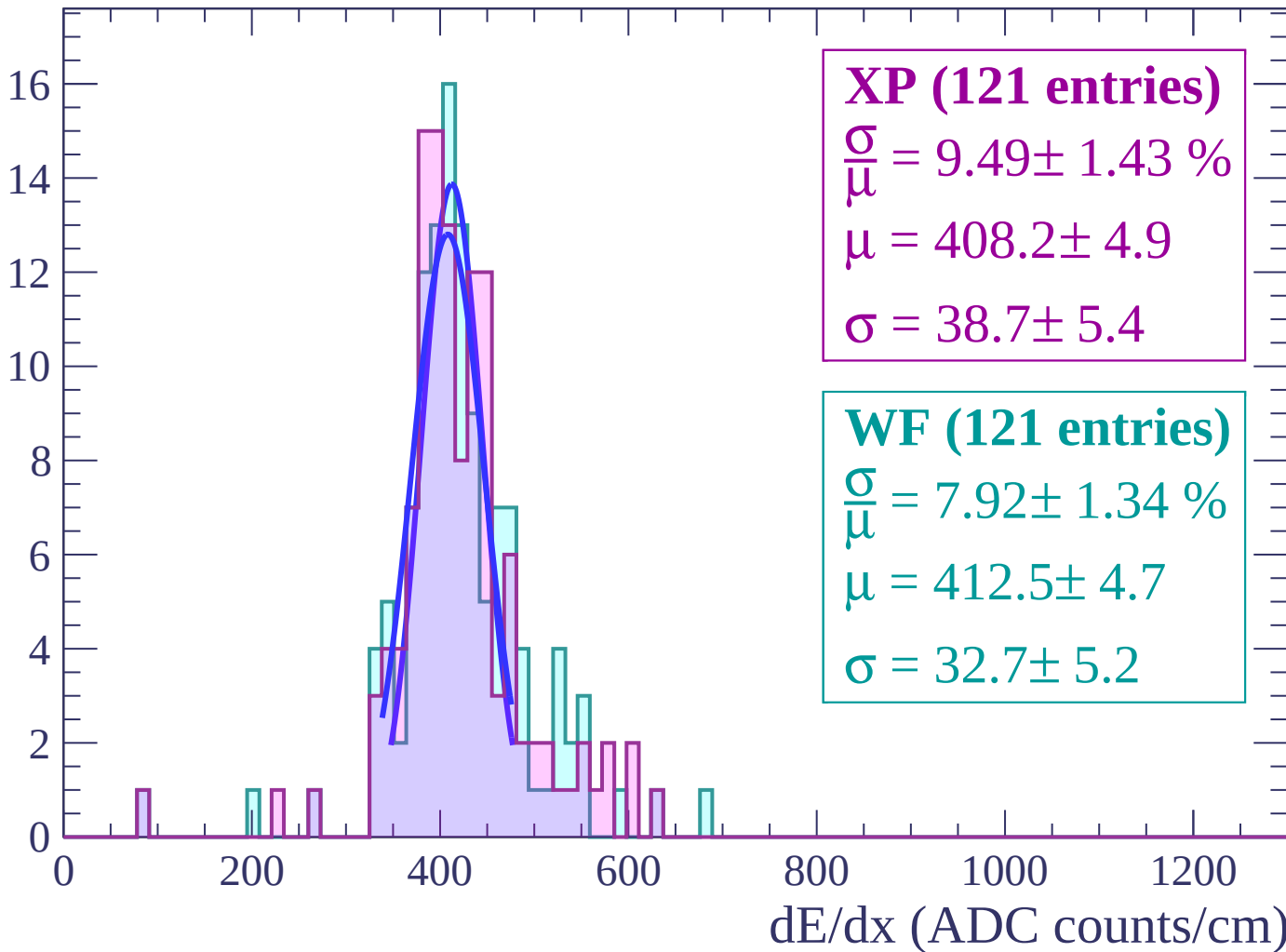
Energy loss | $-52 < \theta < -50$

Count



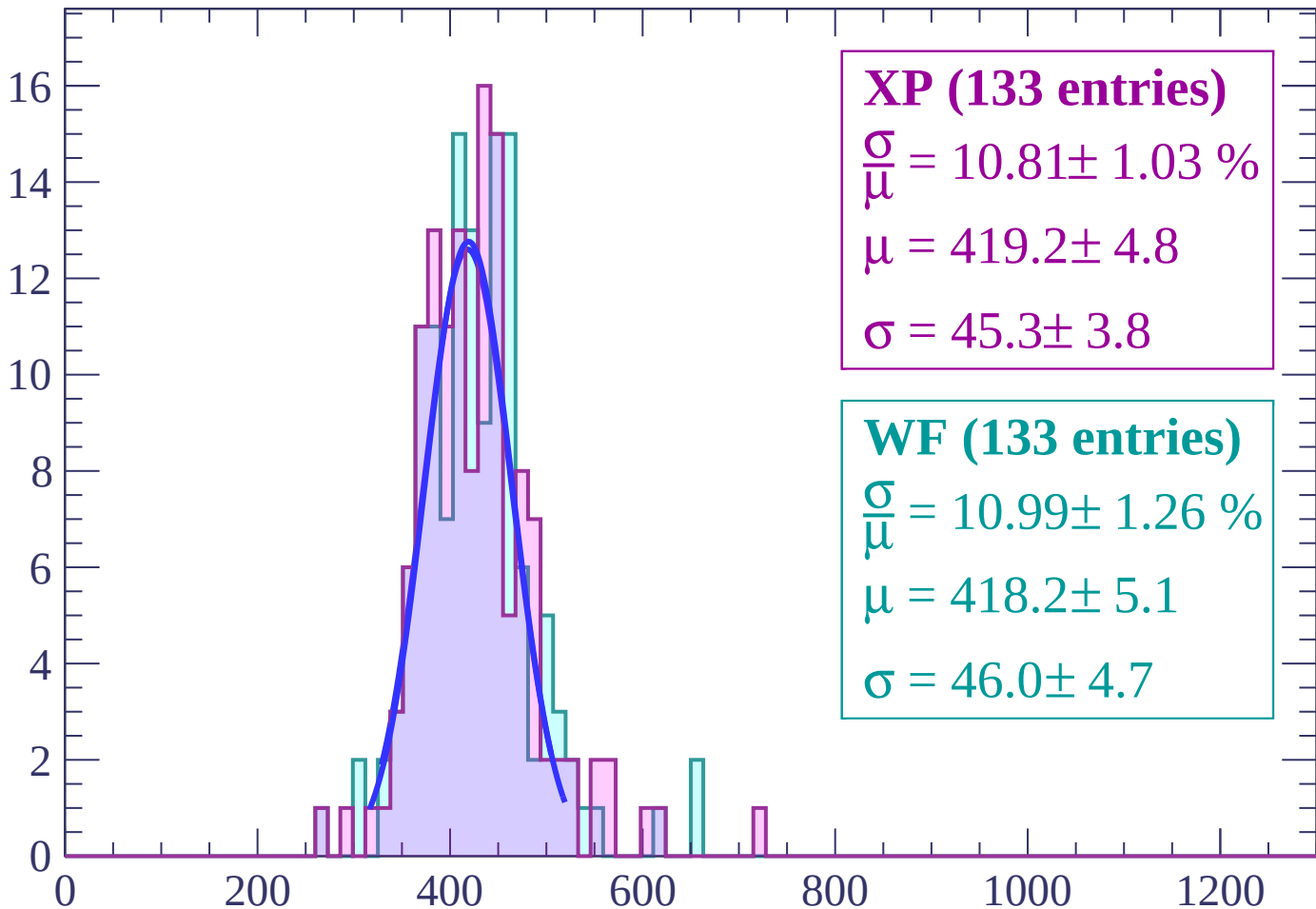
Energy loss | $-50 < \theta < -48$

Count



Energy loss | $-48 < \theta < -46$

Count



XP (133 entries)

$$\frac{\sigma}{\mu} = 10.81 \pm 1.03 \%$$

$$\mu = 419.2 \pm 4.8$$

$$\sigma = 45.3 \pm 3.8$$

WF (133 entries)

$$\frac{\sigma}{\mu} = 10.99 \pm 1.26 \%$$

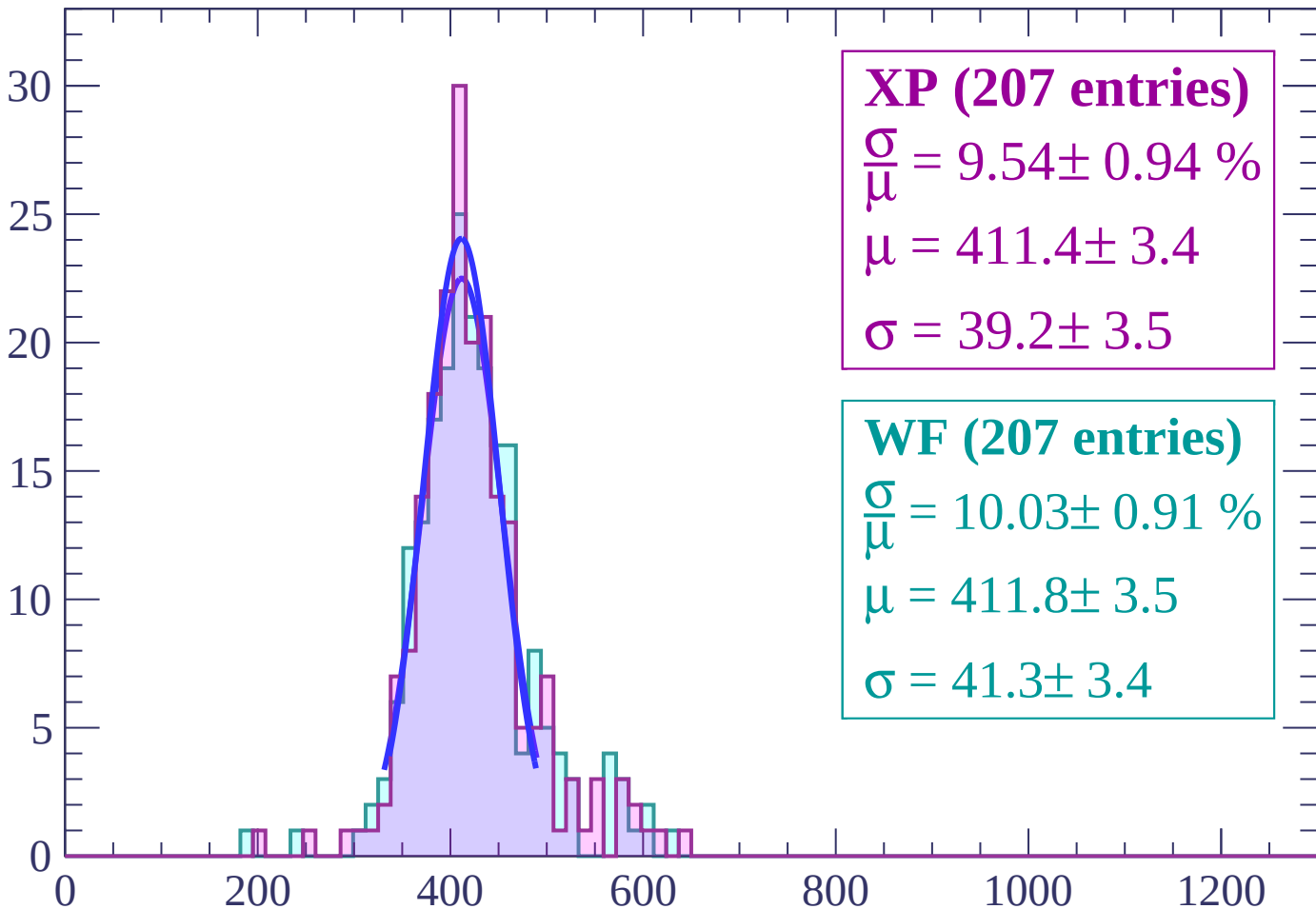
$$\mu = 418.2 \pm 5.1$$

$$\sigma = 46.0 \pm 4.7$$

dE/dx (ADC counts/cm)

Energy loss | $-46 < \theta < -44$

Count



XP (207 entries)

$\frac{\sigma}{\mu} = 9.54 \pm 0.94 \%$

$\mu = 411.4 \pm 3.4$

$\sigma = 39.2 \pm 3.5$

WF (207 entries)

$\frac{\sigma}{\mu} = 10.03 \pm 0.91 \%$

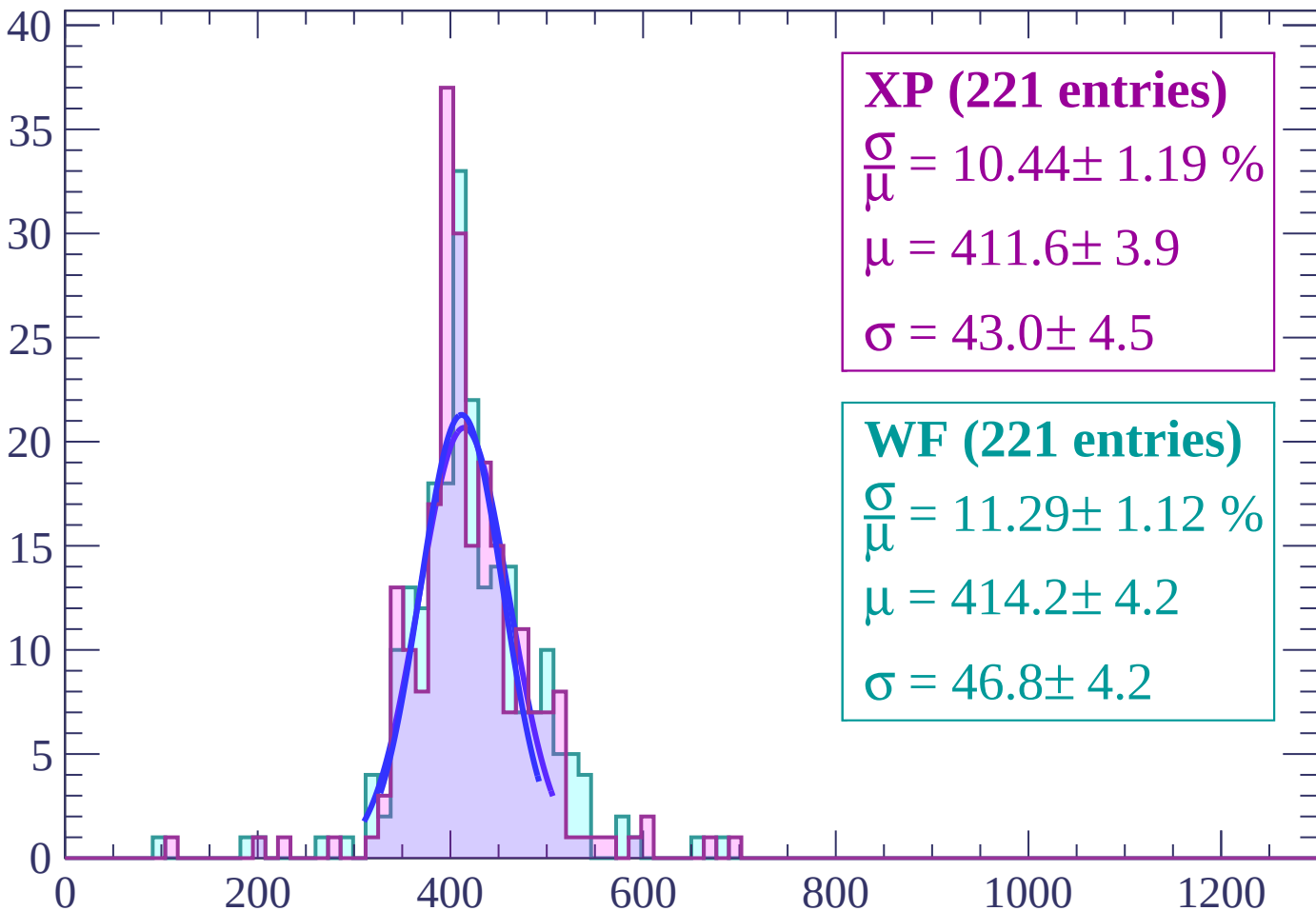
$\mu = 411.8 \pm 3.5$

$\sigma = 41.3 \pm 3.4$

dE/dx (ADC counts/cm)

Energy loss | $-44 < \theta < -42$

Count



XP (221 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 1.19 \%$$

$$\mu = 411.6 \pm 3.9$$

$$\sigma = 43.0 \pm 4.5$$

WF (221 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 1.12 \%$$

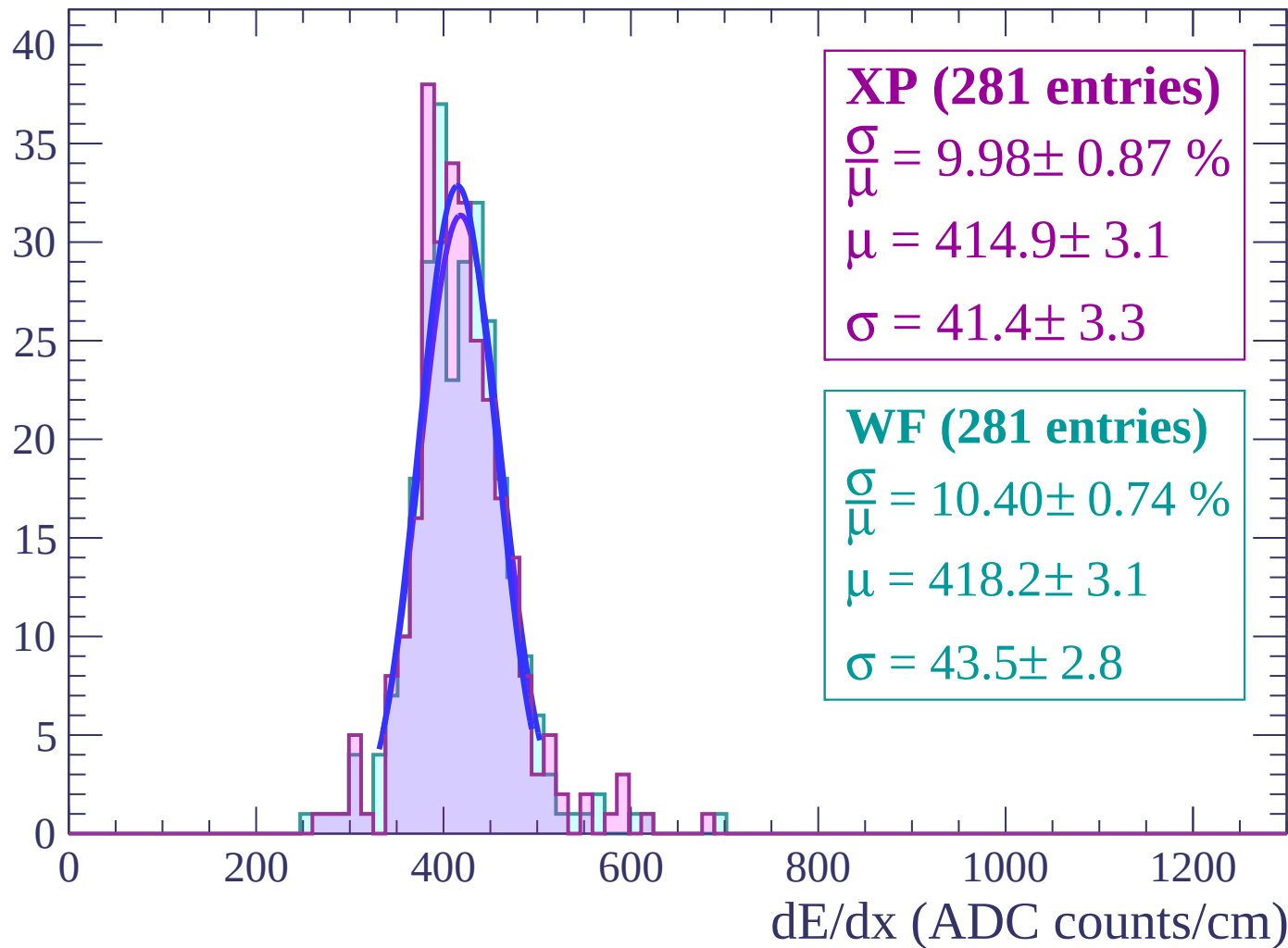
$$\mu = 414.2 \pm 4.2$$

$$\sigma = 46.8 \pm 4.2$$

dE/dx (ADC counts/cm)

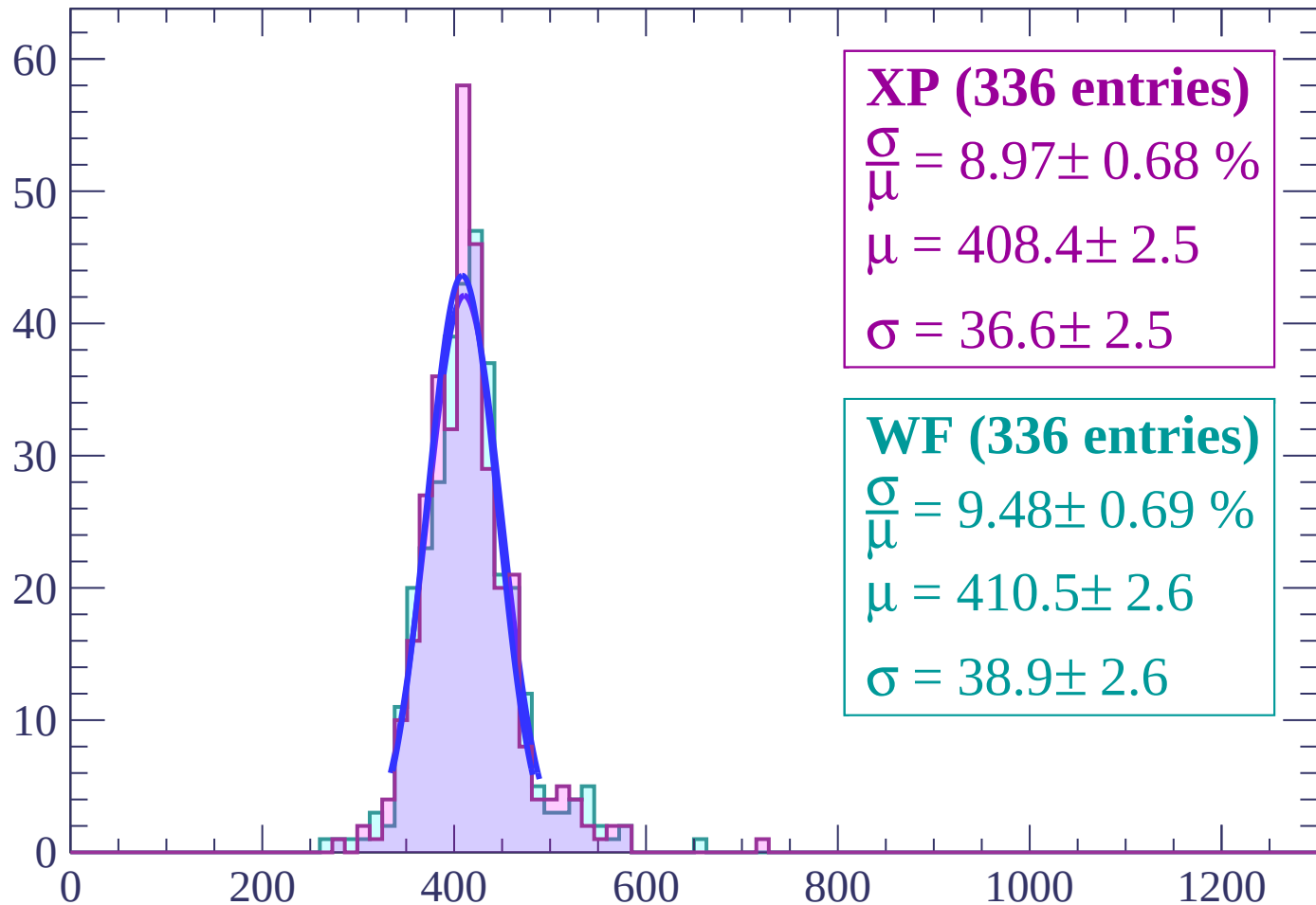
Energy loss | $-42 < \theta < -40$

Count



Energy loss | $-40 < \theta < -38$

Count



XP (336 entries)

$\frac{\sigma}{\mu} = 8.97 \pm 0.68 \%$

$\mu = 408.4 \pm 2.5$

$\sigma = 36.6 \pm 2.5$

WF (336 entries)

$\frac{\sigma}{\mu} = 9.48 \pm 0.69 \%$

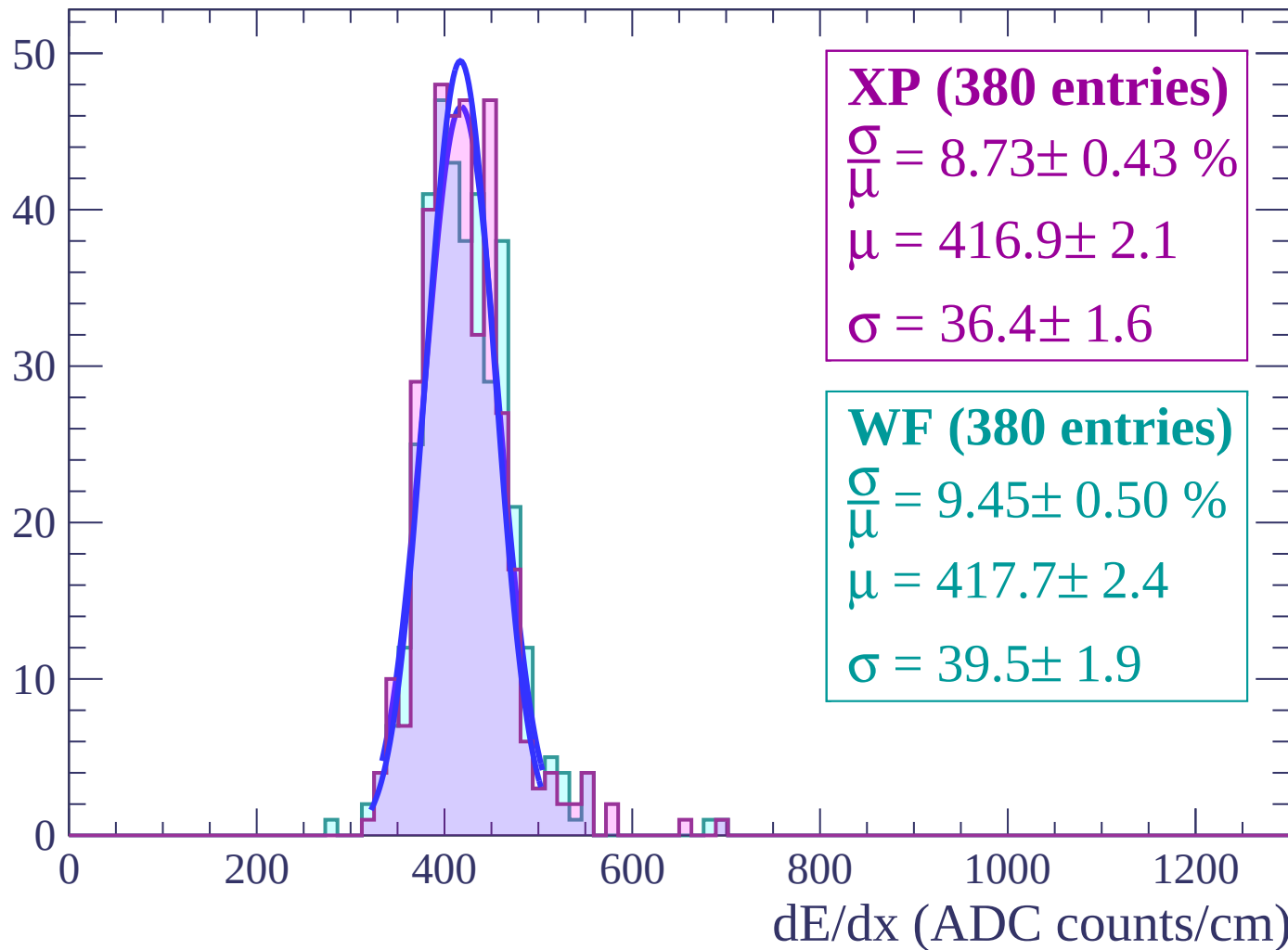
$\mu = 410.5 \pm 2.6$

$\sigma = 38.9 \pm 2.6$

dE/dx (ADC counts/cm)

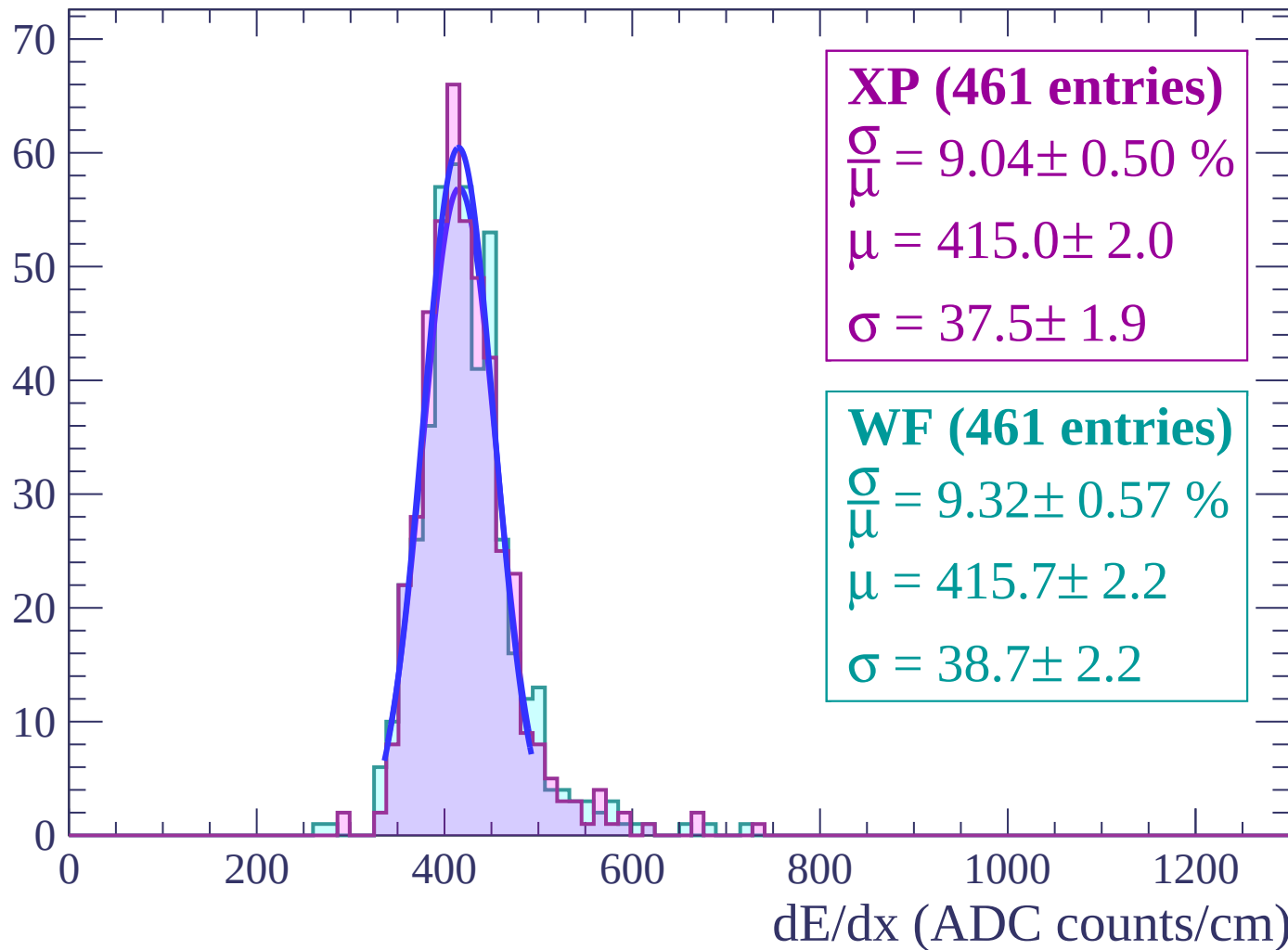
Energy loss | $-38 < \theta < -36$

Count



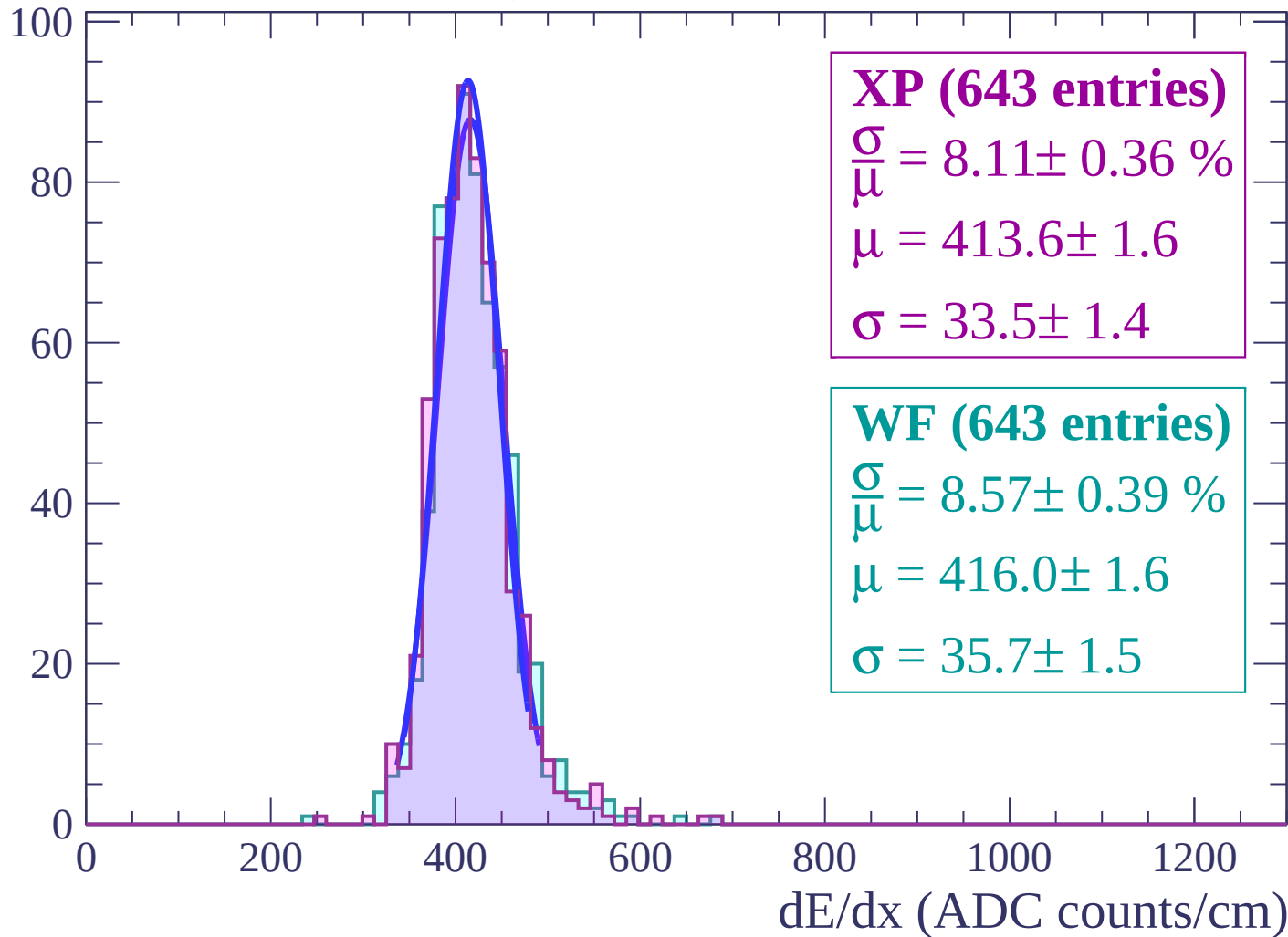
Energy loss | $-36 < \theta < -34$

Count



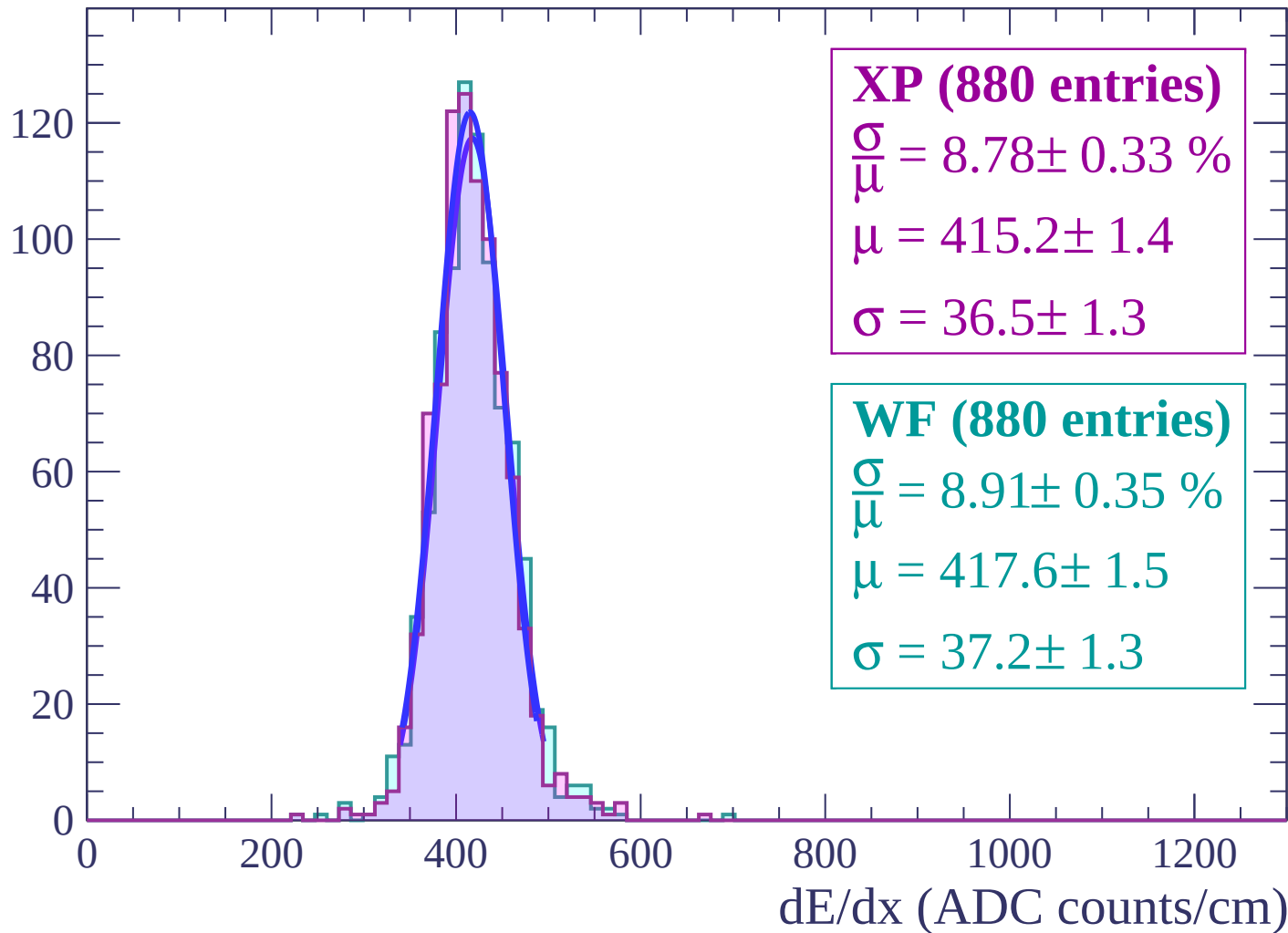
Energy loss | $-34 < \theta < -32$

Count



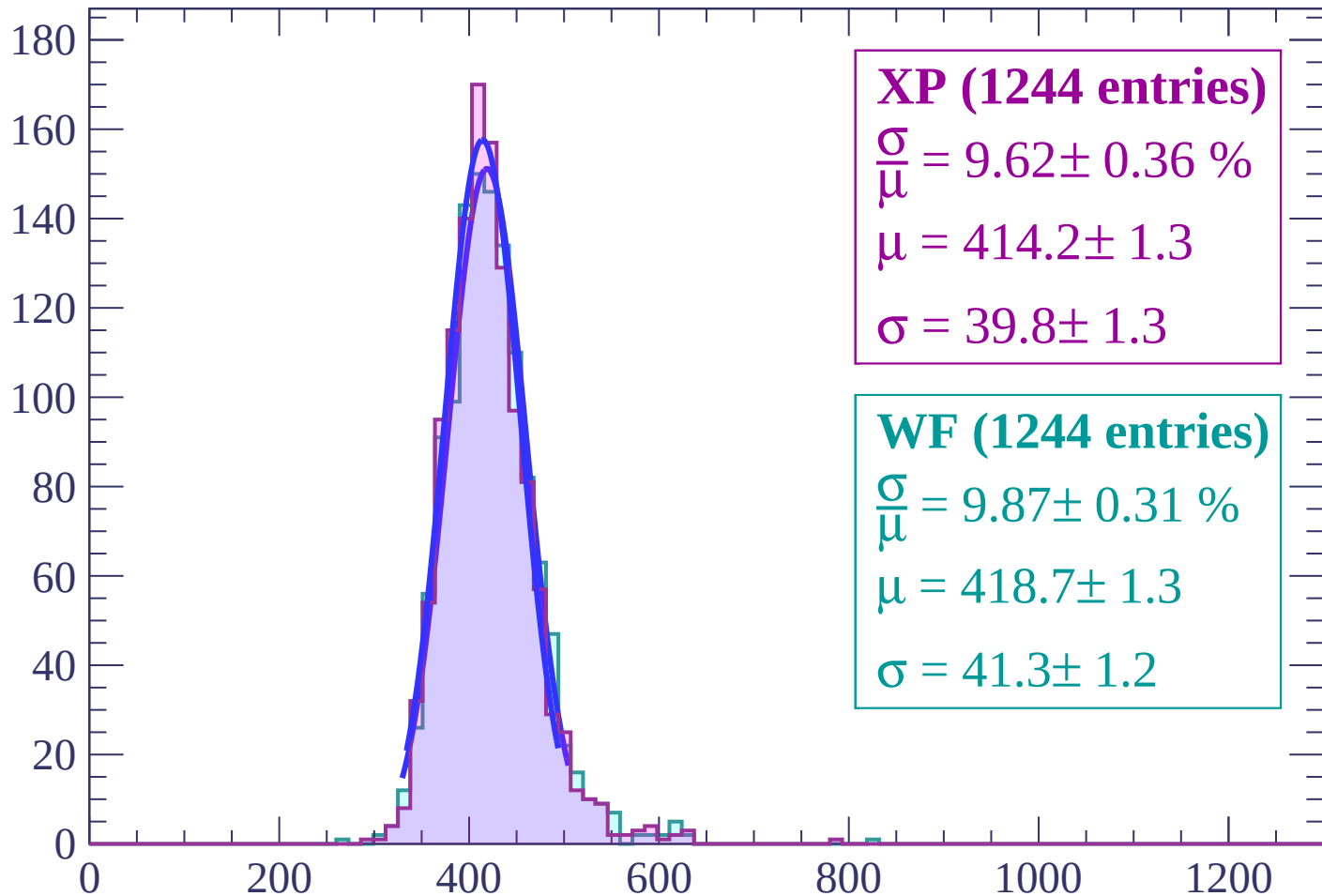
Energy loss | $-32 < \theta < -30$

Count



Energy loss | $-30 < \theta < -28$

Count



XP (1244 entries)

$\frac{\sigma}{\mu} = 9.62 \pm 0.36 \%$

$\mu = 414.2 \pm 1.3$

$\sigma = 39.8 \pm 1.3$

WF (1244 entries)

$\frac{\sigma}{\mu} = 9.87 \pm 0.31 \%$

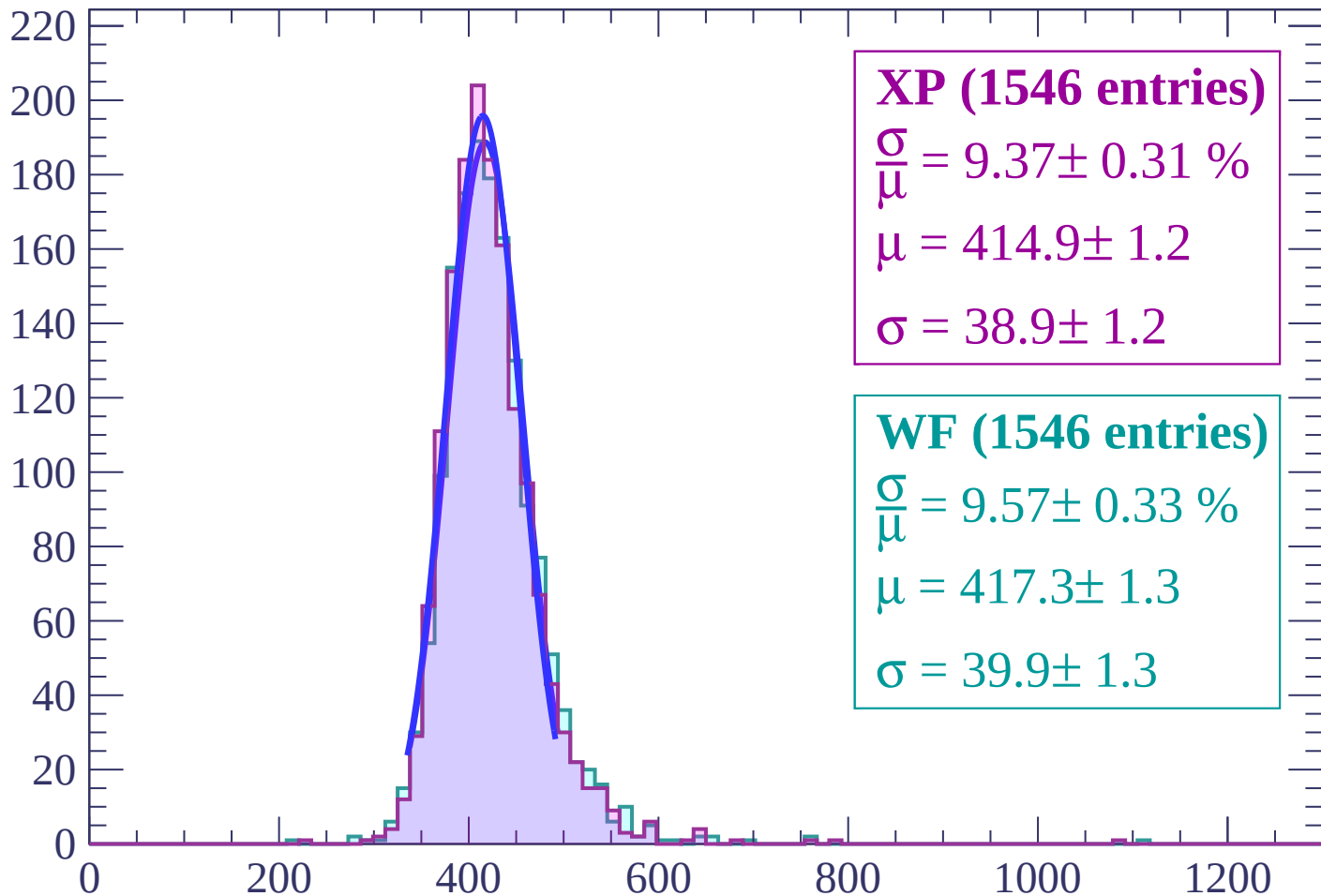
$\mu = 418.7 \pm 1.3$

$\sigma = 41.3 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

Count



XP (1546 entries)

$$\frac{\sigma}{\mu} = 9.37 \pm 0.31 \%$$

$$\mu = 414.9 \pm 1.2$$

$$\sigma = 38.9 \pm 1.2$$

WF (1546 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.33 \%$$

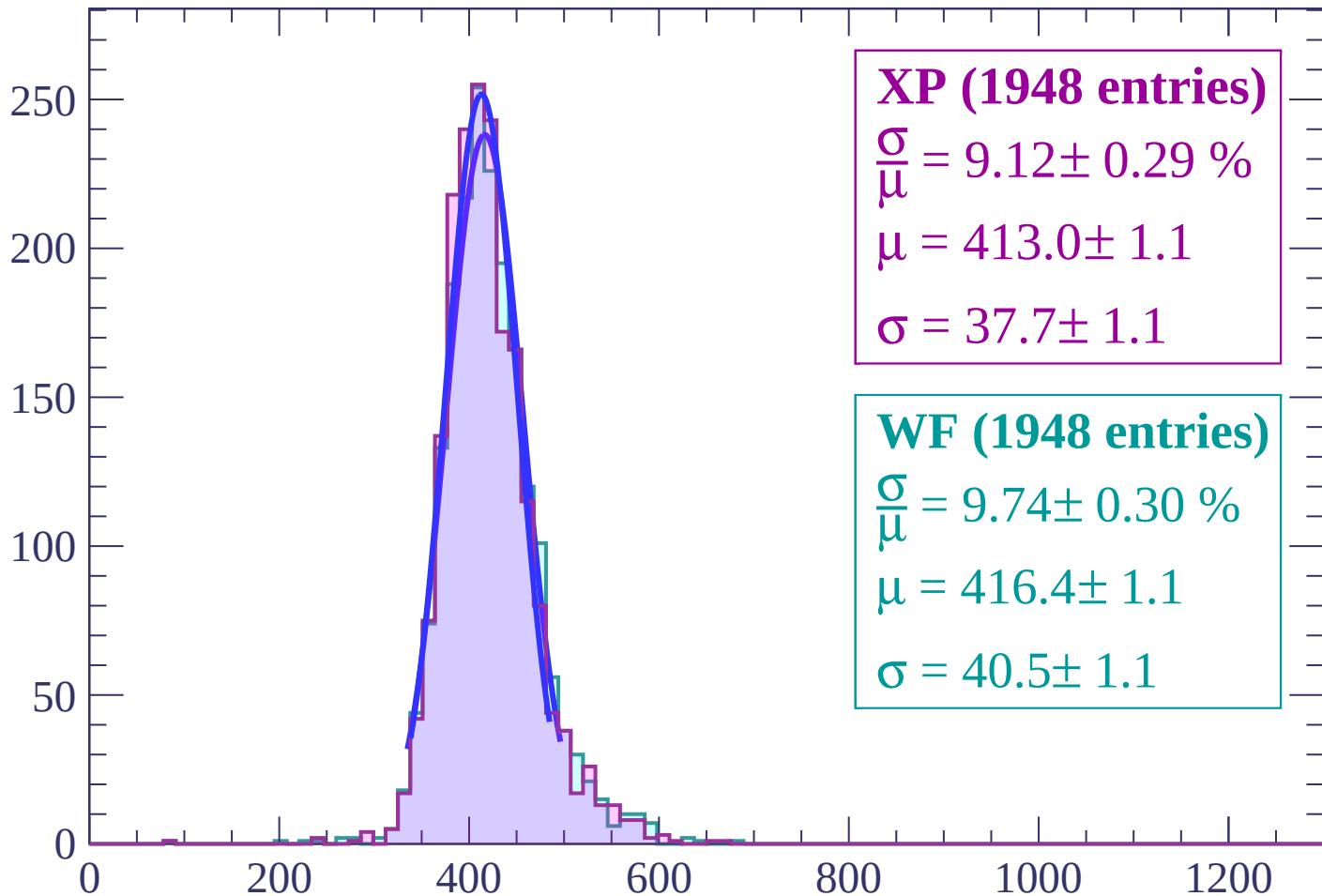
$$\mu = 417.3 \pm 1.3$$

$$\sigma = 39.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

Count



XP (1948 entries)

$$\frac{\sigma}{\mu} = 9.12 \pm 0.29 \%$$

$$\mu = 413.0 \pm 1.1$$

$$\sigma = 37.7 \pm 1.1$$

WF (1948 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.30 \%$$

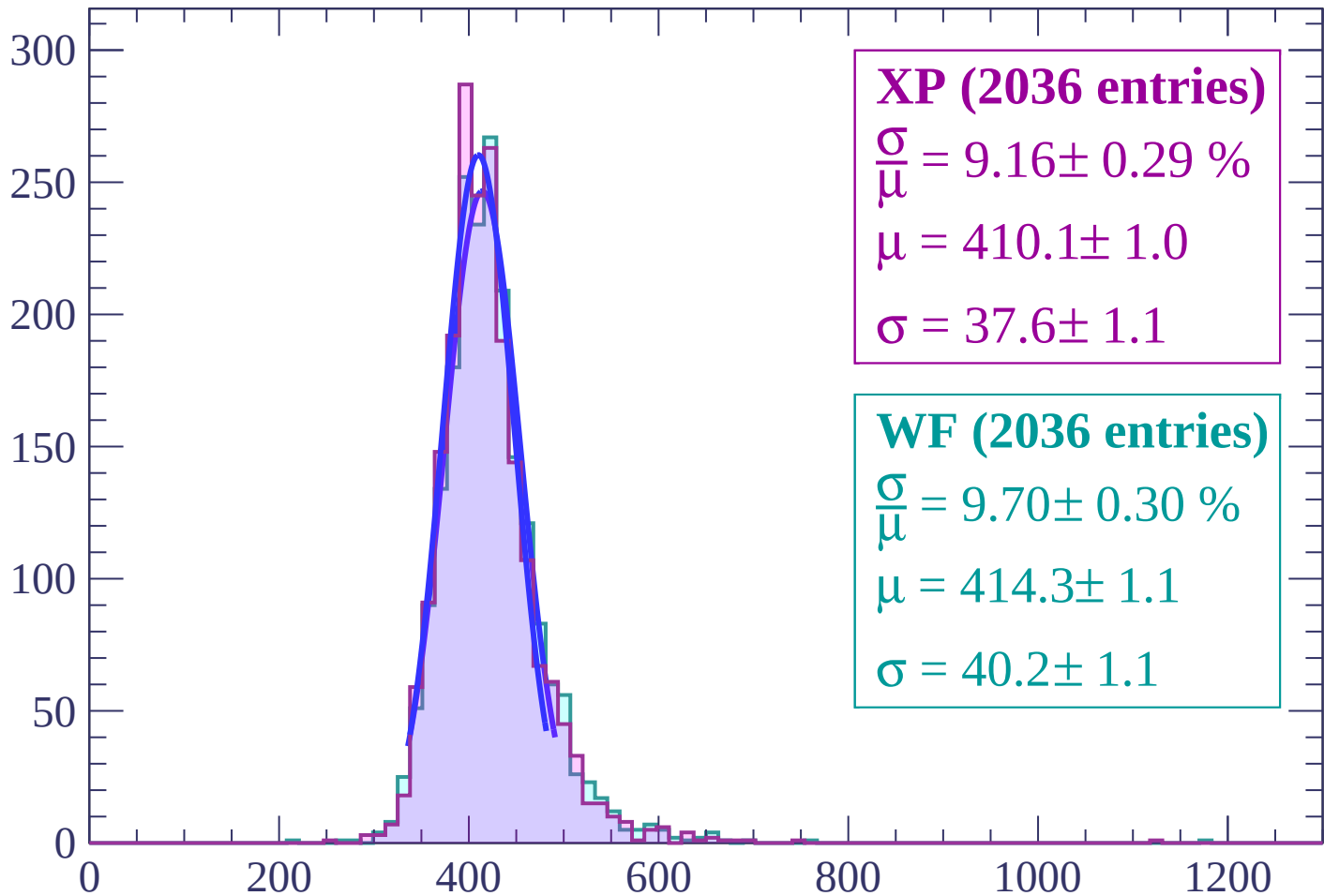
$$\mu = 416.4 \pm 1.1$$

$$\sigma = 40.5 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

Count



XP (2036 entries)

$$\frac{\sigma}{\mu} = 9.16 \pm 0.29 \%$$

$$\mu = 410.1 \pm 1.0$$

$$\sigma = 37.6 \pm 1.1$$

WF (2036 entries)

$$\frac{\sigma}{\mu} = 9.70 \pm 0.30 \%$$

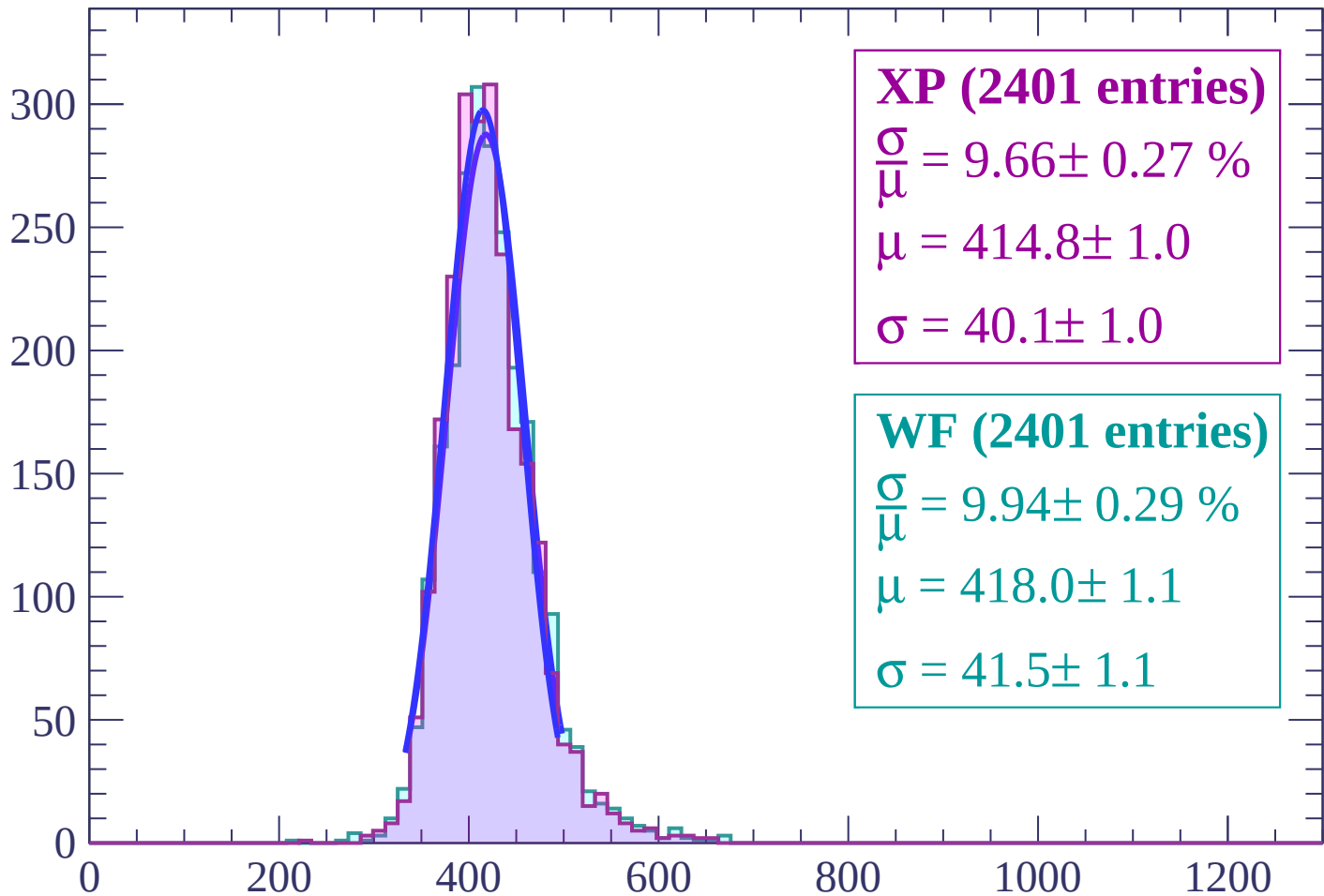
$$\mu = 414.3 \pm 1.1$$

$$\sigma = 40.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-22 < \theta < -20$

Count



XP (2401 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.27 \%$$

$$\mu = 414.8 \pm 1.0$$

$$\sigma = 40.1 \pm 1.0$$

WF (2401 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.29 \%$$

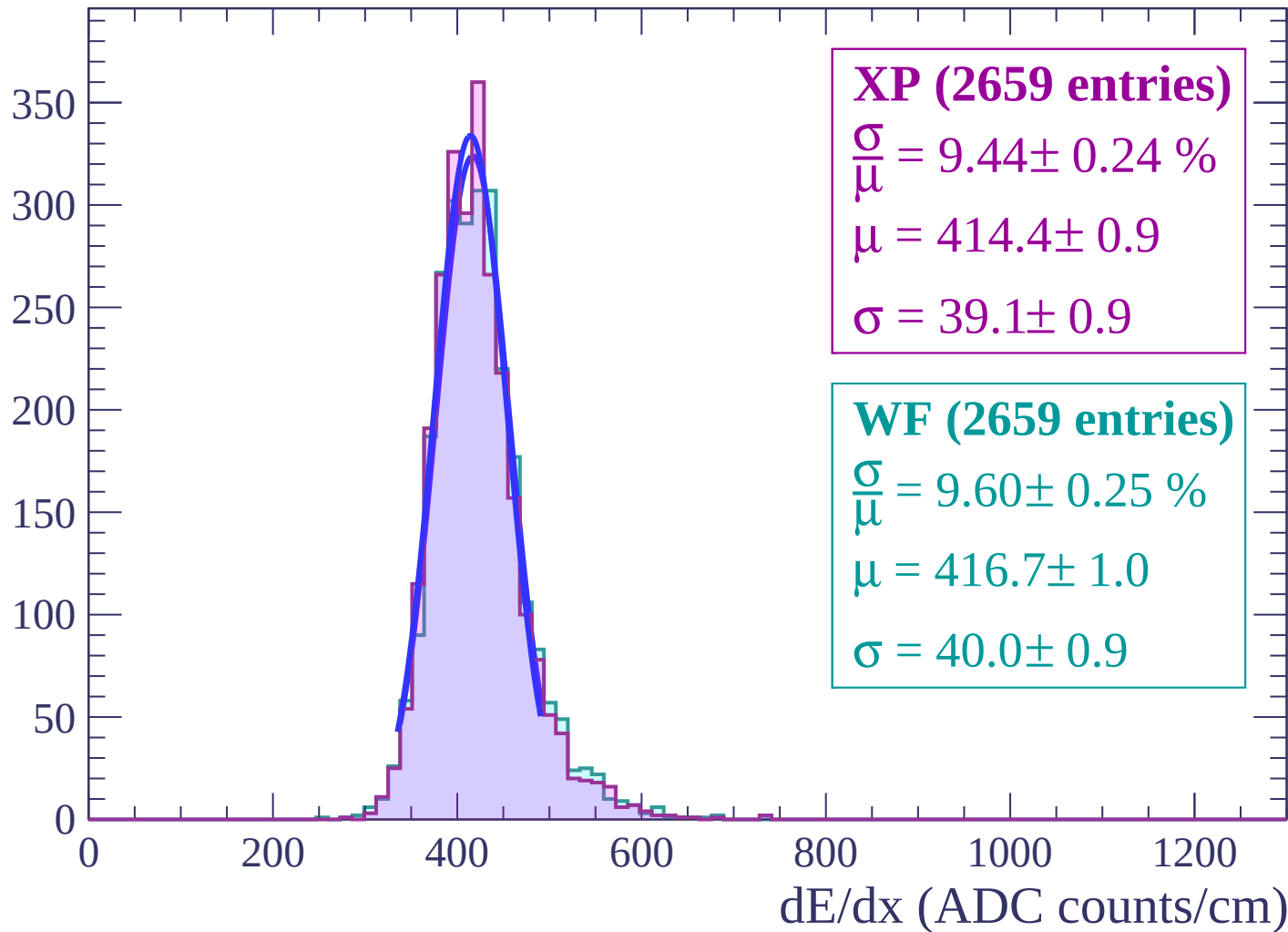
$$\mu = 418.0 \pm 1.1$$

$$\sigma = 41.5 \pm 1.1$$

dE/dx (ADC counts/cm)

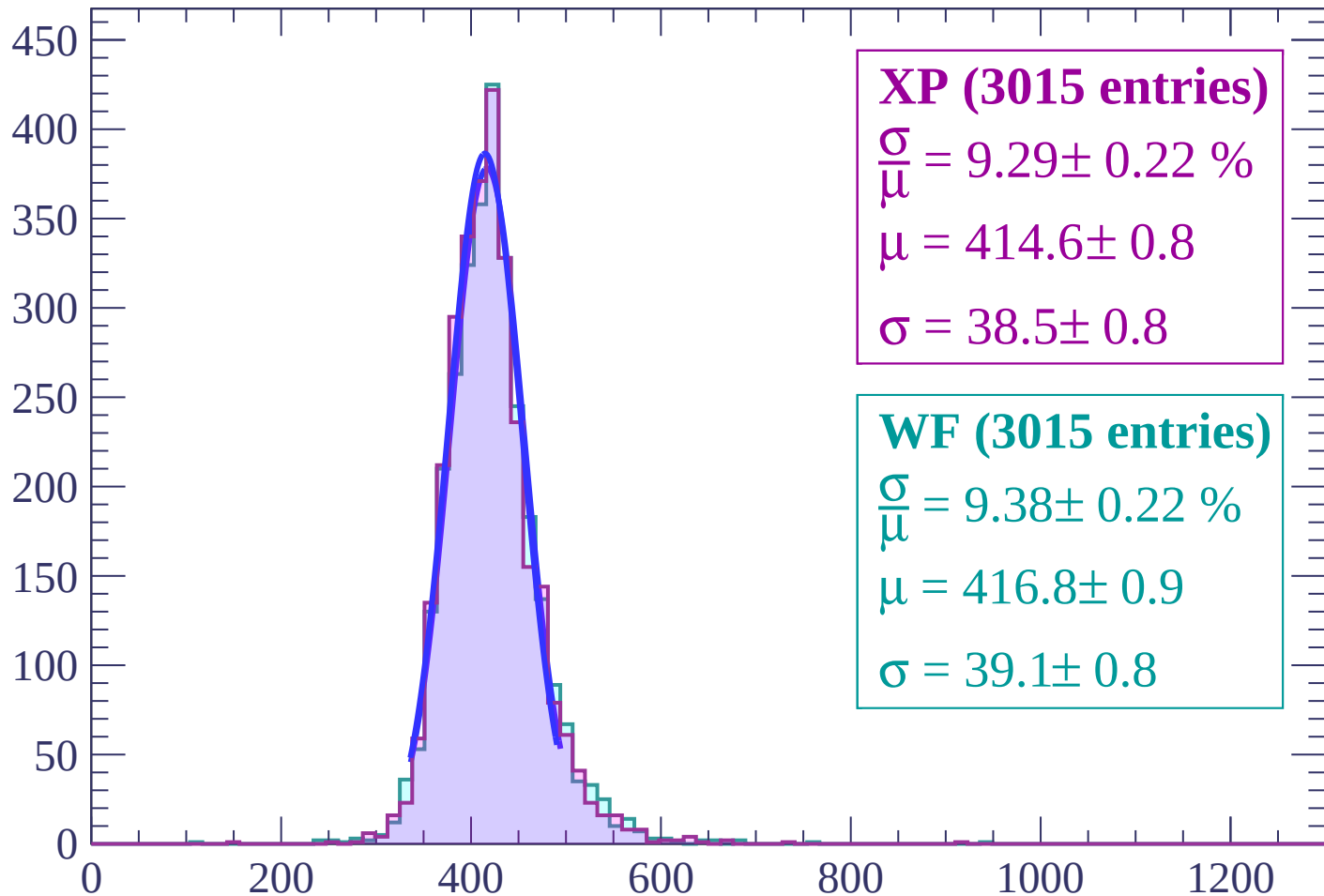
Energy loss | $-20 < \theta < -18$

Count



Energy loss | $-18 < \theta < -16$

Count



XP (3015 entries)

$$\frac{\sigma}{\mu} = 9.29 \pm 0.22 \%$$

$$\mu = 414.6 \pm 0.8$$

$$\sigma = 38.5 \pm 0.8$$

WF (3015 entries)

$$\frac{\sigma}{\mu} = 9.38 \pm 0.22 \%$$

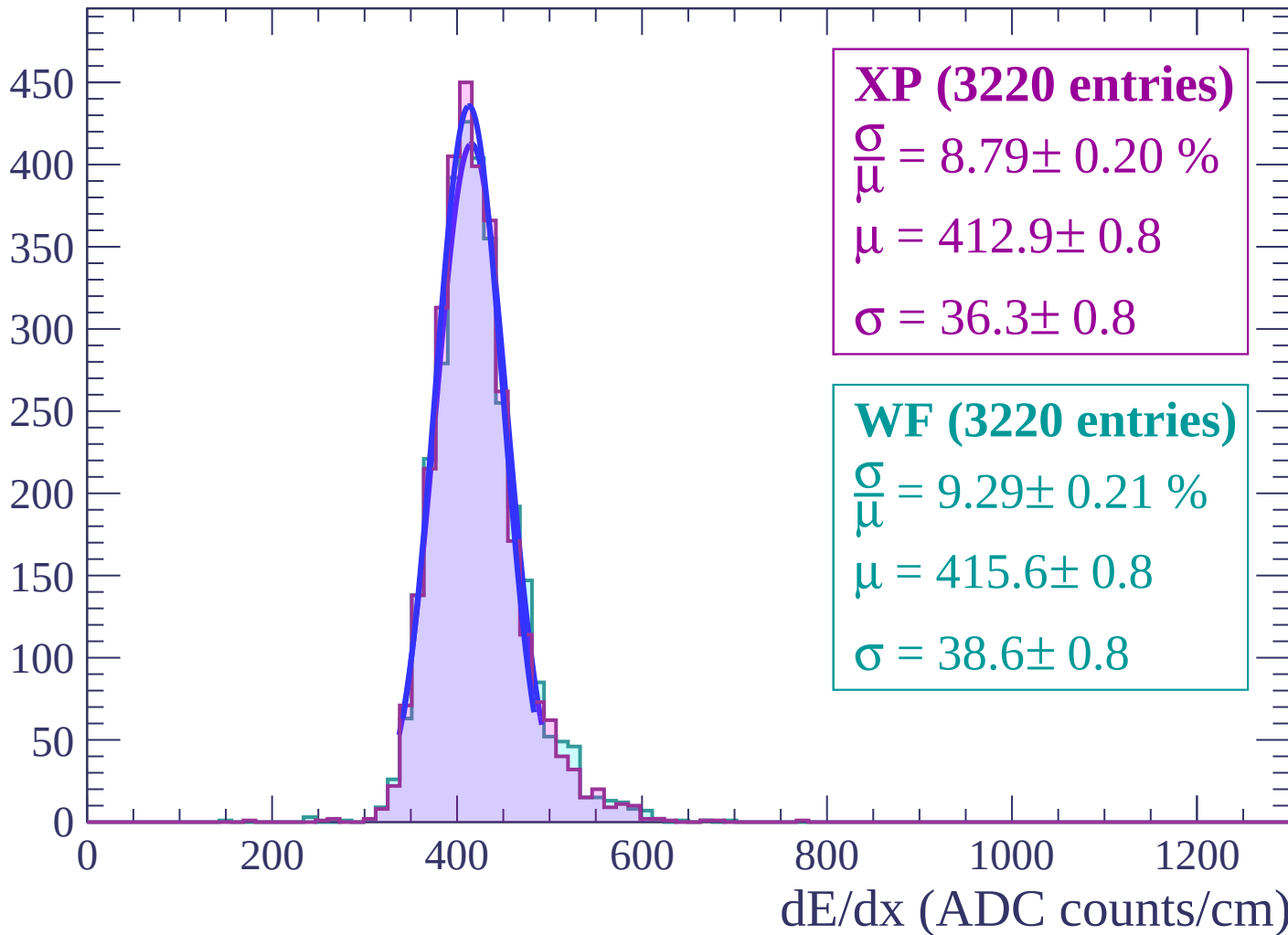
$$\mu = 416.8 \pm 0.9$$

$$\sigma = 39.1 \pm 0.8$$

dE/dx (ADC counts/cm)

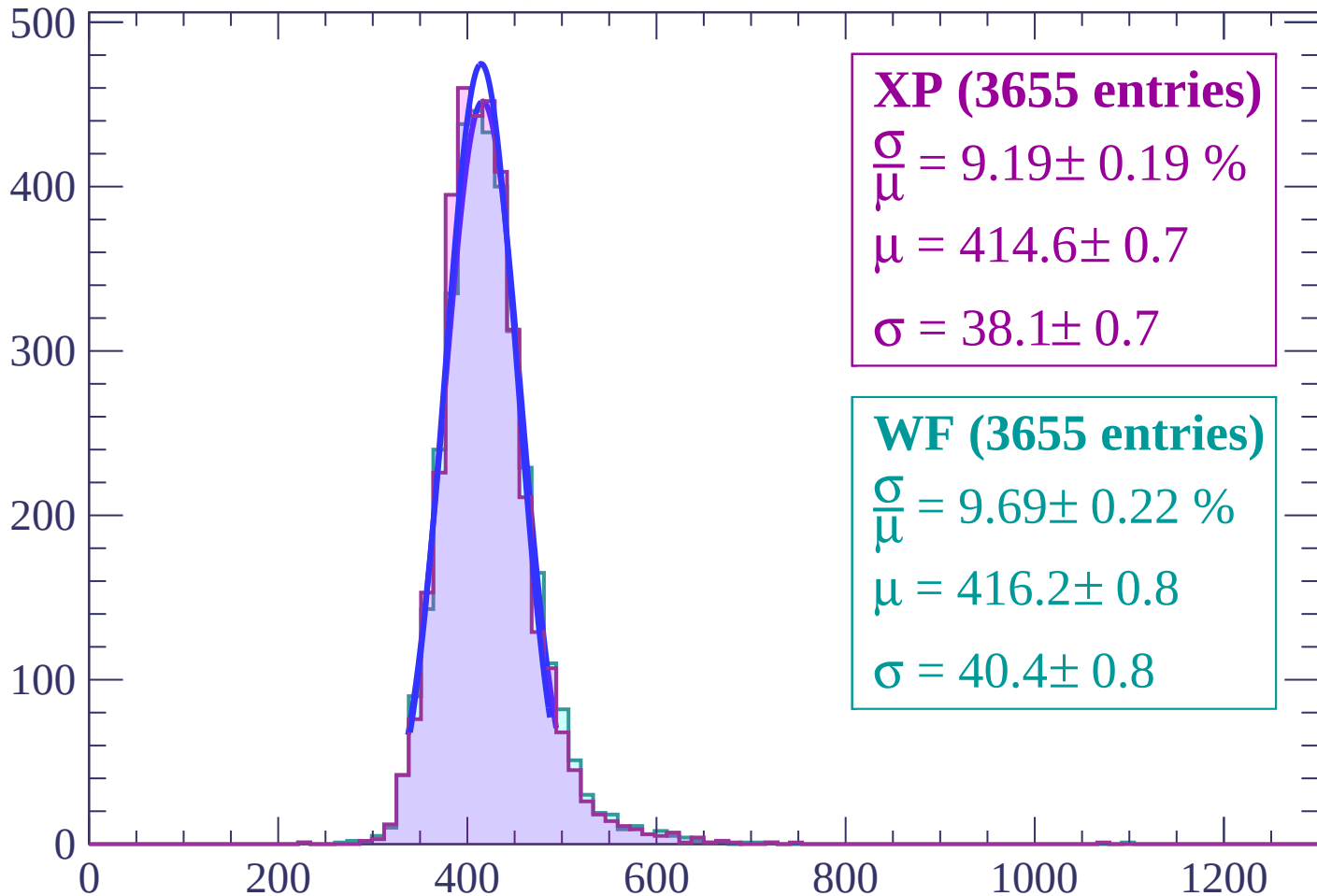
Energy loss | $-16 < \theta < -14$

Count



Energy loss | $-14 < \theta < -12$

Count



XP (3655 entries)

$\frac{\sigma}{\mu} = 9.19 \pm 0.19 \%$

$\mu = 414.6 \pm 0.7$

$\sigma = 38.1 \pm 0.7$

WF (3655 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.22 \%$

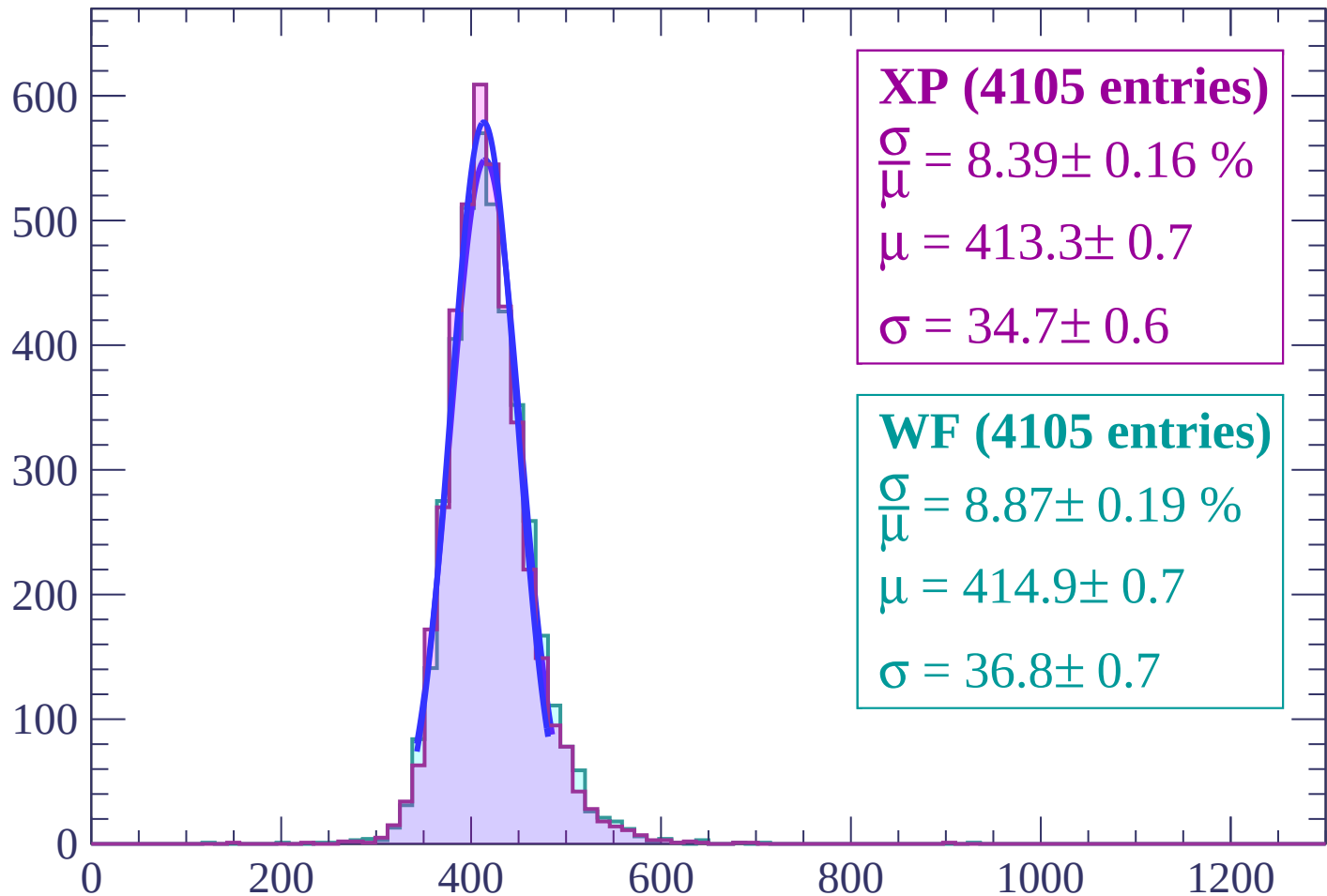
$\mu = 416.2 \pm 0.8$

$\sigma = 40.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

Count



XP (4105 entries)

$\frac{\sigma}{\mu} = 8.39 \pm 0.16 \%$

$\mu = 413.3 \pm 0.7$

$\sigma = 34.7 \pm 0.6$

WF (4105 entries)

$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$

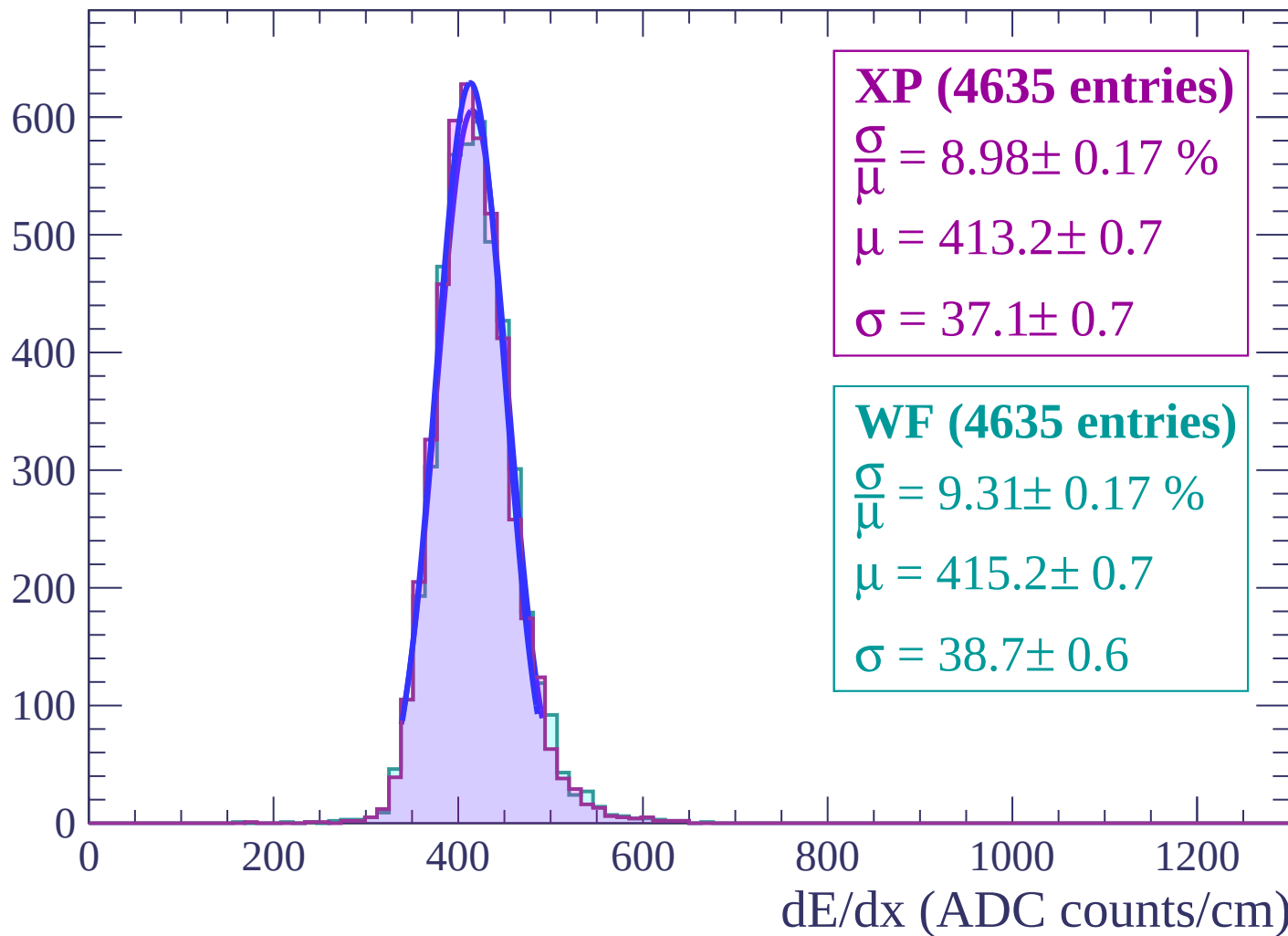
$\mu = 414.9 \pm 0.7$

$\sigma = 36.8 \pm 0.7$

dE/dx (ADC counts/cm)

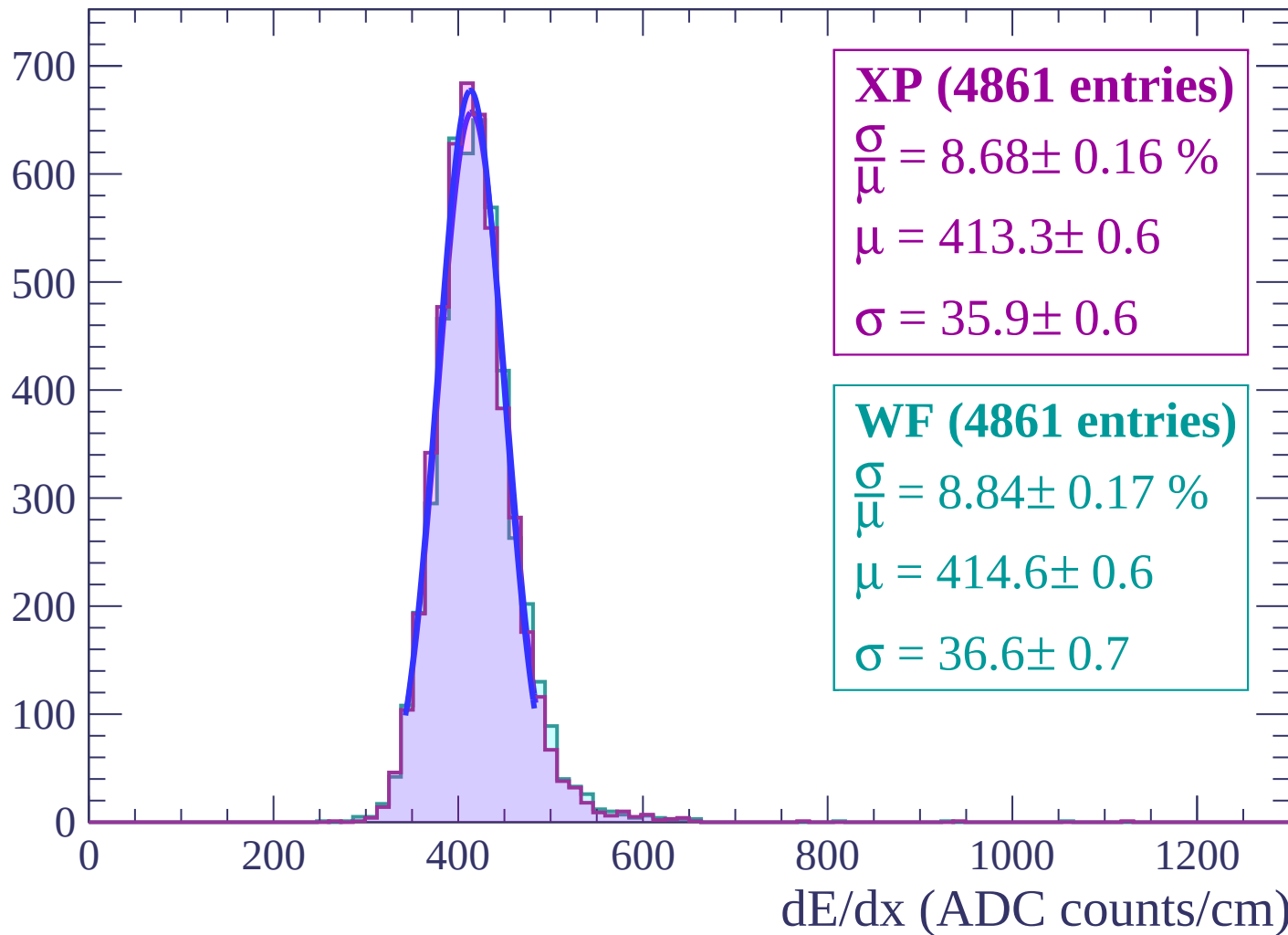
Energy loss | $-10 < \theta < -8$

Count



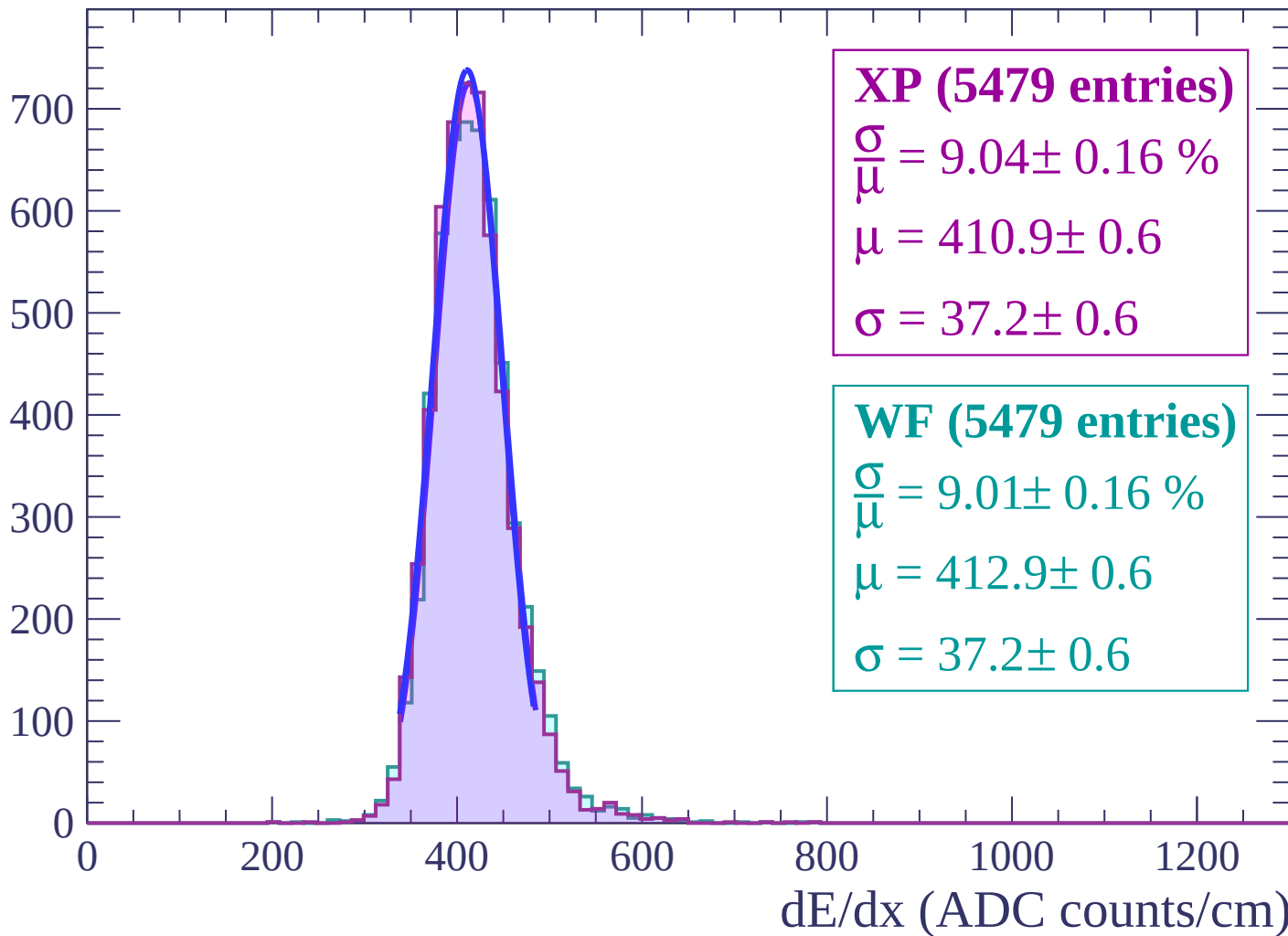
Energy loss | $-8 < \theta < -6$

Count



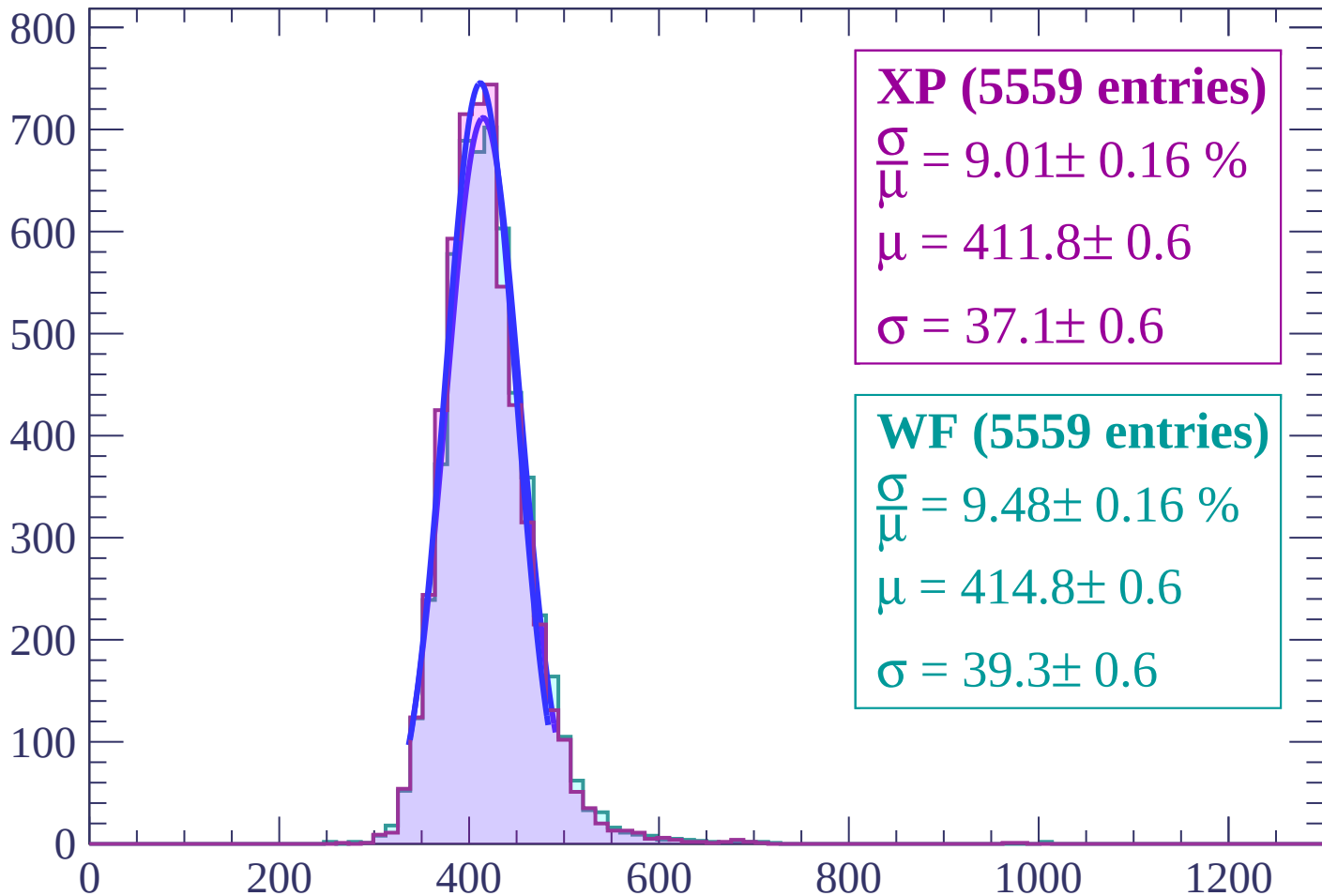
Energy loss | $-6 < \theta < -4$

Count



Energy loss | $-4 < \theta < -2$

Count



XP (5559 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.16 \%$$

$$\mu = 411.8 \pm 0.6$$

$$\sigma = 37.1 \pm 0.6$$

WF (5559 entries)

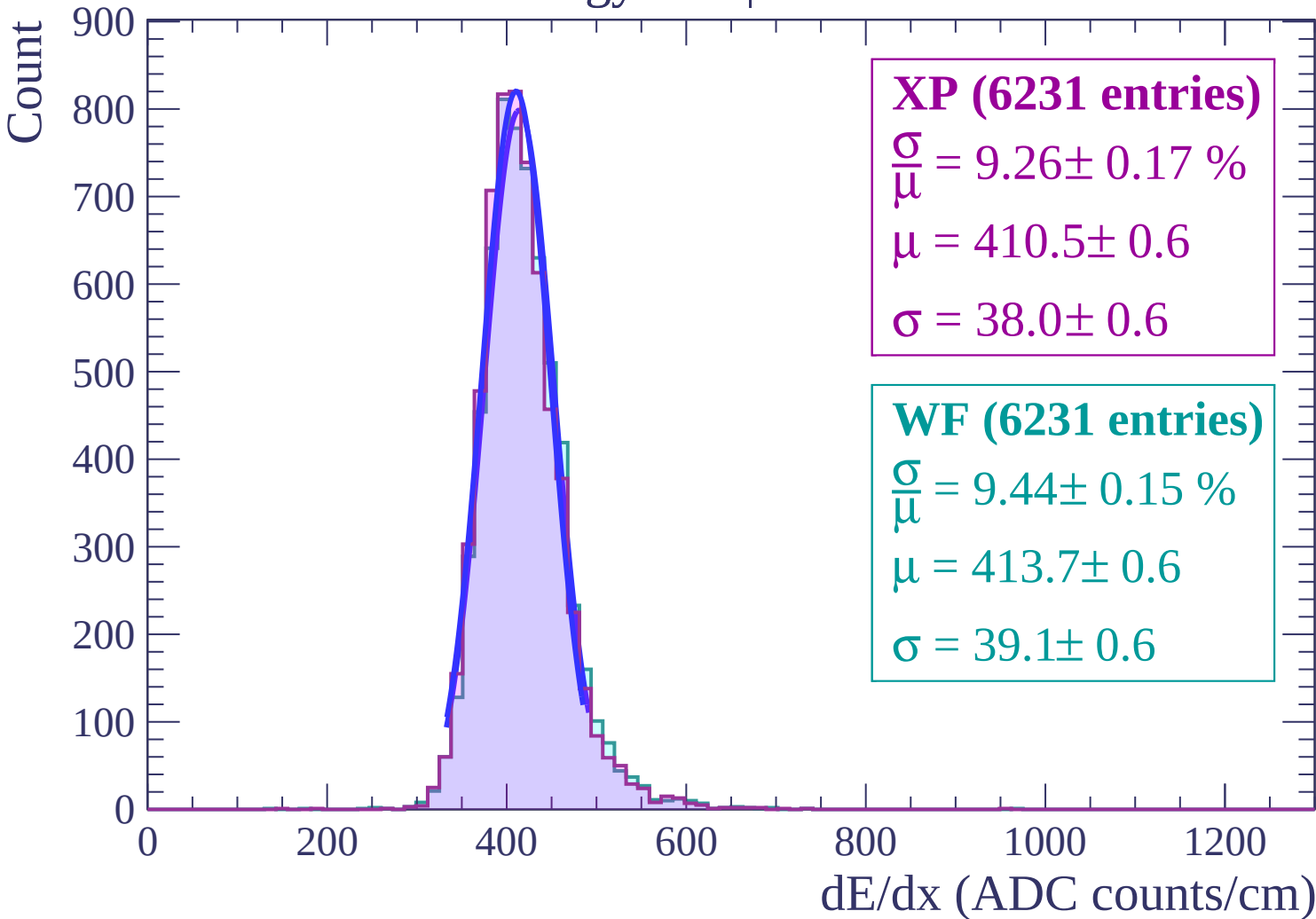
$$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$$

$$\mu = 414.8 \pm 0.6$$

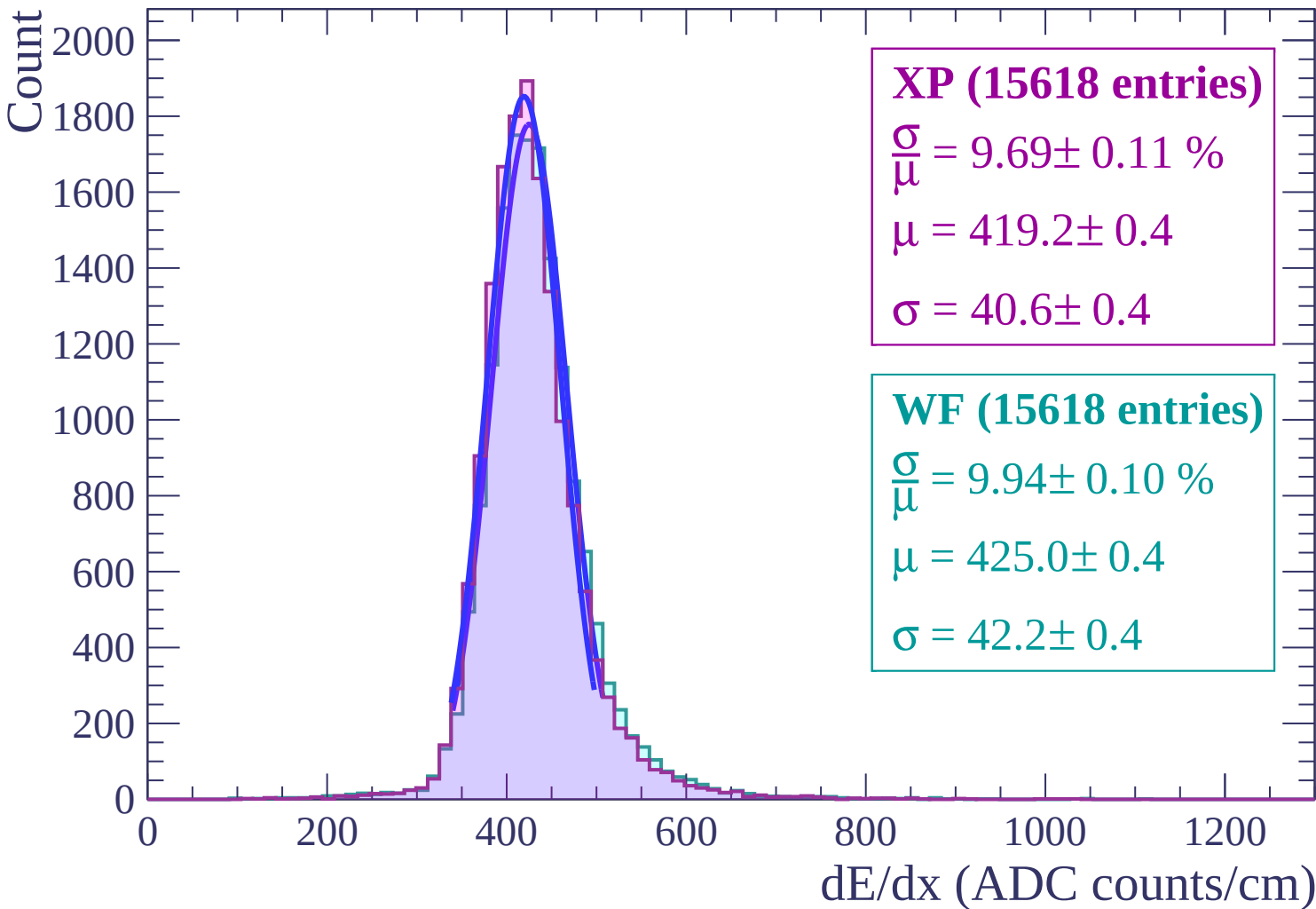
$$\sigma = 39.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-2 < \theta < 0$

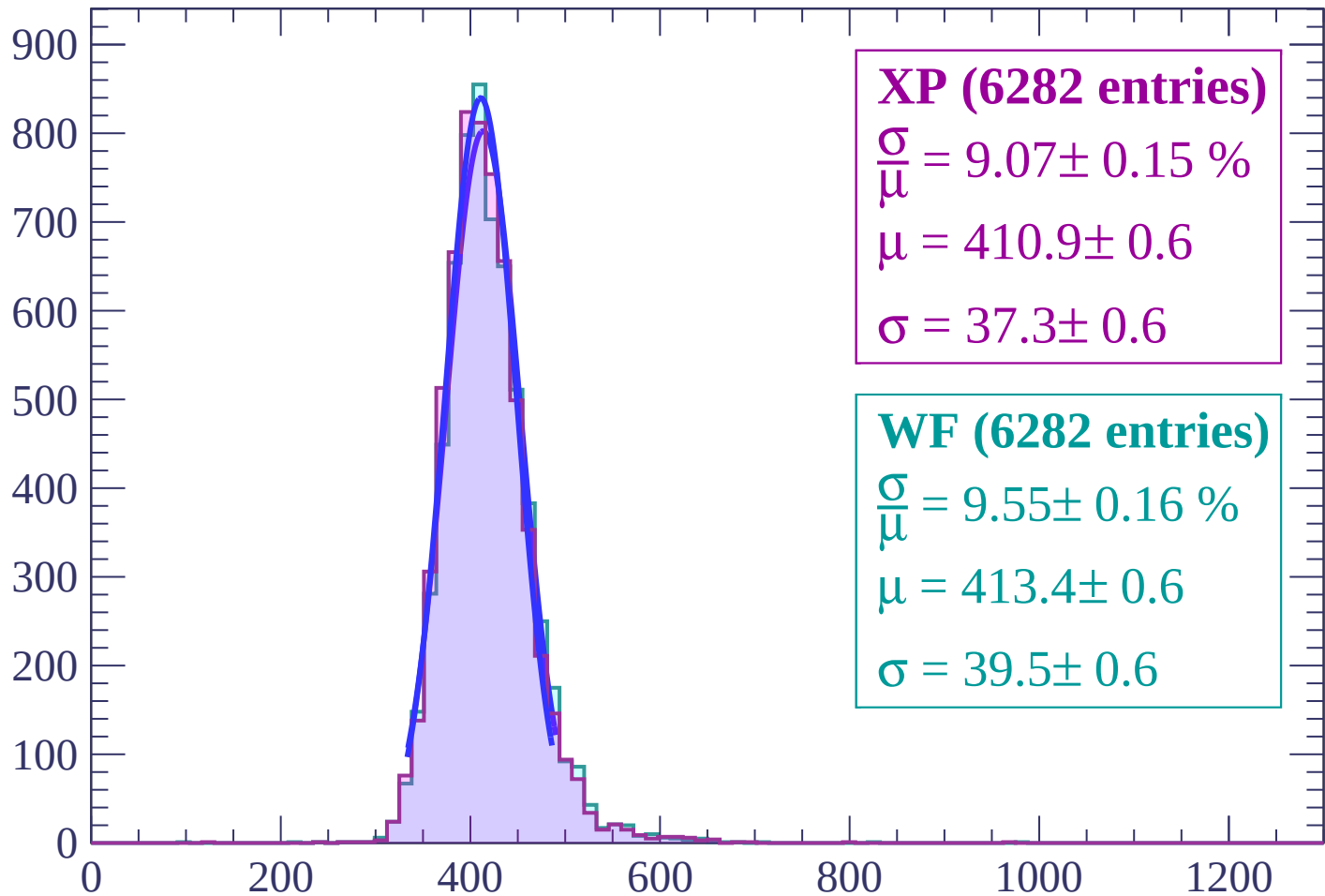


Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

Count



XP (6282 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.15 \%$

$\mu = 410.9 \pm 0.6$

$\sigma = 37.3 \pm 0.6$

WF (6282 entries)

$\frac{\sigma}{\mu} = 9.55 \pm 0.16 \%$

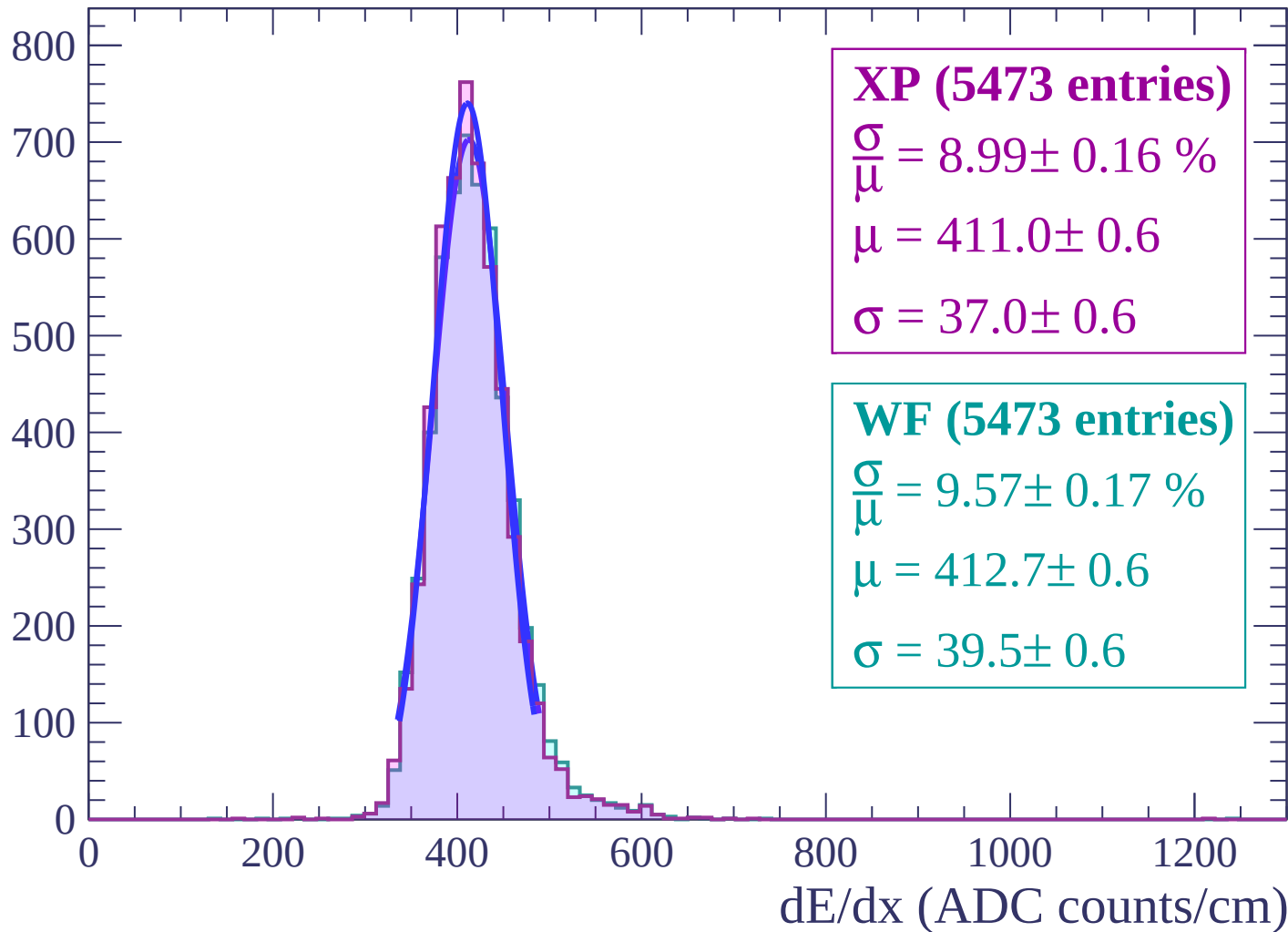
$\mu = 413.4 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

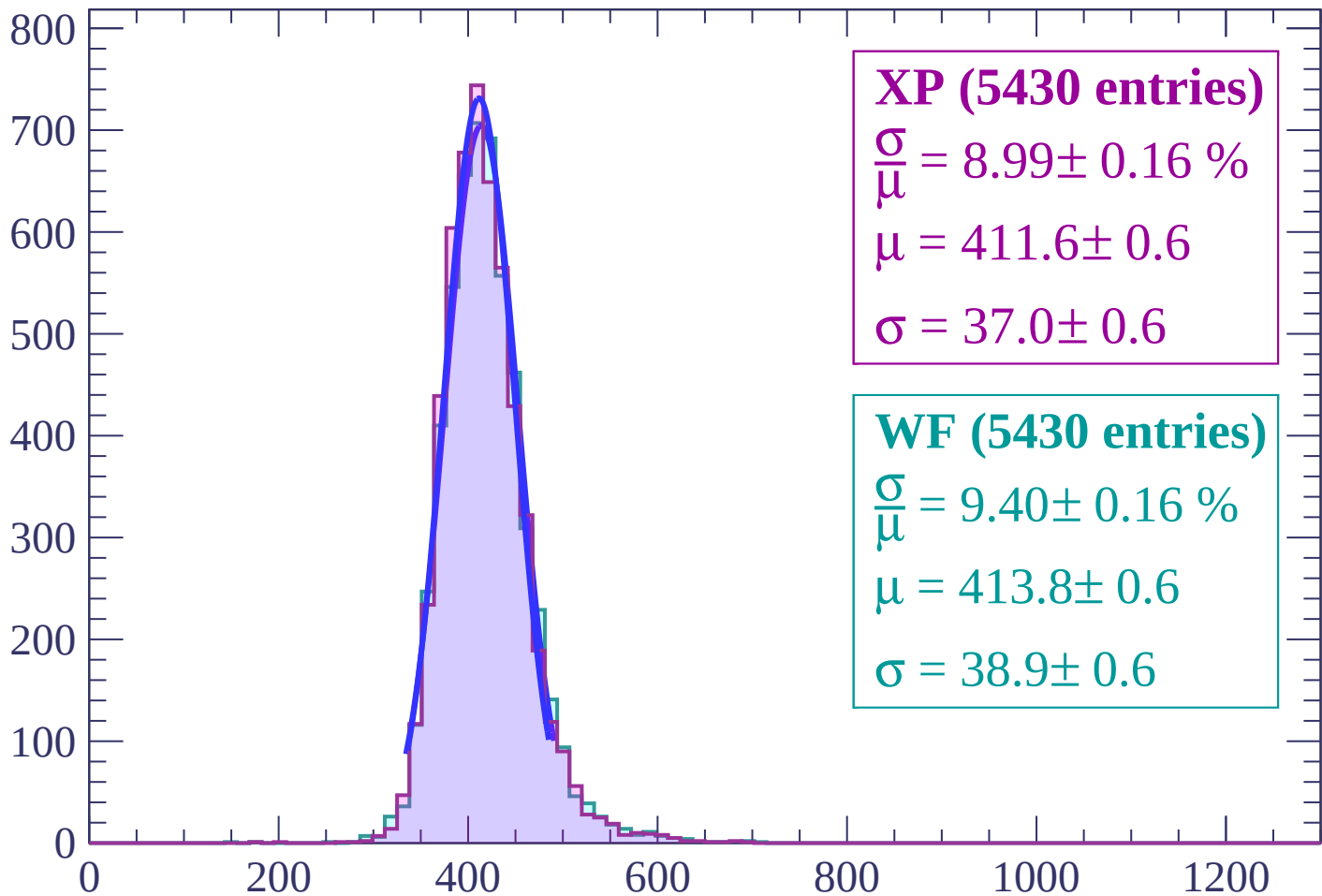
Energy loss | $4 < \theta < 6$

Count



Energy loss | $6 < \theta < 8$

Count



XP (5430 entries)

$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$

$\mu = 411.6 \pm 0.6$

$\sigma = 37.0 \pm 0.6$

WF (5430 entries)

$\frac{\sigma}{\mu} = 9.40 \pm 0.16 \%$

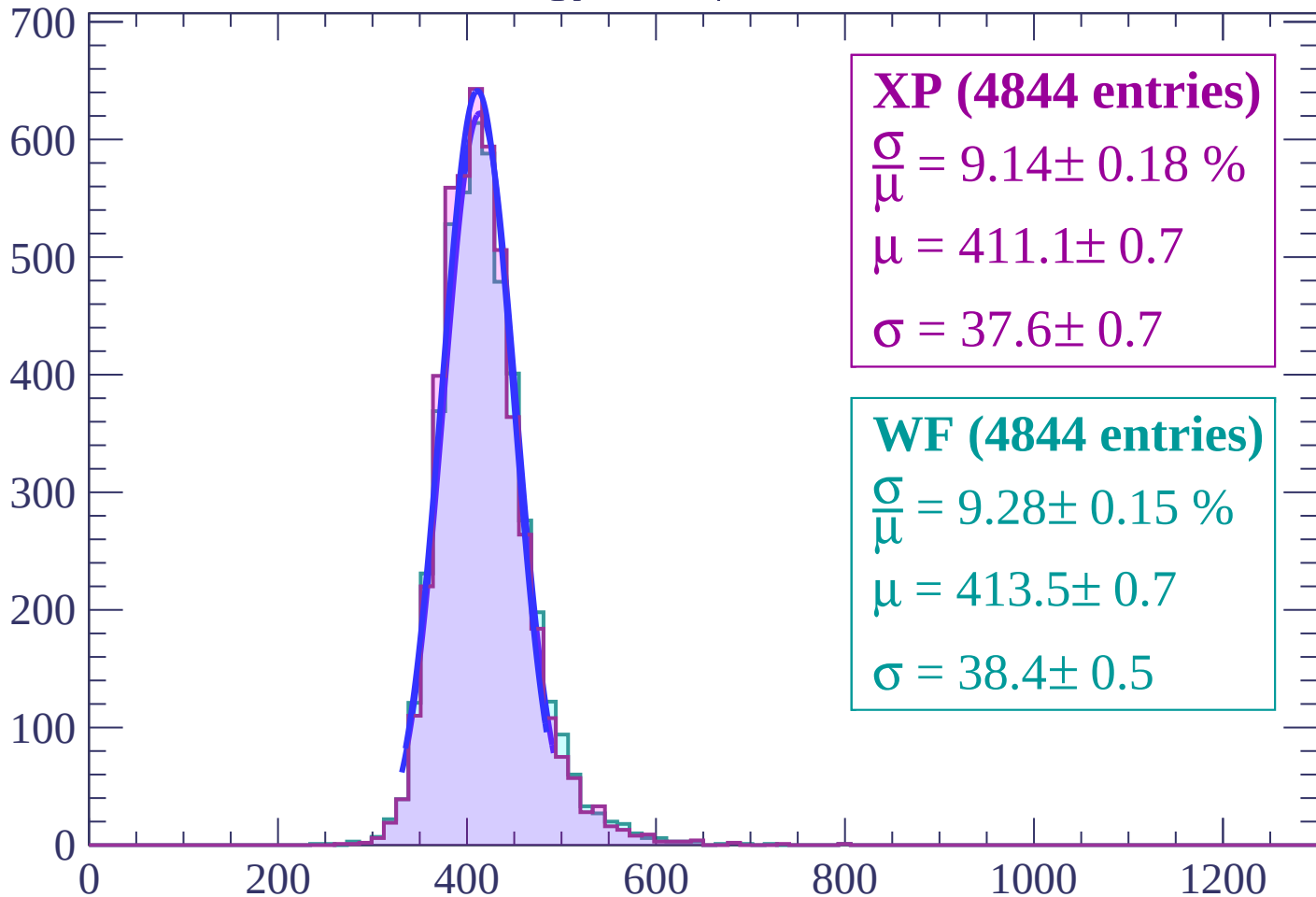
$\mu = 413.8 \pm 0.6$

$\sigma = 38.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $8 < \theta < 10$

Count



XP (4844 entries)

$\frac{\sigma}{\mu} = 9.14 \pm 0.18 \%$

$\mu = 411.1 \pm 0.7$

$\sigma = 37.6 \pm 0.7$

WF (4844 entries)

$\frac{\sigma}{\mu} = 9.28 \pm 0.15 \%$

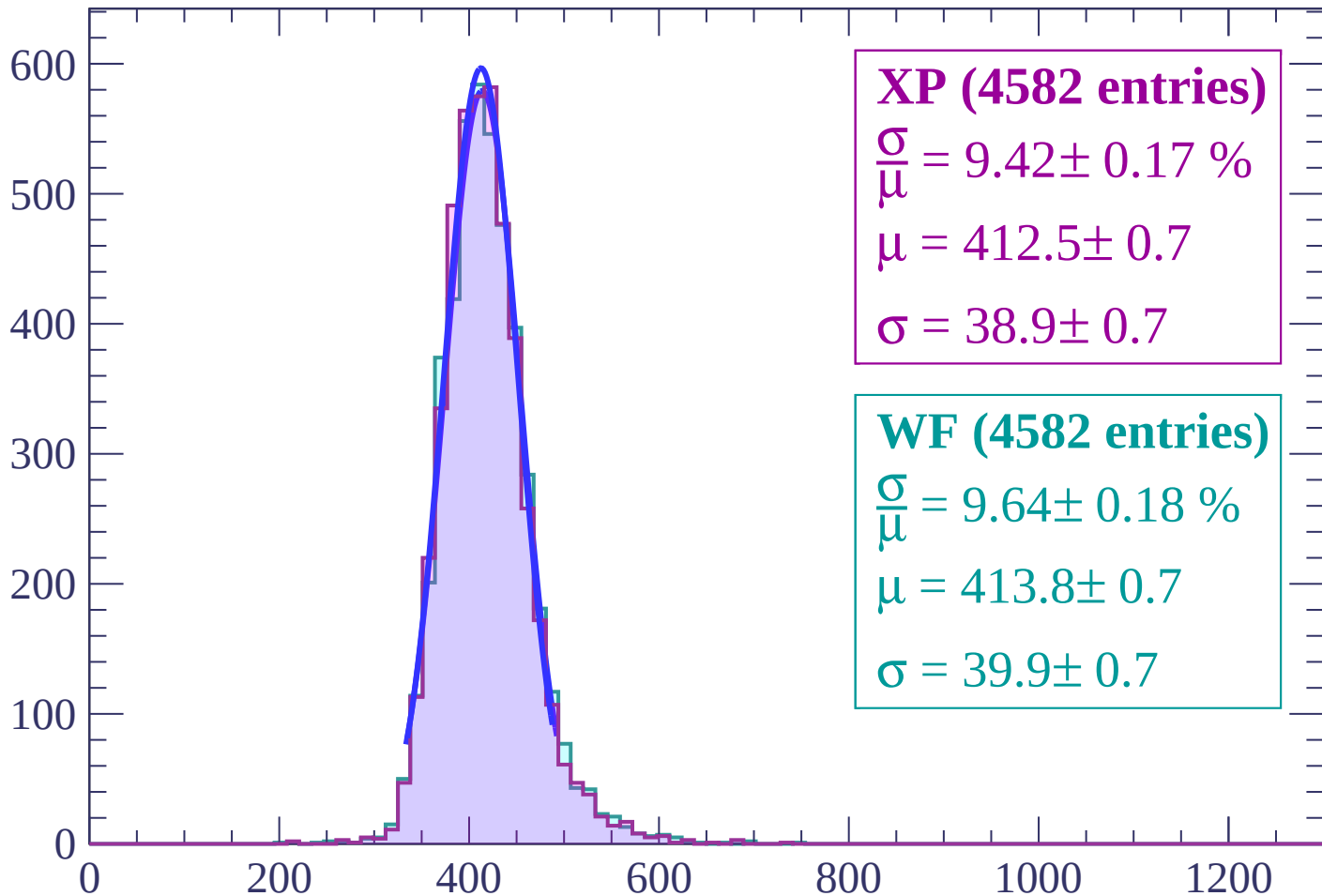
$\mu = 413.5 \pm 0.7$

$\sigma = 38.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $10 < \theta < 12$

Count



XP (4582 entries)

$\frac{\sigma}{\mu} = 9.42 \pm 0.17 \%$

$\mu = 412.5 \pm 0.7$

$\sigma = 38.9 \pm 0.7$

WF (4582 entries)

$\frac{\sigma}{\mu} = 9.64 \pm 0.18 \%$

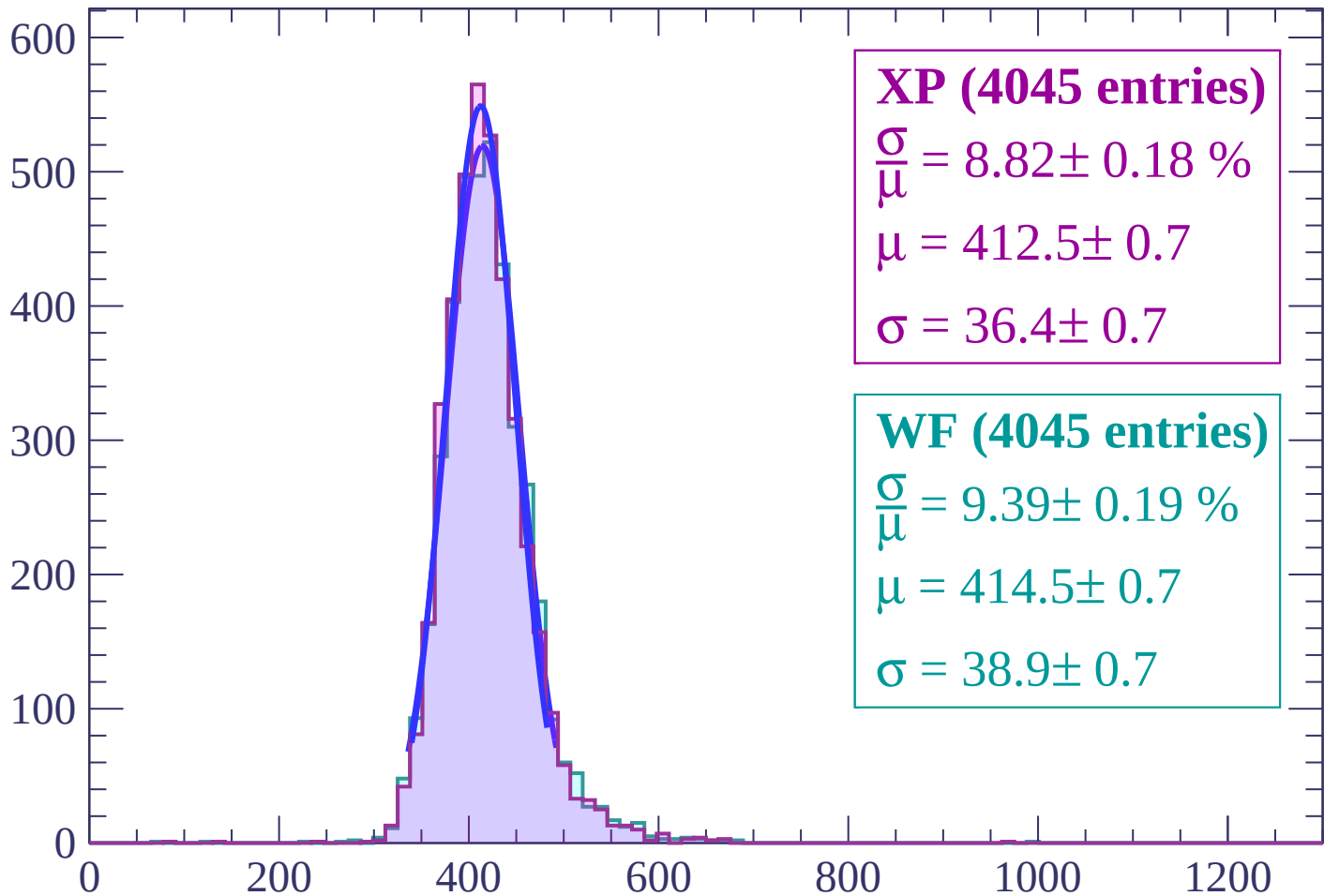
$\mu = 413.8 \pm 0.7$

$\sigma = 39.9 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 14$

Count



XP (4045 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.18 \%$$

$$\mu = 412.5 \pm 0.7$$

$$\sigma = 36.4 \pm 0.7$$

WF (4045 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.19 \%$$

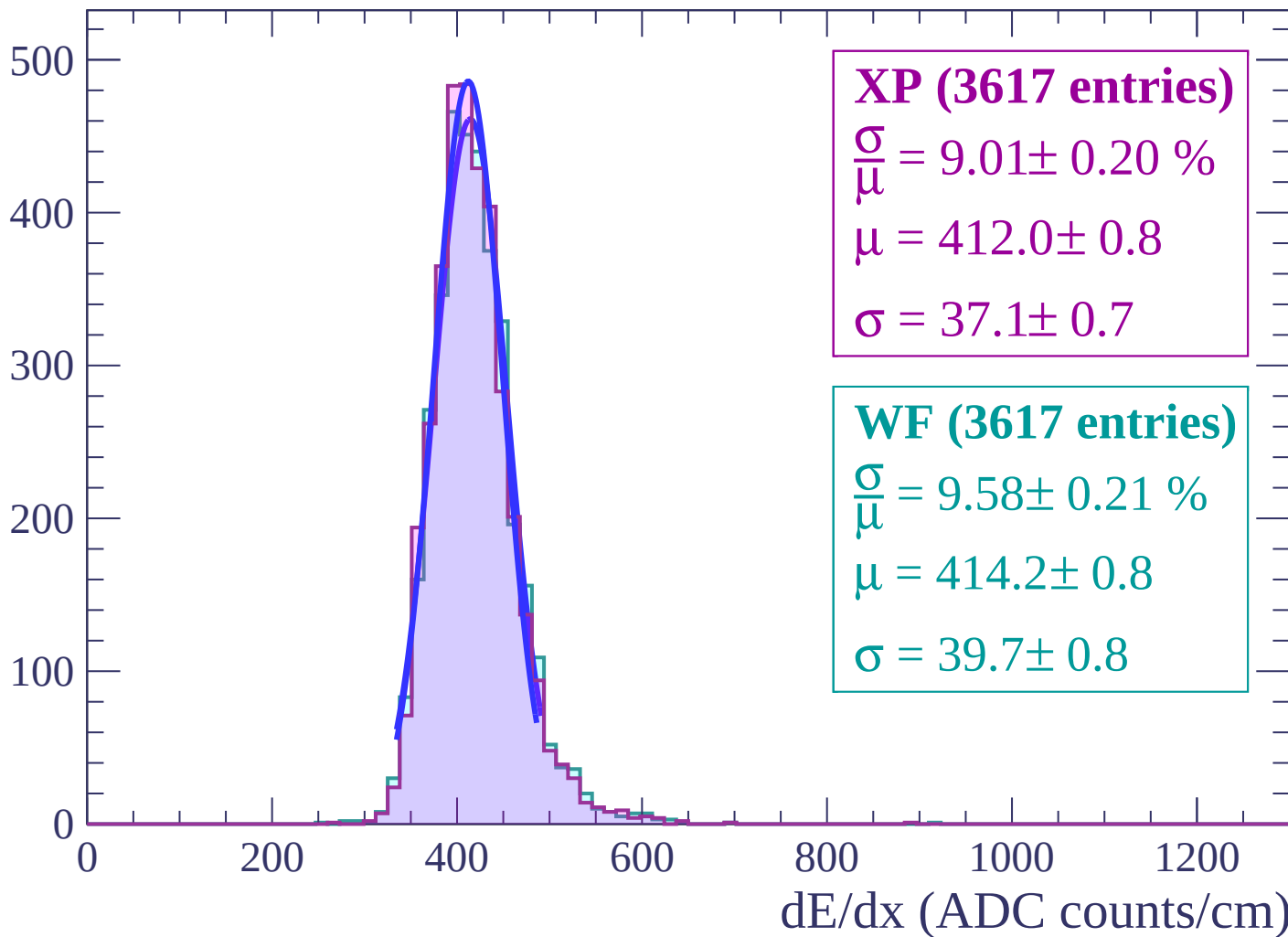
$$\mu = 414.5 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

dE/dx (ADC counts/cm)

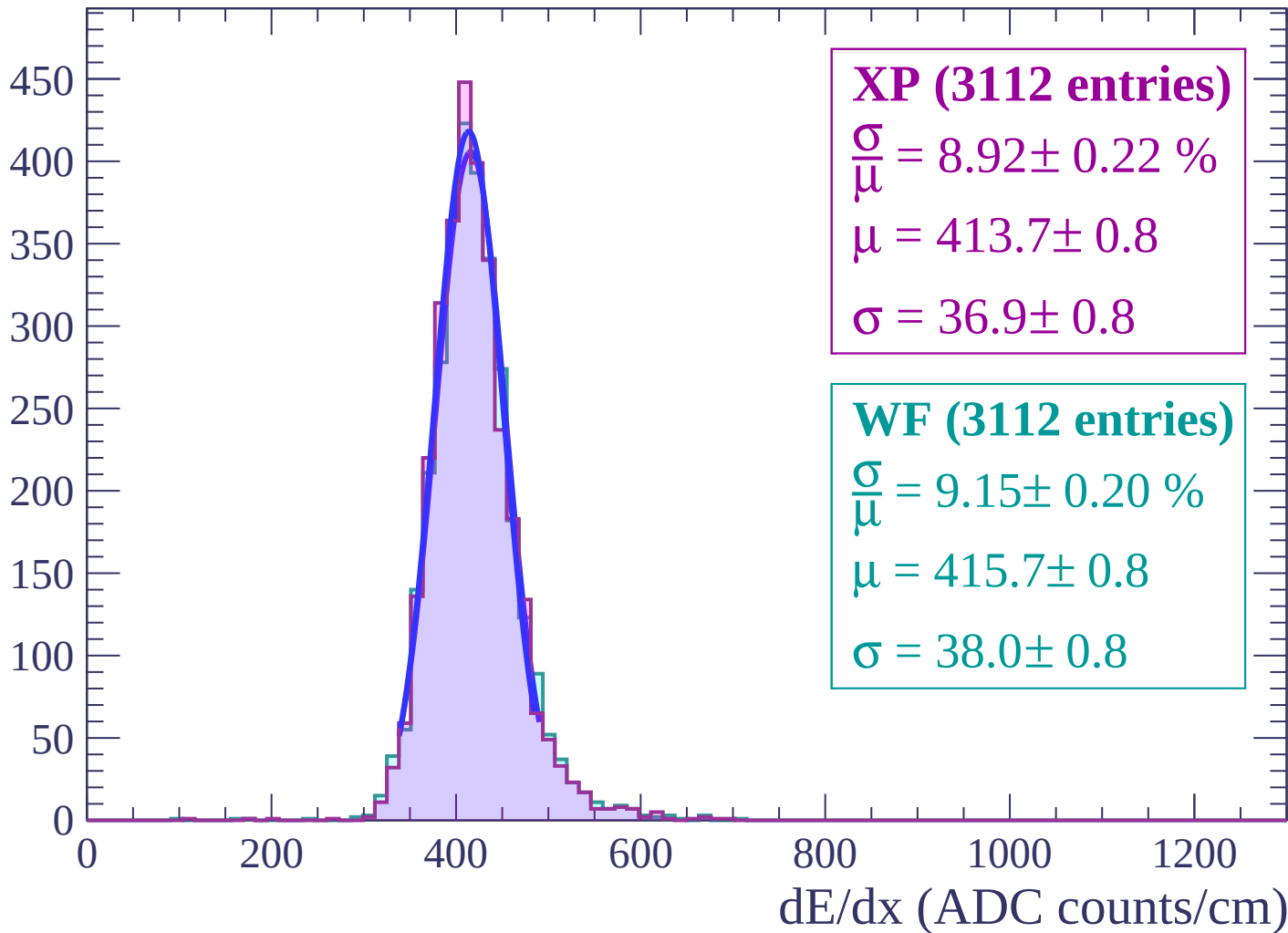
Energy loss | $14 < \theta < 16$

Count



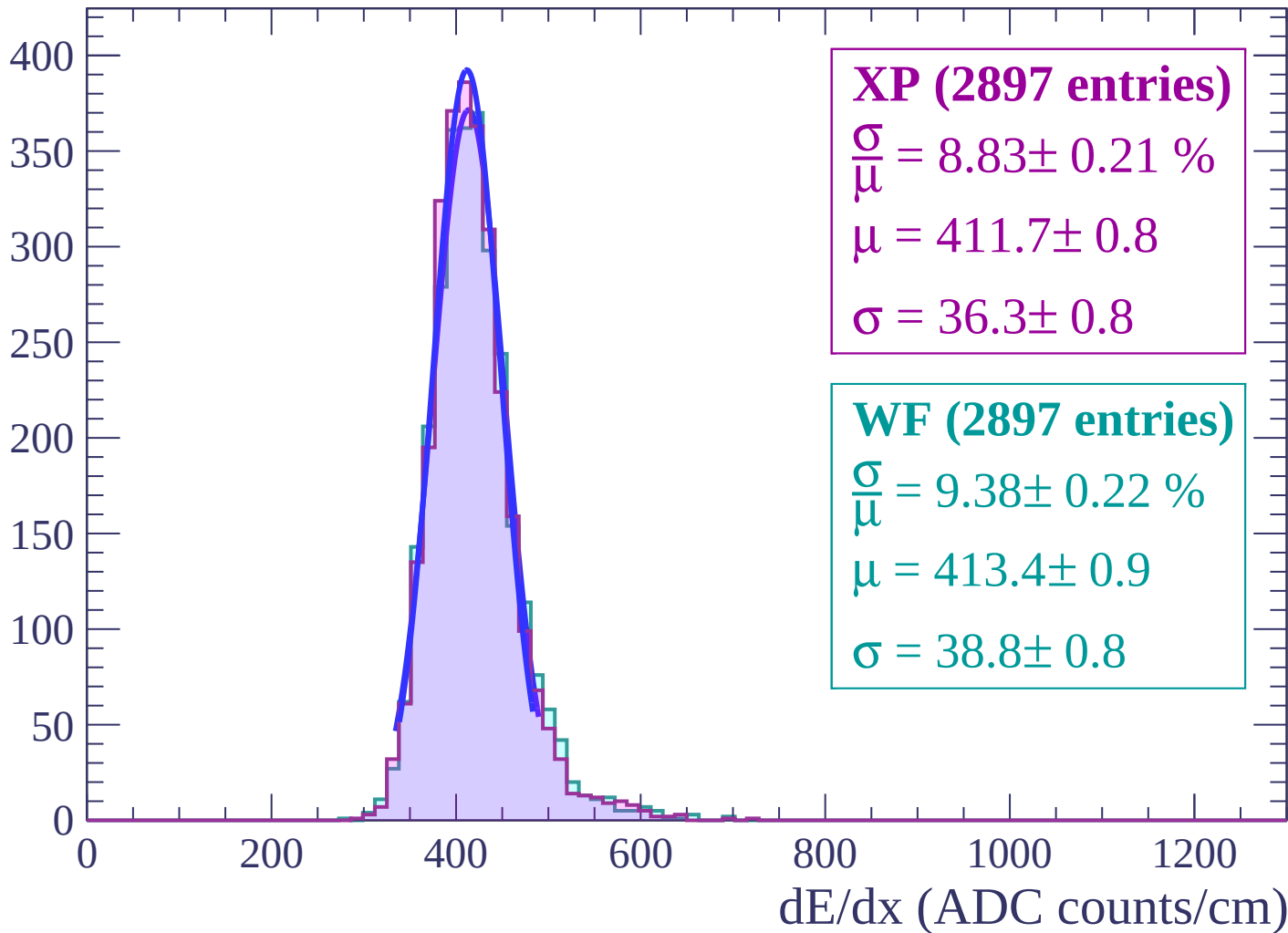
Energy loss | $16 < \theta < 18$

Count



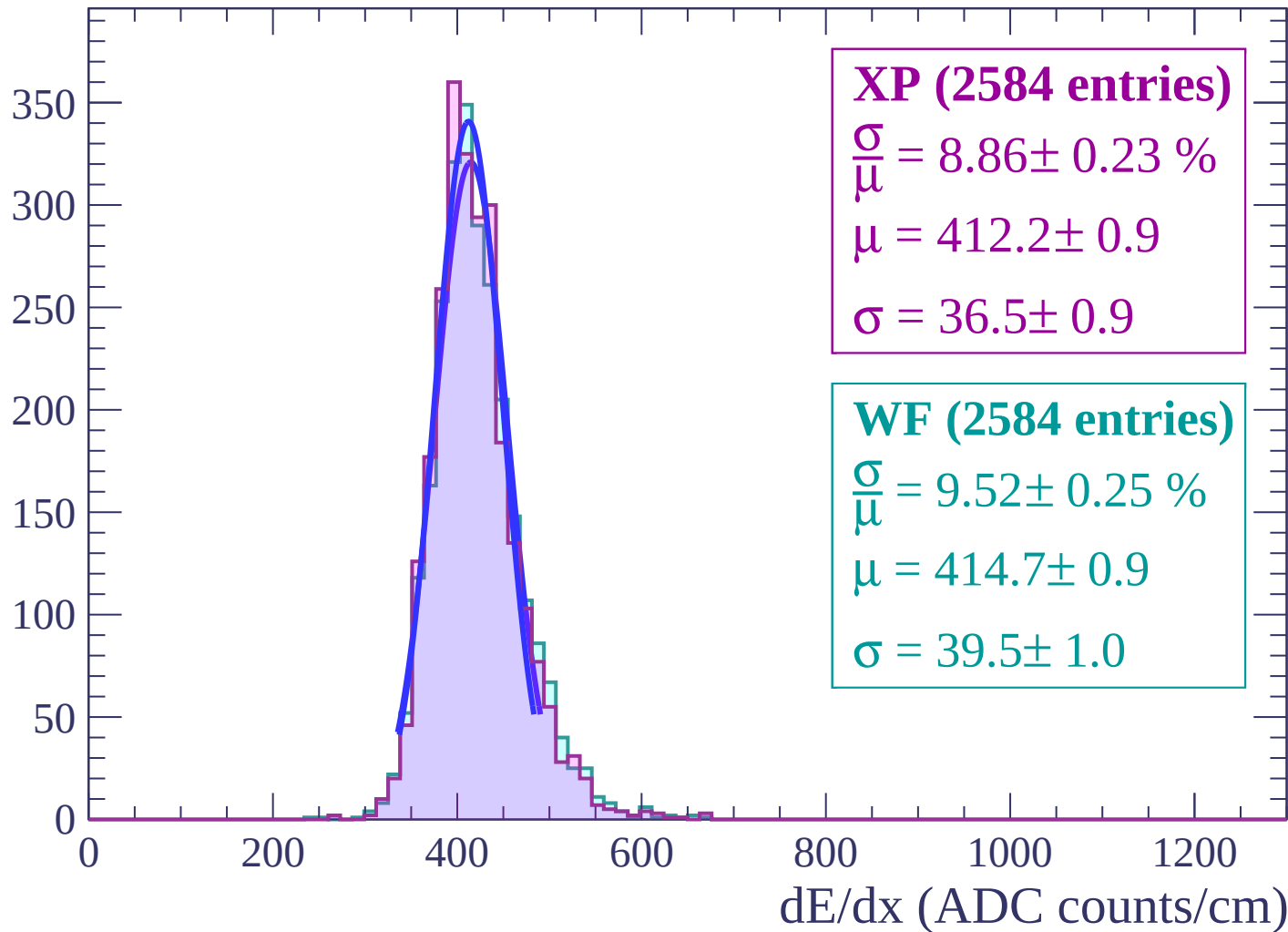
Energy loss | $18 < \theta < 20$

Count



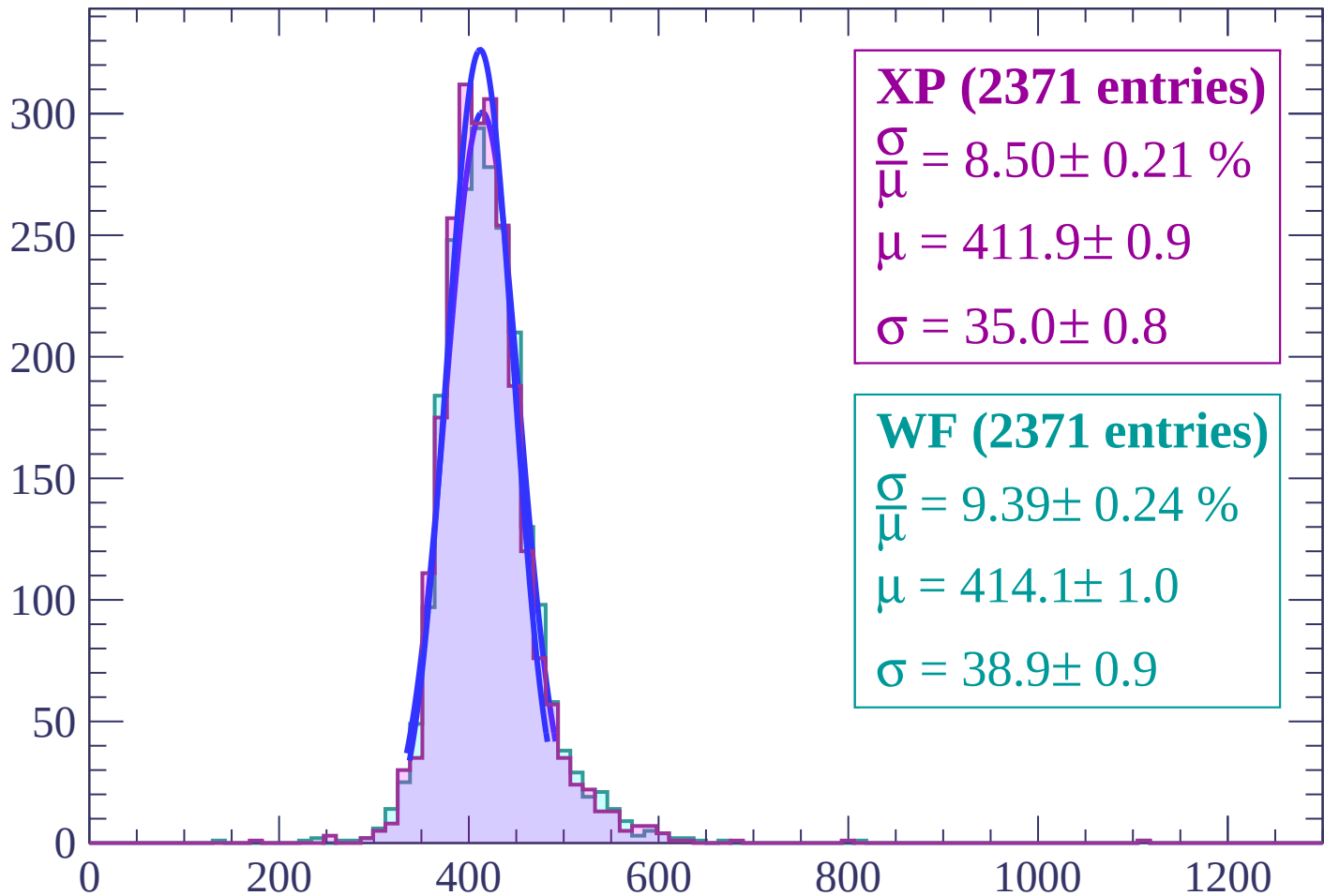
Energy loss | $20 < \theta < 22$

Count



Energy loss | $22 < \theta < 24$

Count



XP (2371 entries)

$\frac{\sigma}{\mu} = 8.50 \pm 0.21 \%$

$\mu = 411.9 \pm 0.9$

$\sigma = 35.0 \pm 0.8$

WF (2371 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.24 \%$

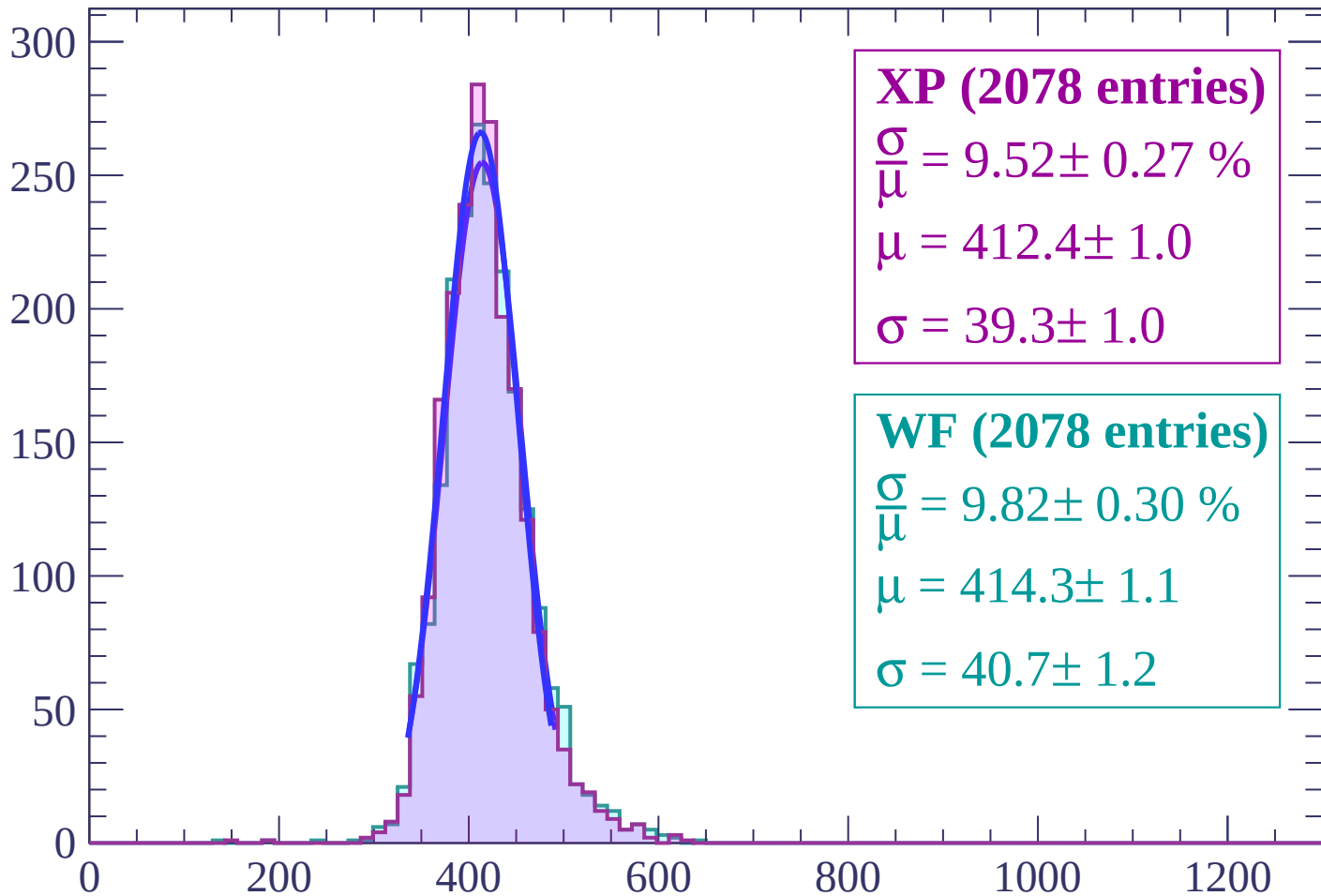
$\mu = 414.1 \pm 1.0$

$\sigma = 38.9 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 26$

Count



XP (2078 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.27 \%$

$\mu = 412.4 \pm 1.0$

$\sigma = 39.3 \pm 1.0$

WF (2078 entries)

$\frac{\sigma}{\mu} = 9.82 \pm 0.30 \%$

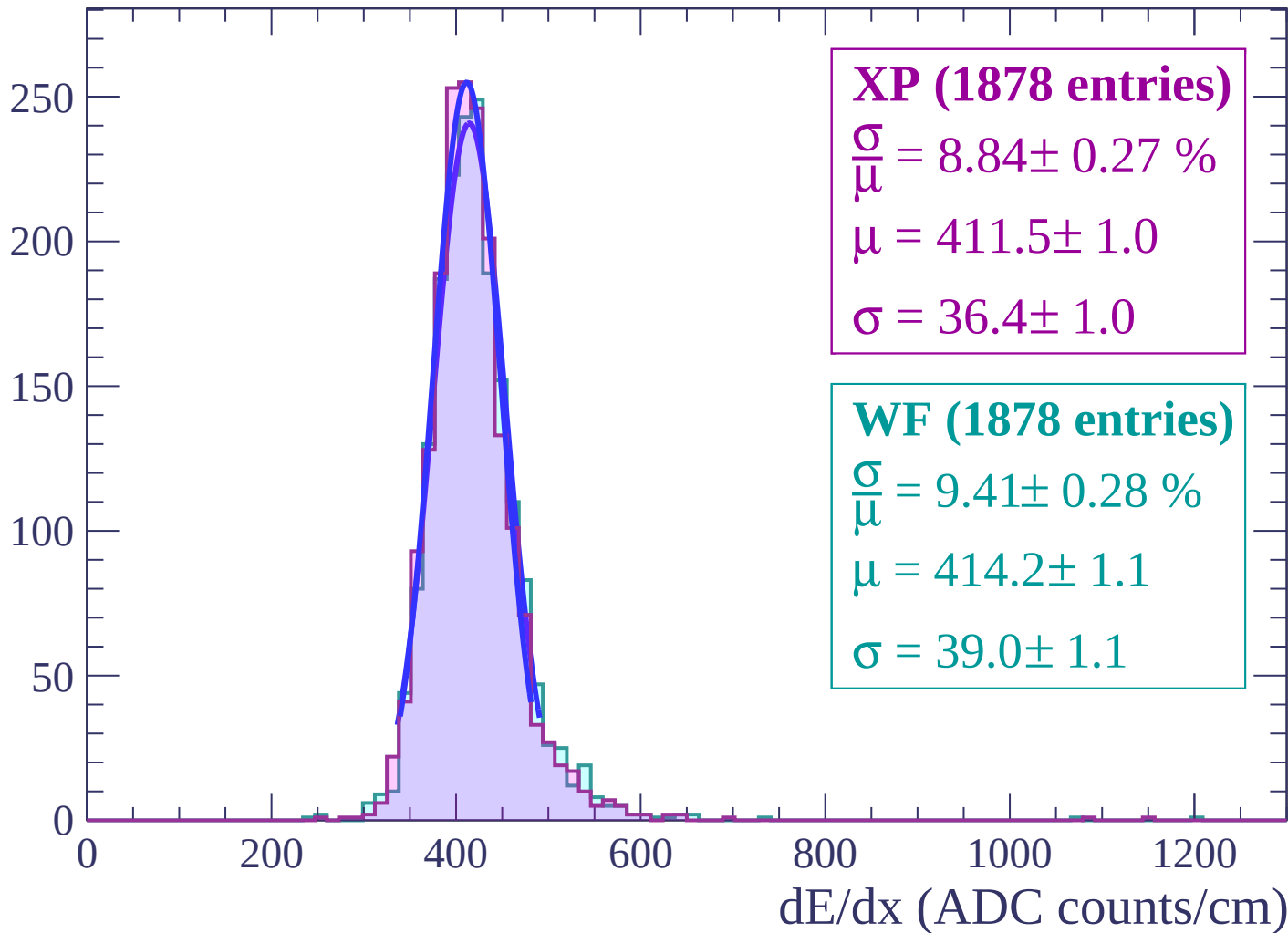
$\mu = 414.3 \pm 1.1$

$\sigma = 40.7 \pm 1.2$

dE/dx (ADC counts/cm)

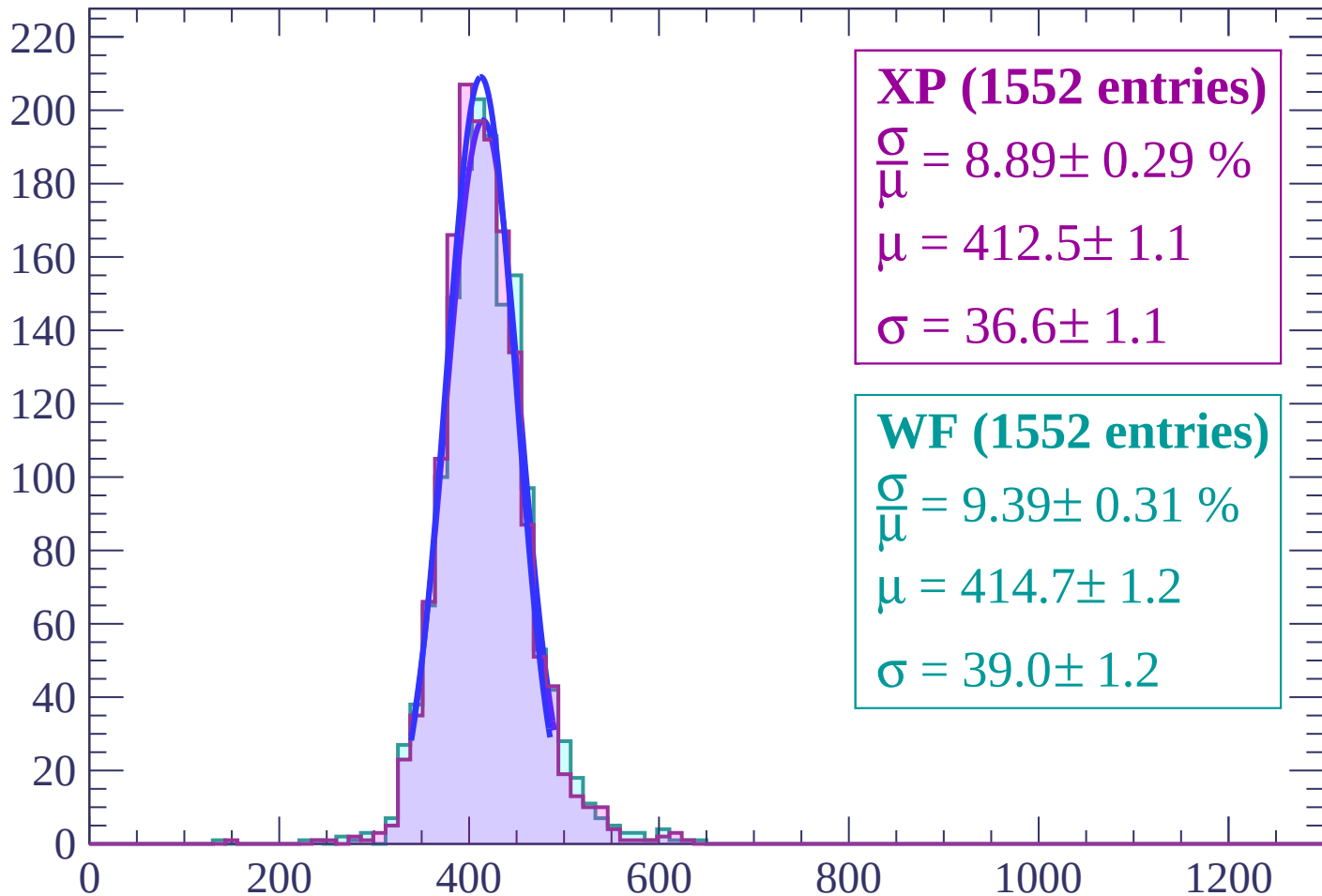
Energy loss | $26 < \theta < 28$

Count



Energy loss | $28 < \theta < 30$

Count



XP (1552 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.29 \%$

$\mu = 412.5 \pm 1.1$

$\sigma = 36.6 \pm 1.1$

WF (1552 entries)

$\frac{\sigma}{\mu} = 9.39 \pm 0.31 \%$

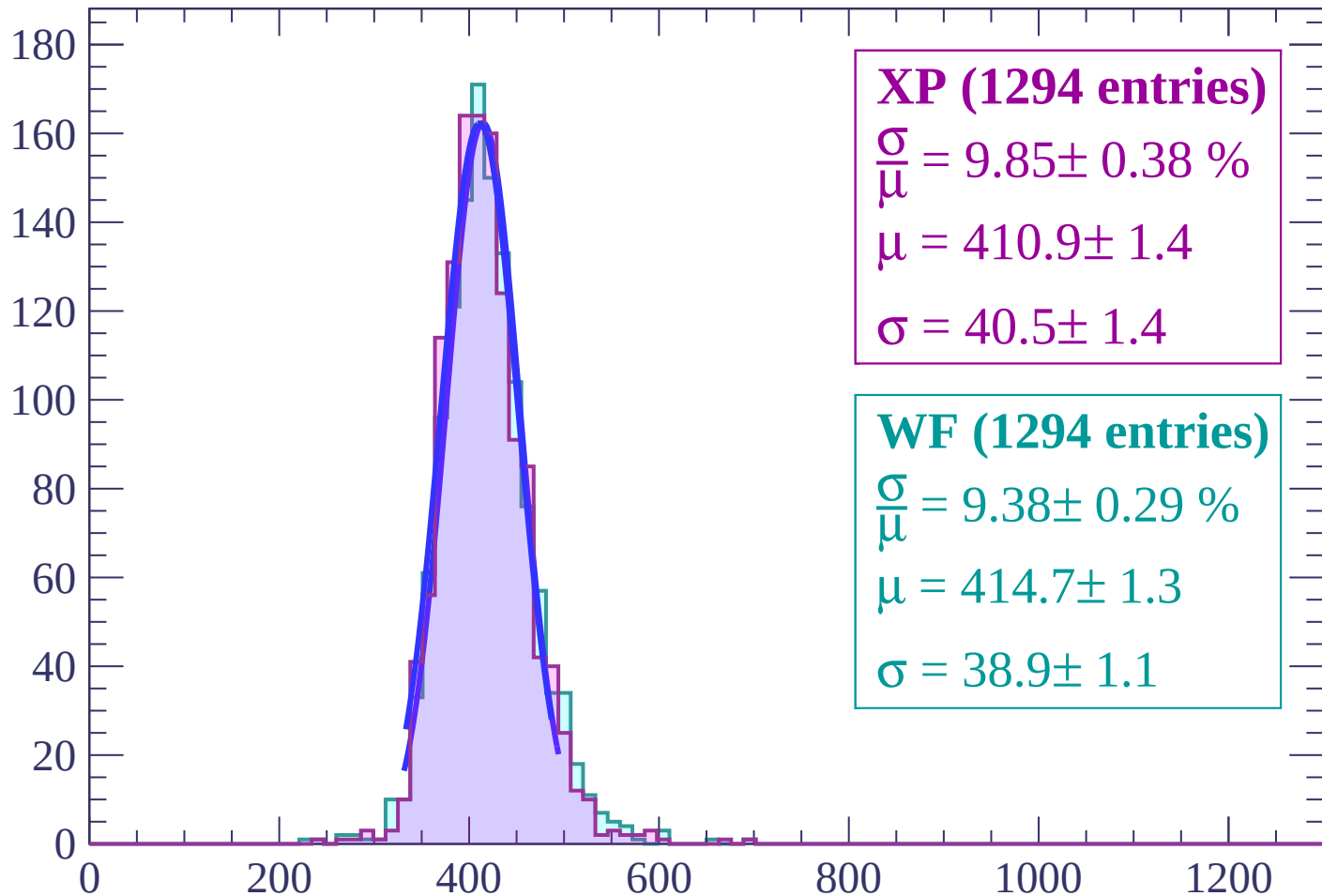
$\mu = 414.7 \pm 1.2$

$\sigma = 39.0 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $30 < \theta < 32$

Count



XP (1294 entries)

$\frac{\sigma}{\mu} = 9.85 \pm 0.38 \%$

$\mu = 410.9 \pm 1.4$

$\sigma = 40.5 \pm 1.4$

WF (1294 entries)

$\frac{\sigma}{\mu} = 9.38 \pm 0.29 \%$

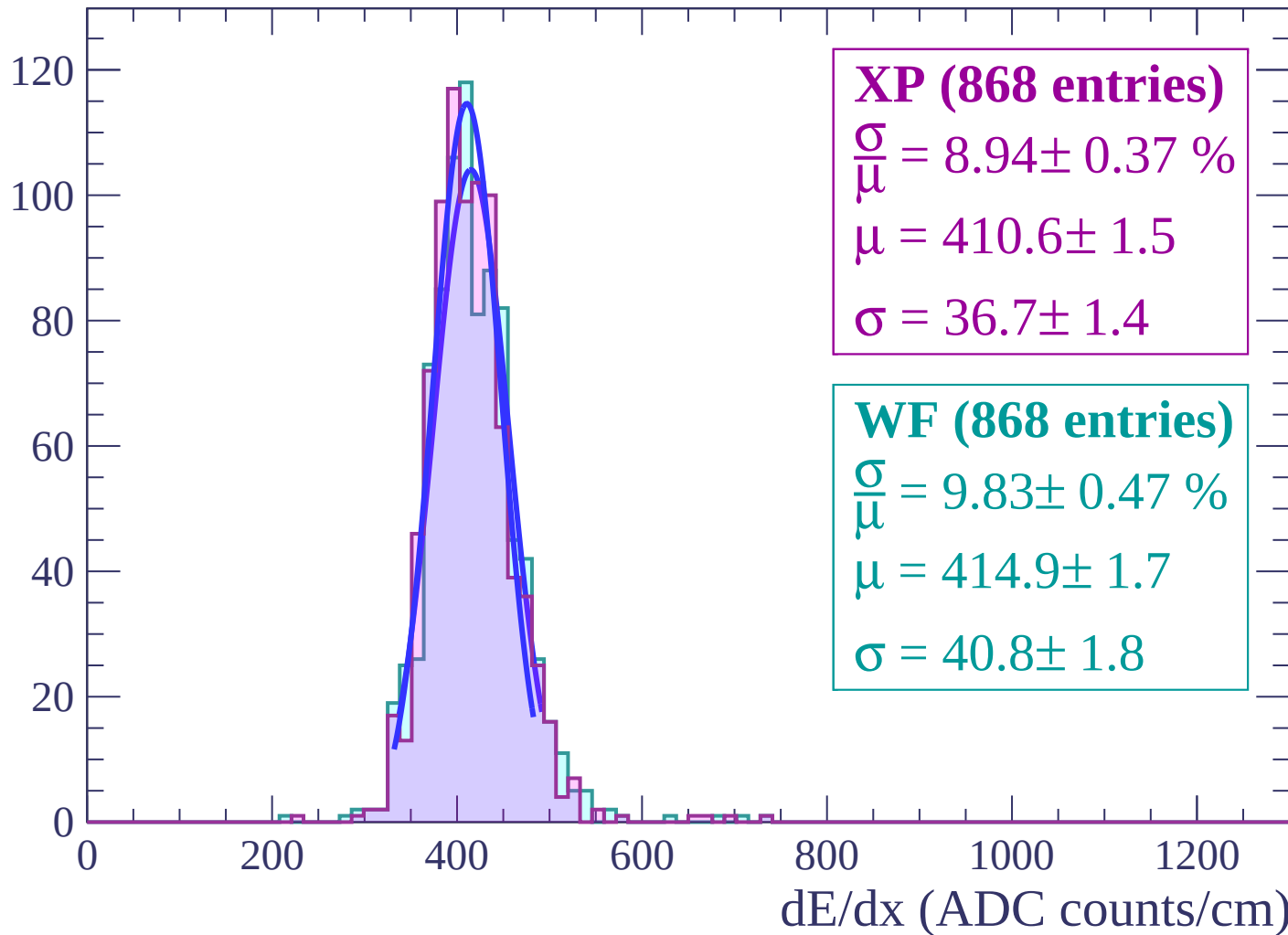
$\mu = 414.7 \pm 1.3$

$\sigma = 38.9 \pm 1.1$

dE/dx (ADC counts/cm)

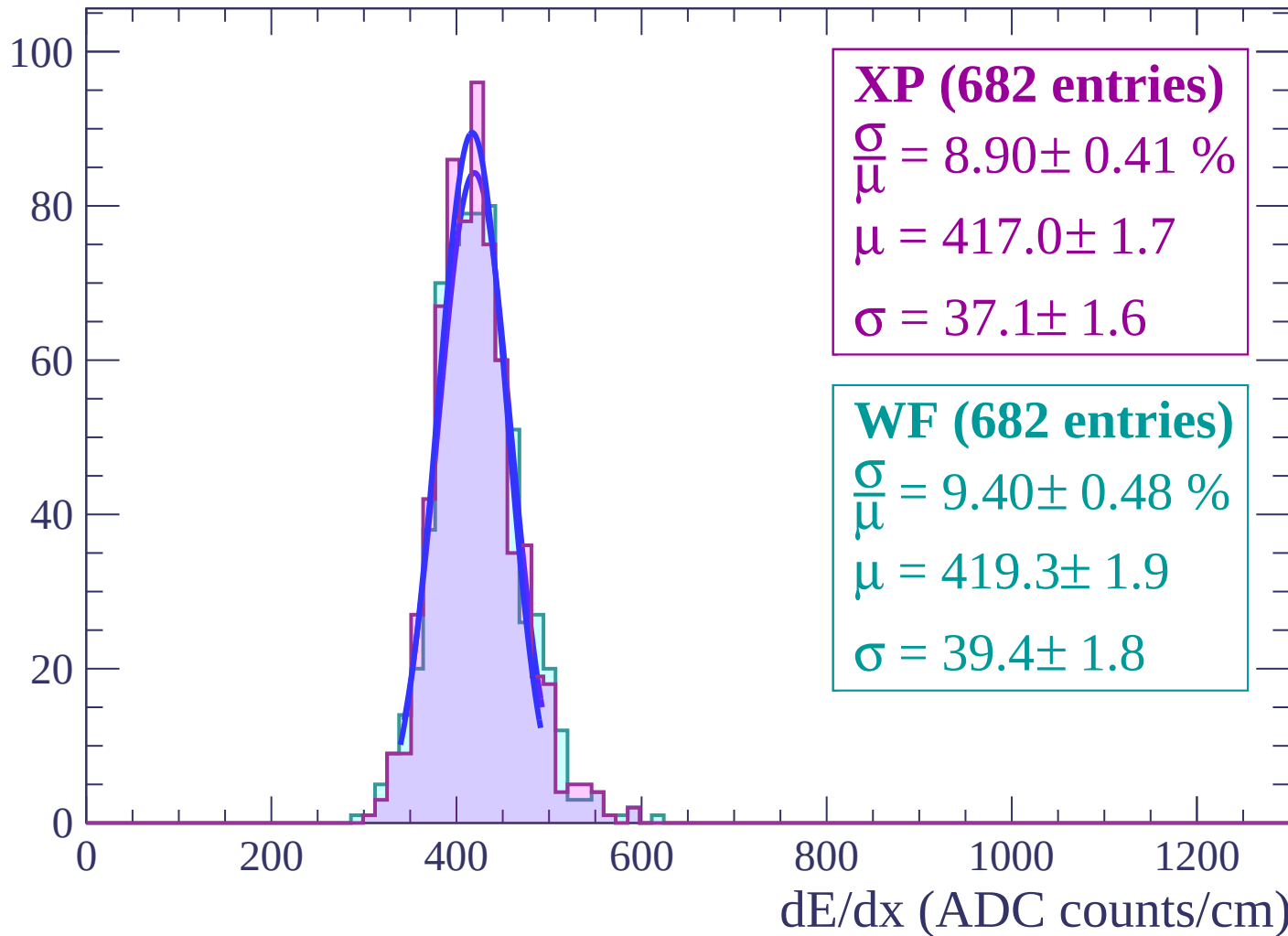
Energy loss | $32 < \theta < 34$

Count



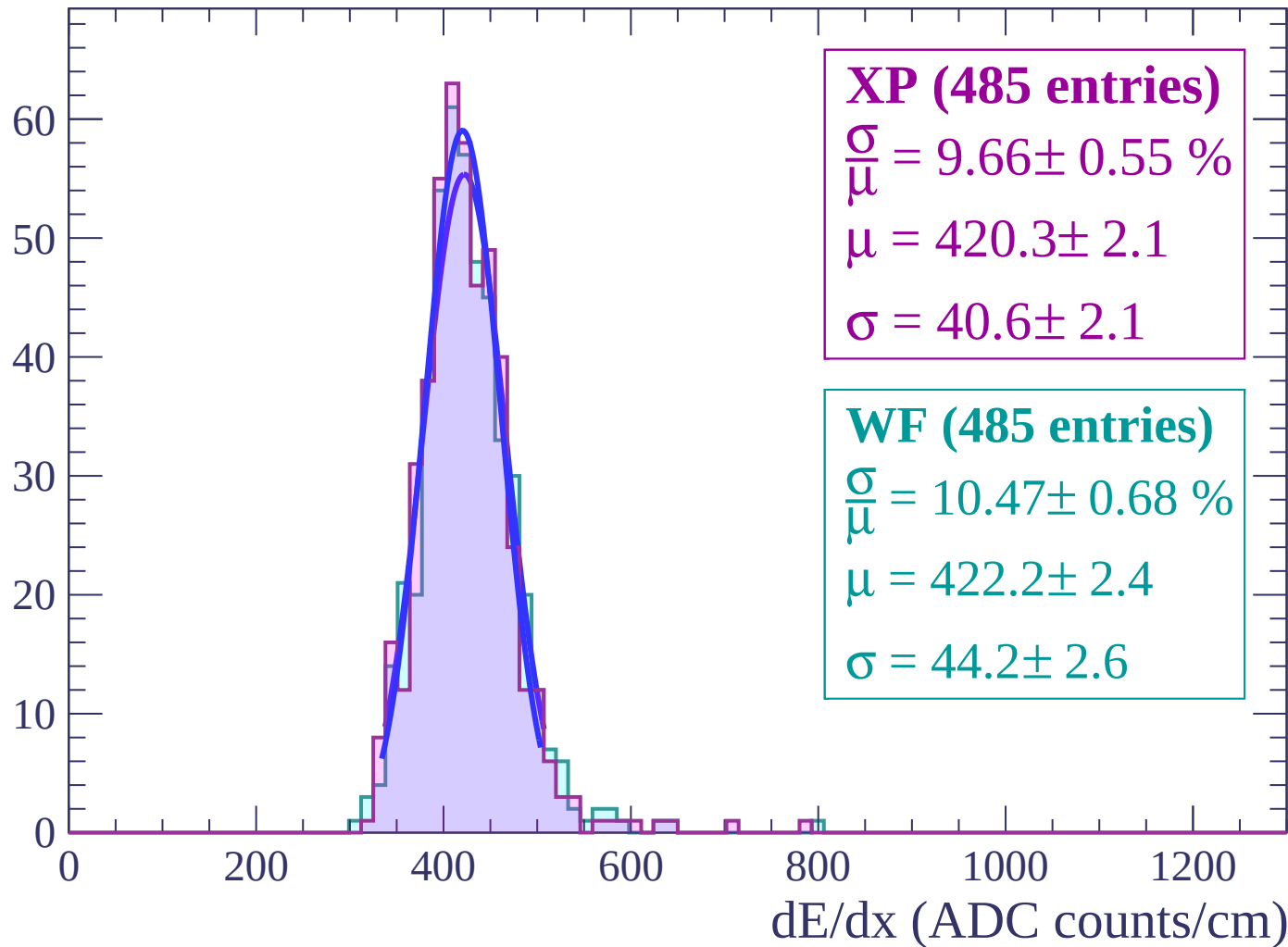
Energy loss | $34 < \theta < 36$

Count



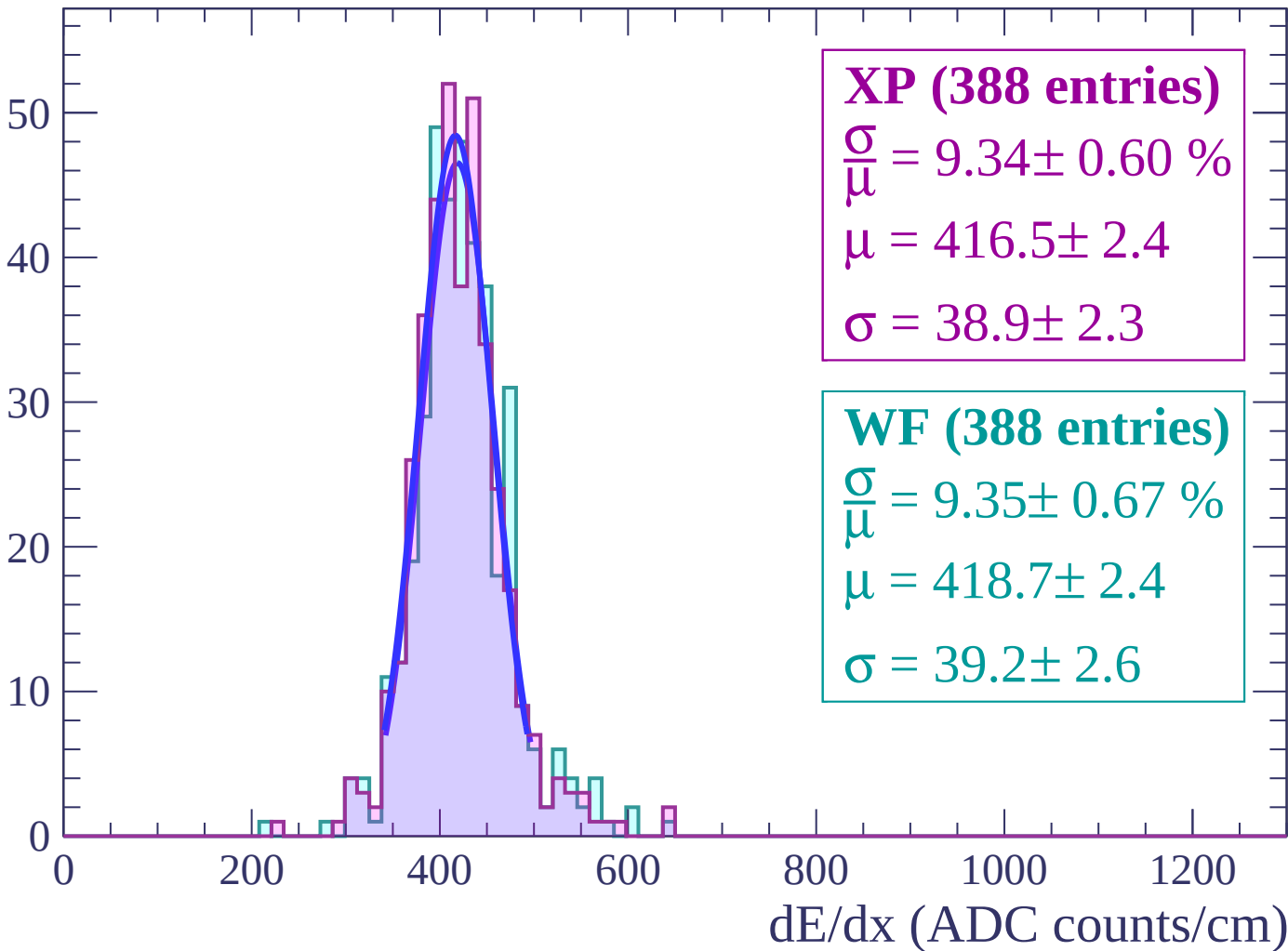
Energy loss | $36 < \theta < 38$

Count



Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

Count

XP (343 entries)

$$\frac{\sigma}{\mu} = 9.07 \pm 0.59 \%$$

$$\mu = 417.2 \pm 2.6$$

$$\sigma = 37.8 \pm 2.2$$

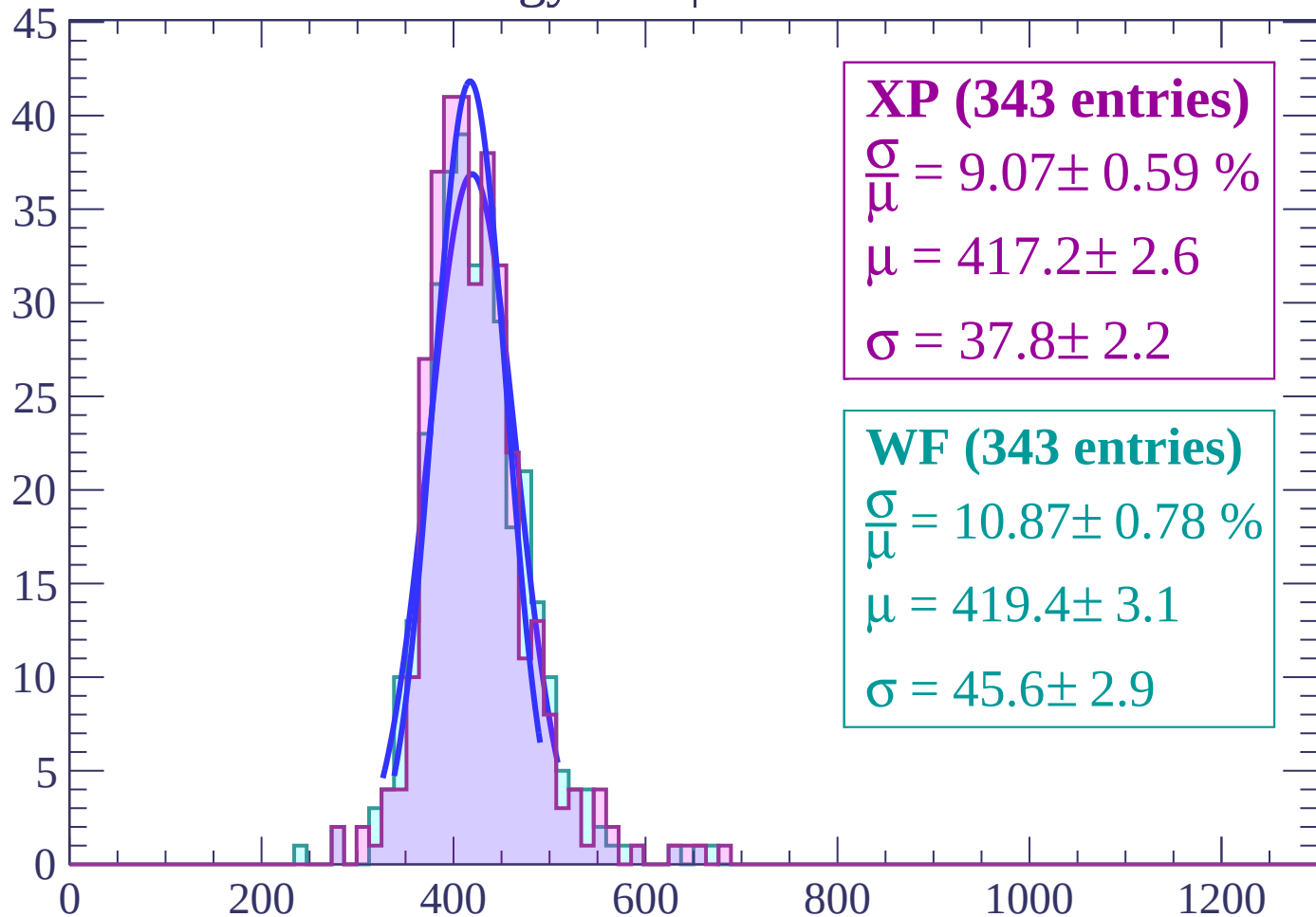
WF (343 entries)

$$\frac{\sigma}{\mu} = 10.87 \pm 0.78 \%$$

$$\mu = 419.4 \pm 3.1$$

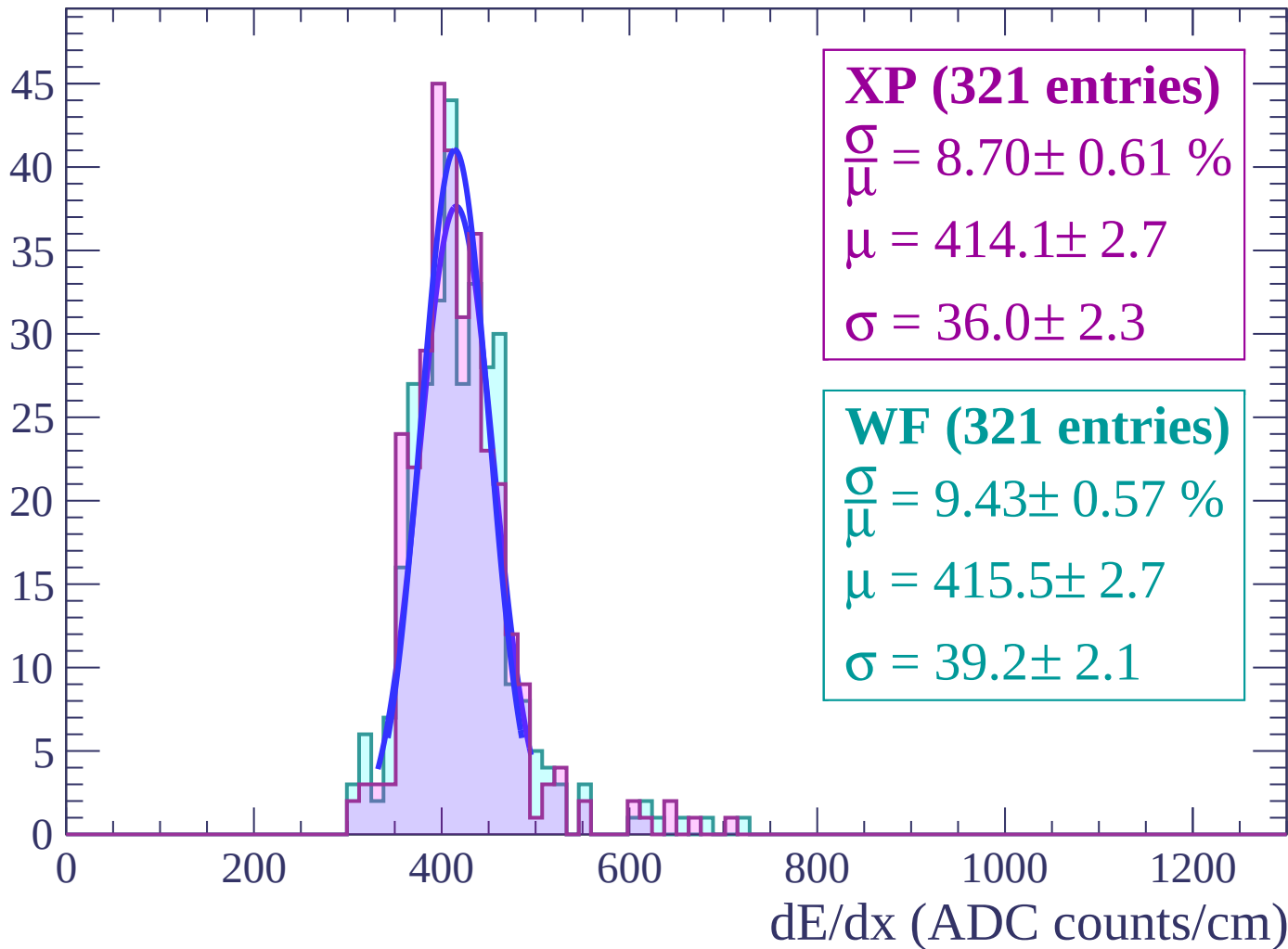
$$\sigma = 45.6 \pm 2.9$$

dE/dx (ADC counts/cm)



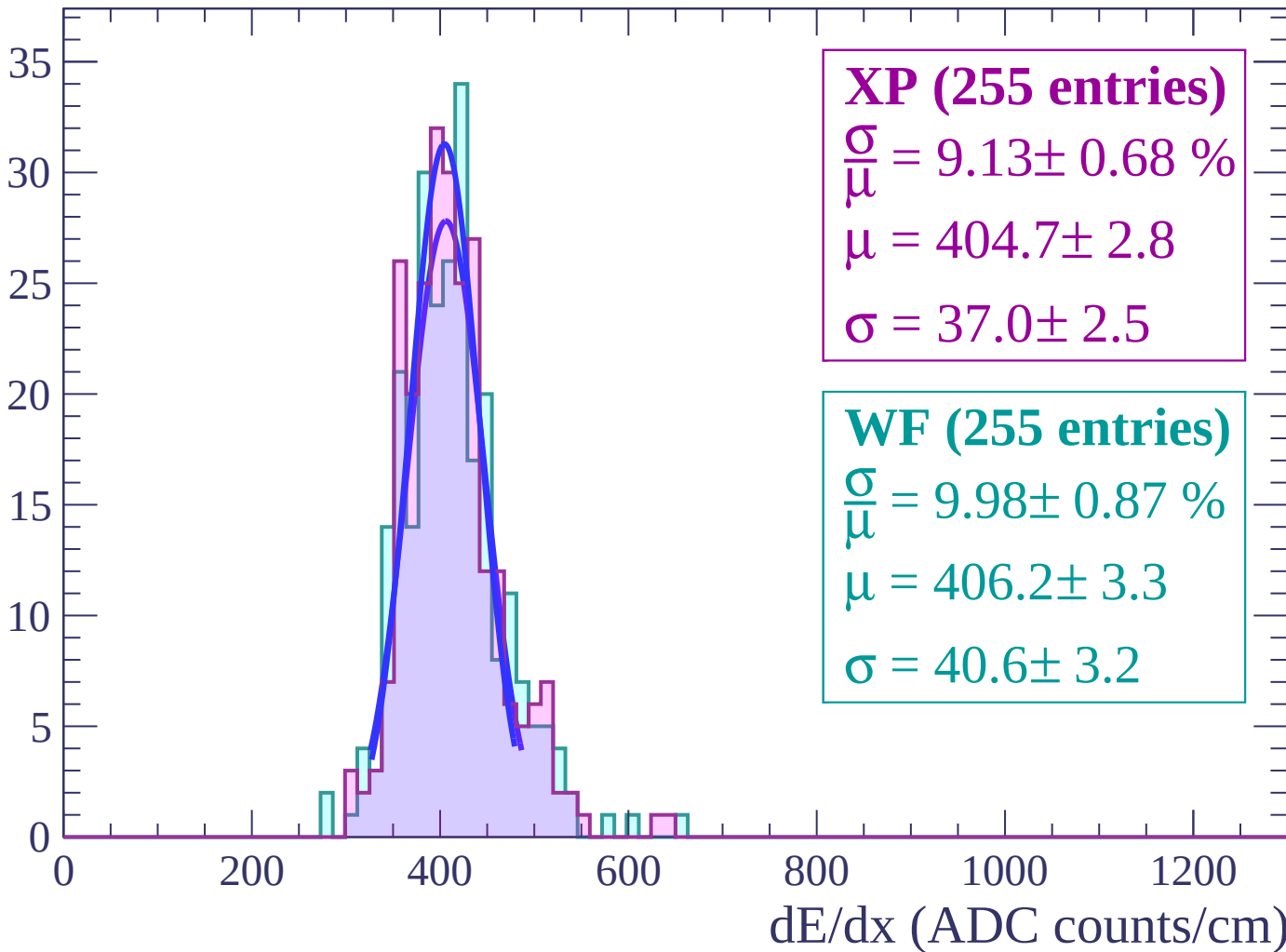
Energy loss | $42 < \theta < 44$

Count



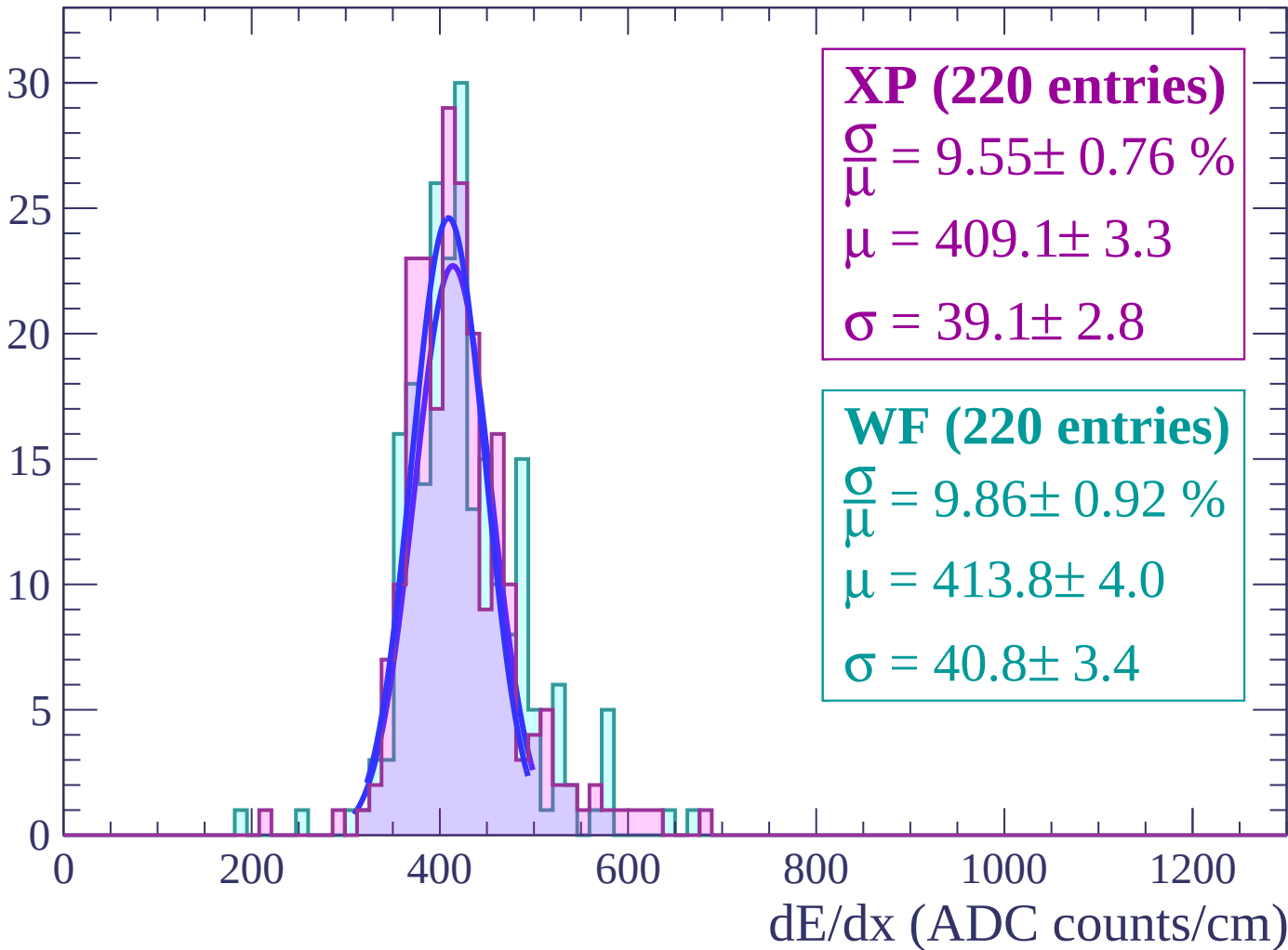
Energy loss | $44 < \theta < 46$

Count



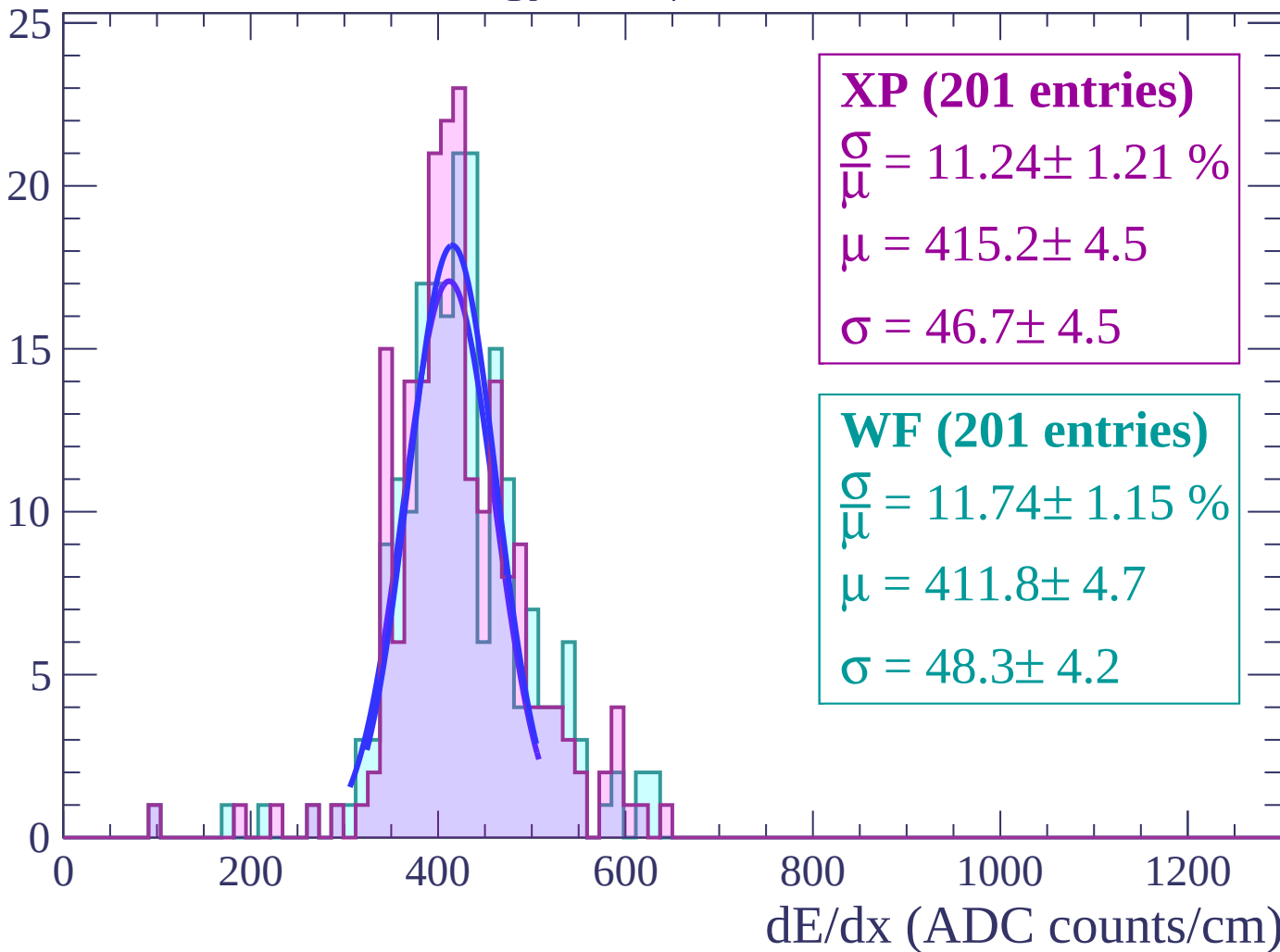
Energy loss | $46 < \theta < 48$

Count



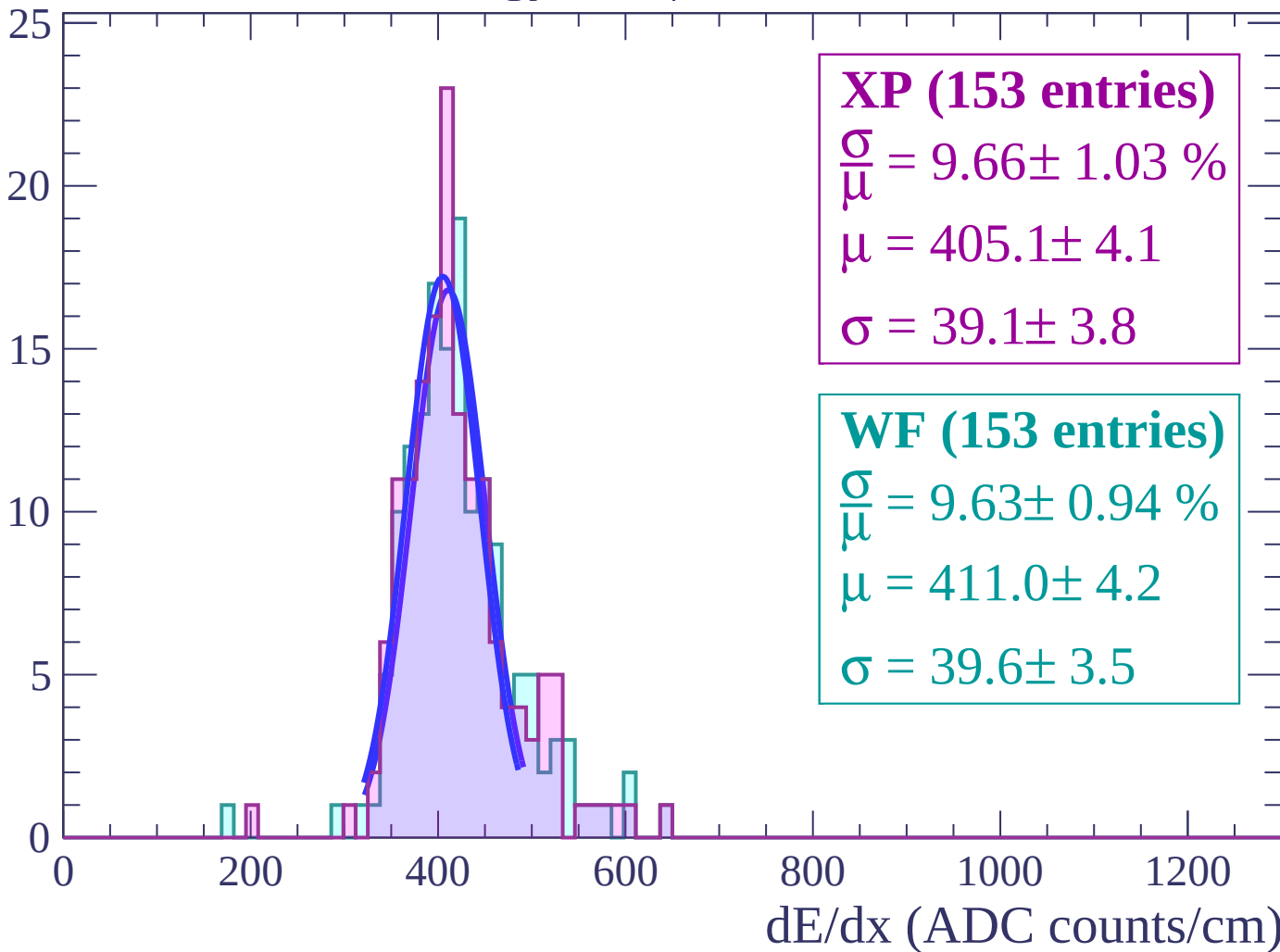
Energy loss | $48 < \theta < 50$

Count



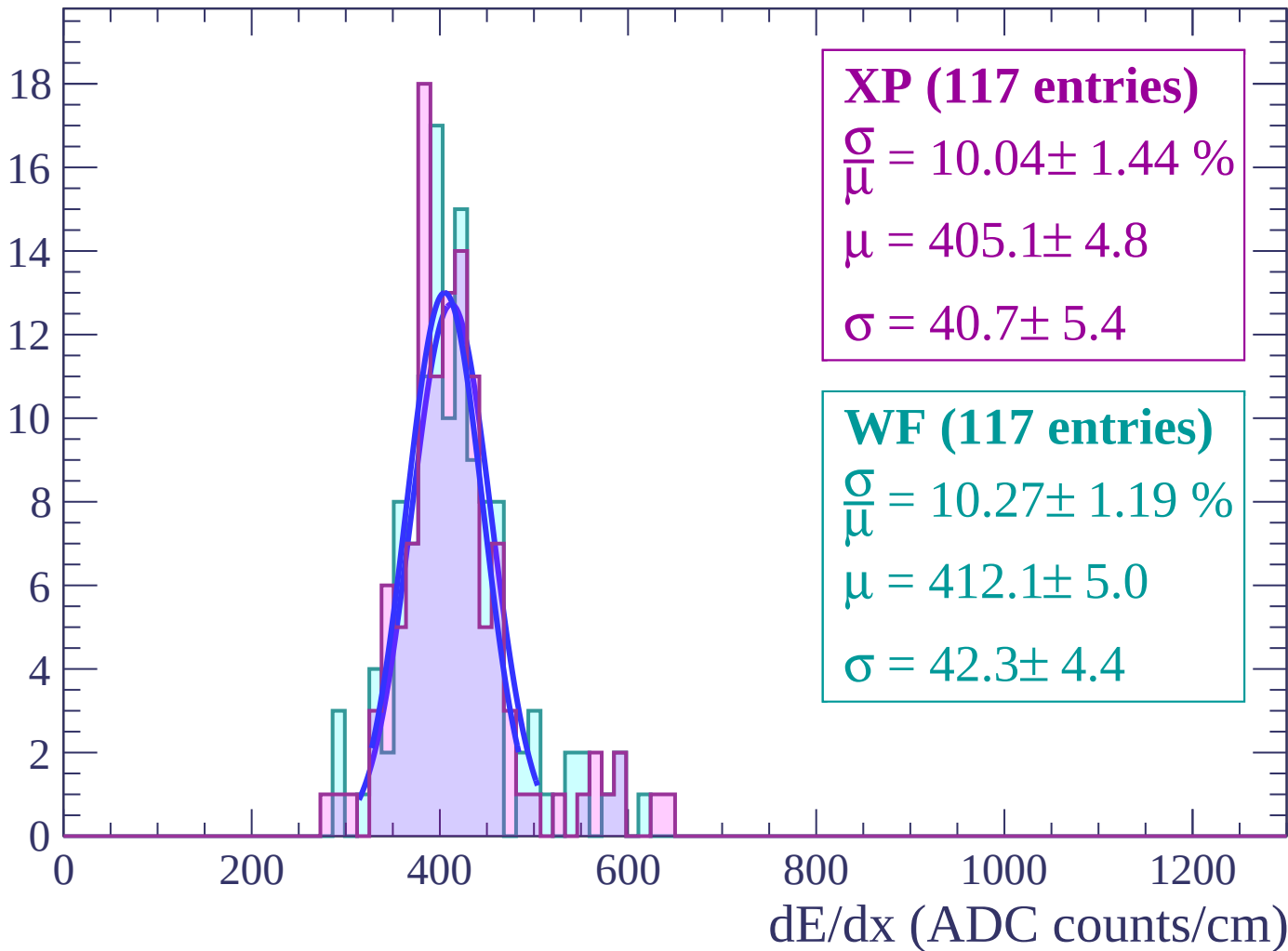
Energy loss | $50 < \theta < 52$

Count



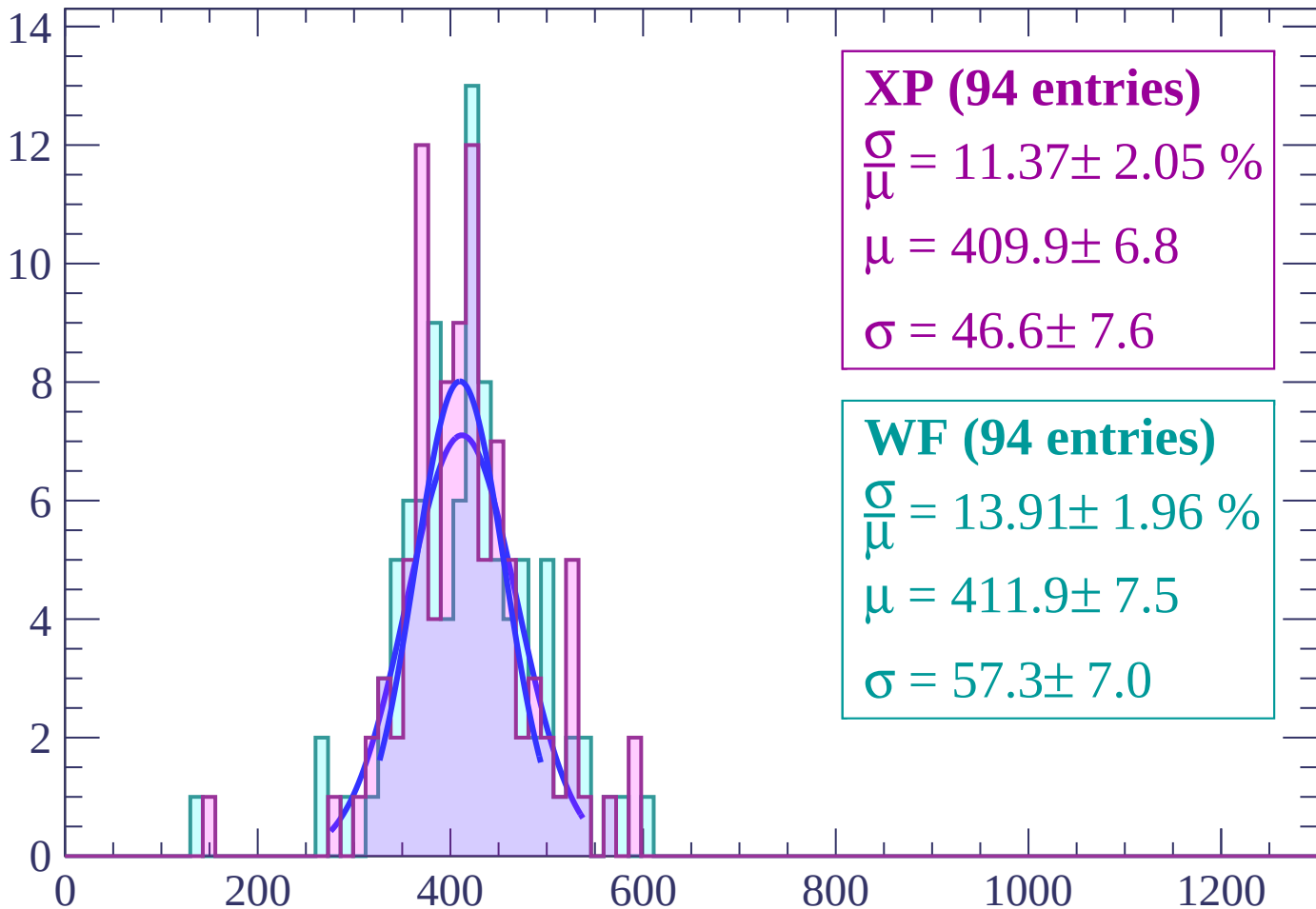
Energy loss | $52 < \theta < 54$

Count



Energy loss | $54 < \theta < 56$

Count



XP (94 entries)

$$\frac{\sigma}{\mu} = 11.37 \pm 2.05 \%$$

$$\mu = 409.9 \pm 6.8$$

$$\sigma = 46.6 \pm 7.6$$

WF (94 entries)

$$\frac{\sigma}{\mu} = 13.91 \pm 1.96 \%$$

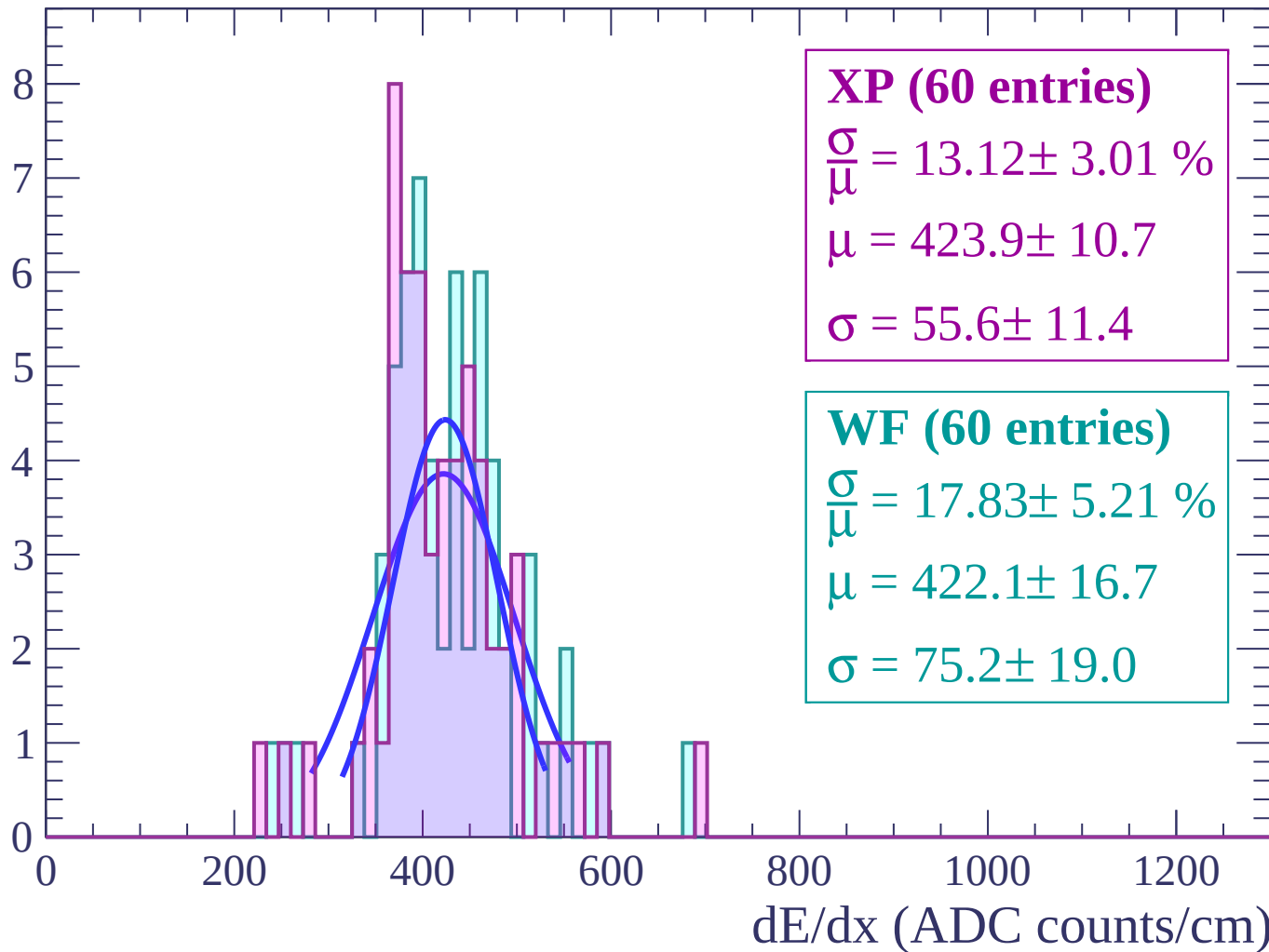
$$\mu = 411.9 \pm 7.5$$

$$\sigma = 57.3 \pm 7.0$$

dE/dx (ADC counts/cm)

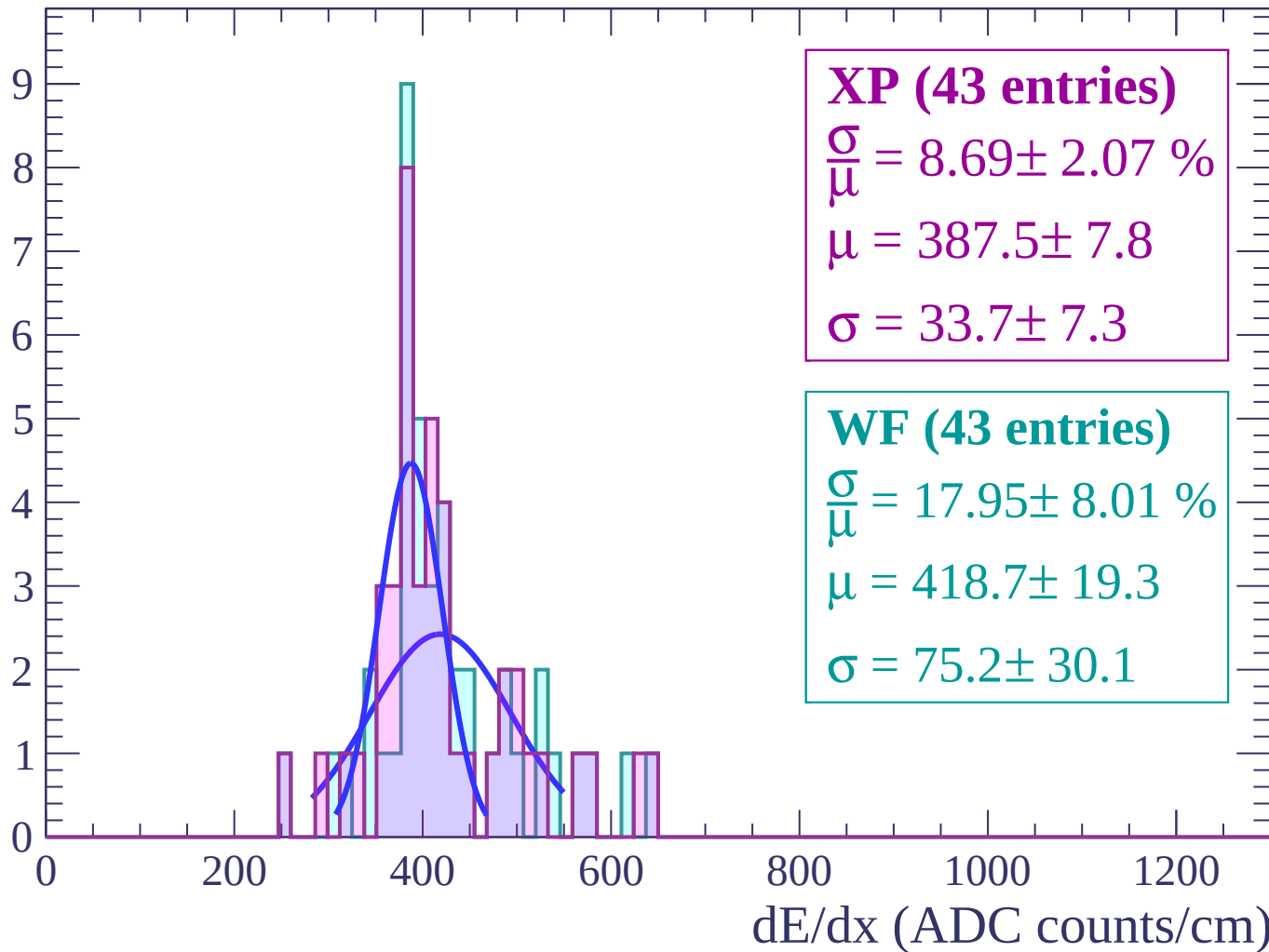
Energy loss | $56 < \theta < 58$

Count



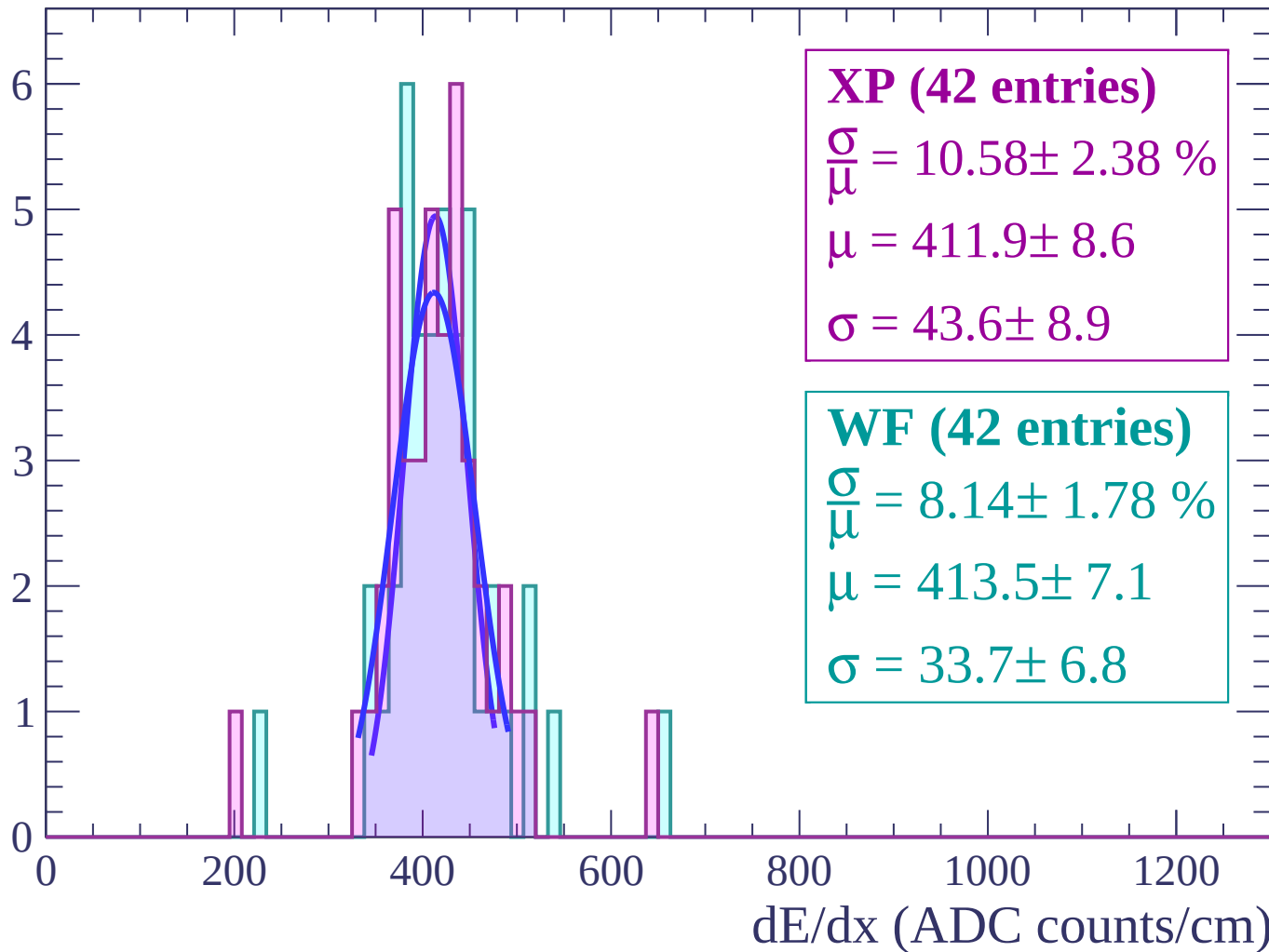
Energy loss | $58 < \theta < 60$

Count



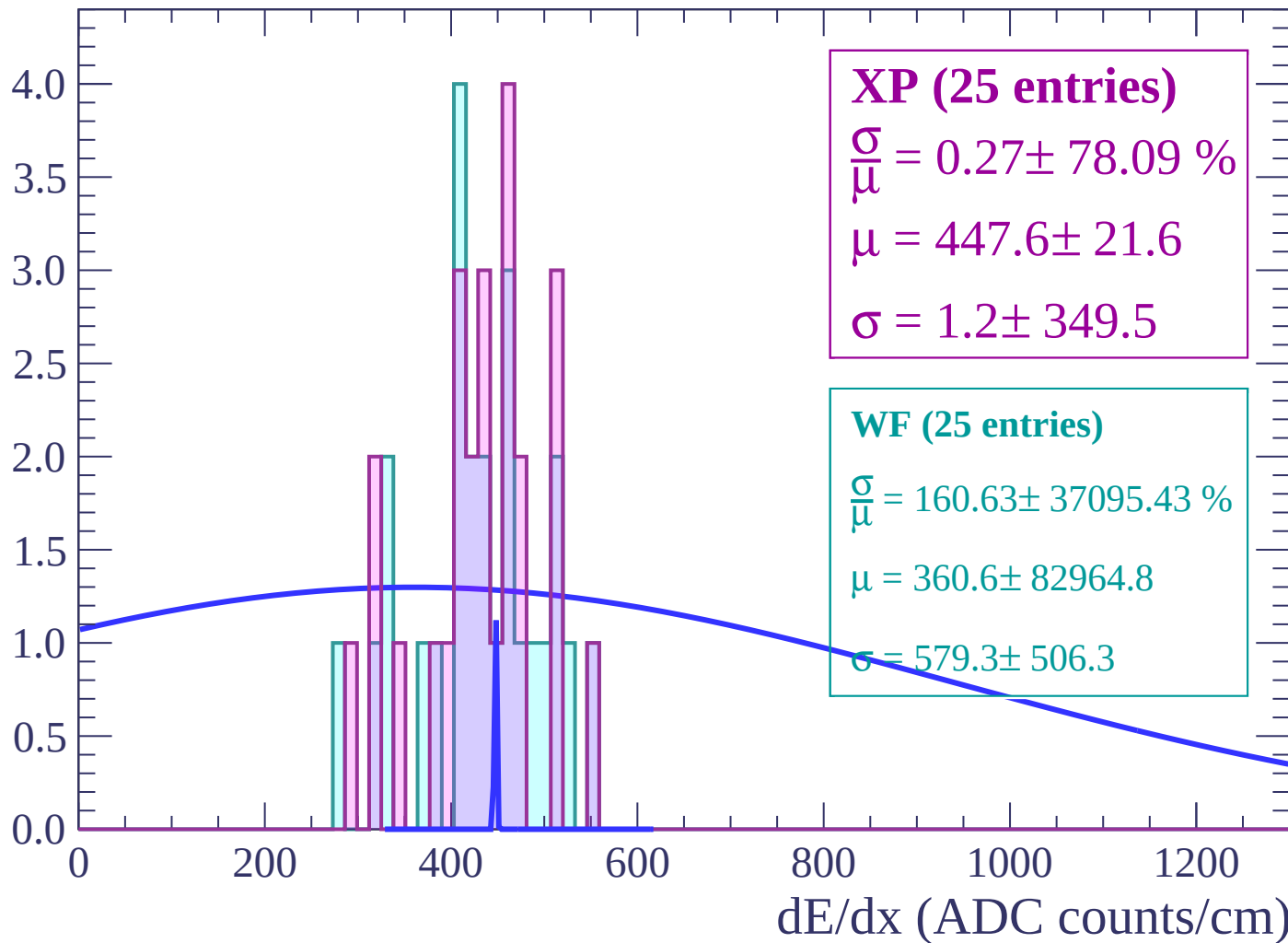
Energy loss | $60 < \theta < 62$

Count



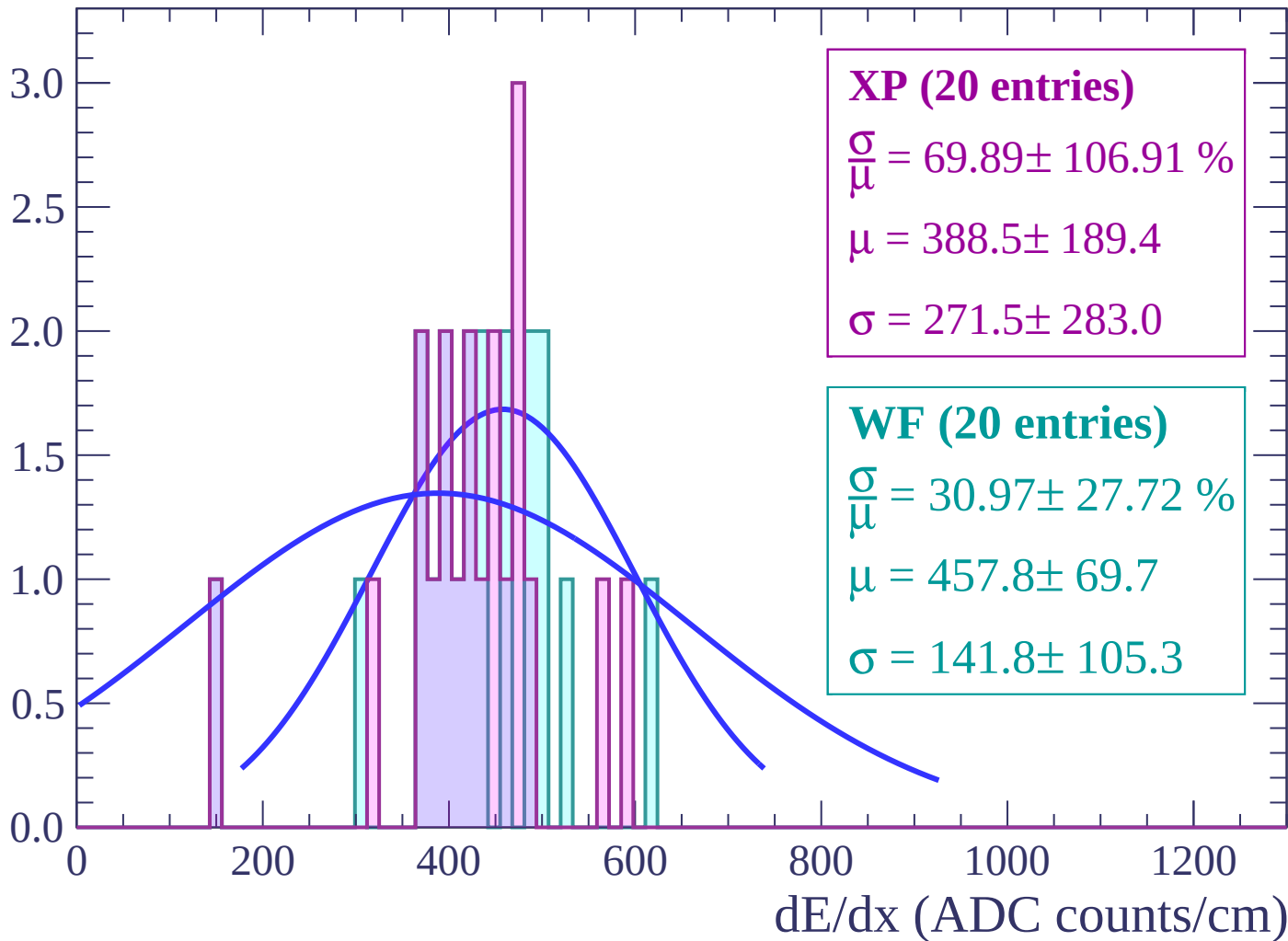
Energy loss | $62 < \theta < 64$

Count



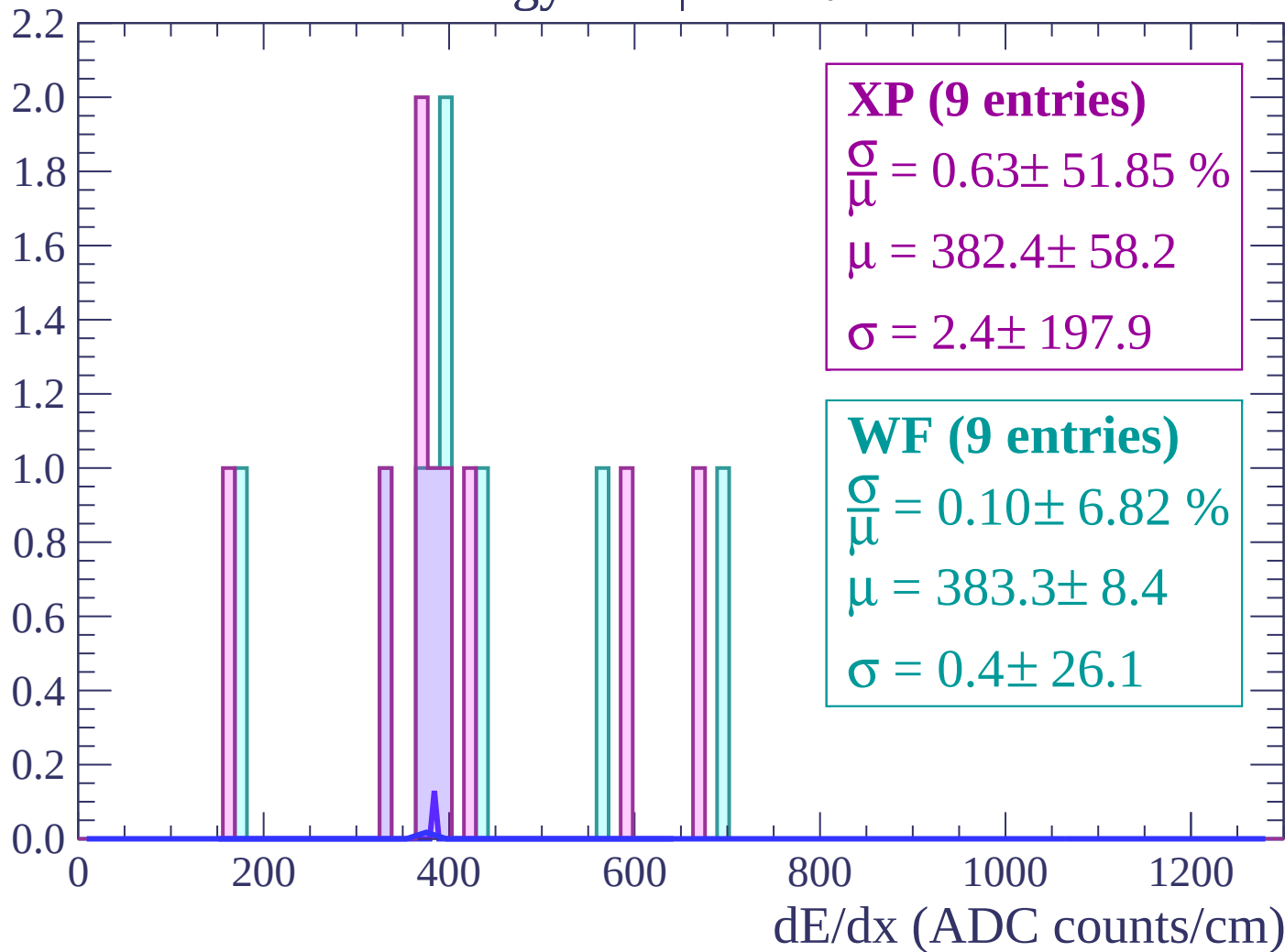
Energy loss | $64 < \theta < 66$

Count



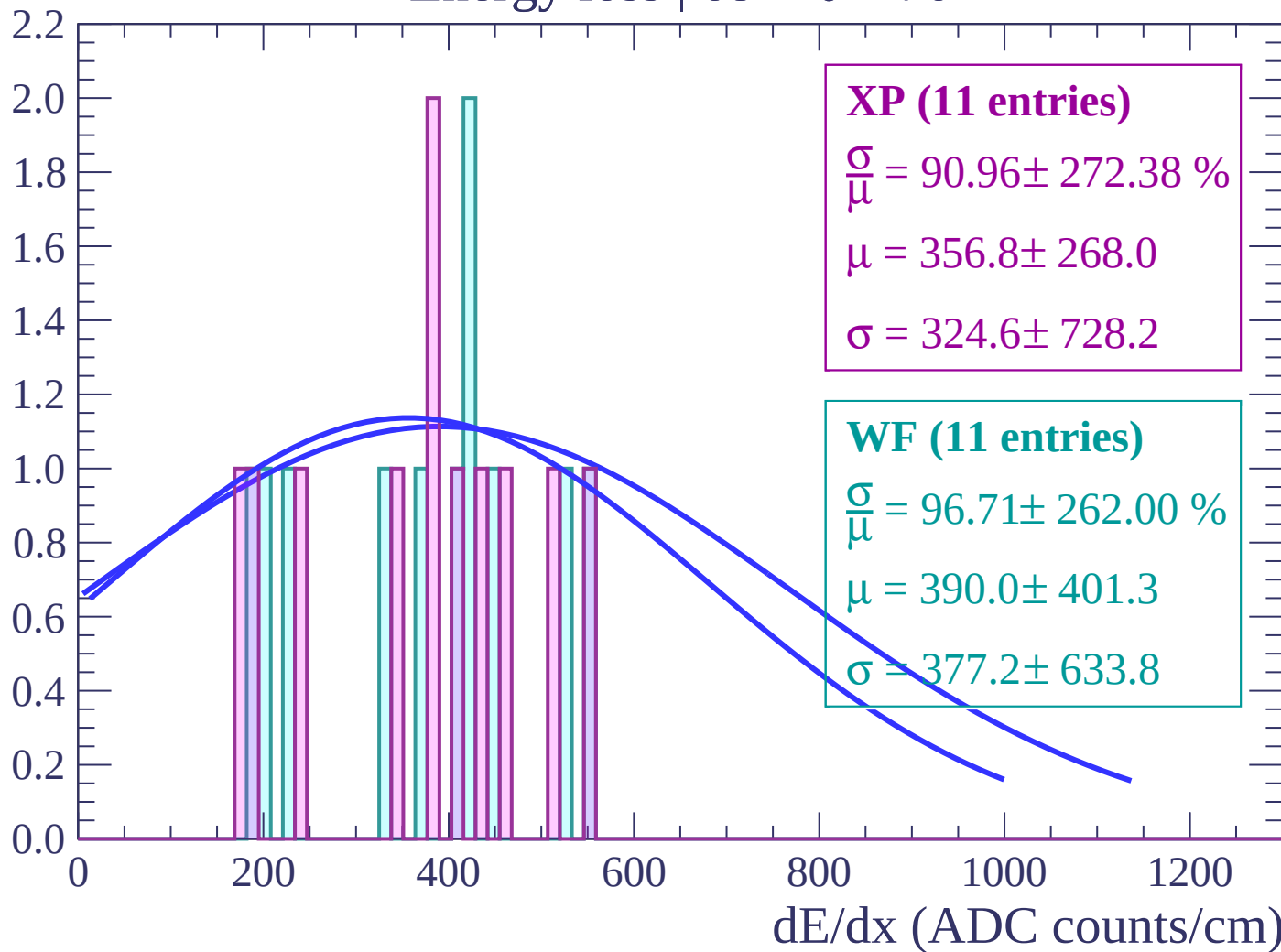
Energy loss | $66 < \theta < 68$

Count



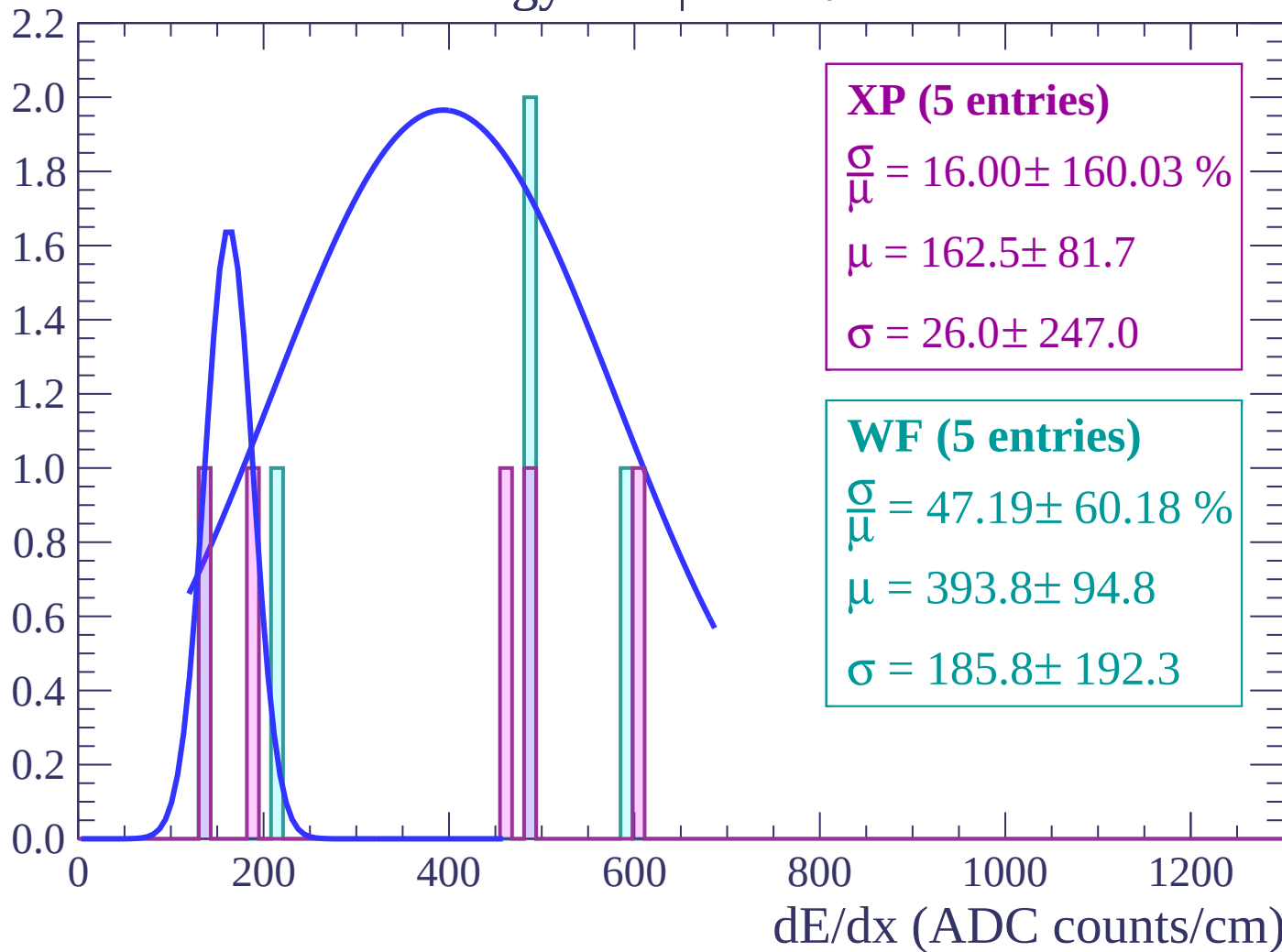
Energy loss | $68 < \theta < 70$

Count



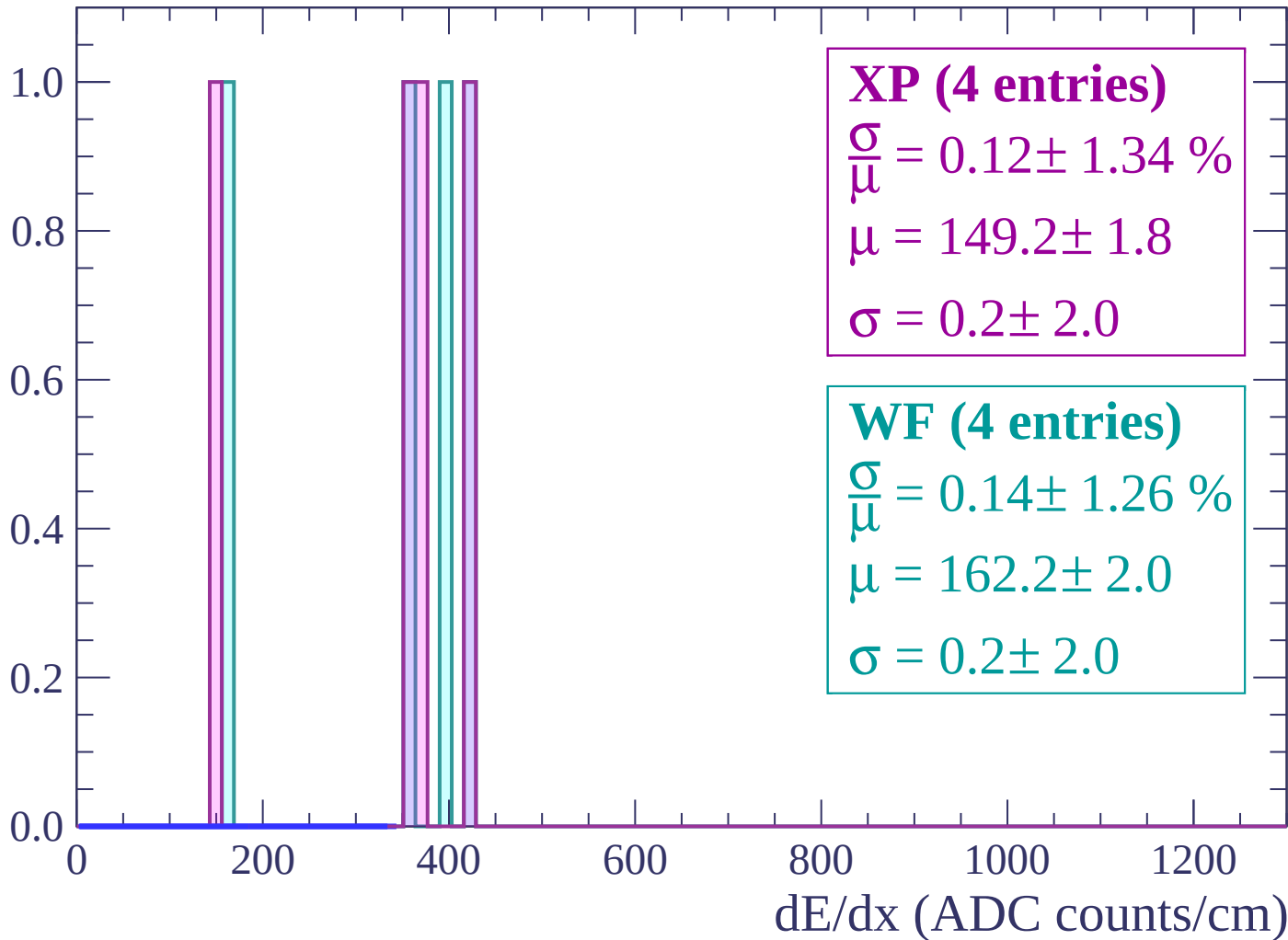
Energy loss | $70 < \theta < 72$

Count



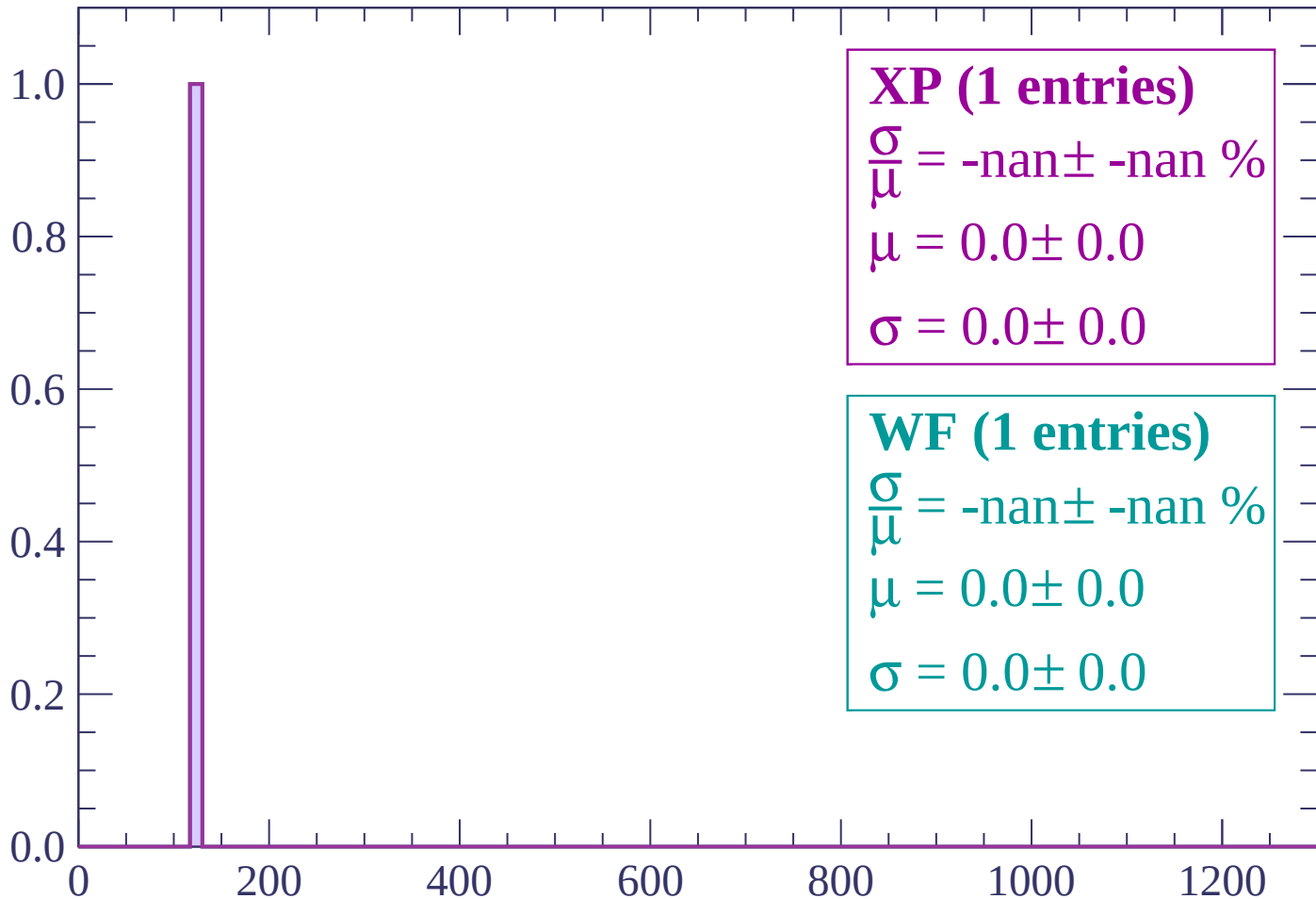
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $78 < \theta < 80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

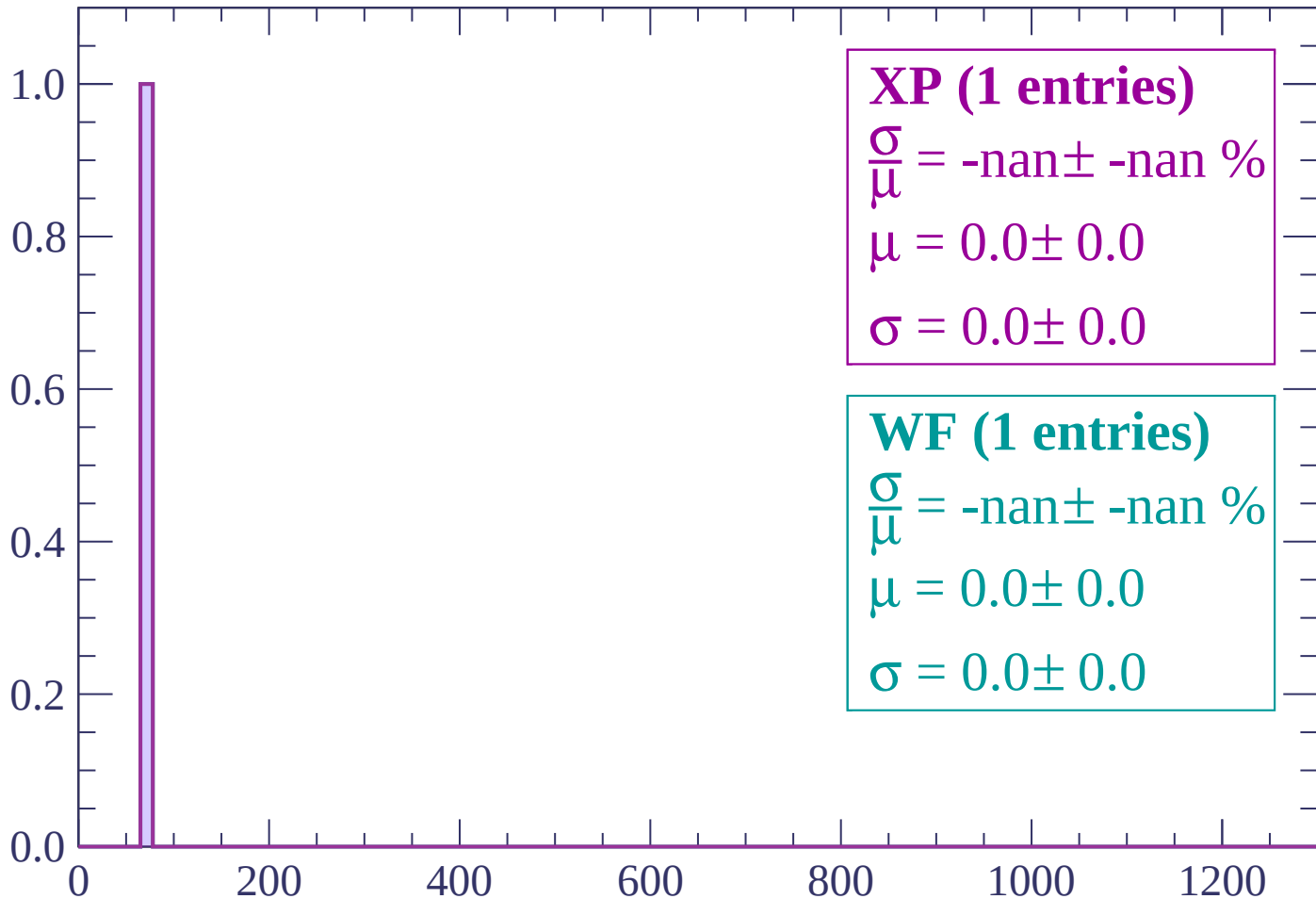
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

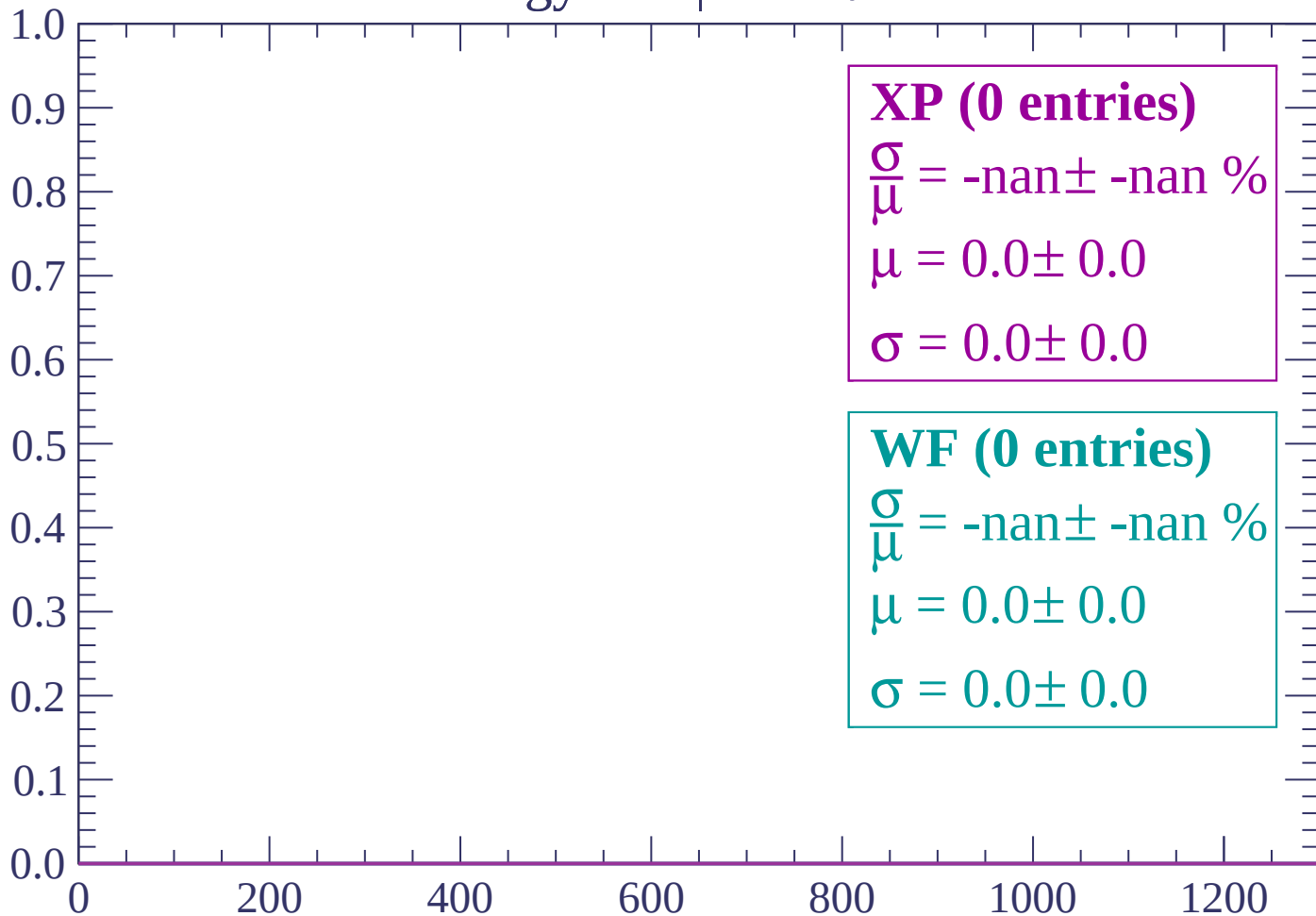
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

